CHMI ATFCM MAP REFERENCE GUIDE

Network Manager

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1 INTRODUCTION

The ATFCM **CHMI** (**Collaboration Human Machine Interface**) is a common interface providing access for all Air Navigation Units to dedicated Network Operations Services. It complements the Network Operations Portal, providing more resilience to the access to Network Operations systems.

CHMI is compliant to the most advanced standards and is based on a tool provided by the industry (Luciad™). An additional effort has been made to comply with the **ICAO** (**International Civil Aviation Organization**) standards.

Security is ensured by Strong User Authentication using the SecurId Software Token. The **'logon procedure'** allows the identification of the Air Navigation Unit and its User Profile. Depending on this profile, CHMI will configure itself to provide the required services.

Offering a great level of versatility, this mechanism allows the sharing of services between AOs, AROs, FMPs, TWRS, AMCs, Handling Companies, etc. based on the Operational Workflow.

This **CHMI MAP Reference Guide** is part of two documents providing Users with detailed description of all ATFCM CHMI functionalities.

2 GENERAL MAP HUMAN MACHINE INTERFACE

2.1 The Map Window

2.1.1 Opening the map

(1) When you open the CHMI (ENV, FLIGHT or ATFCM) the toolbar contains the map button, if there is a map. If there is no map the button is disabled.



Figure 2-1: Map button



Figure 2-2: Disabled Map button

(2) It takes some time before the map is loaded. At the bottom left of the CHMI frame window there is also a map image. While the map is loading the button is animated with blocks. The blocks disappear when the map is loaded.



Figure 2-3: Animated button while loading the map



Figure 2-4: Button when the map is loaded

(3) If you click on the animated button at the bottom of the CHMI frame a window is opened that displays information about the data that is currently retrieved. If you go with the mouse over the text of the window, a list with the data that is retrieved until now appears in a tooltip. When the loading is finished "Done!" appears in the window. If you go with the mouse over the text of the window, you see how many types are loaded in the map. For ATFCM and IFPS there should be 7 types.

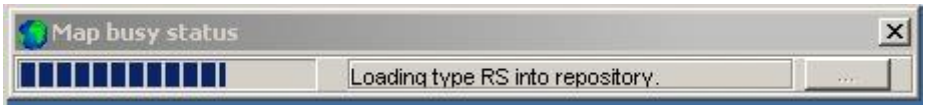


Figure 2-5: Map busy status - Information about the data that is retrieved

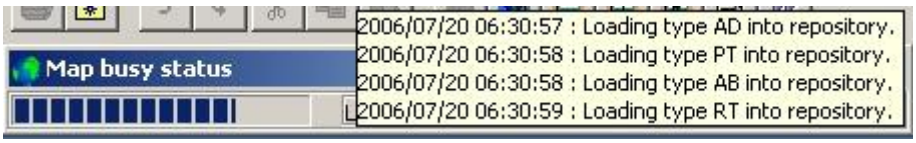


Figure 2-6: Map busy status - Tooltip with a list of all the data that is retrieved



Figure 2-7: Map busy status - Information when the loading is finished

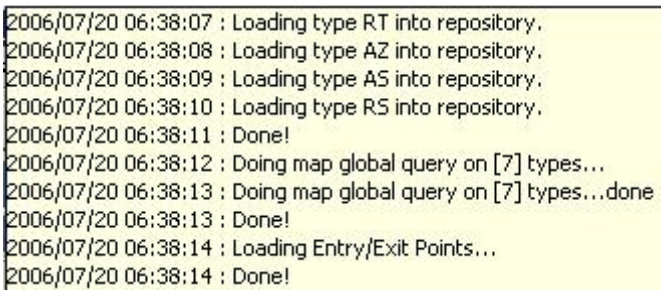


Figure 2-8: Map busy status - Tooltip with number of types loaded for ATFCM or IFPS

(4) When you click again on the button at the bottom of the CHMI frame, the 'Map busy status window' disappears.

2.1.2 Externalise and internalise the map

(1) The map can be an internal frame of the whole CHMI application or it can be an external frame.

(2) Make the frame external by clicking the externalise button:



Figure 2-9: Externalise button

(3) When the map is externalised the buttons of the map are at the top of the map window.

(4) Make the frame internal by clicking the internalise button:



Figure 2-10: Internalise button

(5) When the map is internalised the buttons of the map are at the top of the CHMI window.

(6) It is possible to work with 2 screens. Externalise the map. Drag the map to the other screen. Maximise the map.

2.1.3 Opening the map in the same state as it was closed

(1) The map should open in the same state as it was closed. This means:

a) If it was closed as external frame, it opens as external frame.

b) If it was closed as internal frame, it opens as internal frame.

c) The position is the same as it was when it was closed.

d) The size of the map is the same as it was when it was closed.

(2) There can be exceptions. If the map would not be visible anymore if it would open in the same state, then the position and the size of the map are adapted that it is completely visible.

2.2 General Map interaction

2.2.1 Map controllers

(1) There are 5 different controllers for the map. Only one controller can be active on the map. The controller decides what the mouse interactions with the map mean.

(2) A controller can be activated

a) by clicking the menu item for the controller

b) by clicking the button in the toolbar

c) by typing a shortcut

(3) You can see the popup menu when you right click on the map. Select the menu "Mode".

(4) The Cursor is changing depending on the activated controller.

(5) The table describes the different options for activating the controllers and the cursor associated with the active controller.

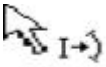
	Menu option	Tooltip	Toolbar	Shortcut Key	Cursor	Remark
A	General	General mouse behaviour		G		
B	Pan	Panning mouse behaviour		P		
D	Distance Calculation	Distance Calculation Mouse behaviour (click to start/ double click to stop)		R		
E	Centre			C		Not in toolbar
F	Azimuth	Azimuth range mouse behaviour		A		

Table 2-1: Map controllers

2.2.1.1 General Controller

2.2.1.1.1 Description

(1) The general controller:

a) Button



Figure 2-11: General Controller - Button

b) Cursor shape



Figure 2-12: General Controller - Cursor shape

c) Keyboard shortcut: **G**

(2) This mode offers a lot of possibilities.

2.2.1.1.2 Left mouse button: Zoom in / zoom out.

(1) Simply clicking will not do anything good. You must press the button, keep it pressed and drag your cursor across the map to draw a white rectangle. Then only release the button.

(2) You will notice that the mouse cursor takes briefly the following aspect:



Figure 2-13: Mouse Cursor - Zoom Action

(3) This means that zoom actions are being performed.

a) Zoom in: draw your rectangle DOWN (left or right does not matter). The content of the rectangle will fit the whole map when button is released.

b) Zoom out: draw your rectangle UP (left or right does not matter). The content of the whole map box will fit the rectangle.

2.2.1.1.3 Middle click: re-centre

(1) The wheel is used as a button for middle clicks. Middle clicking will set the position of the mouse cursor to the centre of the map. Zoom is not affected.

(2) Briefly, a pan symbol will be shown:



Figure 2-14: Mouse Cursor - Pan symbol

2.2.1.1.4 CTR+ Middle click: Azimuth filter

(1) When you press the CTRL key and the middle mouse button, you can set the azimuth filtering.

(2) See azimuth controller.

2.2.1.1.5 Middle scroll: Zoom in / zoom out.

(1) You can zoom in and out by scrolling.

a) Zoom in: scroll up.

b) Zoom out: scroll down.

2.2.1.2 Pan controller

(1) The pan controller:

a) Button



Figure 2-15: Pan Controller - Button

b) Cursor shape



Figure 2-16: Pan Controller - Cursor shape

c) Keyboard shortcut: **P**

(2) With this controller you can shift the map. Press the left mouse button, keep it pressed and drag your cursor across the map. Then release. The map will shift along the segment you drew.

2.2.1.3 Distance Calculation controller

(1) The distance calculation controller:

a) Button



Figure 2-17: Distance Calculation Controller - Button

b) Cursor shape



Figure 2-18: Distance Calculation Controller - Cursor shape

c) Keyboard shortcut: **R**

(2) Select the distance calculation controller. The 'Distance Calculation' dialog is displayed:

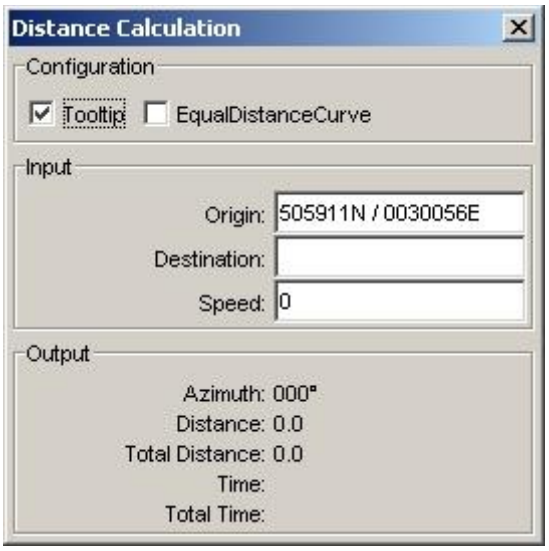


Figure 2-19: Distance Calculation dialog

(3) Move with the mouse over the map. In the origin field you see the coordinates of the current mouse position.

(4) When you come in the neighbourhood of a point or an aerodrome the name is displayed in the origin field.

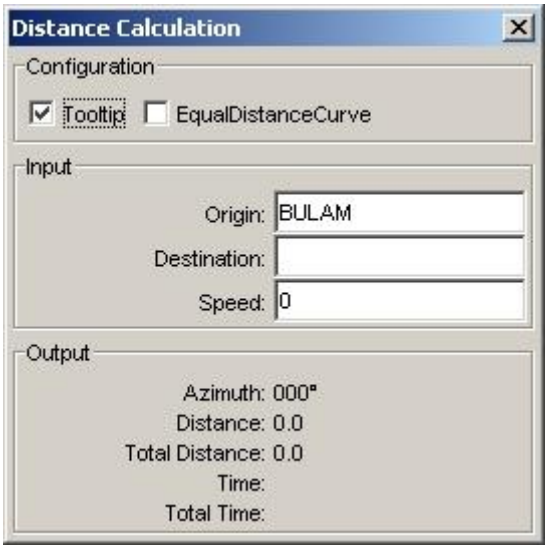


Figure 2-20: Distance Calculation dialog - Origin selected

(5) When you click this point is selected as origin for the distance calculation.

(6) Move the cursor. When you click again his point is added as step.

(7) When you double click this point is taken as destination point.

(8) When there is an origin entered, you see in the dialog the following output:

a) Azimuth: the azimuth of the last segment.

b) Distance: the distance of the last segment.

c) Total distance: the sum of the distances of all the segments.

(9) When there is no destination, these values change while moving the cursor over the map.

(10) When the destination is selected the values for the last origin and destination stay displayed.

(11) In the speed field you can enter the speed. This should be a double.

(12) When there is an origin entered, you see in the dialog also the following output:

a) Time: the time for the last segment with the given speed.

b) Total time: the time for the all the segments with the given speed.

(13) If the speed is valid no values are displayed for time and total time in the dialog.

(14) When the Tooltip property is checked in the dialog, there is also a tooltip shown after the first click or double click with the azimuth, distance and total distance.

(15) When the tooltip property is not checked, there is no tooltip.

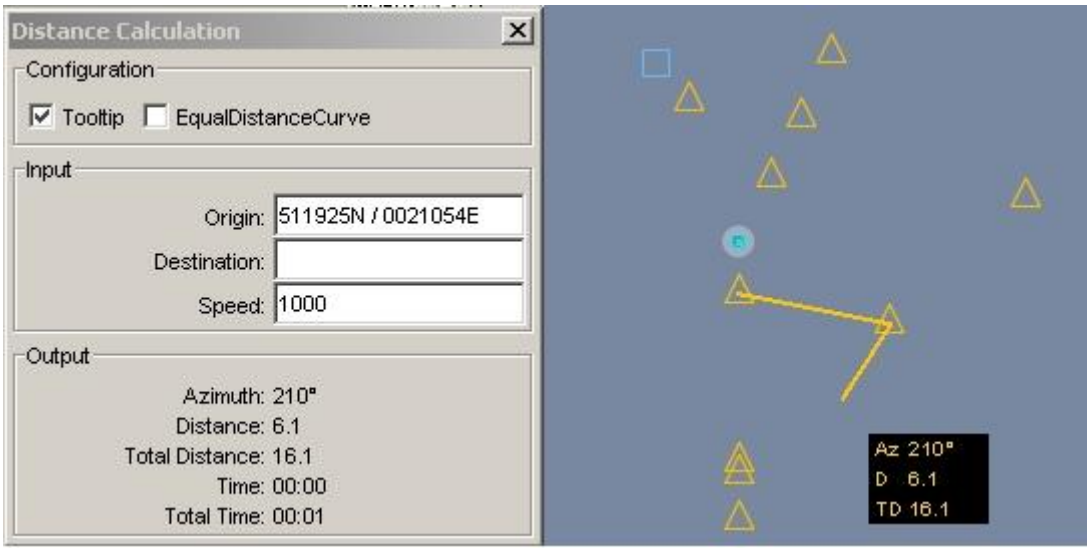


Figure 2-21: Distance Calculation dialog - Tooltip

(16) When the EqualDistanceCurve property is checked in the dialog, there is also a circle shown around the last clicked point and with the cursor on the border.

(17) When the EqualDistanceCurve property is not checked, there is no circle.

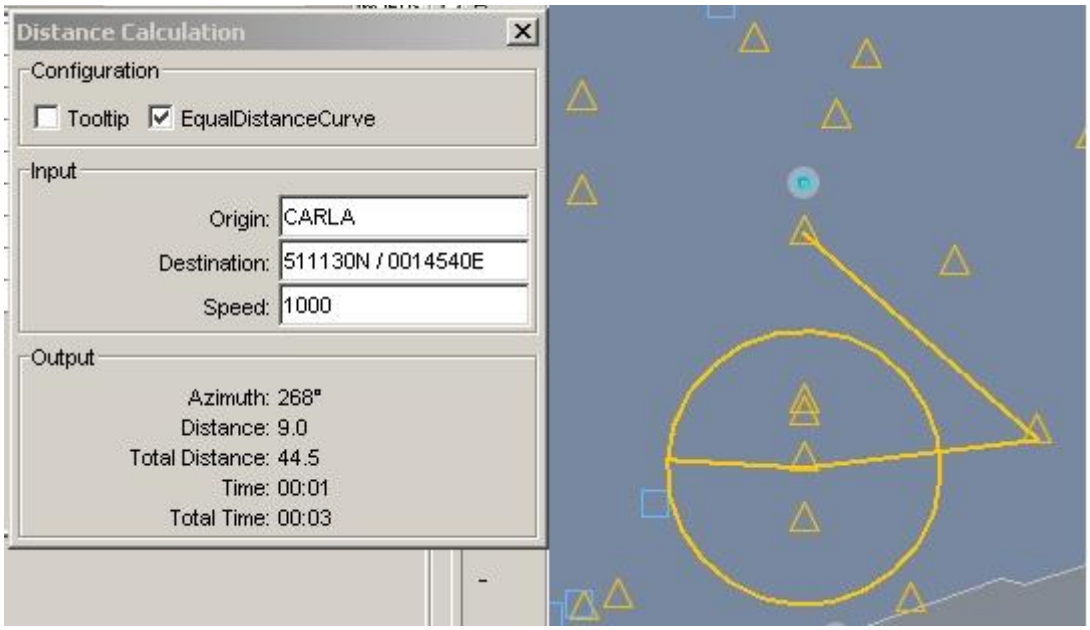


Figure 2-22: Distance Calculation dialog - EqualDistanceCurve

2.2.1.4 Azimuth controller

(1) The azimuth controller:
Although the azimuth filter can be activated directly from the general mode, an azimuth controller has been implemented.



Figure 2-23: Azimuth Controller - Button



Figure 2-24: Azimuth Controller - Cursor shape

(2) With the azimuth filter, you can set a filter for a specific angle and direction.

(3) What does that mean?

(4) Basically, it will display routes or routes of restrictions in a certain direction, and hide others:

a) An azimuth tool enables you to determine both at the same time:

i) A direction axis;

ii) An aperture angle around it.

b) The routes or routes of restrictions that match these two direction criteria will be displayed. The other ones will be hidden.

(5) The two figures below will help you understand:

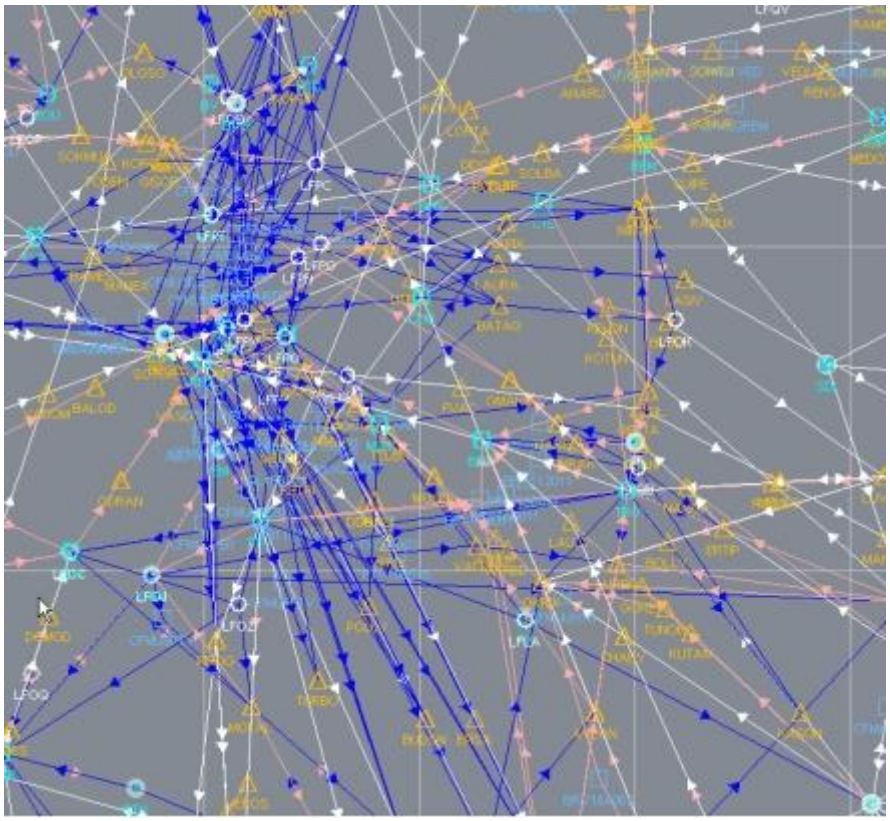


Figure 2-25: route network above Paris FL240 no filter

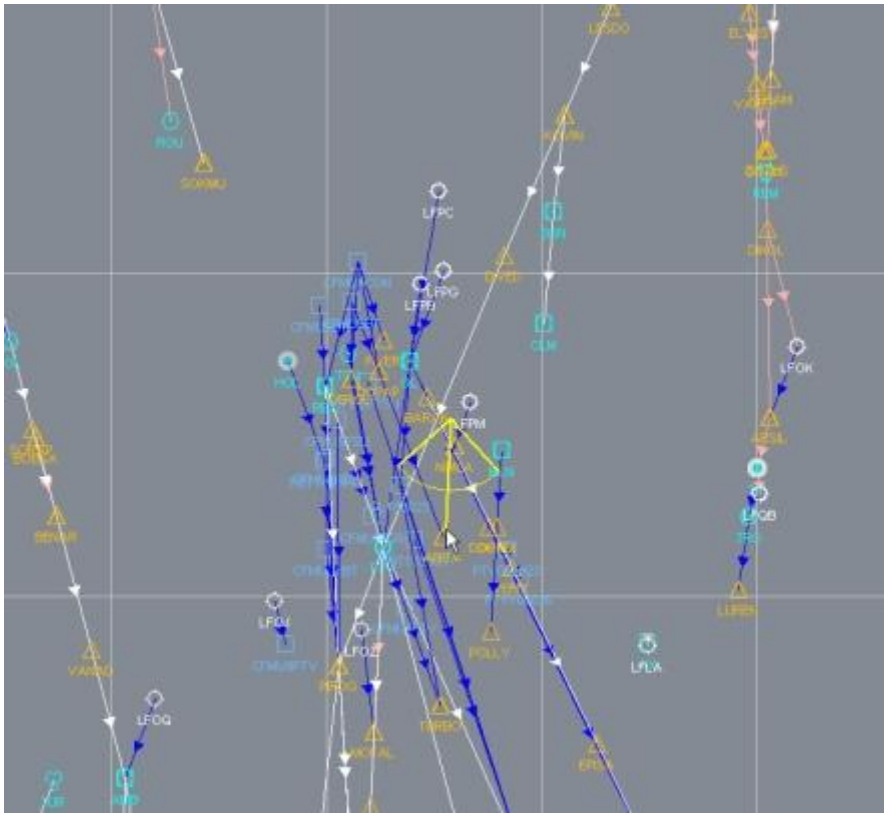


Figure 2-26: same azimuth south/90 degrees (in yellow)

(6) These two figures represent the route network (with AP/DP's) above Paris at FL240.

a) The first one is not filtered: everything is represented.

b) The second one is filtered. You can see the azimuth tool in yellow. It is oriented south (axis), and has an aperture angle of approximately 90 degrees. Therefore, only the route tracks that are between roughly radial 160 and 200 are represented. The rest is hidden.

(7) The angle can vary between 360 and 1 degree.

(8) Therefore, you can imagine that coming back to unfiltered view can be done by setting the angle to 360 degrees.

(9) But there is an easier solution to enable and disable azimuth filtering: the  button (toolbar).

(10) This will disable/able the azimuth filter.

(11) Note that this button is synchronised with the corresponding box on the map configuration Filtering Tab.



Figure 2-27: Azimuth Filter - Button

(12) **Watch out for the azimuth symbol  at the top right corner of the map. When this symbol is shown, the filter is active. Otherwise it is not active.**



Figure 2-28: Azimuth Controller - Azimuth symbol

2.2.1.5 Centre controller

(1) Although the centre controller can be activated directly from the general mode, a centre controller has been implemented.



Figure 2-29: Centre Controller - Button



Figure 2-30: Centre Controller - Cursor shape

(2) Clicking will set the position of the mouse cursor to the centre of the map. Zoom is not affected.

2.2.2 Map actions

(1) Different actions can be executed on the map. A controller can be activated

a) by clicking the menu item for the action

b) by clicking the button in the toolbar

c) by typing a shortcut

(2) When you go with the mouse over a button for an action in the toolbar a tooltip appears.

(3) The popup actions appear when you right click on the map:

a) immediately in the popup menu

b) in the popup menu under the item "Map Controls"

c) in the popup menu under the item "View"

d) in the popup menu under the item "Refresh"

e) in the popup menu under the item "Set zoom"

(4) The table describes the different actions:

	Menu option	Tooltip	Toolbar	Shortcut Key	Remark
1	Clear all dossier elements	Clear all dossier elements		SHIFT+C	
2	Save as image	Save as image		CTRL+G	
3	Show altitude meter	Show altitude meter		M	
4	Show time slider	Show time slider		T	
5	Show Overview	Show Overview and properties		V	
6	Resize (Initial range)	Reset zoom to initial range		CTLR+N	
7	Resize (Default range)	Reset zoom to default range		CTLR+D	
8	Set default range	Set default zoom to actual range		CTRL+E	
9	Azimuth filter active	Enable azimuth filtering		CTLR+O	
10	Point icon panting active	Enable point icon panting			
11	Point label painting active	Enable point label painting			
12	Fit on layers	Fit to one of the layers		CTLR+L	

13	Map configuration	Map configuration		ALT+Z	
14	Map legend	Map legend		ALT+L	
15	Geographical distance calculation	Geographical distance calculation		CTRL + SHIFT+D	
16	Repair local data	Repair local data			Only IFPS, ATFCM
17	Dossier explorer	Dossier explorer		CTRL+X	Not in toolbar, Only ENV
18	Save dossier	Save dossier		CTRL + SHIFT + S	Only ENV
19	Save dossier as…	Save dossier as…			Only ENV
20	Refresh the data currently in the view	Refresh the data currently in the view			Only ENV
21	Refresh all the data	Refresh all the data			Only ENV
22	File/Print view	Print view			General CHMI
23	File/Preferences…	Open preferences dialog			General CHMI
24	Set Zoom/All			SHIFT + A	
25	Set Zoom/ All default			ALT + A	
26	Set Zoom/ AERODROME			SHIFT + D	
27	Set Zoom/ AERODROME default			ALT + D	
28	Set Zoom/ SETOFAERODROME			SHIFT + O	
29	Set Zoom/ SETOFAERODROME default			ALT + O	
30	Set Zoom/ AIRBLOCK			SHIFT +B	
31	Set Zoom/ AIRBLOCK default			ALT + B	

32	Set Zoom/ AIRSPACE			SHIFT + S	
33	Set Zoom/ AIRSPACE default			ALT + S	
34	Set Zoom/ POINT			SHIFT + P	
35	Set Zoom/ POINT default			ALT + P	
36	Set Zoom/ ROUTE			SHIFT + R	
37	Set Zoom/ ROUTE default			ALT + R	
38	Set Zoom/ RESTRICTION			SHIFT + I	
39	Set Zoom/ RESTRICTION default			ALT + I	

Table 2-2: Map actions

2.2.2.1 Clear all dossier elements action

(1) Action: Clear all dossier elements

(2) Button:



Figure 2-31: Clear all dossier elements - Button

(3) Shortcut: **SHIFT + C**

(4) Click on the button in the toolbar or the popup menu 'Clear all'.

a) All the elements that are in dossier layers are removed.

b) The dossier panel is repainted and the map is repainted.

2.2.2.2 Save as Image action

(1) Action: Save as Image

(2) Button



Figure 2-32: Save as Image - Button

(3) Shortcut: **CTRL + G**

(4) Click on the button in the toolbar or the popup menu 'Save As Image…'.

a) A file chooser is shown. The personal directory ('My Documents') of the user is opened.

b) Enter the name of the image. Choose the type GIF. Click 'Save'.

c) The image of the map is saved as GIF. The following message is shown.



Figure 2-33: Save as Image - Save of gif successful

(5) Click on the button in the toolbar or the popup menu 'Save As Image…'.

a) A file chooser is shown. The personal directory ('My Documents') of the user is opened.

b) Enter the name of the image. Choose the type JPEG. Click 'Save'.

c) The image of the map is saved as JPEG. The following message is shown.



Figure 2-34: Save as Image - Save of jpeg successful

2.2.2.3 Show/hide altitude meter action

(1) Action: Show/hide altitude meter

(2) Button



Figure 2-35: Show/hide altitude meter - Button

(3) Shortcut: **M**

(4) Click on the button in the toolbar or the popup menu 'View/Show Altitude'.

a) The panel with the altitude controls is shown or hidden.

b) When the panel is hidden, the altitude filter is not deactivated.

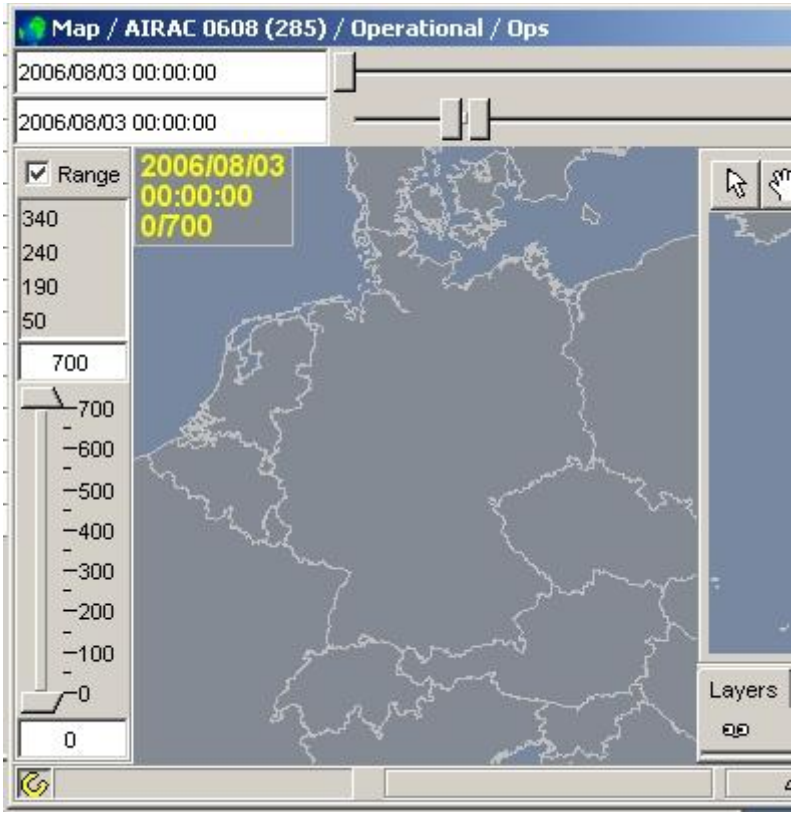


Figure 2-36: Show/hide altitude meter - Altitude meter shown



Figure 2-37: Show/hide altitude meter - Altitude meter hidden

2.2.2.4 Show/hide time slider action

(1) Action: Show/hide time slider

(2) Button



Figure 2-38: Show/hide time slider - Button

(3) Shortcut: **T**

(4) Click on the button in the toolbar or the popup menu 'View/Show Timeslider'.

a) The panel with the time controls is shown or hidden.

b) Remember that the time filter cannot be deactivated.



Figure 2-39: Show/hide time slider - Time meter shown



Figure 2-40: Show/hide time slider - Time meter hidden

2.2.2.5 Show/hide overview and properties

(1) Action: Show/hide overview and properties

(2) Button



Figure 2-41: Show/hide overview and properties - Button

(3) Shortcut: **V**

(4) Click on the button in the toolbar or the popup menu 'View/Show Overview'.

a) The right map window is shown or removed.

b) This right window can contain the overview.

c) It contains the tab with the layers, the tab with the find panel, the tab with the features panel and other tabs.

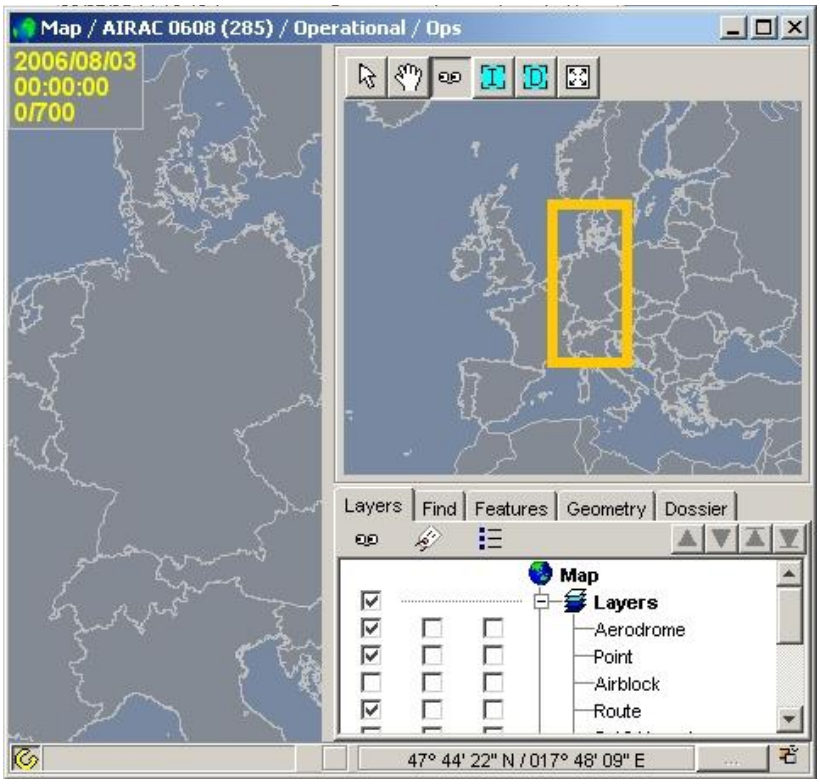


Figure 2-42: Show/hide overview and properties - Overview and properties shown

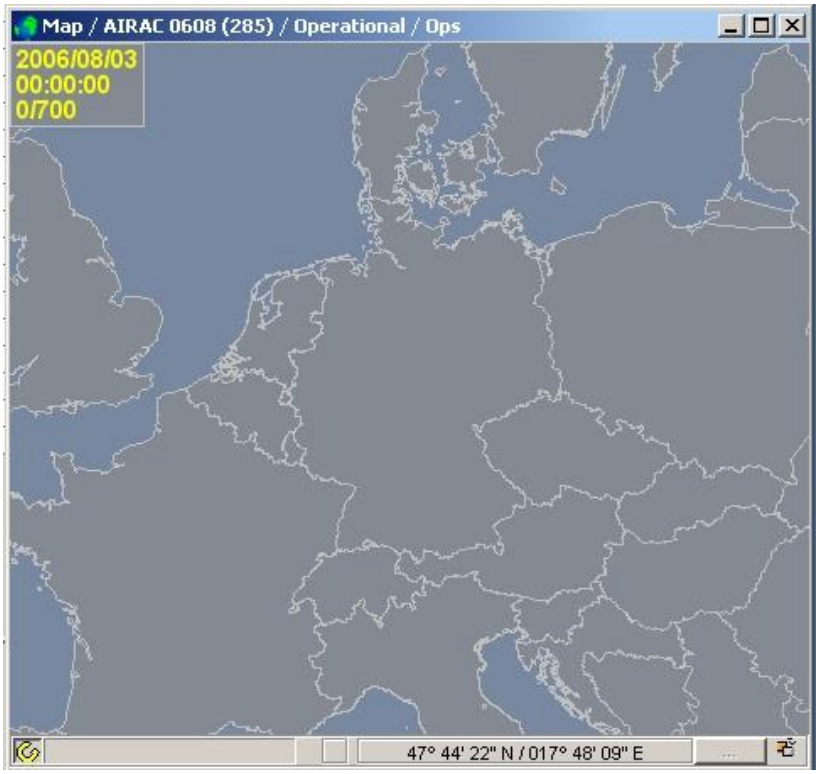


Figure 2-43: Show/hide overview and properties - Overview and properties hidden

2.2.2.6 Reset zoom to initial range action

(1) Action: Reset zoom to initial range

(2) Button



Figure 2-44: Reset zoom to initial range - Button

(3) Shortcut: **CTRL + N**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Resize(Initial Range)'.

a) The scale of the map is reset to the initial scale. With this scale Europe is visible.



Figure 2-45: Reset zoom to initial range - Initial map scale

2.2.2.7 Reset zoom to default range action

(1) Action: Reset zoom to default range

(2) Button



Figure 2-46: Reset zoom to default range - Button

(3) Shortcut: **CTRL + D**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Resize(Default Range)'.

a) The scale of the map is reset to the default scale.

2.2.2.8 Set default zoom to actual range action

(1) Action: Set default zoom to actual range

(2) Button



Figure 2-47: Set default zoom to actual range - Button

(3) Shortcut: **CTRL + E**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Set default range'.

a) The current scale of the map is taken as the default scale.

b) Nothing changes on the visible map.

2.2.2.9 Enable azimuth filtering action

(1) Action: Enable azimuth filtering

(2) Button



Figure 2-48: Enable azimuth filtering - Button

3. Shortcut: **CTRL + O**

(3) Click on the button in the toolbar or the popup menu 'Map Controls/Azimuth filter active'.

a) Azimuth filtering is made active on the map. The angle and axis are defined by the Azimuth controller.

b) When azimuth filtering is active there is always an icon in the top right corner of the map that shows the direction and angle.



Figure 2-49: Enable azimuth filtering - Azimuth filtering active

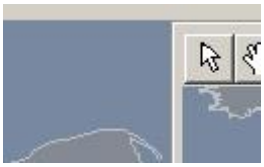


Figure 2-50: Enable azimuth filtering - Azimuth filtering not active

2.2.2.10 Enable point icon panting action

(1) Action: Enable point icon panting

(2) Button



Figure 2-51: Enable point icon panting - Button

(3) Click on the button in the toolbar or the popup menu 'Map Controls/Point icon painting active'.

a) If this is active no point icons and point labels are painted on the map.

b) The layers that have points are: Aerodrome, Point, Route, SetOfAerodrome and restriction.

2.2.2.11 Enable point label painting action

(1) Action: Enable point label painting

(2) Button



Figure 2-52: Enable point label painting - Button

(3) Click on the button in the toolbar or the popup menu 'Map Controls/Point label painting active'.

a) If this is active no point labels are painted on the map.

b) The layers that have points are: Aerodrome, Point, Route, SetOfAerodrome and restriction.

2.2.2.12 Fit to one of the layers

(1) Action: Fit to one of the layers

(2) Button



Figure 2-53: Fit to one of the layers - Button

(3) Shortcut: **CTRL + L**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Fit on layer.

a) A dialog appears with the possible layers on which you can fit.

b) Select a layer.

c) Click 'Fit.

d) The range of the map will be set that all the elements of the selected layer fit in the map.

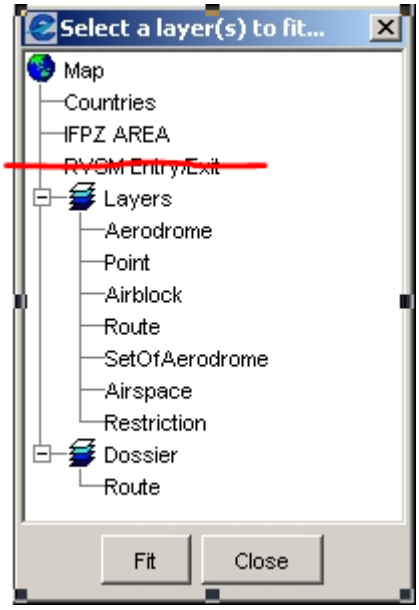


Figure 2-54: Fit to one of the layers - Select layer dialog

(5) The map before fitting, the layer Dossier/Route has one route: UW70.

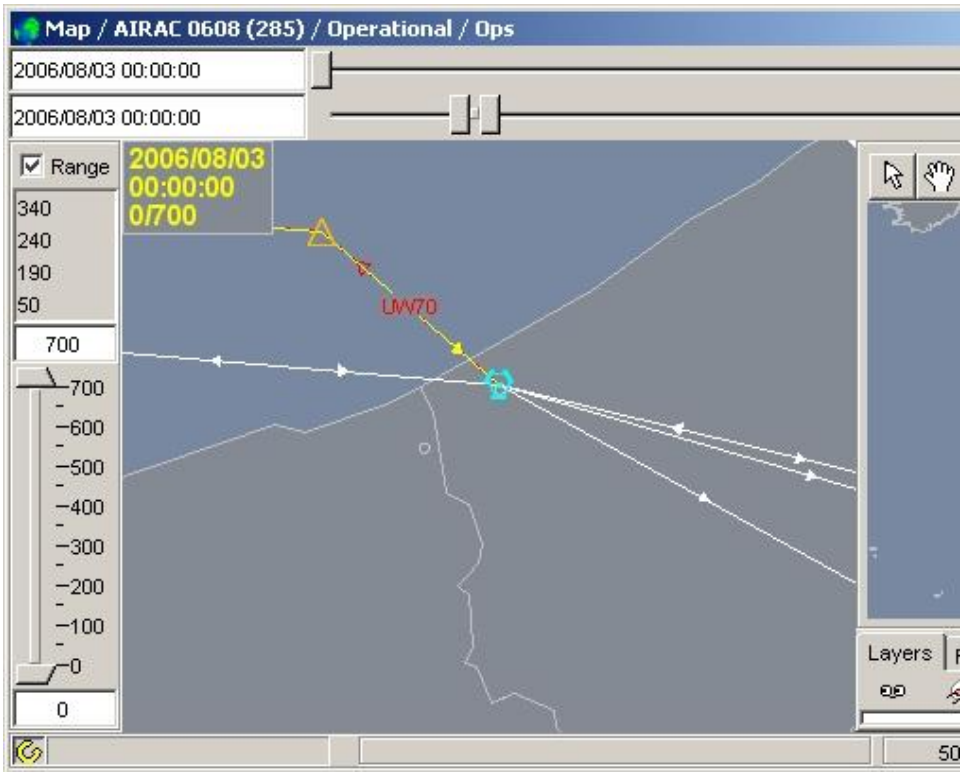


Figure 2-55: Fit to one of the layers - Map before fitting

(6) The map after fitting on layer Dossier/Route:

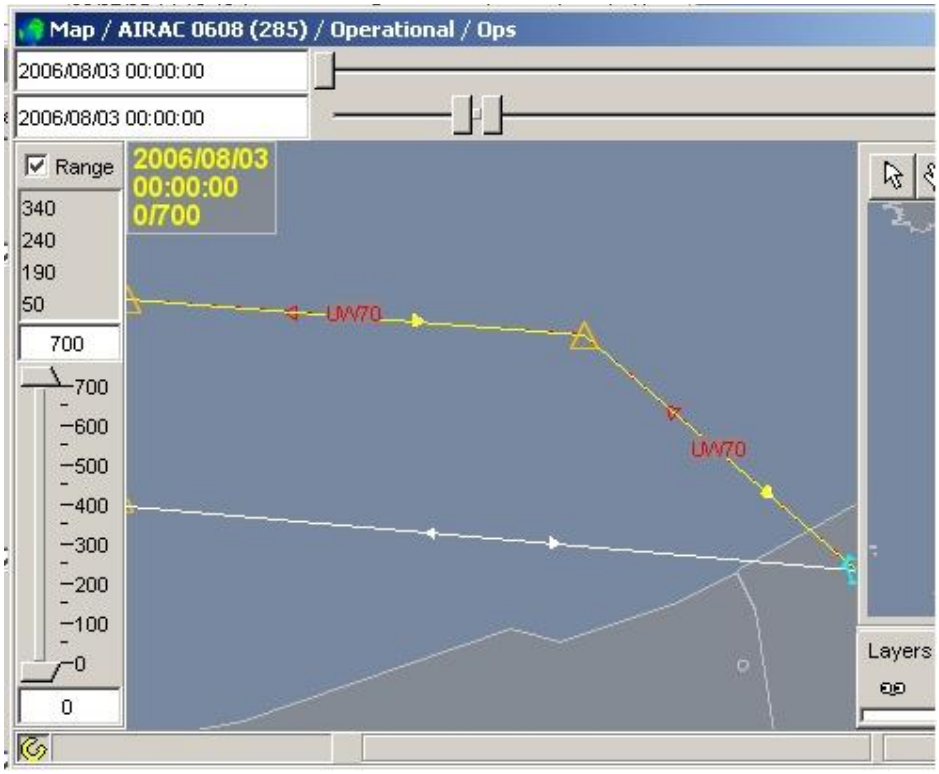


Figure 2-56: Fit to one of the layers - Map after fitting

(7) Click on 'Close' in the layer fit dialog.

(8) The dialog is closed.

2.2.2.13 Show map configuration action

(1) Action: Show map configuration

(2) Button



Figure 2-57: Show map configuration - Button

(3) Shortcut: **ALT + Z**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Map Configuration.

a) The window with the Map Configuration is shown. If the window was already visible, the window is moved to the front.

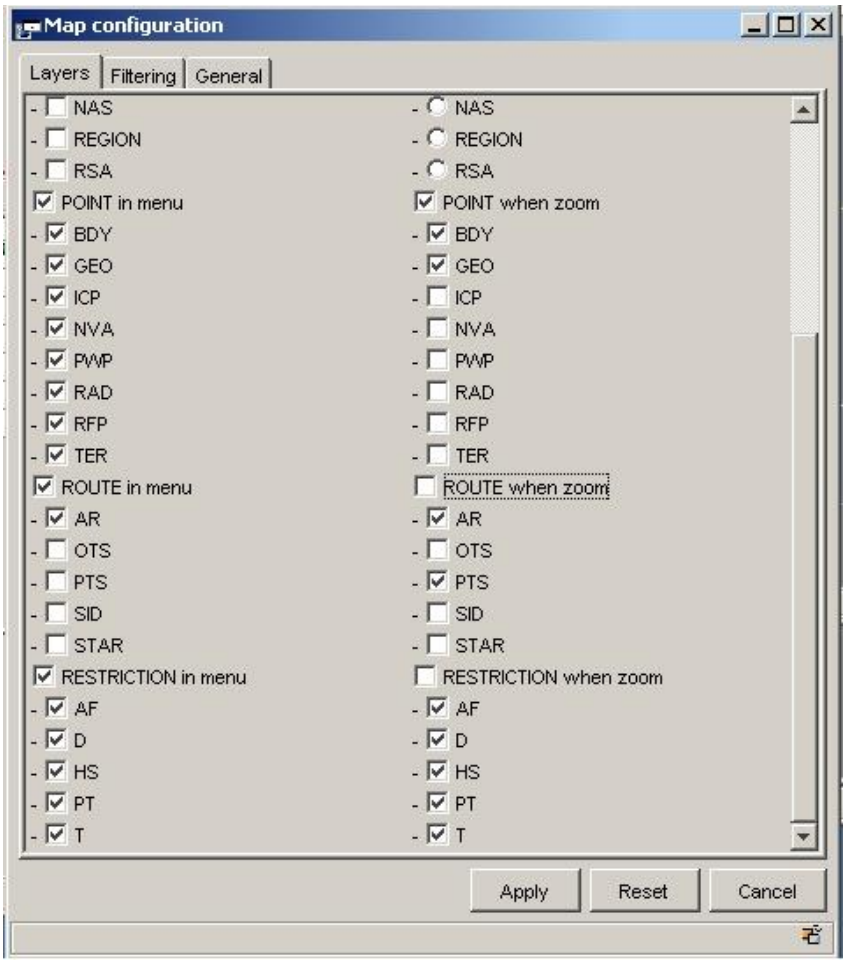


Figure 2-58: Show map configuration - Dialog

2.2.2.14 Map legend action

(1) Action: Map legend

(2) Button



Figure 2-59: Map legend - Button

(3) Shortcut: **ALT + L**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Map Legend.

a) The window with the Map Legend is shown. If the window was already visible, the window is moved to the front.

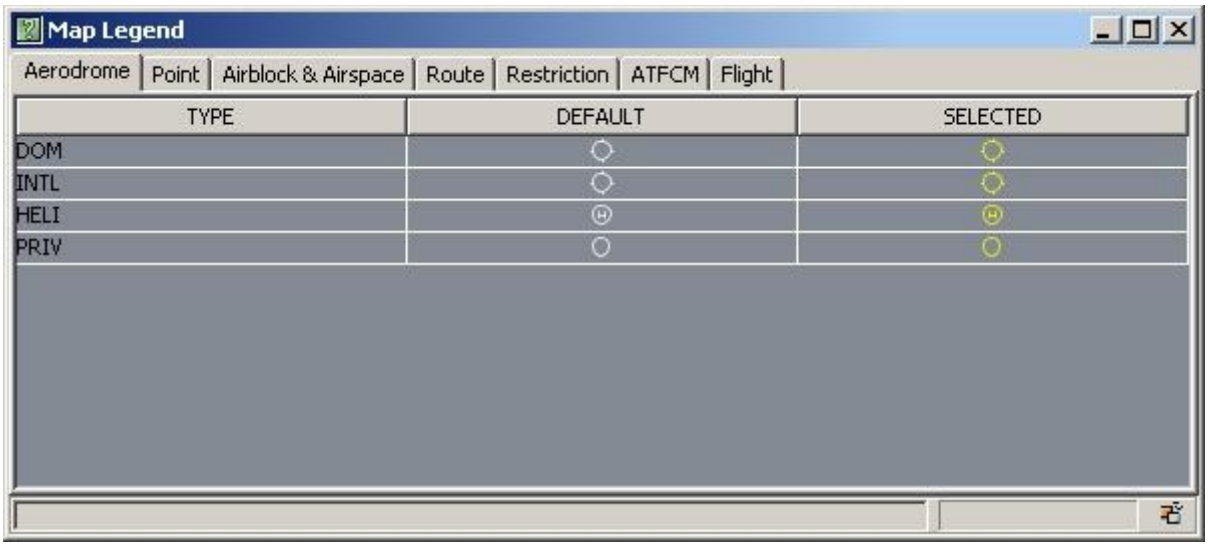


Figure 2-60: Map legend window

2.2.2.15 Geographical distance calculation action

(1) Action: Geographical distance calculation

(2) Button



Figure 2-61: Geographical distance calculation - Button

(3) Shortcut: **SHIFT + D**

(4) Click on the button in the toolbar or the popup menu 'Map Controls/Geographical Distance Calculation.

a) The window with the Map Distance Calculation is shown. If the window was already visible, the window is moved to the front.



Figure 2-62: Geographical distance calculation - Dialog

2.2.2.16 Repair local data action

(1) Action: Repair local data

(2) Button



Figure 2-63: Repair local data - Button

(3) Click on the button in the toolbar or the popup menu 'Refresh/Repair local data.

(4) The action 'Repair local data' is only visible for flight and ATFCM.

2.2.2.17 Dossier explorer

(1) Action: Dossier explorer

(2) Button



Figure 2-64: Dossier explorer - Button

(3) Click on the button in the toolbar or on the menu 'Application/Map/Dossier Explorer' or in the popup menu 'Map Controls/Dossier Explorer'.

a) The dossier explorer is shown. If the window was already visible, the window is moved to the front.

(4) The 'Dossier explorer' action is only visible for ENV.

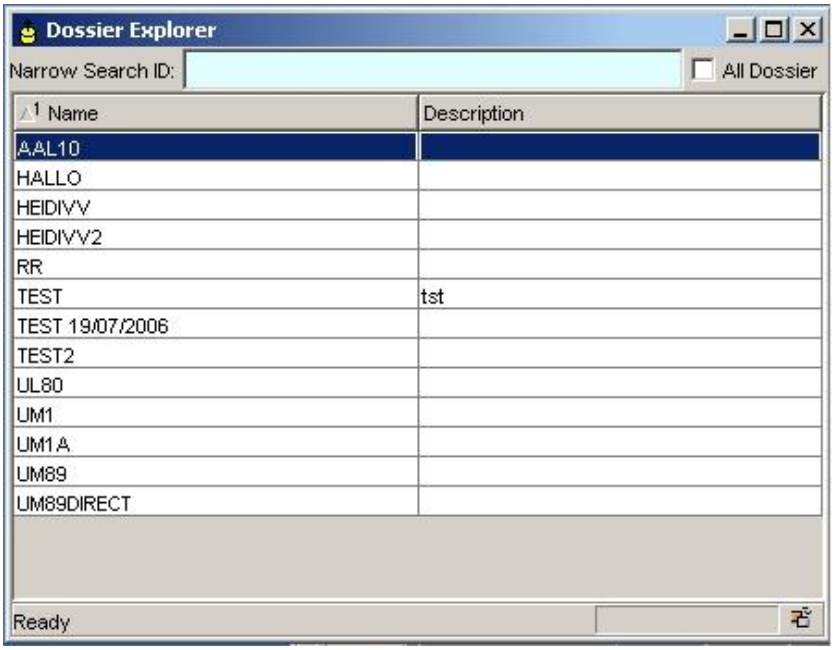


Figure 2-65: Dossier explorer window

2.2.2.18 Save Dossier action

(1) Action: Save Dossier

(2) Button



Figure 2-66: Save Dossier - Button

(3) Click on the button in the toolbar or the popup menu 'Map Controls/Save Dossier.

a) The dossier is saved.

b) If the dossier is saved for the first time, the window for saving the dossier is shown.

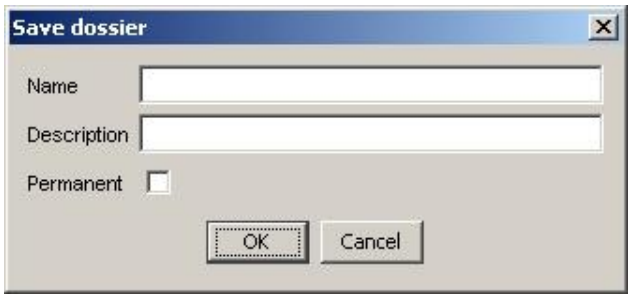


Figure 2-67: Save Dossier - Dialog

(4) The 'Save dossier' action is only visible for ENV.

2.2.2.19 Save dossier as action

(1) Action: Save dossier as

(2) Button



Figure 2-68: Save dossier as - Button

(3) Click on the button in the toolbar or the popup menu 'Map Controls/Save Dossier As.. '.

a) The dialog for saving the dossier is shown.

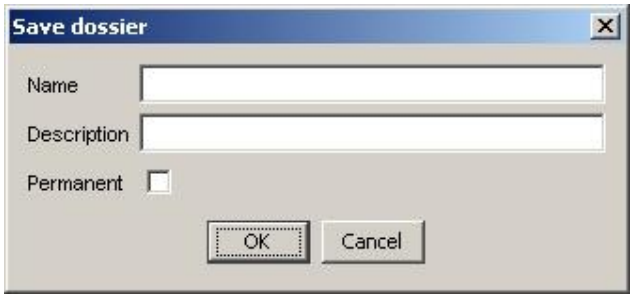


Figure 2-69: Save dossier as - Dialog

(4) The 'Save dossier as' action is only visible for ENV.

2.2.2.20 Refresh the data currently in the view action

(1) Action: Refresh the data currently in the view

(2) Button



Figure 2-70: Refresh data currently in view - Button

(3) Click on the button in the toolbar or the popup menu 'Refresh/Currently in view'.

a) The data in the bounds that are currently in the view is refreshed.

(4) At the bottom of the map you see information about the data that is currently retrieved:



Figure 2-71: Refresh data currently in view - Retrieving information

(5) If you move the mouse over this field you can see the actions that are already performed as tooltip:

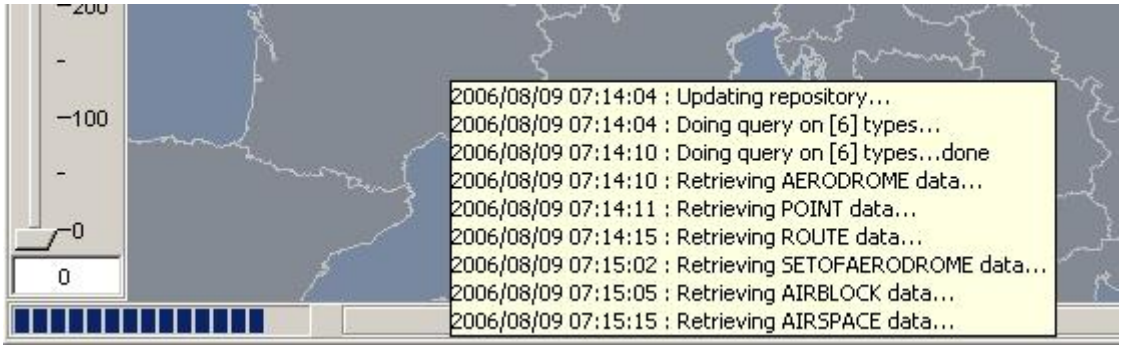


Figure 2-72: Refresh data currently in view - Retrieving information tooltip

(6) 6 types should be retrieved:

a) AERODROME,

b) POINT,

c) ROUTE,

d) SETOFAERODROME,

e) AIRBLOCK and

f) AIRSPACE.

(7) The RESTRICTION elements are not retrieved because that would take too much time.

(8) On the bottom left of the map you see a progress bar. This bar restarts with the retrieve of every type.

(9) The 'Refresh the data currently in the view' action is only visible for ENV.

2.2.2.21 Refresh all the data action

(1) Action: Refresh all the data

(2) Button



Figure 2-73: Refresh all the data - Button

(3) Click on the button in the toolbar or the popup menu 'Refresh/World.

a) All the data of the map is refreshed.

(4) At the bottom of the map you see information about the data that is currently retrieved:



Figure 2-74: Refresh all the data - Retrieving information

(5) If you move the mouse over this field you can see the actions that are already performed as tooltip:

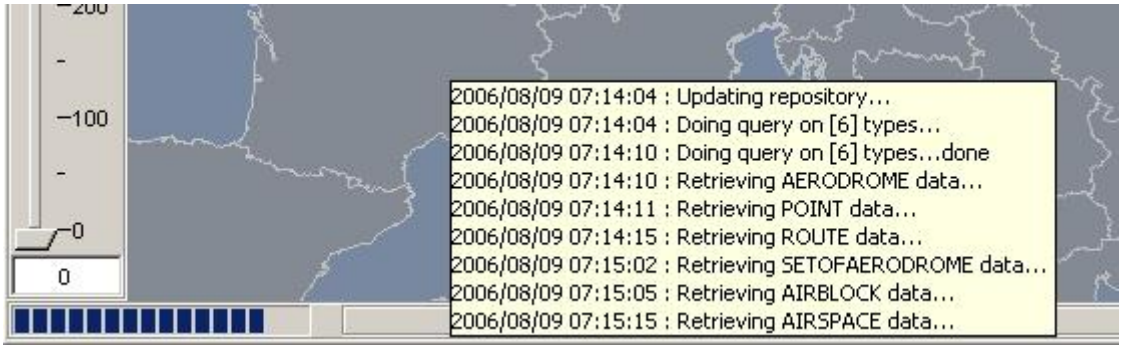


Figure 2-75: Refresh all the data - Retrieving information tooltip

(6) 6 types should be retrieved:

a) AERODROME,

b) POINT,

c) ROUTE,

d) SETOFAERODROME,

e) AIRBLOCK and

f) AIRSPACE.

(7) The RESTRICTION elements are not retrieved because that would take too much time.

(8) On the bottom left of the map you see a progress bar. This bar restarts with the retrieve of every type.

(9) The 'Refresh all the data' action is only visible for ENV.

2.2.2.22 Print view action

(1) Action: Print view

(2) Button



Figure 2-76: Print view - Button

(3) Click on the button in the toolbar or the menu 'File/Print'.

a) If the map is the active window, the map print dialog should be opened.

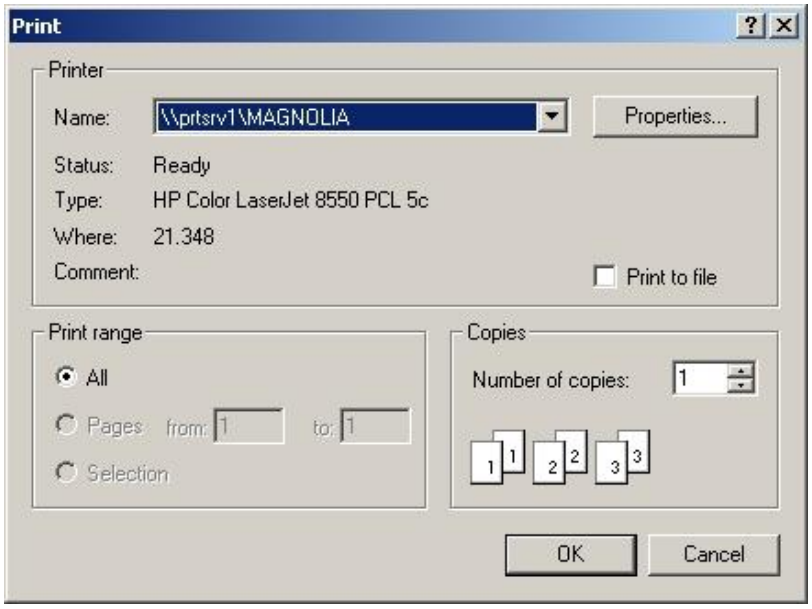


Figure 2-77: Print view - Print Dialog

b) Click 'OK'.

c) The Page Setup dialog is displayed.

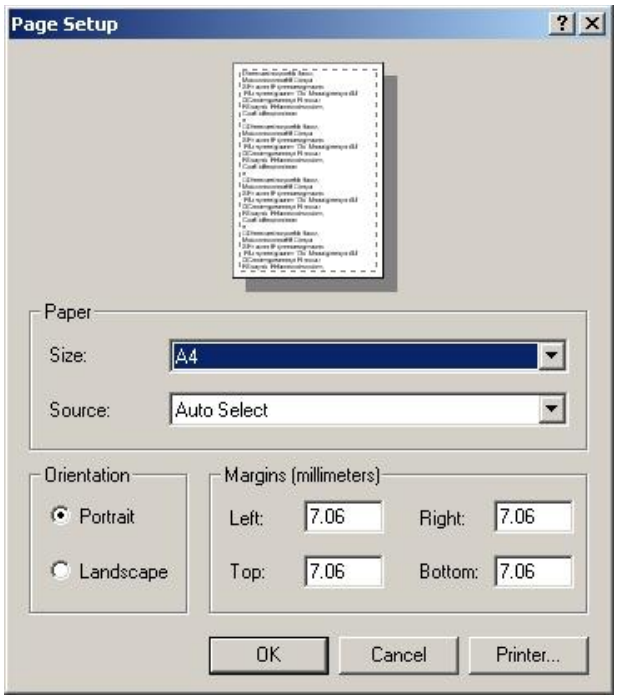


Figure 2-78: Print view - Page Setup Dialog

d) Click OK.

e) The map should be printed. Only the map image should be printed and the map is resized to fit maximal on the print page. The colour of the Country layer when printing comes from the system preference (default=white i.e. RGB=255,255,255). The background colour (Water colour) when printing comes from the system preference (default= light grey i.e. RGB=225,225,225)



Figure 2-79: Print view - Printed Map Image

2.2.2.23 Open preferences dialog action

(1) Action: Open preferences dialog

(2) Button



Figure 2-80: Open preferences dialog - Button

(3) Click on the button in the toolbar or in the menu 'File/Preferences'.

a) The preferences dialog is opened.

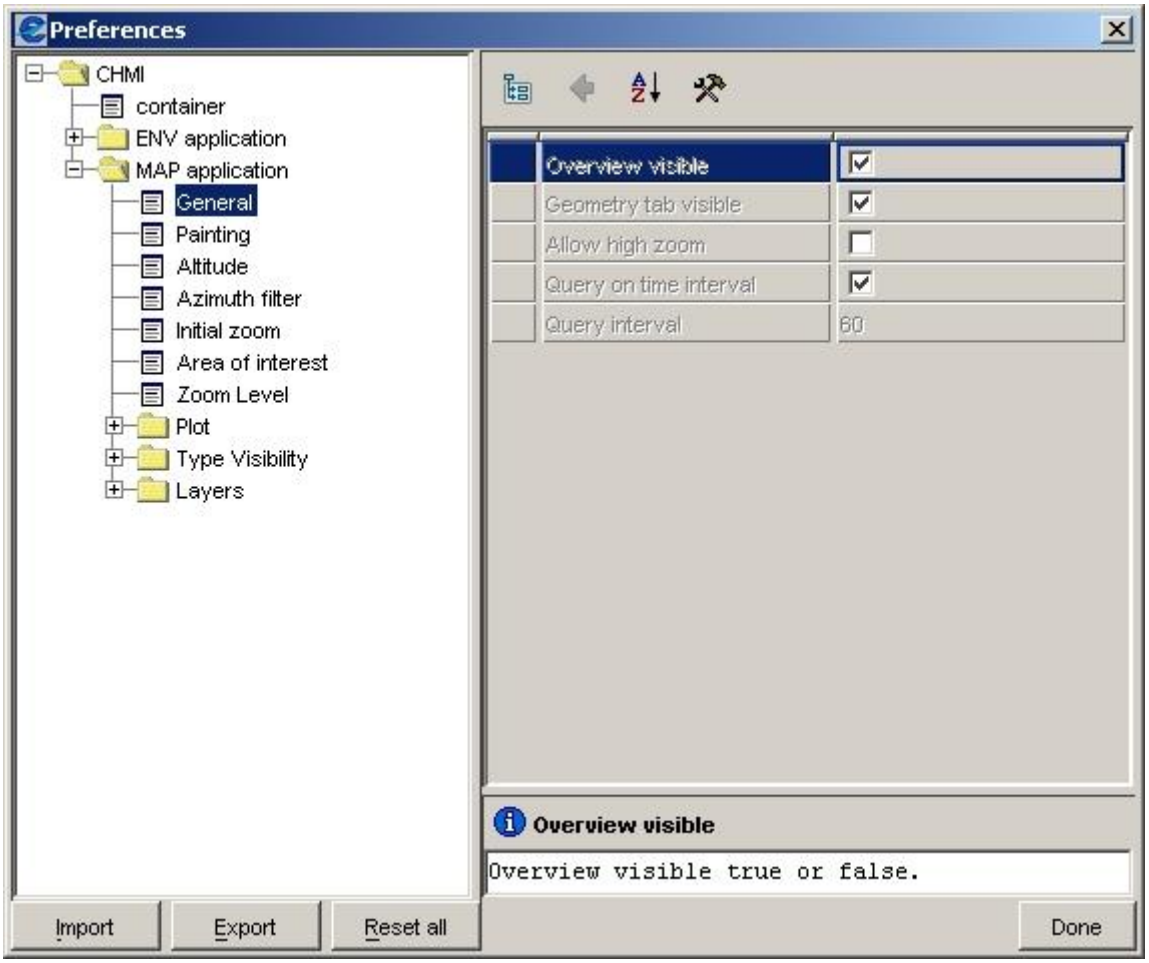


Figure 2-81: Open preferences dialog - Preferences Dialog

2.2.2.24 Set Zoom / All

(1) Action: All the reference layers will be displayed from the current zoom level

(2) Shortcut: **SHIFT + A**

2.2.2.25 Set Zoom / All Default

(1) Action: All the reference layers will be displayed from the default zoom level

(2) Shortcut: **ALT + A**

2.2.2.26 Set Zoom / AERODROME

(1) Action: The reference aerodromes will be displayed from the current zoom level

(2) Shortcut: **SHIFT + D**

2.2.2.27 Set Zoom / AERODROME Default

(1) Action: The reference aerodromes will be displayed from the default zoom level for aerodromes

(2) Shortcut: **ALT + D**

2.2.2.28 Set Zoom / SETOFAERODROME

(1) Action: The reference setofaerodromes will be displayed from the current zoom level

(2) Shortcut: **SHIFT + O**

2.2.2.29 Set Zoom / SETOFAERODROME Default

(1) Action: The reference setofaerodromes will be displayed from the default zoom level for setofaerodromes

(2) Shortcut: **ALT + O**

2.2.2.30 Set Zoom / AIRBLOCK

(1) Action: The reference airblocks will be displayed from the current zoom level

(2) Shortcut: **SHIFT + B**

2.2.2.31 Set Zoom / AIRBLOCK Default

(1) Action: The reference airblocks will be displayed from the default zoom level for airblocks

(2) Shortcut: **ALT + B**

2.2.2.32 Set Zoom / AIRSPACE

(1) Action: The reference airspaces will be displayed from the current zoom level

(2) Shortcut: **SHIFT + S**

2.2.2.33 Set Zoom / AIRSPACE Default

(1) Action: The reference airspaces will be displayed from the default zoom level for airspaces

(2) Shortcut: **ALT + S**

2.2.2.34 Set Zoom / POINT

(1) Action: The reference points will be displayed from the current zoom level

(2) Shortcut: **SHIFT + P**

2.2.2.35 Set Zoom / POINT Default

(1) Action: The reference points will be displayed from the default zoom level for points

(2) Shortcut: **ALT + P**

2.2.2.36 Set Zoom / ROUTE

(1) Action: The reference routes will be displayed from the current zoom level

(2) Shortcut: **SHIFT + R**

2.2.2.37 Set Zoom / ROUTE Default

(1) Action: The reference routes will be displayed from the default zoom level for routes

(2) Shortcut: **ALT + R**

2.2.2.38 Set Zoom / RESTRICTION

(1) Action: The reference restrictions will be displayed from the current zoom level

(2) Shortcut: **SHIFT + I**

2.2.2.39 Set Zoom / RESTRICTION Default

(1) Action: The reference restrictions will be displayed from the default zoom level for restrictions

(2) Shortcut: **ALT + I**

2.3 Map Overview

(1) Open the Map Configuration window.

(2) Check the property 'General/Painting/Overview visible'.

a) You see the overview at the top right of the map window.

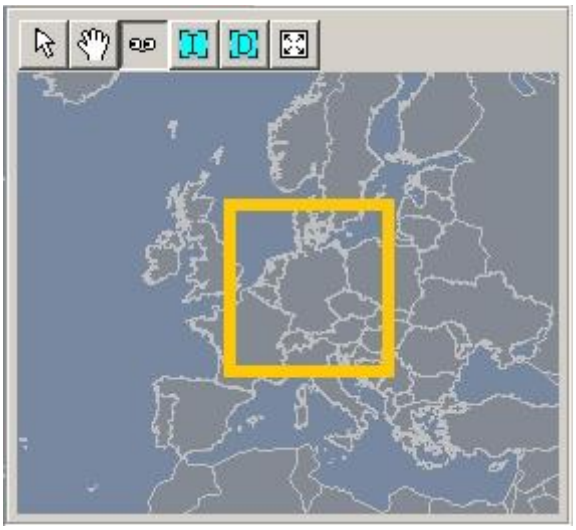


Figure 2-82: Map Overview Window

2.3.1 Overview controllers

(1) There are 3 different controllers for the map overview.

(2) Only one controller can be active on the map overview.

(3) The controller decides what the mouse interactions with the overview map mean.

(4) There are no shortcuts for the controllers.

(5) The 3 controllers are displayed above the overview at the left side:



Figure 2-83: Map Overview Controllers

(6) The table describes the toolbar button for activating the controllers and the cursor associated with the active controller.

	Toolbar	Tooltip	Cursor
1		General mouse behaviour	
2		Panning mouse behaviour	
3		Overview mouse behaviour	

Table 2-3: Map Overview Controllers

2.3.1.1 General controller

(1) The general controller for the overview has the same behaviour as the general controller for the main map.

2.3.1.2 Pan controller

(1) The general controller for the overview has the same behaviour as the general controller for the main map.

2.3.1.3 Overview controller

(1) The overview controller button:



Figure 2-84: Map Overview Controller - Button

(2) When the overview controller is active you can draw a rectangle on the overview. The rectangle is drawn in yellow.

(3) The map is resized that the bounds of the rectangle of the overview fit maximal on the map.

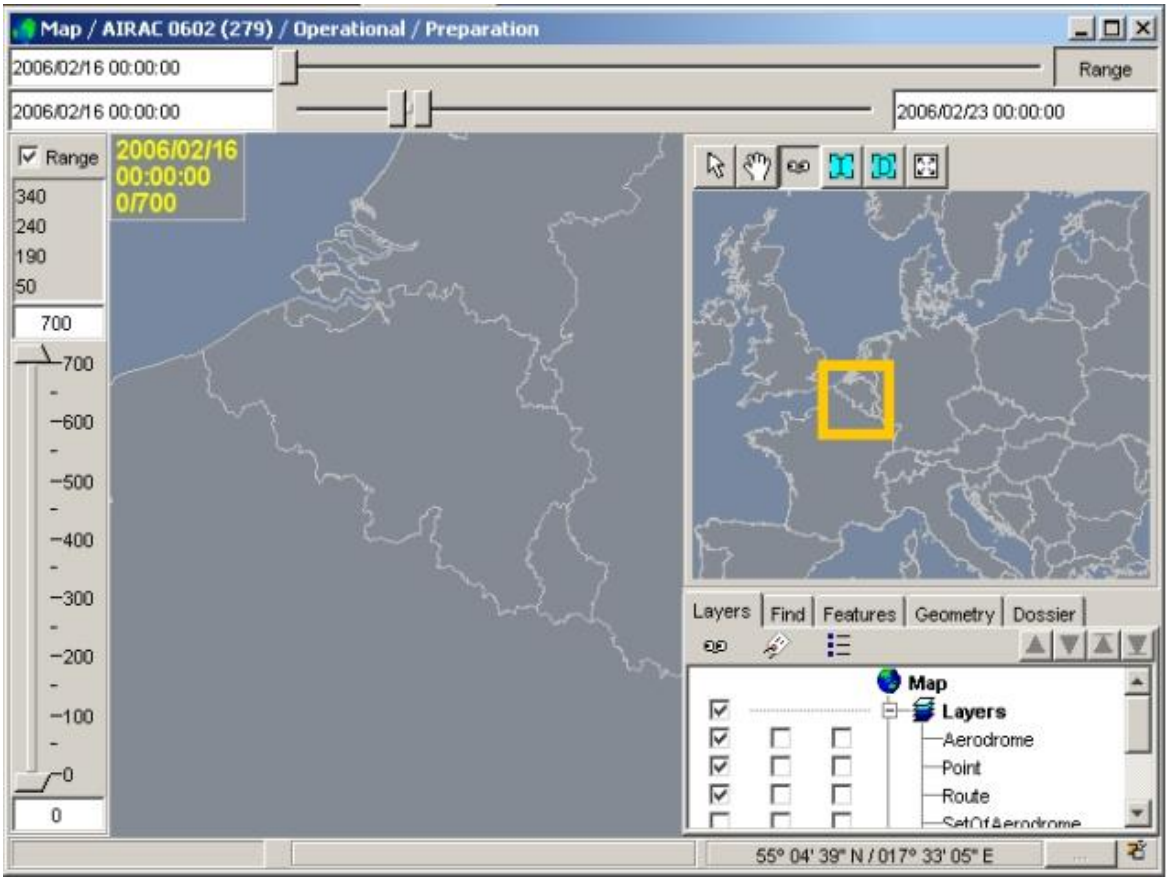


Figure 2-85: Map Overview Controller - Active example

2.3.2 Map overview actions

(1) Different actions can be executed on the overview map.

(2) There are no shortcuts for the actions

(3) The 3 overview actions are displayed above the overview at the right side:



Figure 2-86: Map Overview Actions - Buttons

(4) The actions are:

a) Reset zoom to initial range

b) Reset zoom to default range

c) Fit to one of the layers

(5) The table describes the different actions:

	Menu option	Tooltip	Toolbar	Shortcut Key	Remark
1	Resize (Initial range)	Reset zoom to initial range			
2	Resize (Default range)	Reset zoom to default range			
3	Fit on layers	Fit to one of the layers			

Table 2-4: Map overview actions

(6) The actions of the overview do the same as the correspondent action of the main map.

2.4 Map Layer control

(1) The layer control is displayed as one tab at the right side of the map with name 'Layers when the 'Show overview and properties button in the toolbar is pressed down.

(2) The button:



Figure 2-87: Map Layer Control - Button

(3) The layer control:

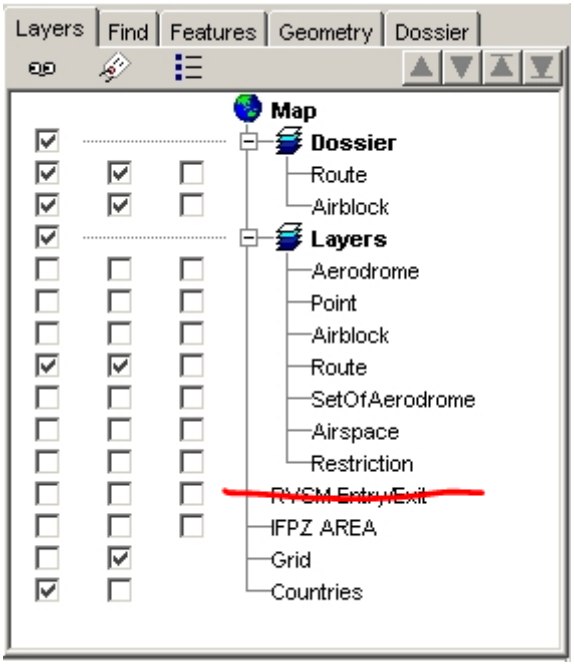


Figure 2-88: Map Layer Control Window

(4) The layer control panel is used to show or hide the different layers and change the properties of the layers.

2.4.1 Show layer, show labels, show point coordinates

(1) Every line in the layer control represents a layer of the map.

(2) The name of the layer is displayed together with two or three checkboxes.

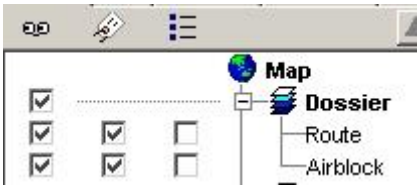


Figure 2-89: Map Layer Control Window - Show Details

a) The first column of checkboxes is to make the layer visible or not.

b) The second column of the checkboxes is to make the labels of the layer visible. The labels will only be visible if the layer is visible.

c) The third column of the checkboxes is to show the coordinates of the point in the labels of points.

2.4.2 Layer groups

(1) Most of the layers are grouped.

(2) When you start the CHMI in FLIGHT or ATFCM you always have one group 'Layers' with 7 layers.

(3) If you add elements to the dossier the 'Dossier' group appears in the layer control.

(4) If you click on the checkbox next to the layer group, you can make all the layers of the group visible or invisible. To see a layer on the map, the checkbox of the layer and its group should be checked.

2.4.3 Select layers

(1) If you click on the name of a layer in the layer control it is selected.

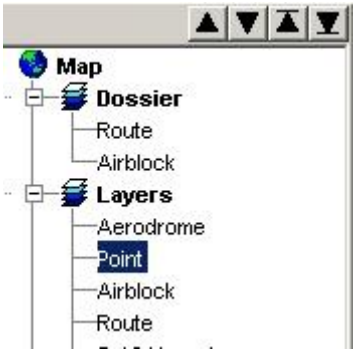


Figure 2-90: Map Layer Control Window - Select layers

2.4.4 Move layers

(1) Select a layer. You can move the layer up or down by clicking the arrows on the top of the layer control.



Figure 2-91: Map Layer Control- Arrows

a)  Move the layer one place up.

b)  Move the layer one place down.

c)  Move the layer to the top of its group.

d)  Move the layer to the bottom of its group.

(2) If it is not possible to move the layer in the layer control in one of the directions, the buttons for that direction should be disabled.

a)  For a top layer of a layer group.

b)  For a bottom layer of a layer group.

2.4.5 Show layer customizer

(1) If you right click on some layers a layer customizer is shown to change some properties of the layer.

a) For route in dossier:



Figure 2-92: Show layer customizer - For route in dossier

b) For route in Layers:

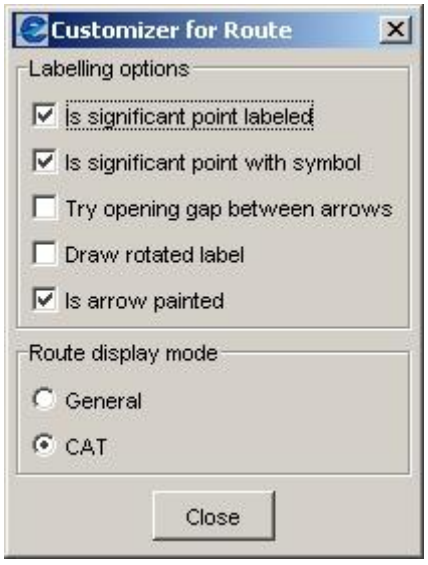


Figure 2-93: Show layer customizer - For route in Layers

c) For airspace in Layers and Dossier:



Figure 2-94: Show layer customizer - For airspace in Layers and Dossier

d) For restriction in dossier:

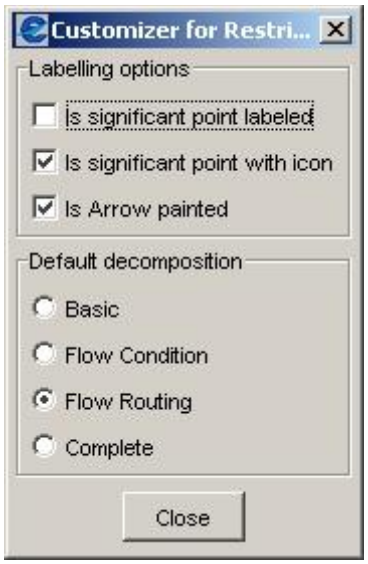


Figure 2-95: Show layer customizer - For restriction in dossier

e) For restriction in Layers:



Figure 2-96: Show layer customizer - For restriction in Layers

2.5 Find Panel

(1) The find panel is displayed as one tab at the right side of the map with name 'Find' when the 'Show overview and properties' button in the toolbar is pressed down.

(2) The button:



Figure 2-97: Find Panel - Button

(3) The find panel:

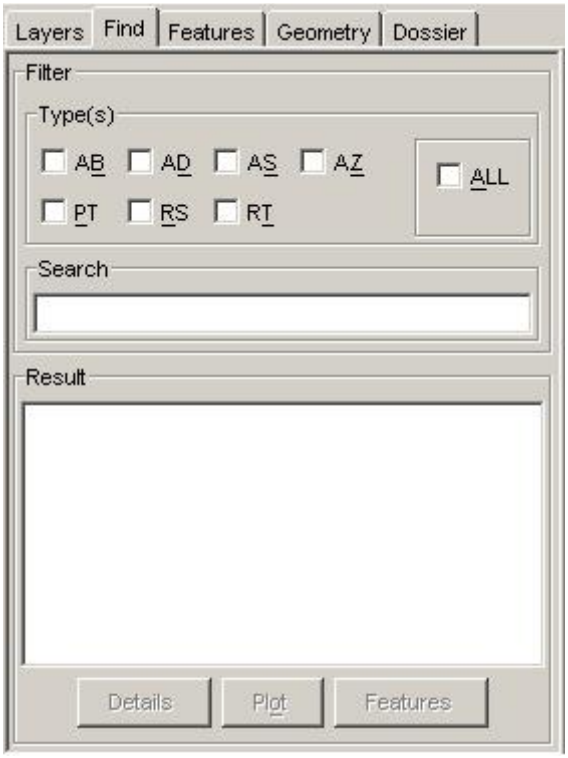


Figure 2-98: Find Panel Window

(4) With the find panel you can search for an object. The object should already be loaded in the map.

a) You have to select a type or multiple types by checking the checkbox of the type. There are 7 types:

i) AB,

ii) AD,

iii) AS,

iv) AZ,

v) PT,

vi) RT and

vii) RS.

b) You can also search in all the types together.

(5) When you check the checkbox of a type, all the elements of that type that are loaded in the map appear in the results list. The first element of the list is selected automatically.

(6) If you move with the cursor over the short description of a type, you see the long description as tooltip.

a) **AB** Airblock

b) **AD** Aerodrome

c) **AS** Airspace

d) **AZ** Set of aerodrome

e) **PT** Point

f) **RT** Route

g) **RS** Restriction

(7) In the search field you can type the name of the searched element. While you are typing the list with the results is adapted.

(8) When one element is selected in the list you can plot it, show the details or show the features.

(9) When multiple elements are selected in the list you can plot them.

2.6 Map Configuration

2.6.1 Introduction

(1) The Map Configuration window has 3 tabs:

a) Layers,

b) Filtering and

c) General.

(2) It contains the settings of the map.

(3) There are 3 buttons at the bottom.

(4) **Apply:**

a) If you check or uncheck checkboxes in the configuration window, nothing happens.

b) When you click 'Apply' the values are applied on the map.

c) The map will be repainted if necessary.

(5) **Reset:**

a) If you check or uncheck checkboxes in the configuration window, nothing happens.

b) When you click 'Reset' the values that are currently applied in the map are put in the configuration window.

(6) **Cancel:**

a) The window is closed without changing anything.

(7) When a setting is changed in another place, this value of the setting will appear immediately in the Map Configuration window, if the window is visible.

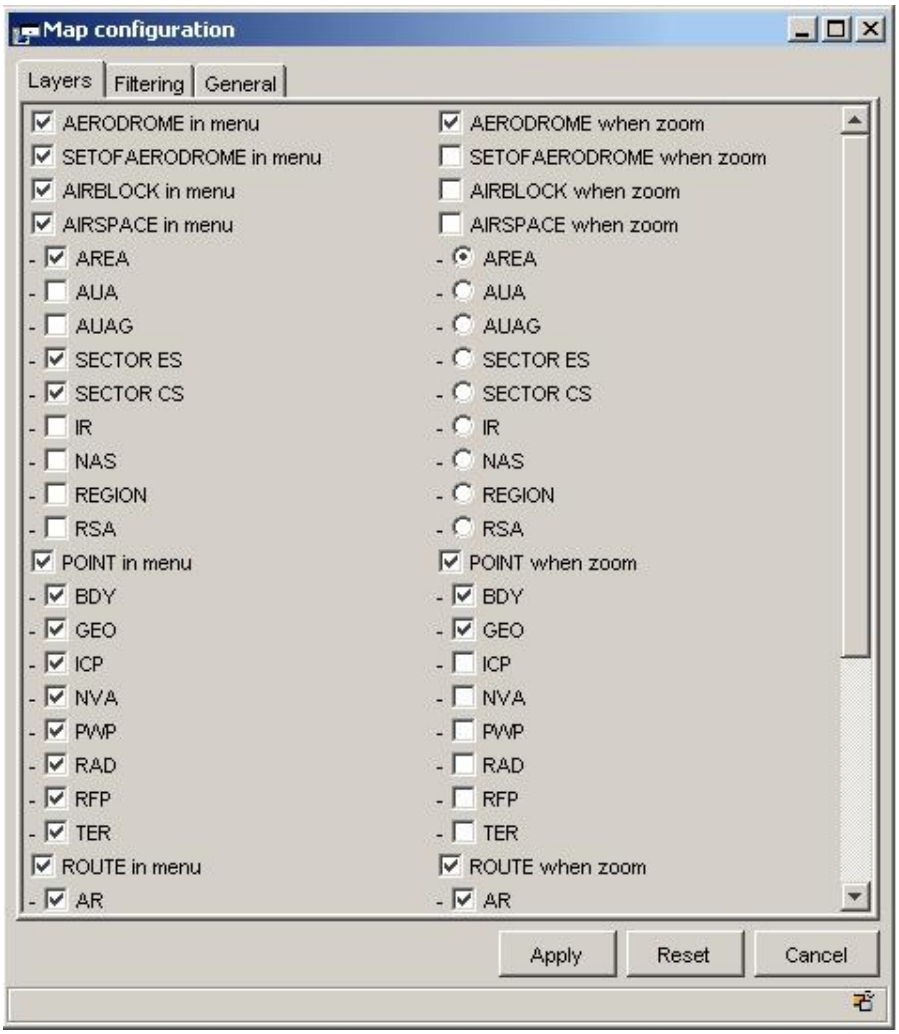
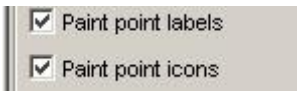


Figure 2-99: Map Configuration - Dialog

(8) Example:

a) Paint Point Labels is currently checked in the configuration/general.



b) Click in the toolbar for the map on:



c) Paint Point Labels will be automatically unchecked in the configuration.



Figure 2-100: Map Configuration - Paint Point Labels example

2.6.2 Layers

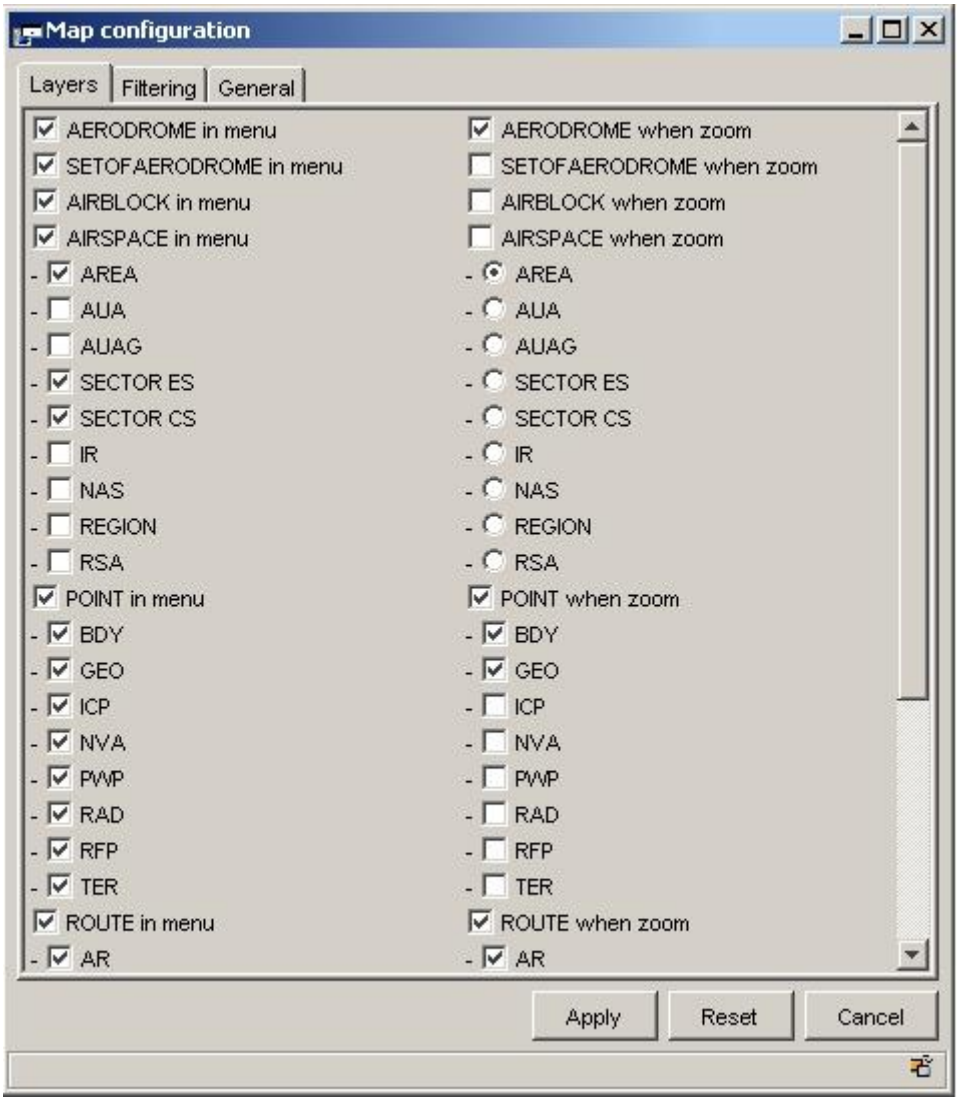


Figure 2-101: Map Configuration - Layers Tab

(1) **Left column:**

a) When you right click on the map the popup menu is shown.

b) The types and subtypes that are checked in the Configuration window and that are under the mouse position will be visible in the popup menu.

(2) Example: We have the following Configuration:

a) Airspace in menu with subtypes:

i) AREA,

ii) SECTOR ES,

iii) REGION.



Figure 2-102: Map Configuration - Layers Tab - Airspace in menu options

b) Right click on the map.

c) There are menus:

i) Layers/Airspace [AREA],

ii) Airspace [REGION],

iii) Airspace [SECTOR ES].



Figure 2-103: Map Configuration - Layers Tab - Airspace menu example

(3) **Right column:**

a) When you right click on the map, the popup menu appears.

b) Select 'Set Zoom/All' or 'Set Zoom/All default'. The range of the types that are selected in the right column of the Configuration is changed.



Figure 2-104: Map Configuration - Layers Tab - Set Zoom example

(4) Move with the mouse over every restriction type in the Map Configuration. You can see the description of these types as tooltip.

a) AF Aerodrome Flight Rule Restriction

b) DDCT Limitation Restriction

c) FFRA DCT Restriction

d) HS Hard / Soft Traffic Flow Restriction

e) PT Profile Tuning Restriction

f) T Flight Property Restriction on Terminal Procedure

g) SC SSR Code Allocation Restriction

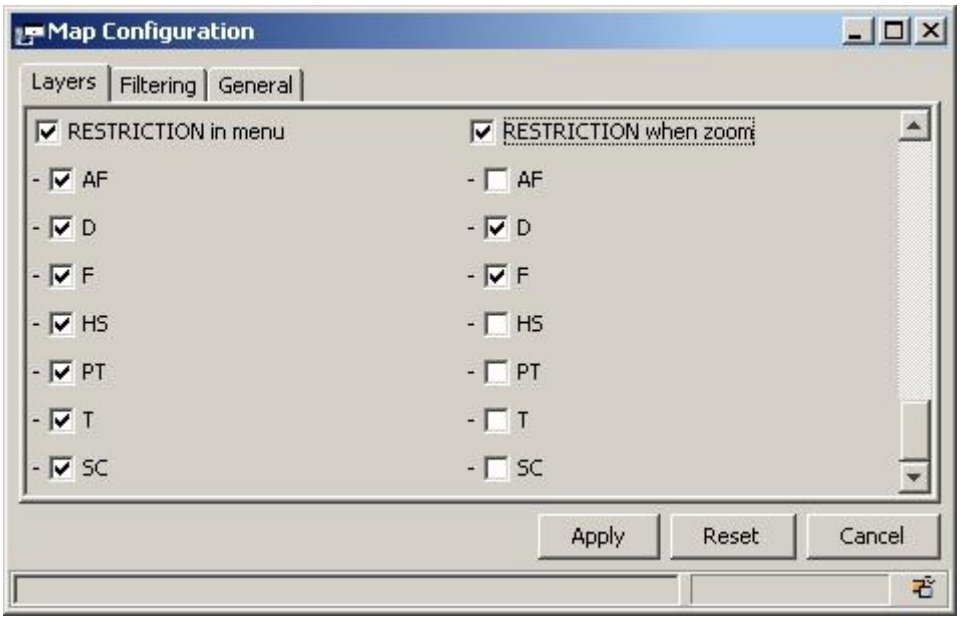


Figure 2-105: Map Configuration - Layers Tab - RESTRICTION options

2.6.3 Filtering

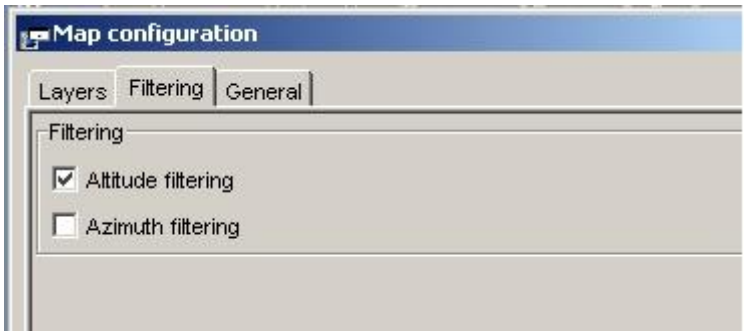


Figure 2-106: Map Configuration - Filtering Tab

2.6.3.1 Altitude filtering

(1) When 'Altitude filtering' is checked in the map configuration there is altitude filtering on the map, otherwise not.

(2) When there is no altitude filtering the panel is disabled:

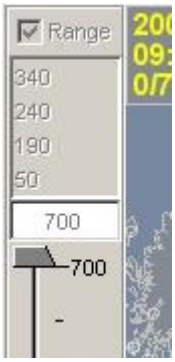


Figure 2-107: Map Configuration - Filtering Tab - Altitude filtering example

2.6.3.2 Azimuth filtering

(1) When 'Azimuth filtering' is checked in the map configuration there is azimuth filtering on the map, otherwise not.

(2) When there is azimuth filtering, you see the azimuth filter icon on the top right of the map:



Figure 2-108: Map Configuration - Filtering Tab - Azimuth filtering example

2.6.4 General

2.6.4.1 Introduction

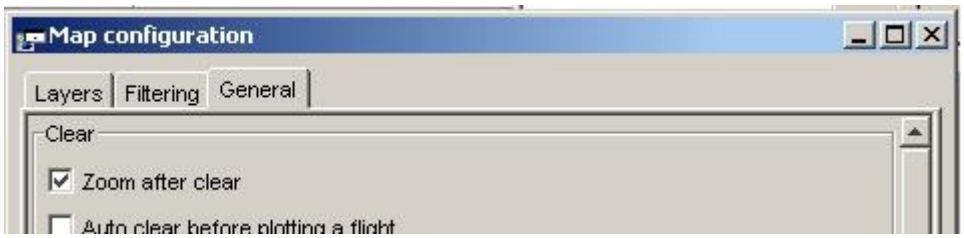


Figure 2-109: Map Configuration - General Tab

(1) The third tab of the Configuration window is 'General'. There are 5 sections:

2.6.4.2 Clear

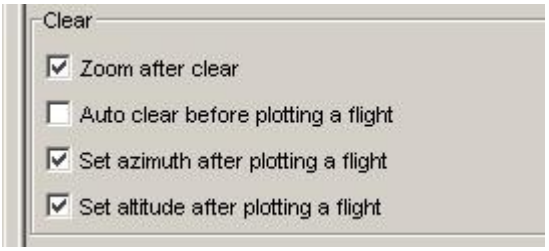


Figure 2-110: Map Configuration - General Tab - Clear Section

2.6.4.3 Flight

(1) When a flight is plotted on the map, the elements the flight goes through are also plotted on the map if they are checked in the Configuration window.

(2) Aerodrome, point, SID/STAR, route, violated restriction and complying restrictions can be plotted with the flight.

(3) If Show features with flight is checked the features of the flight are displayed together with the flight.

(4) If Keep map time when plotting flight data is checked, each time a flight is plotted, the time of the map is not changed. Default value is Unchecked. Other time change manual or coming from monitor plot are still allowed.

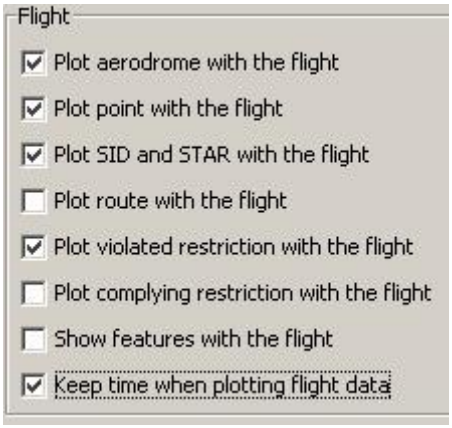


Figure 2-111: Map Configuration - General Tab - Flight Section

(5) Example:

a) Plot a flight.

b) For the Configuration above, the aerodromes and the violated restrictions are also plotted and added to the dossier.



Figure 2-112: Map Configuration - General Tab - Flight Section example

2.6.4.4 Zoom after plot

(1) The bounds of the map can be changed after plotting an object.

(2) Following possibilities are available, only one should be chosen:

a) **None:** The zoom is not changed after plotting an object.

b) **Dossier:** The bounds are set that all the elements of the dossier fit in the map.

c) **Objects:** The bounds set that all the plotted elements fit in the map.

d) **Only on flight:** The bounds set that only the plotted flight fits in the map. If not a flight is plotted the zoom is not changed.

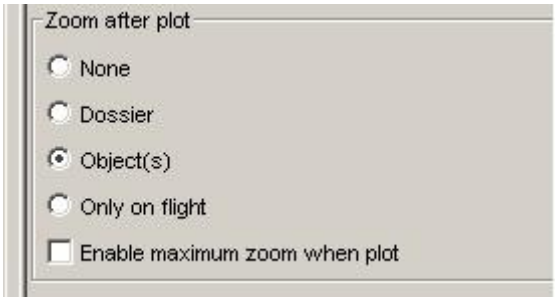


Figure 2-113: Map Configuration - General Tab - Zoom after plot Section

2.6.4.5 Painting



Figure 2-114: Map Configuration - General Tab - Painting Section

(1) **Draw geodesic lines:**

a) Unchecked: lines are drawn as a straight line on the map.



b) Checked: the real shape of the line is drawn.



Figure 2-115: Map Configuration - Geodesic lines example

(2) **Draw circle around points and aerodromes:**

a) Unchecked: no circles around points and aerodromes of the dossier.



b) Checked: yellow circles around points and aerodromes of the dossier.



Figure 2-116: Map Configuration - Draw circle around points and aerodromes example

(3) **Paint labels filled:**

a) Unchecked: The labels have no background.



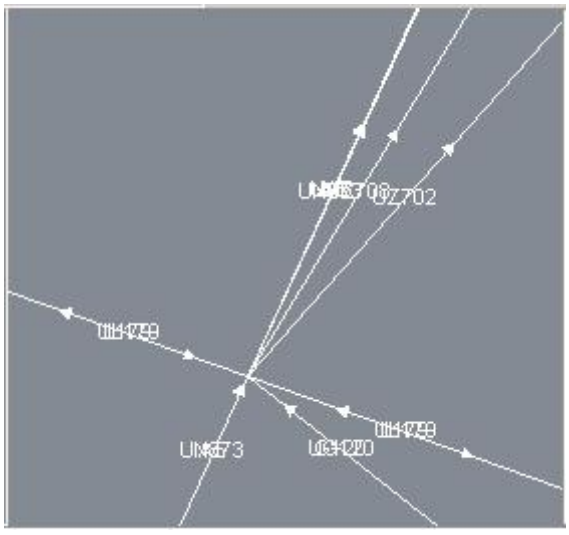
b) Checked: The labels have a grey background (the colour of the countries).



Figure 2-117: Map Configuration - Paint labels filled example

(4) **Label displace:**

a) Unchecked: labels are painted in there preferred position, but it is possible that multiple labels overlap.



b) Checked: if multiple labels are painted, the application tries to find a free place to paint the label.

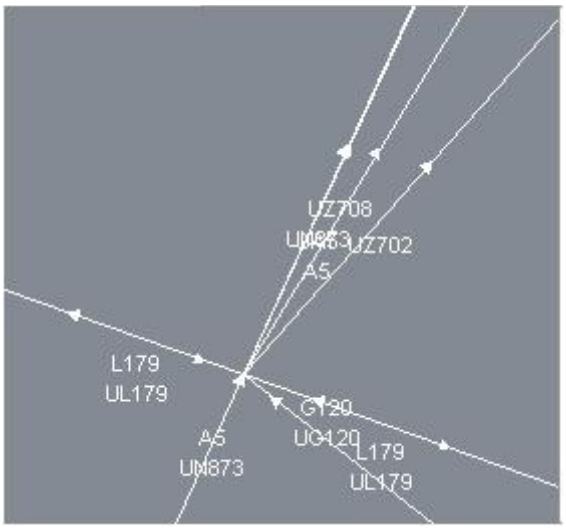


Figure 2-118: Map Configuration - Label displaced example

(5) **Paint short labels:**

a) Unchecked: the whole label is painted.



b) Checked: a shorter version of the label is painted.



Figure 2-119: Map Configuration - Paint short labels example

(6) **Paint point labels:**

a) Unchecked: in every layer there are no labels for points painted, even if the labels of the layer are visible.

b) Checked: the labels of the points can be painted if the labels of the layer are visible.

(7) **Paint point icons:**

a) Unchecked: in every layer there are no points painted, even if the layer is visible.

b) Checked: the points can be painted if the layer is visible.

(8) **Paint small point and aerodrome symbols:**

a) Unchecked: the point and aerodrome icons are painted big.

Layer:  Dossier: 

b) Checked: the point and aerodrome icons are painted small.

Layer:  Dossier: 

Figure 2-120: Map Configuration - Paint small point and aerodrome symbols example

(9) **Show time altitude icon:**

a) Unchecked: No icon with time and altitude on the map.

b) Checked: An icon with time and altitude is displayed in the top left corner of the map

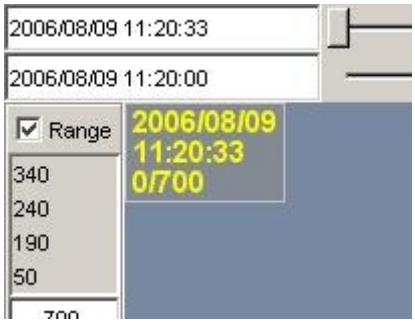
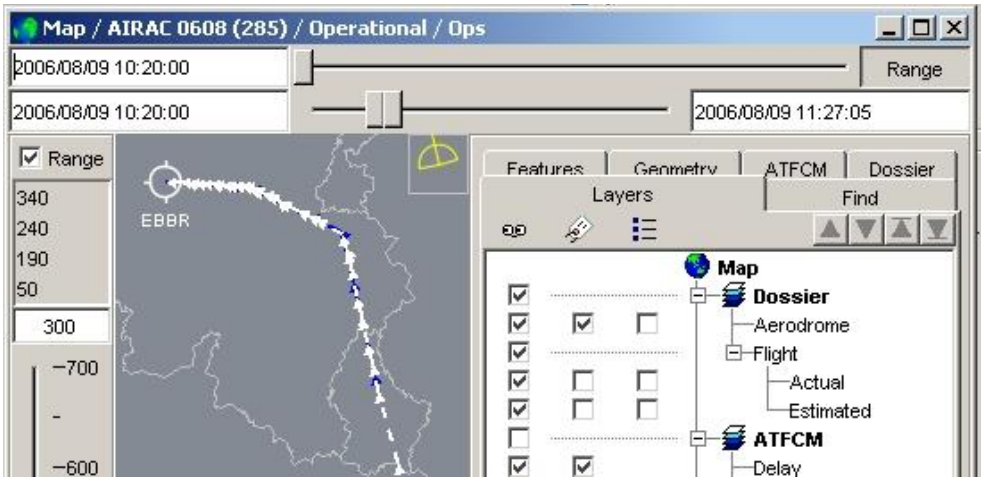


Figure 2-121: Map Configuration - Show time altitude icon example

(10) **Overview visible:**

a) Unchecked: The overview is not visible.



b) Checked: The overview is displayed on the right top of the map.

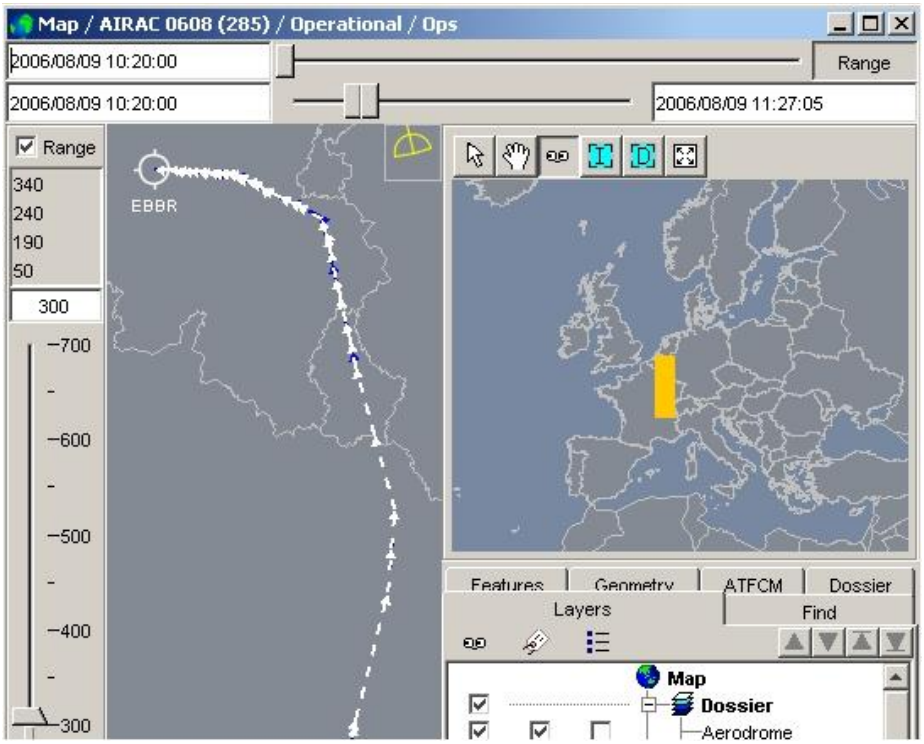


Figure 2-122: Map Configuration - Overview visible example

(11) **Geometry tab visible:**

a) Unchecked: the geometry tab is not displayed.

b) Checked: the geometry tab is displayed.

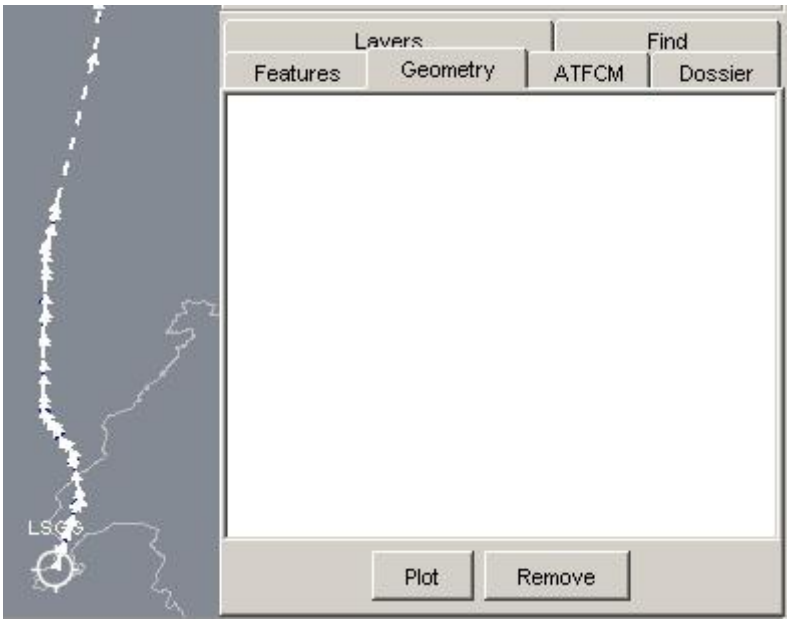


Figure 2-123: Map Configuration - Geometry tab visible example

(12) **Allow high zoom:**

a) Unchecked: there is a limit for the zoom of the map.

b) Checked: you can zoom deeper into the map.

2.6.4.6 Projection

2.6.4.6.1 Introduction

(1) There are different projections possible for the map.

(2) The default is Mercator.

(3) You can choose other projections in the combo box. The others are:

a) Equidistant cylindrical

b) Mercator

c) Lambert Conformal

d) Gnomic

e) Orthographic

f) Polar Stereographic

g) Stereographic

h) Vertical Perspective

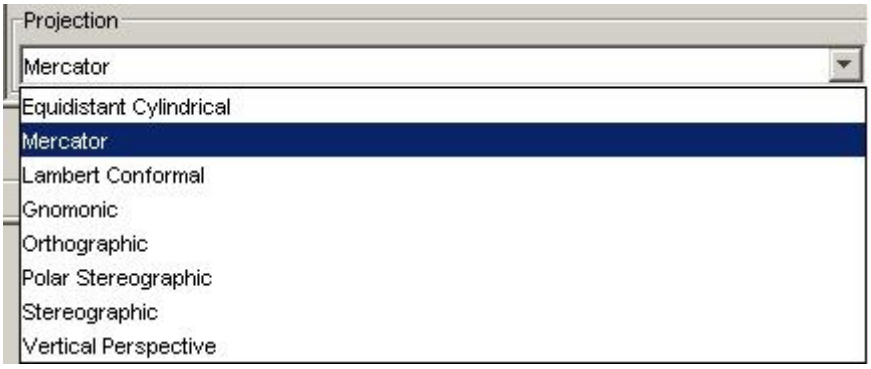


Figure 2-124: Map Configuration - Projection Section

2.6.4.6.2 Equidistant cylindrical



Figure 2-125: Map Configuration - Equidistant cylindrical Projection example

2.6.4.6.3 Mercator



Figure 2-126: Map Configuration - Mercator Projection example

2.6.4.6.4 Lambert Conformal



Figure 2-127: Map Configuration - Lambert Conformal Projection example

2.6.4.6.5 Gnomic



Figure 2-128: Map Configuration - Gnomic Projection example

2.6.4.6.6 Orthographic



Figure 2-129: Map Configuration - Orthographic Projection example

2.6.4.6.7 Polar Stereographic



Figure 2-130: Map Configuration - Polar Stereographic Projection example

2.6.4.6.8 Stereographic



Figure 2-131: Map Configuration - Stereographic Projection example

2.6.4.6.9 Vertical Perspective

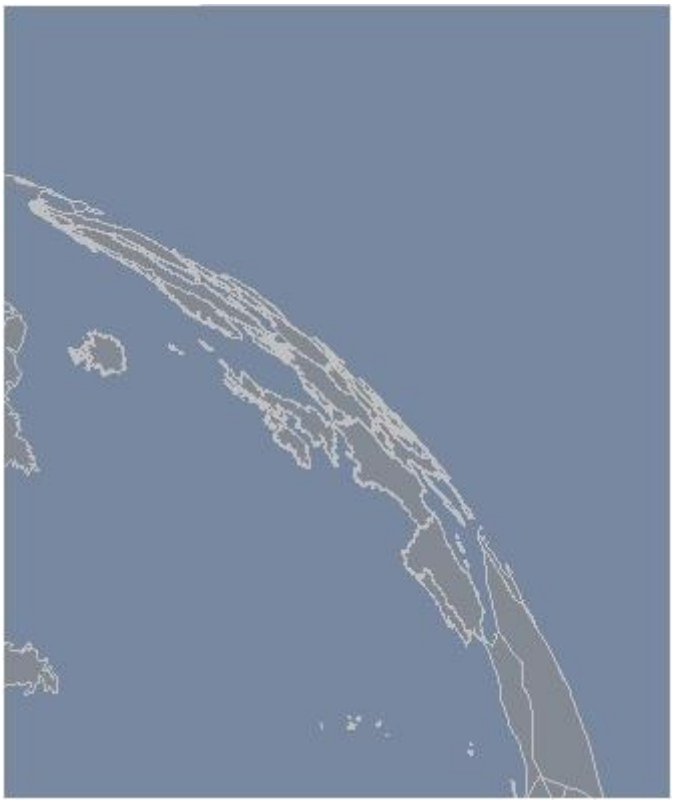


Figure 2-132: Map Configuration - Vertical Perspective Projection example

2.7 Map filter

2.7.1 Time filtering

(1) The time filter is displayed at the top of the map:



Figure 2-133: Map filter - Time filter control

(2) There are two sliders:

a) The lower ones will define a time period, with a start date and an end date.

b) The top one is for fine-tuning within the selected time period.

(3) You can disable the time range by clicking the range button on the right top of the filter.

(4) Click the button and the range slider disappears:



Figure 2-134: Time filter control - Disabled time range

(5) The position of each slider determines a date & time value.

(6) The correspondence between slider & value is as shown here:

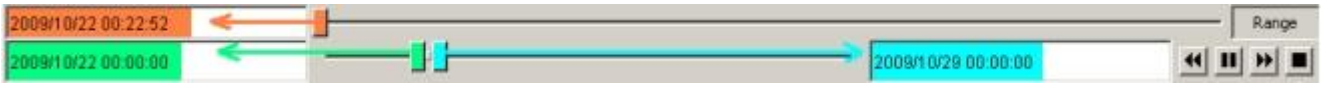


Figure 2-135: Time filter control - Date and time indicator

(7) On the lower slider, double-clicking on a cursor will send it to its extreme positions (left end for left cursor, right end for right cursor):



Figure 2-136: Time filter control - double click example

(8) A 'time range' can be moved about, in a drag & drop fashion, using the left mouse button:



Figure 2-137: Time filter control - Drag and drop of time range

(9) Double clicking on the band will send both cursor to their respective end and therefore maximize the time band.

(10) Date & time can also be typed directly in the white boxes showing cursor value for greater accuracy.

(11) The display will change dynamically in function of the selected time.

(12) You can force alignment of the current MAP time on the current UTC by clicking the **Set current time** button on the right top of the filter, next to the Range button.

(13) Clicking the **Set current time** button might update the current MAP period setting (i.e. if the current time is outside the current time boundaries currently set on the map)



Figure 2-138: Time filter control - Set current time button

(14) To show / hide the **Set current time** button:

a) Open the Map Configuration window.

b) Check / un-check the property **General / Layout / 'Set current time' button visible**.

(15) The time filter is always active.

(16) The time of the horizontal view and the vertical view is always the same.

(17) If the time in one view is changed, it will change also in the other view to the same time.

2.7.2 Time simulator

(1) On the right of the lower slider the time simulator can be activated by clicking on the single arrow button. When simulator is activated, other time changes cannot be done. But the data can be updated (E.g. Flight data or monitor update)



Figure 2-139: Time simulator-Activate button

(2) On the right of the lower slider the time simulator can be paused by clicking on the pause button (i.e. 2 vertical bars). When simulator is paused, other time changes cannot be done. But the data can be updated (E.g. Flight data or monitor update)



Figure 2-140: Time simulator-Pause button

(3) On the right of the lower slider the time simulator can be stopped by clicking on the square button. When simulator is stopped other time changes can be done.



Figure 2-141: Time simulator-Stop button

(4) On the right of the lower slider the time simulator can be accelerated by clicking on the double right arrow button. When simulator is active and accelerated, other time changes cannot be done. But the data can be updated (E.g. Flight data or monitor update).



Figure 2-142: Time simulator-Accelerate button

(5) On the right of the lower slider the time simulator can be decelerated by clicking on the double left arrow button. When simulator is active and decelerated, other time changes cannot be done. But the data can be updated (E.g. Flight data or monitor update).



Figure 2-143: Time simulator-Decelerate button

(6) To show / hide the **Time simulator** control:

a) Open the Map Configuration window.

b) Check / un-check the property **General / Layout / Time simulator visible**.

2.7.3 Altitude filtering

(1) You can enable or disable altitude filtering in the map configuration.

(2) Only entities that contain level information in their own definitions will be affected by altitude filtering. Typically:

a) AS,

b) RT,

c) RS.

(3) The other entities are virtually available at all levels.

(4) You find the altitude filter at the left side of the map.

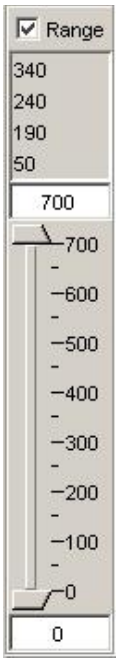


Figure 2-144: Map filter - Altitude filter control

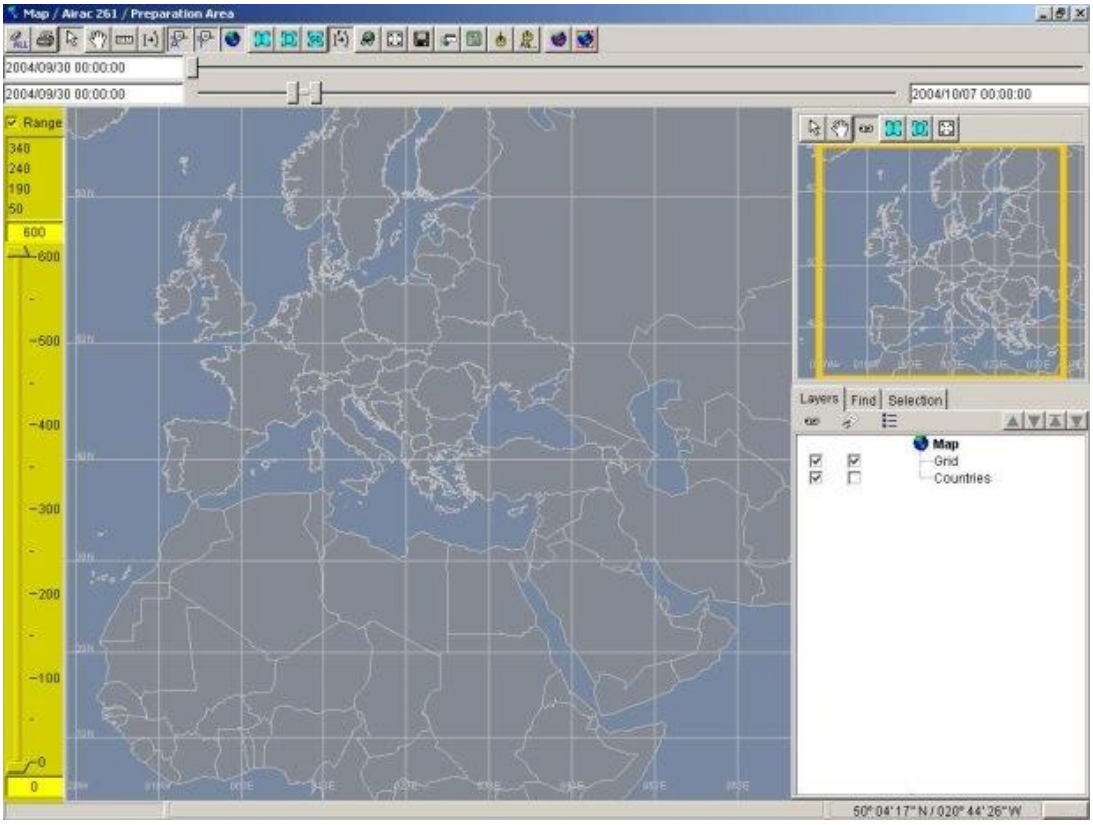


Figure 2-145: Map filter - Altitude filter control overview

(5) Be aware that the default Level band on opening a new map is GND to UNL.

(6) Two modes are possible:

a) single level or

b) range.

2.7.3.1 Single level mode

(1) For single level mode the 'range' button must be deselected.

(2) Only one cursor is shown on the slider.



Figure 2-146: Altitude filter control - Single level mode

(3) Pre-sets are available (FL340, 240, 190, 50).

(4) Click on the altitude and the slider is adapted. You can also move the slider.

(5) The data on your screen will be shown dynamically at the selected level. Depending on the type of data, this will have or not have an impact.

2.7.3.2 Range mode

(1) The range mode is the default mode.

(2) The 'range' button must be selected.

(3) Two cursors are shown.

(4) In range mode, pre-sets are available but they will join the two cursors together, resulting in a single level selection;

(5) Moving them about is not possible unless you press the SHIFT button at the same time;

(6) Levels can be typed directly in the white boxes;

(7) The display will change dynamically to show all the data that exists anywhere INSIDE the range;

(8) Like for the time, you can select a 'level band', and move it up & down, in a drag & drop fashion, using the left mouse button.

(9) Double clicking on a cursor will 'send it to its extreme position, like on the time filter.

(10) Let's note here that selecting a level range GND-UNL (CHMI values: 0-700) will have the same effect as deactivating the 'altitude filter' in the map configuration.

2.7.4 Azimuth filter

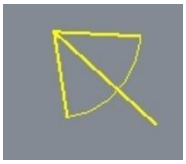


Figure 2-147: Map filter - Azimuth filter control

(1) The Azimuth filter is composed of an angle and direction.

(2) To set the azimuth angle and direction, use the azimuth controller (see controllers).

(3) When azimuth filtering is active all routes (also routes of restrictions) in that direction and angle will be displayed. The other ones are not displayed.

(4) Flights are not filtered with azimuth filtering.

(5) The routes and restrictions in the reference layers and in the dossier layers are filtered.

(6) You can enable azimuth filtering in the Map Configuration or with the button of the toolbar.



Figure 2-148: Map filter - Azimuth filter button

(7) Azimuth filtering is active when you see the azimuth filter icon at the top right of the map, otherwise it is not active:



Figure 2-149: Map filter - Azimuth filter active icon

2.8 Popup menu and popup dialog

2.8.1 Popup Menu

(1) When you right click on the map the popup menu is shown.



Figure 2-150: Popup menu

(2) If you click on 'Show in dialog' all the actions that are in the popup are shown in a dialog.

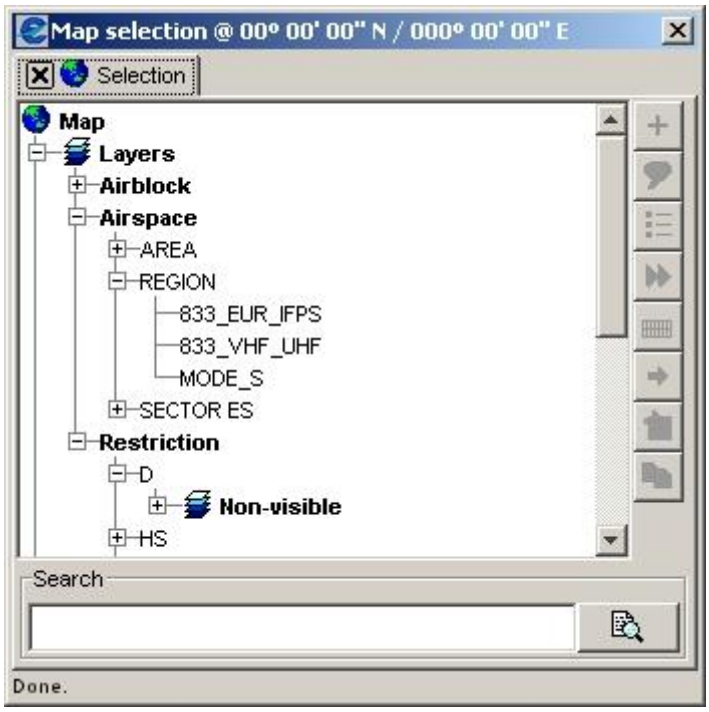


Figure 2-151: Popup menu - Show in dialog

(3) A big red point on the map shows for which position the dialog is.

2.8.2 Popup Dialog

(1) You can show the popup dialog by right clicking on the map and selecting 'Show in dialog'.

(2) You can also show it by right clicking on the map while the CTRL key is pressed.

2.8.3 Last Click

(1) Show the popup dialog by right clicking on the map and selecting 'Show in dialog'.

(2) On the position where you clicked on the map there should be a red point.

a) When you pan the map you should still see the point.

b) When you zoom in or out, you should still see the point.

(3) Close the popup dialog. The red point should disappear.

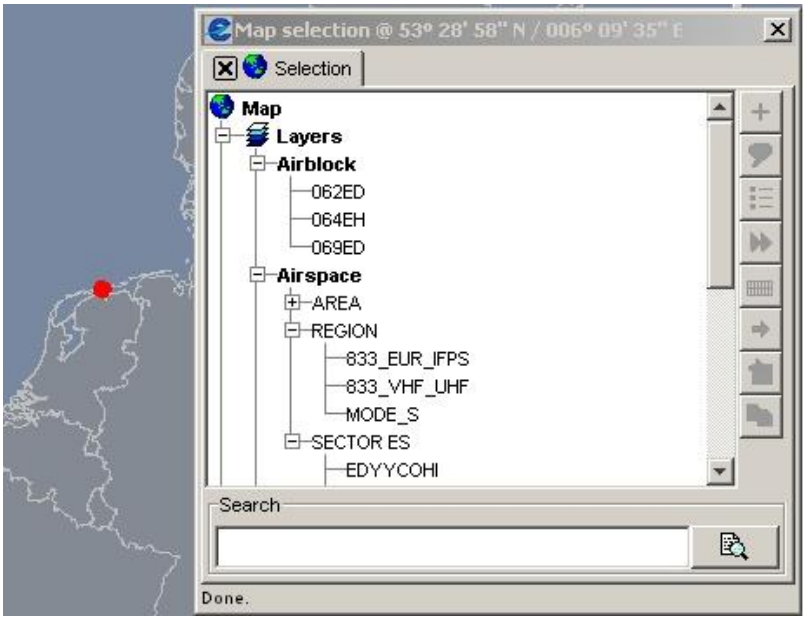


Figure 2-152: Popup menu - Last click red point

2.9 Distance Calculation

(1) Click on  to open the 'Distance Calculation' window.



Figure 2-153: Distance Calculation Button

(2) With the distance calculation tool, it is possible to calculate the distance and the time needed to go from one point (aerodrome, point or coordinate) to another in a given speed.

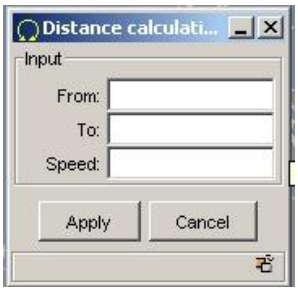


Figure 2-154: Distance Calculation Dialog

(3) You have to fill in the dialog:

a) **Origin (From) and destination (To)** should be a known point, aerodrome or coordinate.

i) **Point:** KOK

ii) **Aerodrome**: EHVB

iii) **Coordinate:** 515408N0043217E

51 degrees
54 minutes
08 seconds
N for north
S for south
004 degrees
43 minutes
21 seconds
E for east
W for west

b) **Speed**: in ICAO format:

100	Nautical miles
N100	Nautical miles
M100	Mach
K100	Knots

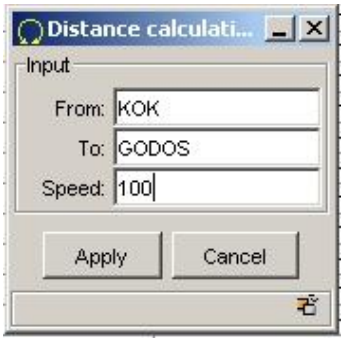


Figure 2-155: Distance Calculation Dialog - Example

(4) Click 'Apply'. The results are displayed.

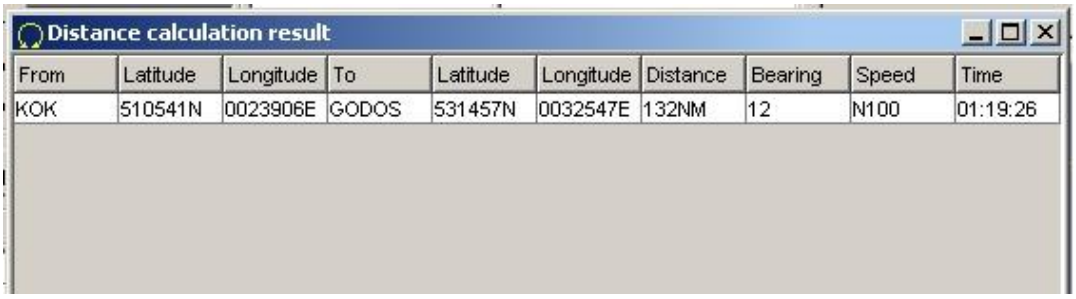


Figure 2-156: Distance Calculation - Single Result Window

(5) It is possible to enter more origins and destinations. They should be divided by a ' '.

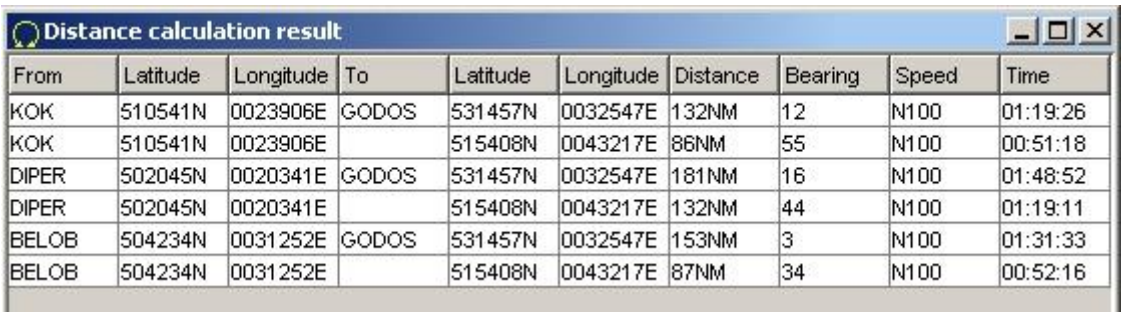
(6) Enter:

a) Origin: KOK DIPER BELOB

b) Destination: GODOS 515408N0043217E

c) Speed: M100

(7) Click 'Apply'.



From	Latitude	Longitude	To	Latitude	Longitude	Distance	Bearing	Speed	Time
KOK	510541N	0023906E	GODOS	531457N	0032547E	132NM	12	N100	01:19:26
KOK	510541N	0023906E		515408N	0043217E	86NM	55	N100	00:51:18
DIPER	502045N	0020341E	GODOS	531457N	0032547E	181NM	16	N100	01:48:52
DIPER	502045N	0020341E		515408N	0043217E	132NM	44	N100	01:19:11
BELOB	504234N	0031252E	GODOS	531457N	0032547E	153NM	3	N100	01:31:33
BELOB	504234N	0031252E		515408N	0043217E	87NM	34	N100	00:52:16

Figure 2-157: Distance Calculation - Multiple Results Window

(8) Enter a wrong Origin (KOOK).

(9) Click 'Apply'.

(10) An error message '[KOOK] is not a valid point.' is displayed.



Figure 2-158: Distance Calculation - Invalid Origin Point Message

(11) Enter a wrong Destination (GRODOS).

(12) Click 'Apply'.

(13) An error message '[GRODOS] is not a valid point.' is displayed.



Figure 2-159: Distance Calculation - Invalid Destination Point Message

(14) Enter a wrong Origin coordinate (515408N0043217).

(15) Click 'Apply'.

(16) An error message '[515408N0043217] is not a valid point.' is displayed.



Figure 2-160: Distance Calculation - Invalid Coordinate Message

(17) Enter a wrong speed (100M).

(18) Click 'Apply'.

(19) An error message 'The speed is not a valid number.' is displayed.



Figure 2-161: Distance Calculation - Invalid Speed Message

(20) Click 'Cancel'.

(21) The Distance Calculation dialog disappears.

2.10 Countries

(1) A common layer is the country layer.

(2) Like all other layers, it can be hidden or shown.

(3) It is always shown in the layer control.

(4) It is visible by default.

2.10.1 Geographical Display

2.10.1.1 Country

(1) The country border is in white.

(2) The country is filled with grey.



Figure 2-162: Countries Geographical Display - Country

2.10.1.2 Labels

(1) Country labels are painted in black.



Figure 2-163: Countries Geographical Display - Labels

2.10.1.3 Selected country

(1) A country can be selected by right clicking on the country.

(2) The popup menu appears.

(3) Select Countries and then the name of the country.

(4) When you go with the cursor over the name of the country the country is selected and painted in yellow.

(5) The name of the country is painted as label.

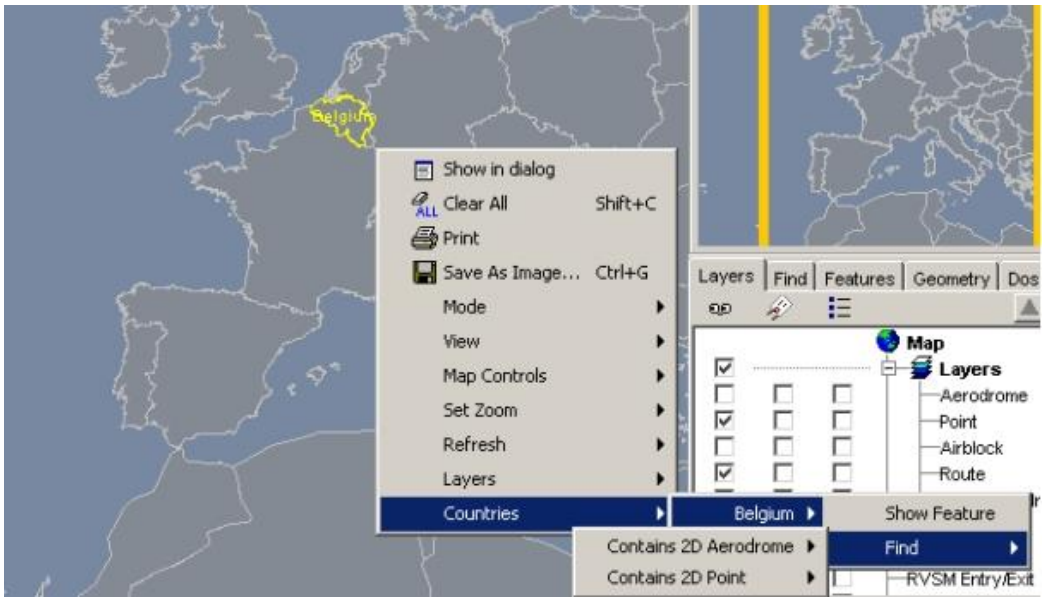


Figure 2-164: Countries Geographical Display - Selected Country

2.10.1.4 Show feature

(1) Select 'Show Feature' in the popup menu.

(2) The features of the country are displayed in the Feature tab.

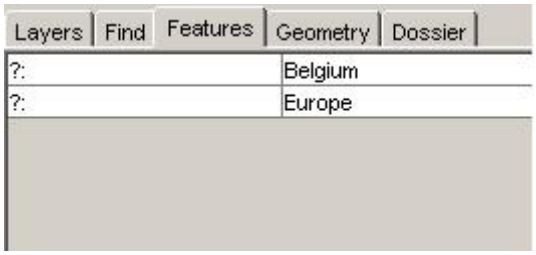


Figure 2-165: Countries Geographical Display - Selected Country / Show Feature

2.10.1.5 Find / Contains 2D Aerodrome

(1) Select 'Contains 2D Aerodrome'.

(2) This finds all the aerodromes in the country.

(3) All the aerodromes can be added to the dossier.

(4) For every aerodrome actions can be done (see the popup for aerodrome).

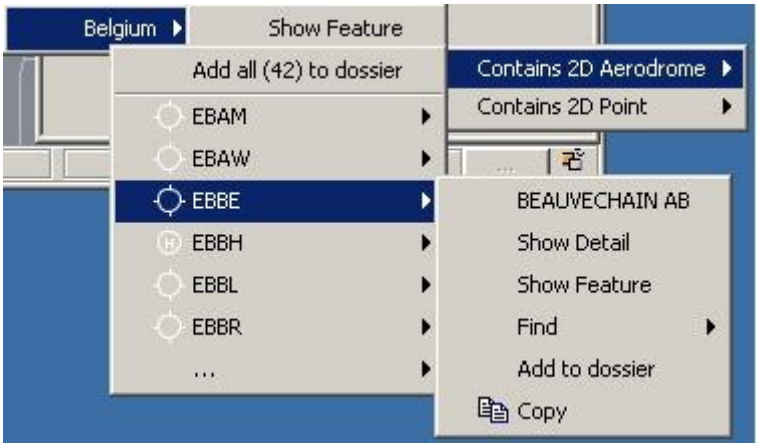


Figure 2-166: Countries Geographical Display - Find / Contains 2D Aerodrome

2.10.1.6 Find / Contains 2D Point

(1) Select 'Contains 2D Point'.

(2) This finds all the points in the country.

(3) All the points can be added to the dossier.

(4) For every point actions can be done (see the popup for point).

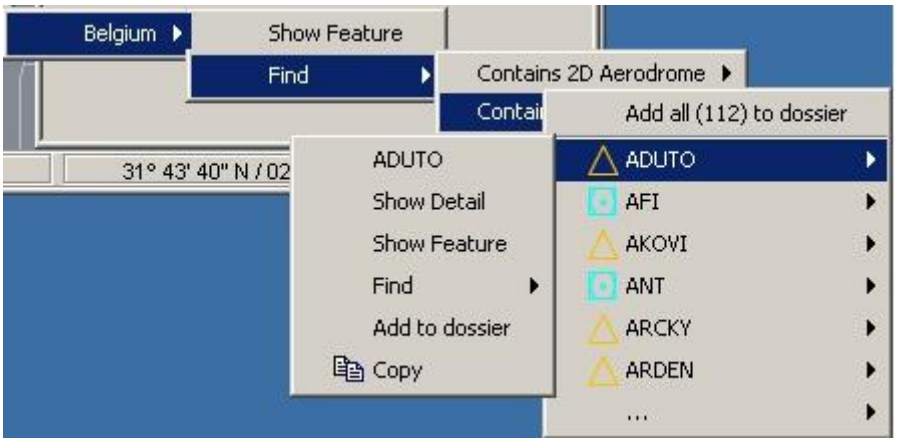


Figure 2-167: Countries Geographical Display - Find / Contains 2D Points

2.11 Grid

2.11.1 Introduction

(1) A common layer is the grid layer.

(2) Like all other layers it can be hidden or shown.

(3) It is not visible by default.

2.11.2 Geographical Display

2.11.2.1 Grid

(1) The grid is painted in white.



Figure 2-168: Grid Geographical Display - Grid

2.11.2.2 Labels

(1) Labels are painted in white.

2.12 IFPZ AREA

2.12.1 Introduction

(1) A layer that is available when the map is loaded for FLIGHT and ATFCM is the layer 'IFPZ AREA'.

(2) Like all other layers it can be hidden or shown.

(3) The layer is not visible by default.

2.12.2 Geographical Display

2.12.2.1 AREA

(1) The IFPZ AREA has a white border and it is filled with transparent white.



Figure 2-173: IFPZ AREA Geographical Display- Area

2.12.2.2 Label

(1) The label IFPZ is painted in the middle of the area in white.

(2) The labels with the coordinates of the points of the polygon can also be shown if the 'coordinate' property in the layer control is checked.



Figure 2-174: IFPZ AREA Geographical Display- Label

(3) The labels with the coordinates of the polygon points:



Figure 2-175: IFPZ AREA Geographical Display- Label with coordinate

2.12.2.3 Selected IFPZ AREA

(1) An IFPZ AREA can be selected.

(2) Open first the 'Map Configuration'.

(3) Check the property 'Airspace in menu' and 'AREA'.

(4) Right click on the map above the area.

(5) The popup menu appears with an item IFPZ AREA. Select IFPZ AREA. It has one sub item IFPZ.

(6) Go with the mouse over IFPZ.

(7) The area is selected: the border is painted in yellow and it is filled with transparent white.

(8) The label is painted in yellow. If the coordinate property of the layer is checked, than also the labels with the coordinates of the points are painted in yellow.

(9) If you uncheck the property 'AREA' in the 'Map Configuration' than the item IFPZ AREA does not appear in the popup.

(10) Selected IFPZ AREA:

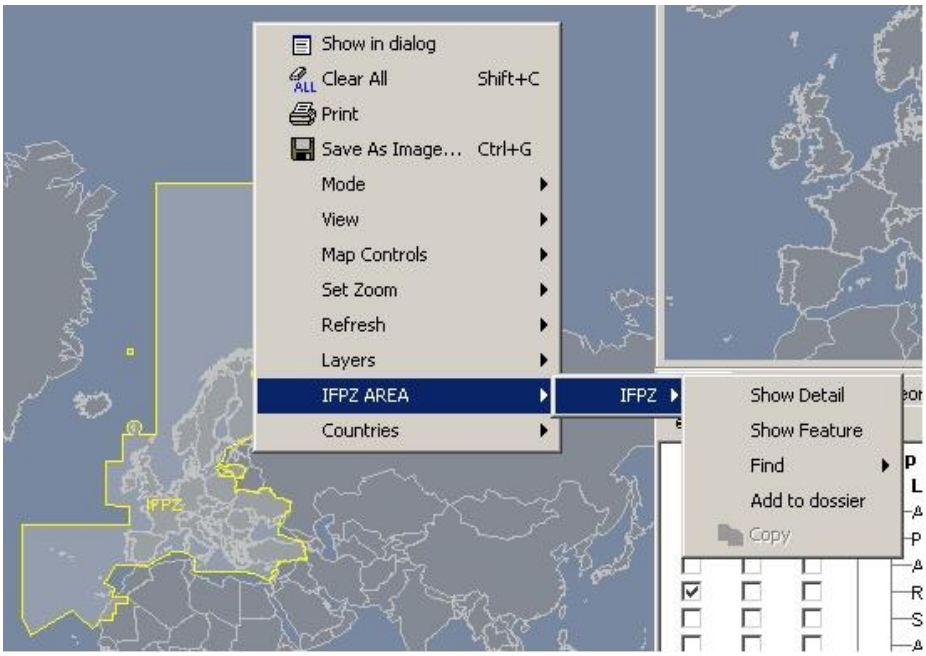


Figure 2-176: IFPZ AREA Geographical Display- Selected IFPZ AREA

(11) Selected IFPZ AREA with coordinates:

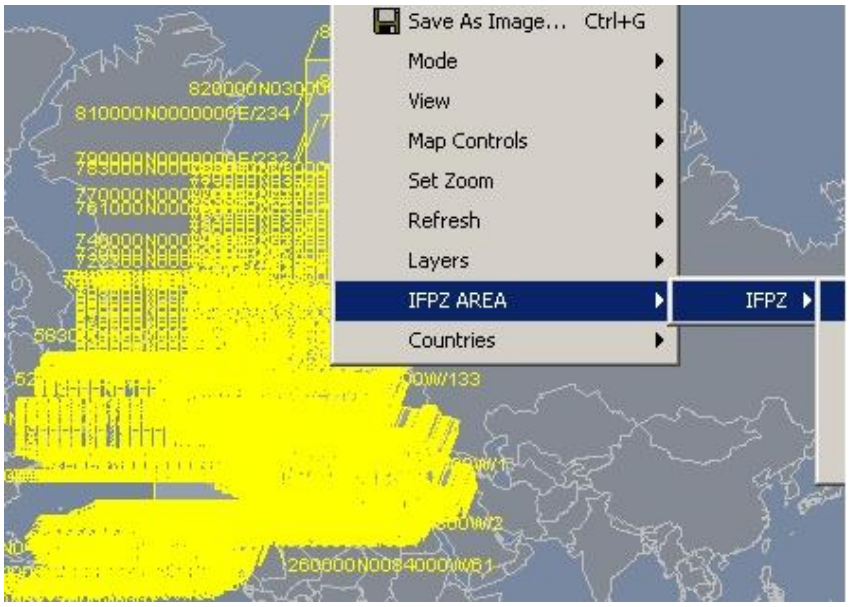


Figure 2-177: IFPZ AREA Geographical Display- Selected IFPZ AREA with coordinate

(12) When you select the IFPZ AREA in the popup menu you see 4 enabled submenus:

a) 'Show Detail'

b) 'Show Feature'

c) 'Find'

d) 'Add to dossier'

2.13 Feature Panel

(1) The feature panel is displayed as one tab at the right side of the map with name 'Feature' when the 'Show overview and properties' button in the toolbar is pressed down.

(2) When no features are displayed the panel is grey.

(3) The button:



Figure 2-178: Feature Panel Button

(4) The feature panel:

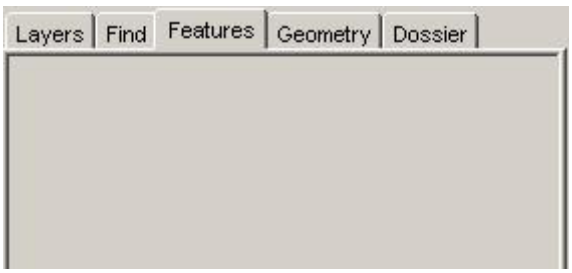


Figure 2-179: Feature Panel

(5) You can show the features of an object in different ways:

a) Select an object in the popup menu of the horizontal view. Select 'Show Feature'.

b) Select an object in the popup menu of the dossier explorer. Select 'Show Feature'.

c) Select an object in the popup window. Select the 'Show Feature' button.

2.14 Legend

2.14.1 Introduction

(1) Click on  to open the legend window.



Figure 2-180: Map Legend Button

(2) In the legend you can see the representation of the different types and subtypes on the map.

(3) You can see also the representation when the type or subtype is selected.

(4) There are 5 tabs:

a) Aerodrome,

b) Point,

c) Airblock & Airspace,

d) Route,

e) Restriction.

2.14.2 Aerodrome

(1) You see the representation of the different types of aerodromes in default state and selected state.

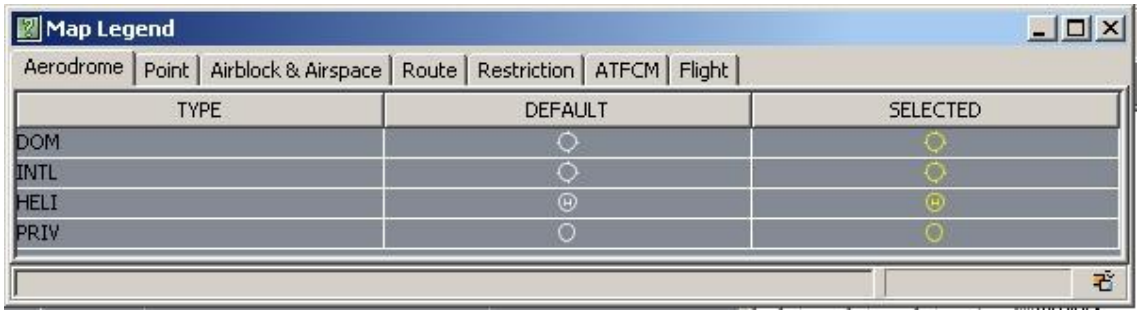


Figure 2-181: Map Legend - Aerodrome Tab

(2) Move with the mouse over every aerodrome type. You can see the description of these types as tooltip.

a) DOM Domestic Aerodrome

b) INTL International Aerodrome

c) HELI Aerodrome for Helicopters Only

d) PRIV Private Aerodrome

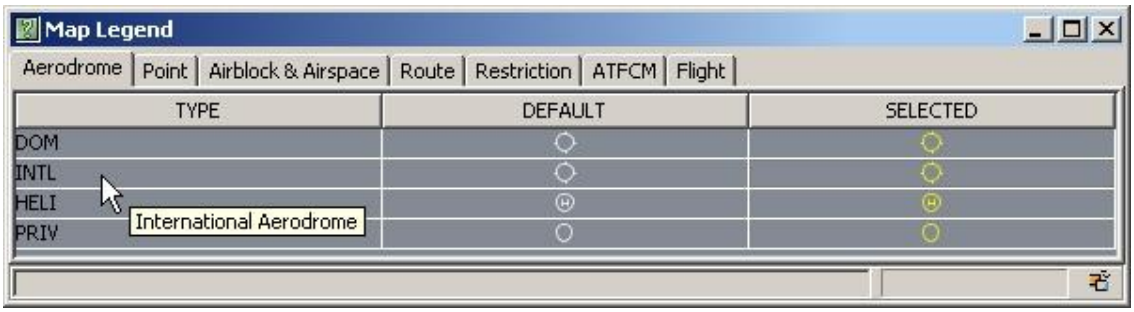


Figure 2-182: Map Legend - Aerodrome Tab Tooltip

2.14.3 Point

(1) On the Point tab legend, you see the following representations:

a) The different ENV types and subtypes of points in normal state, selected state, as RVSM Entry/Exit and as IFPS Entry/Exit in the first table.

b) The different IFPS/ETFMS point types in the second table.

c) The different IFPS point types in the third table.

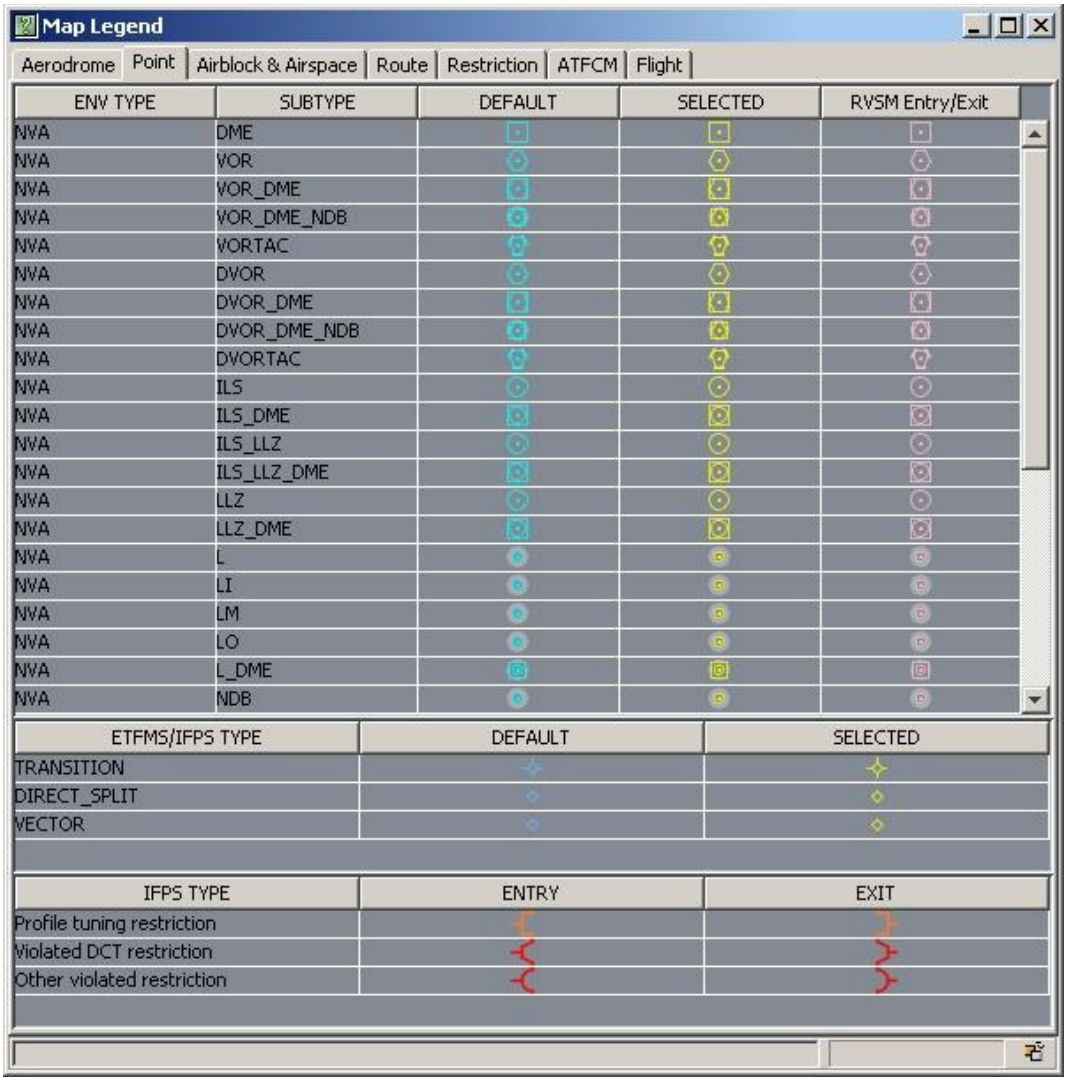


Figure 2-183: Map Legend - Point Tab

(2) Scroll down the first table. Then you see points without subtypes.

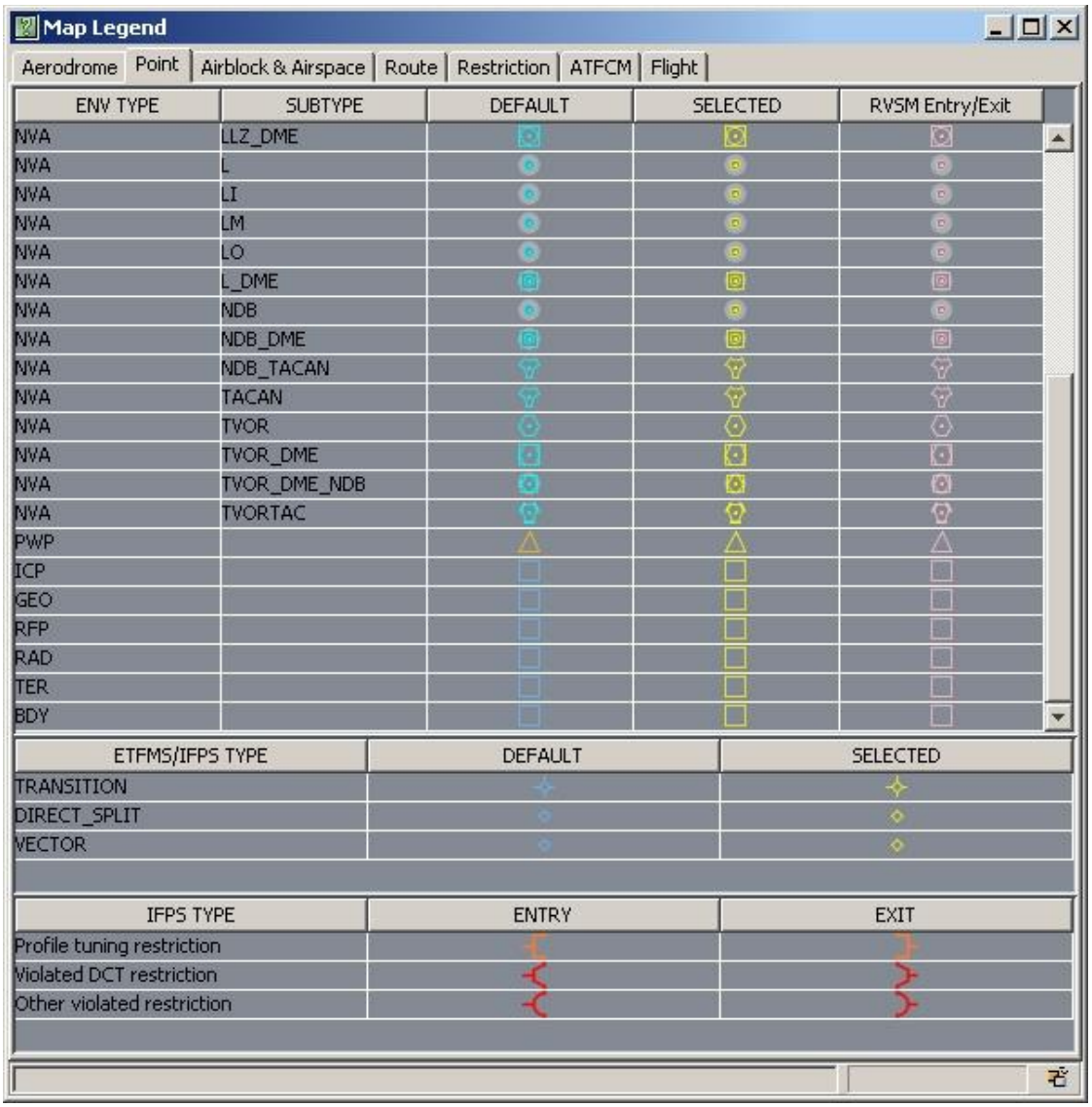


Figure 2-184: Map Legend - Point Tab - First table scrolled down

(3) Move with the mouse over every abbreviated type and every abbreviated subtype. You can see the description of these subtypes as tooltip.

a) NVA Navigation Aid

b) PWP Published Way Point

c) ICP Internal Point

d) GEO GEO Point

e) RFP Reference Point

f) RAD RADAR Point

g) TER Terminal Point

h) BDY Boundary Point

i) DME Distance Measuring Equipment

j) VOR VHF Omni-directional radio Range

k) VOR_DMEco-located VHF Omni-directional radio Range and Distance Measuring Equipment

l) VOR_DME_NDB co-located VHF Omni-directional radio Range, Distance Measuring Equipment and Non-Directional Beacon

m) VORTAC co-located VHF Omni-directional radio Range and TACtical Air Navigator

n) DVOR Doppler VHF Omni-directional radio Range

o) DVOR_DME co-located Doppler VHF Omni-directional radio Range and Distance Measuring Equipment

p) DVOR_DME_NDBco-located Doppler VHF Omni-directional radio Range, Distance Measuring Equipment and Non-Directional Beacon)

q) DVORTAC co-located Doppler VHF Omni-directional radio Range and TACtical Air Navigator

r) ILS Instrument Landing System

s) ILS_DME co-located Instrument Landing System and Distance Measuring Equipment

t) ILS_LLZ co-located Instrument Landing System and Localizer

u) ILS_LLZ_DME co-located Instrument Landing System, Localizer and Distance Measuring Equipment

v) LLZ Localizer

w) LLZ_DME co-located Localizer and Distance Measuring Equipment

x) L Locator

y) LI Inner Locator

z) LM Middle Locator

aa) LO Outer Locator

bb) L_DME co-located Locator and Distance Measuring Equipment

cc) MLS Microwave Landing System

dd) NDB Non-Directional Beacon

ee) NDB_DME co-located Non-Directional Beacon and Distance Measuring Equipment

ff) NDB_TACAN co-located Non-Directional Beacon and TACtical Air Navigator

gg) TACAN TACtical Air Navigator

hh) TVOR Terminal VHF Omnidirectional radio Range

ii) TVOR_DME co-located Terminal VHF Omnidirectional radio Range and Distance Measuring Equipment

jj) TVOR_DME_NDB co-located Terminal VHF Omnidirectional radio Range, Distance Measuring Equipment and Non-Directional Beacon

kk) TVORTAC co-located Terminal VHF Omnidirectional radio Range and TACtical Air Navigator

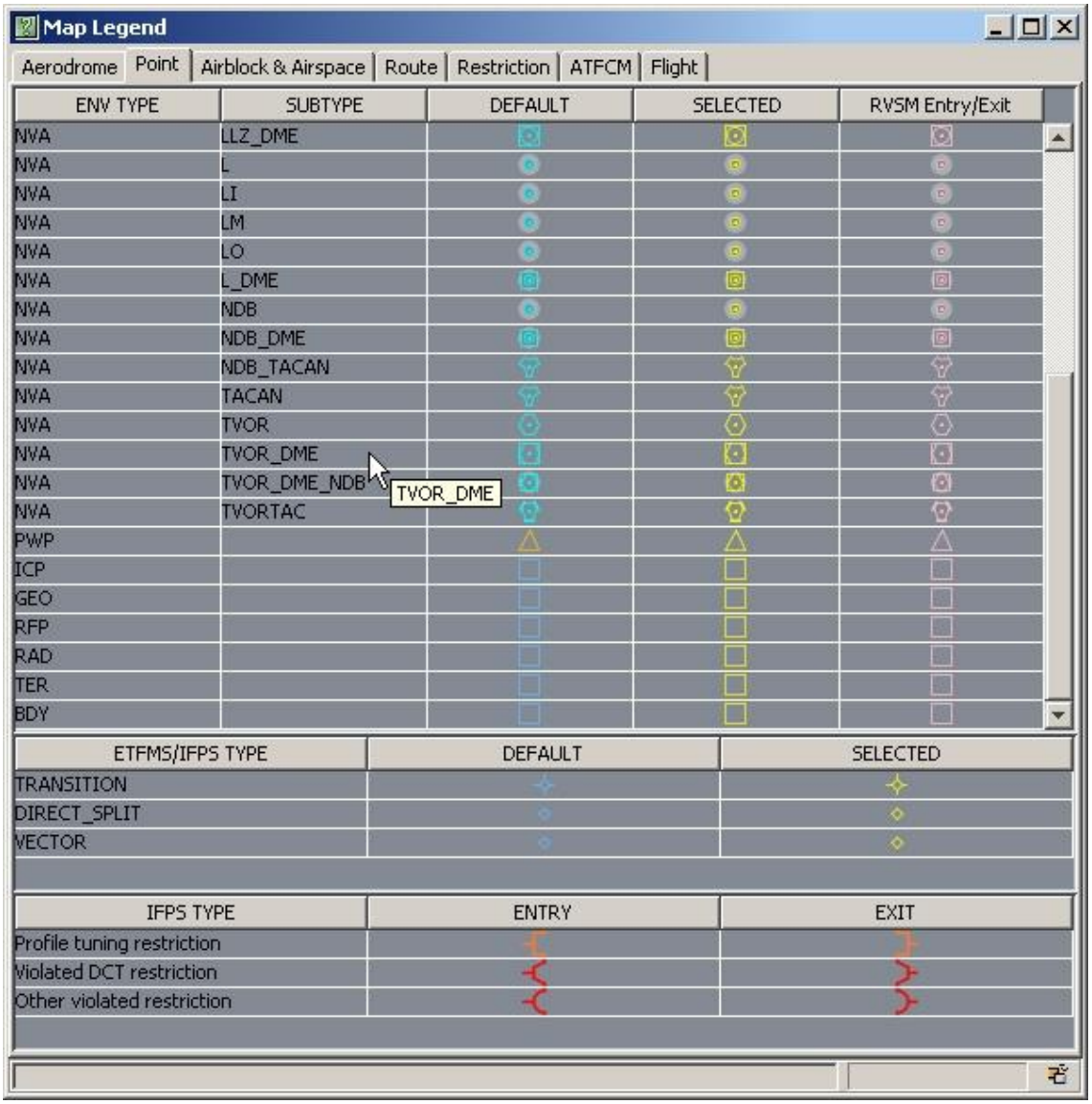


Figure 2-185: Map Legend - Point Tab Tooltip

2.14.4 Airblock and airspace

(1) You see the representation of the different types of airspaces and the representation of the airblocks in default state and selected state.

(2) When they are selected, the border is yellow.

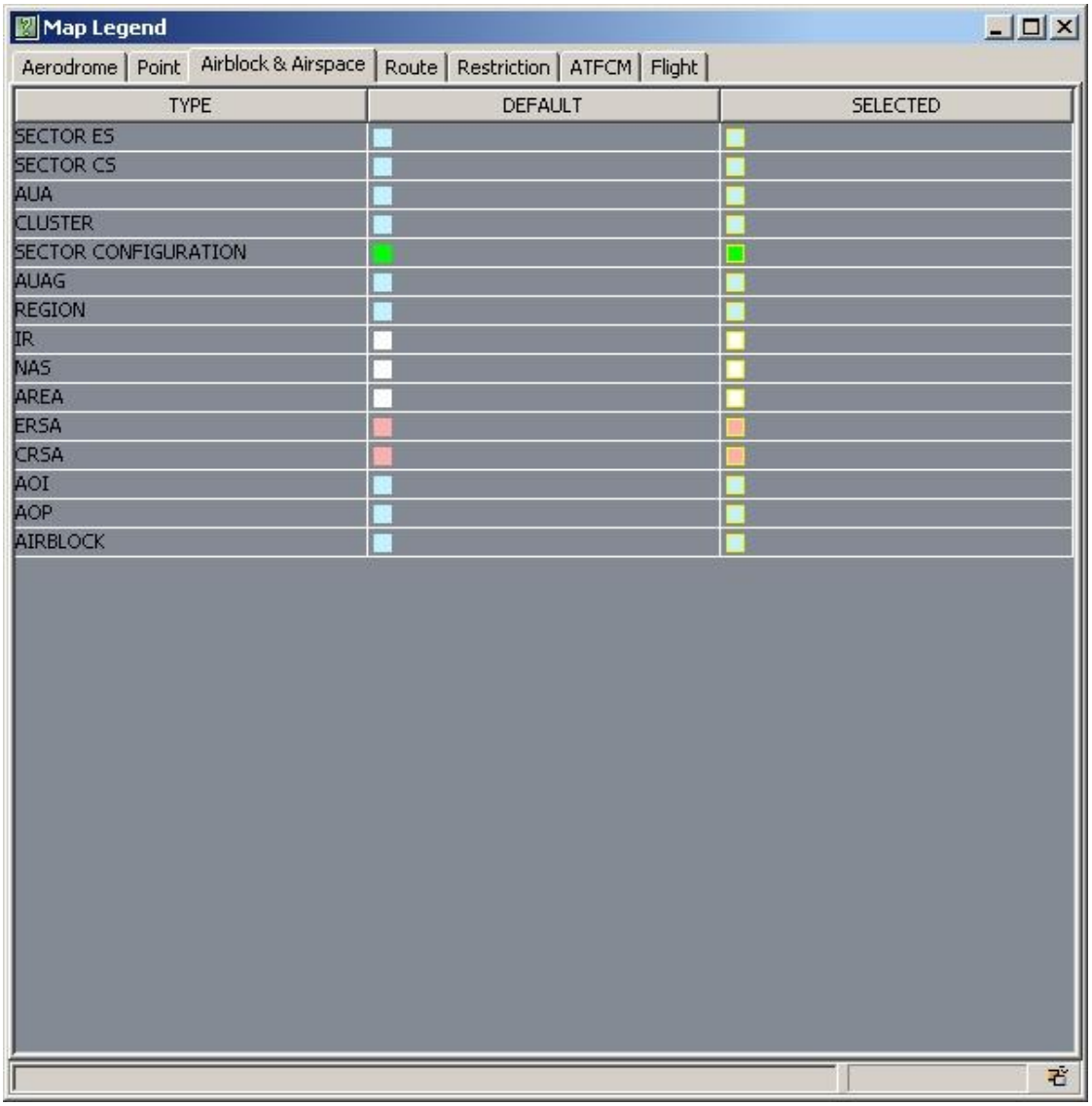


Figure 2-186: Map Legend - Airblock & Airspace Tab

(3) Move with the mouse over every Airblock and Airspace type. You can see the description of these types as tooltip.

a) SECTOR ES Elementary Airspace Sector

b) SECTOR CS Collapsed Sector

c) AUAATC Unit Airspace

d) CLUSTER Airspace Cluster

e) SECTOR CONFIGURATION Sector Configuration

f) AUAGATC Unit Airspace Group

g) REGION Region

h) IR Information Region

i) NAS National Airspace

j) AREA Area

k) ERSA Elementary Restricted Airspace

l) CRSA Composed Restricted Airspace

m) AOI Area of Interest

n) AOP Area of Protection

o) ERAS Elementary Regulated Airspace

p) CRAS Composed Regulated Airspace

q) AIRBLOCK Airblock

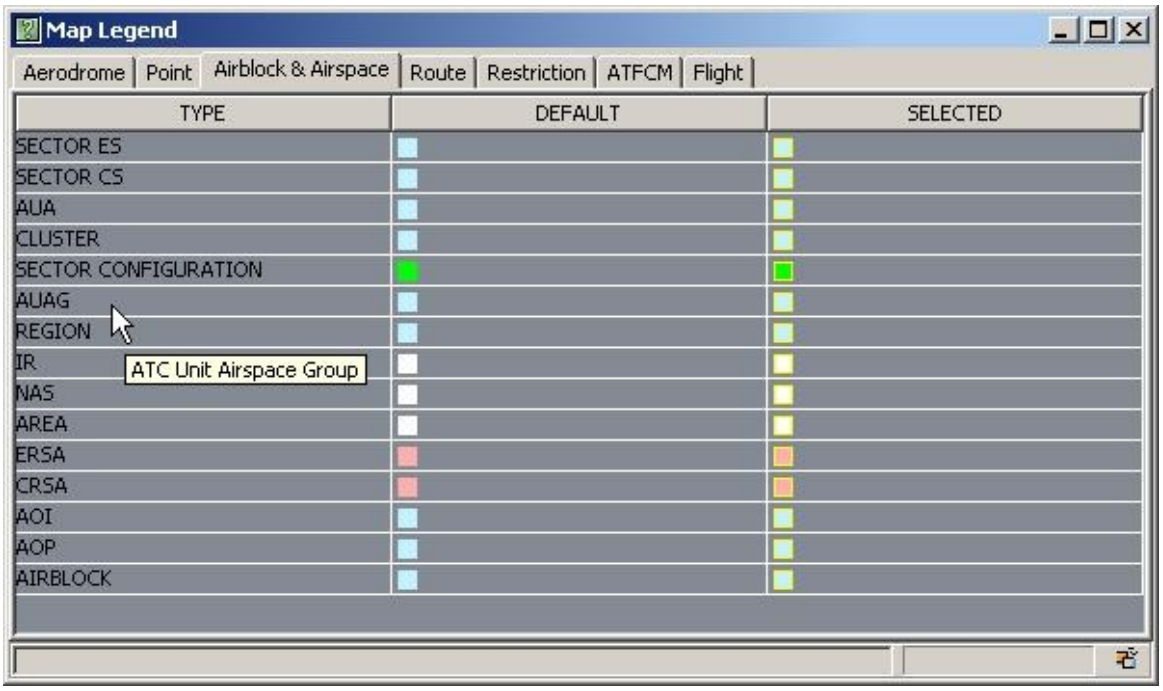


Figure 2-187: Map Legend - Airblock & Airspace Tab Tooltip

2.14.5 Route

(1) You see the representation of the different types of routes in default state (General mode), selected state.

(2) You can also see the representation in CAT mode:

a) ATS Route / CDR 0,

b) CDR 1,

c) CDR 2,

d) CDR 3,

e) NOT AVAILABLE,

f) CDR U

g) Mixed CDR.

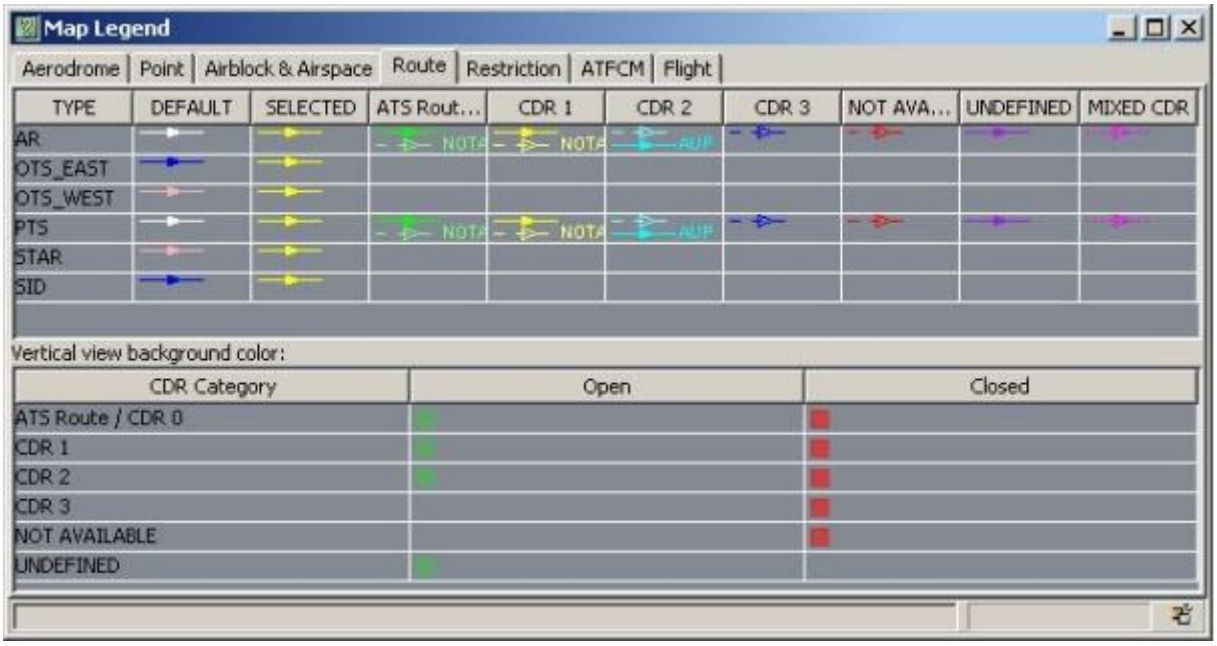


Figure 2-188: Map Legend - Route Tab

(3) Move with the mouse over every Route type. You can see the description of these types as tooltip.

a) AR Air Route

b) OTS East Organised Track System Eastbound Route

c) OTS West Organised Track System Westbound Route

d) PTS Polar Track Structure Route

e) STAR STandard instrument Arrival Procedure

f) SID Standard Instrument Departure Procedure

(4) Move with the mouse over every CDR Category. You can see the description of these categories as tooltip.

a) ATS Route / CDR 0ATS Route

b) CDR 1Conditional Route Type 1

c) CDR 2Conditional Route Type 2

d) CDR 3Conditional Route Type 3

e) CDR U Undefined

f) NOT_AVAILABLE Not Available

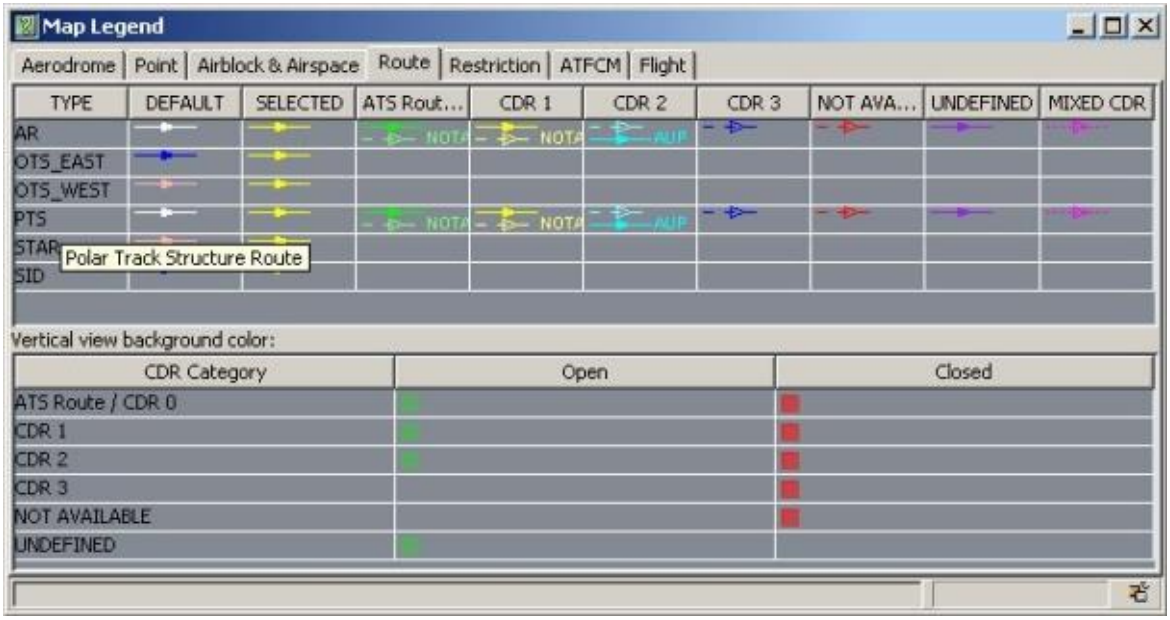


Figure 2-189: Map Legend - Route Tab Tooltip

(5) CDR U must be displayed or not according to a preference called "show CDR U".

2.14.6 Restriction

(1) There is one tab for the representation of the **Flow Routings** of a restriction.

(2) You see the representation of the different types of flow routing restrictions in selected state, as mandatory restriction and as forbidden restriction.

(3) You can also see the specific representation of FRA point flow routings: FRA intermediate points, FRA border entry points, FRA border entry- exit points and FRA border exit points.

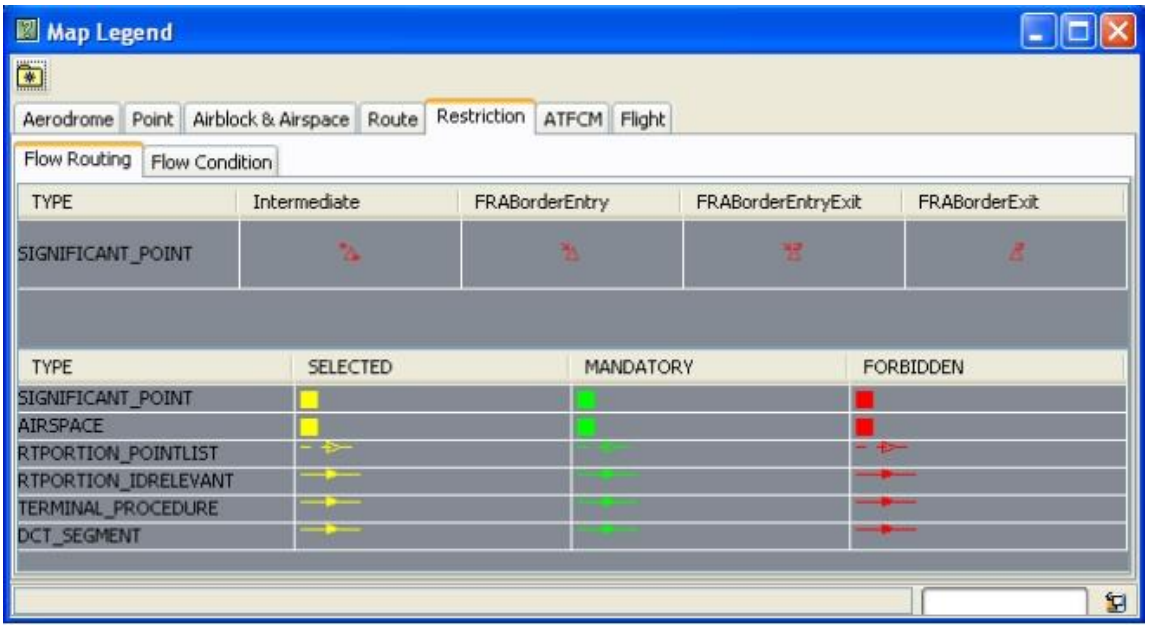


Figure 2-190: Map Legend - Restriction / Flow Routing Tab

(4) Move with the mouse over every type of flow routing. You can see the description of these types as tooltip.

(5) General flow routings:

a) SIGNIFICANT_POINT Significant Point Routing

b) AIRSPACE Airspace Routing

c) RTPORTION_POINTLIST Route Portion Point List Routing

d) RTPORTION_IDRELEVANT Route Portion ID Relevant Routing

e) TERMINAL_PROCEDURE Terminal Procedure Routing

f) DCT_SEGMENTDCT Segment Routing

(6) FRA point flow routings:

a) SIGNIFICANT_POINTFRA Point Routing

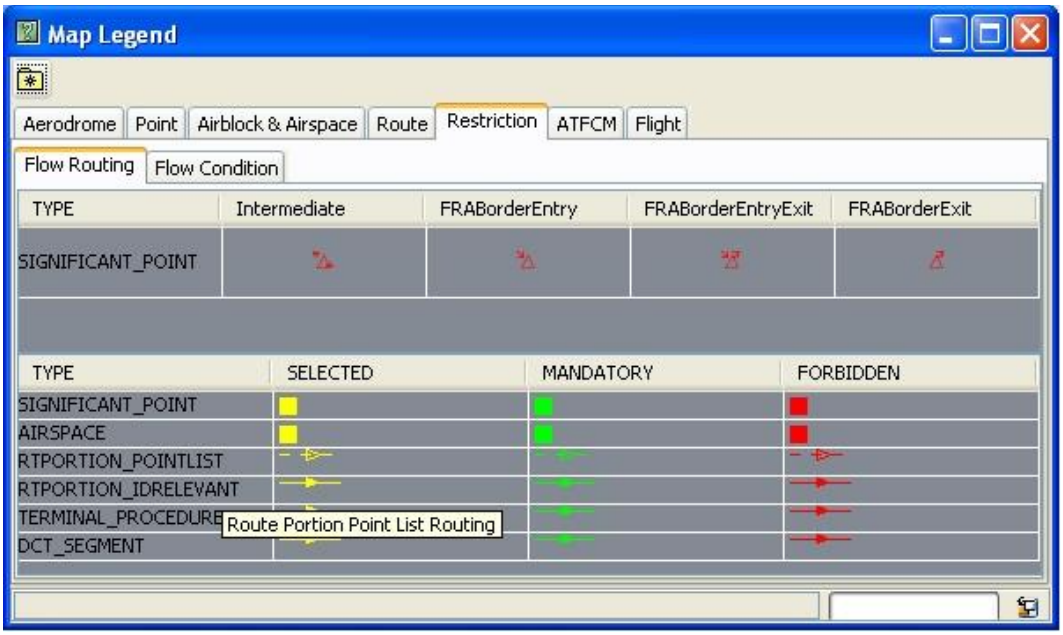


Figure 2-191: Map Legend - Restriction / Flow Routing Tab Tooltip

(7) There is one tab for the representation of the **Flow Conditions** of a restriction.

(8) You see the symbols of the different types of flow condition restrictions as:

a) DEPARTING,

b) ARRIVAL,

c) DEPARTING_ARRIVAL and

d) CROSSING restriction.

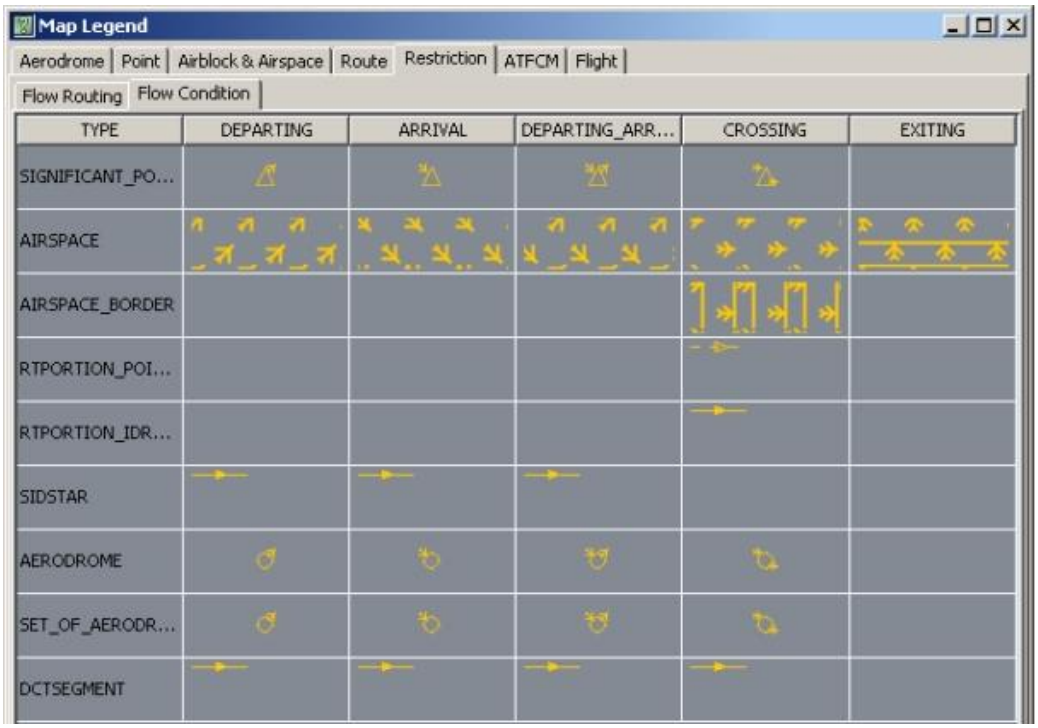


Figure 2-192: Map Legend - Restriction / Flow Condition Tab

(9) You can see also the colours that are used for

a) SELECTED,

b) REFERENCE LOCATION,

c) INCLUDED and

d) NOT INCLUDED restrictions.



Figure 2-193: Map Legend - Restriction Colours

(10) Move with the mouse over every type of flow condition. You can see the description of these types as tooltip.

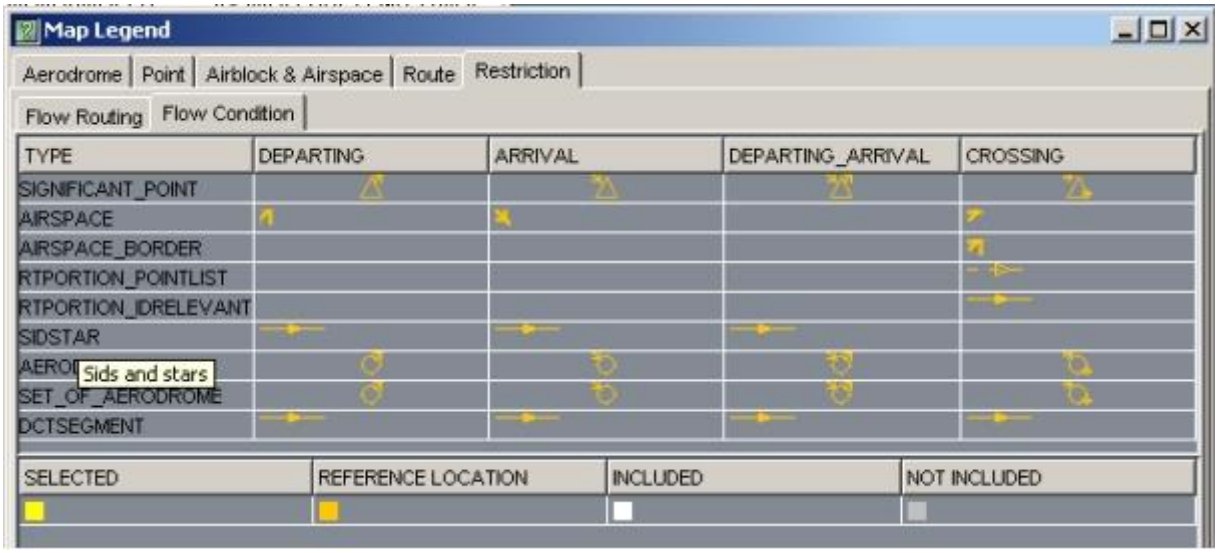


Figure 2-194: Map Legend - Restriction / Flow Condition Tab Tooltip

2.14.7 ATFCM

(1) You see the representation of the different types of traffic volume reference location.

(2) You see the possible colours for those traffic volume reference locations.

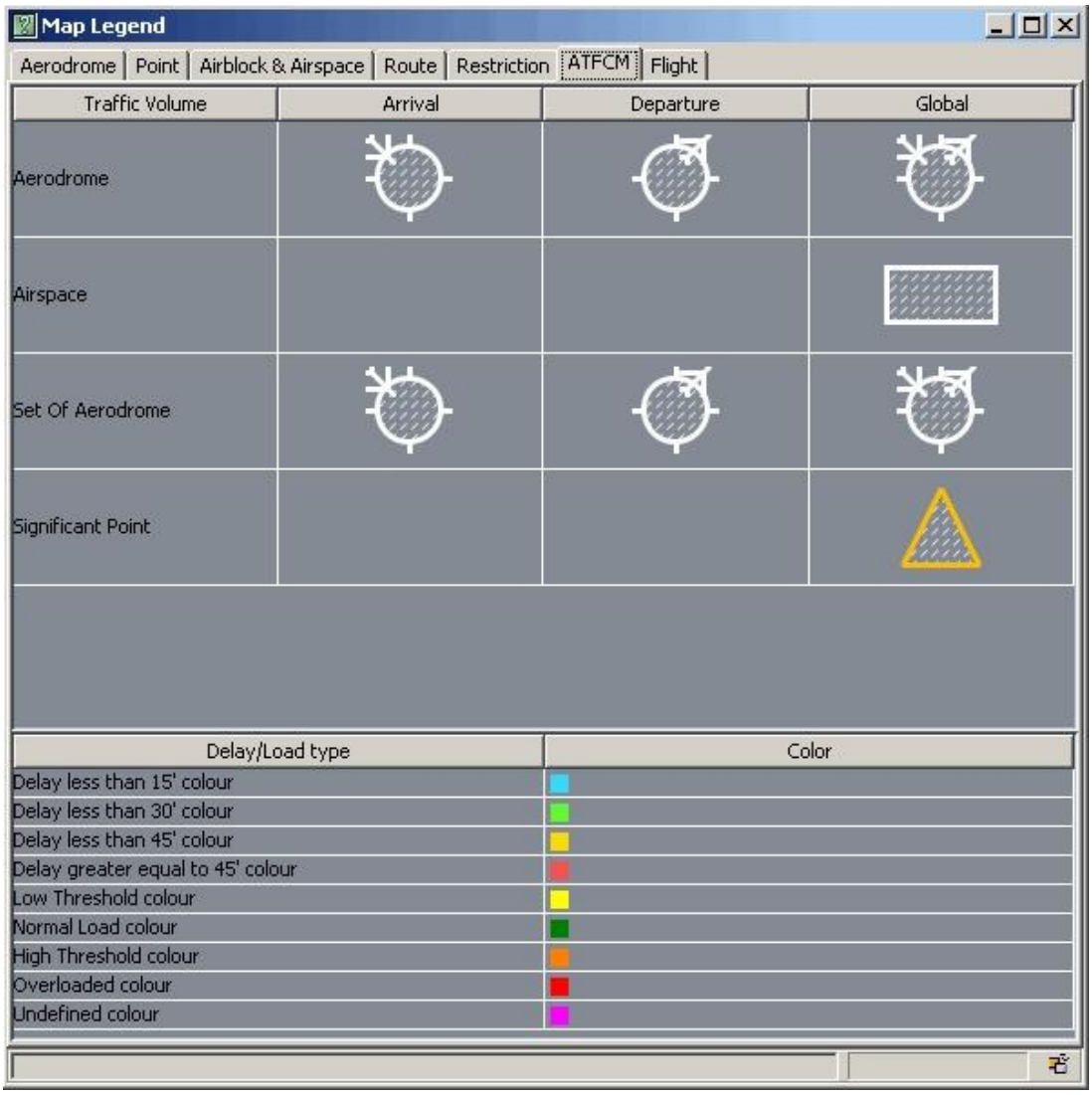


Figure 2-195: Map Legend - ATFCM Tab

2.14.8 Flight

(1) You see the representation of the flight profiles for ATFCM (including Predict and Archive) and IFPS.

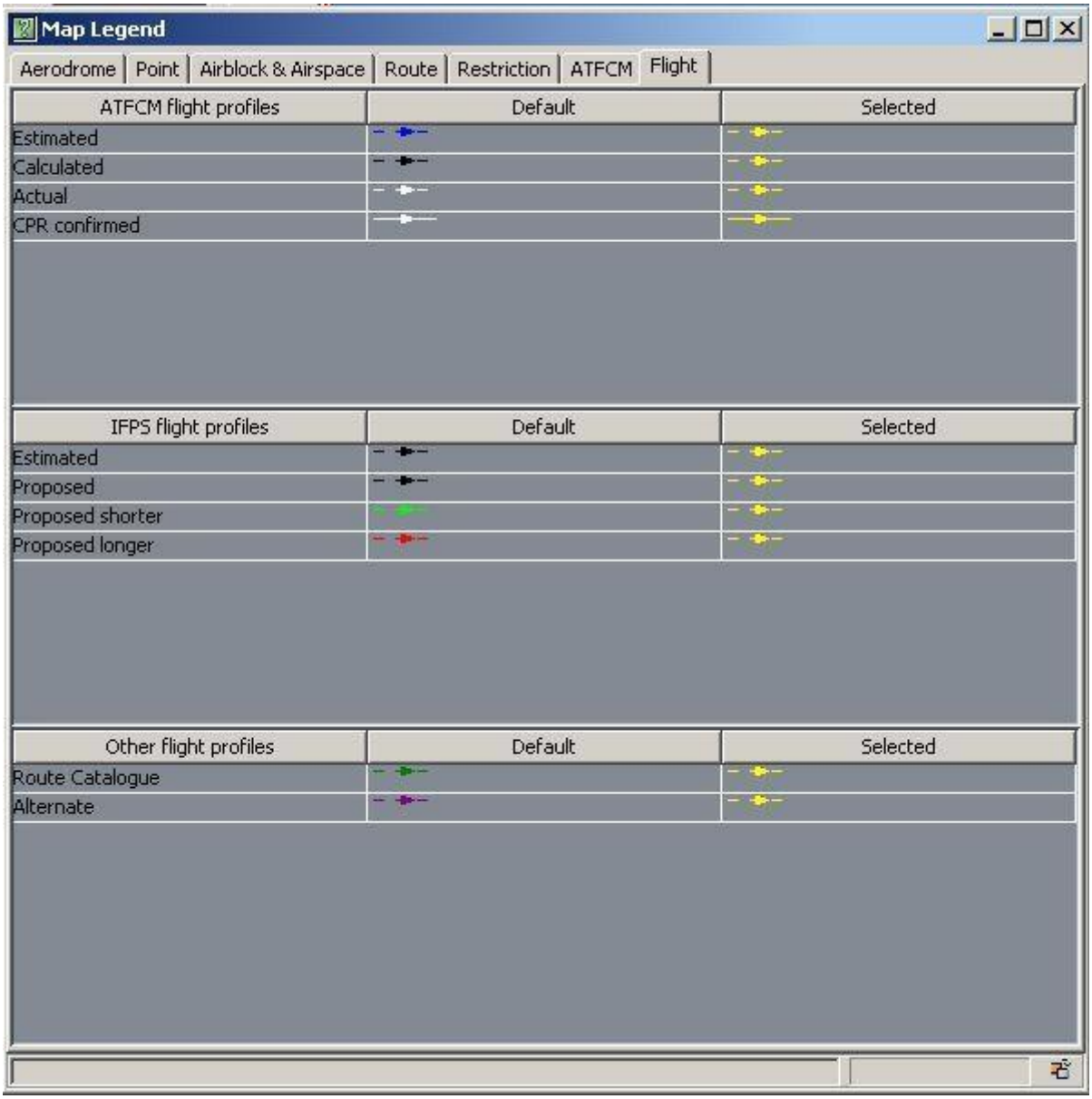


Figure 2-196: Map Legend - Flight Tab

2.15 Geometry

2.15.1 Geometry display

(1) Open the 'Map Configuration' window.

(2) Select the tab General.

(3) Check the item 'Painting/Geometry tab visible'.

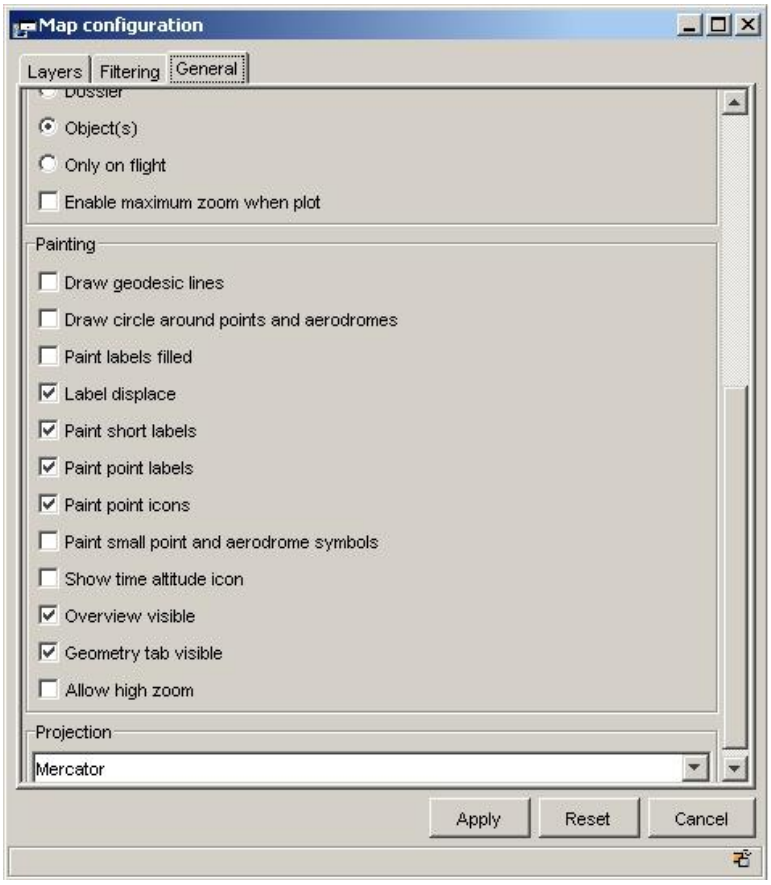


Figure 2-197: Map Configuration - General Tab - Geometry tab visible option

(4) Click 'Apply'.

(5) On the right side of the map, the Geometry tab appears.

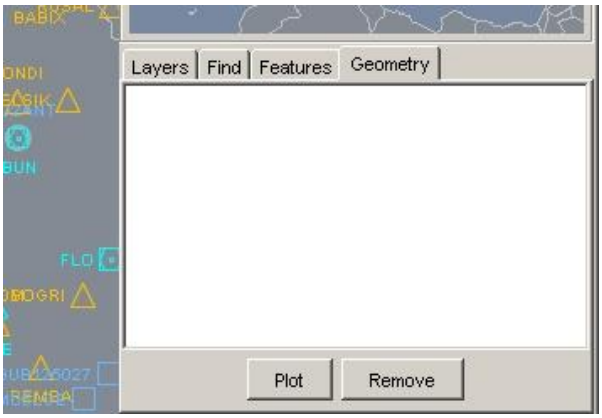


Figure 2-198: Geometry tab visible

(6) Enter in the text area of the geometry tab the coordinates of the points of a polygon: 515408N0043217E, 521429N0033911E, 502841N0038911E

(7) Click 'Plot'.

(8) The triangle with points is displayed on the map.

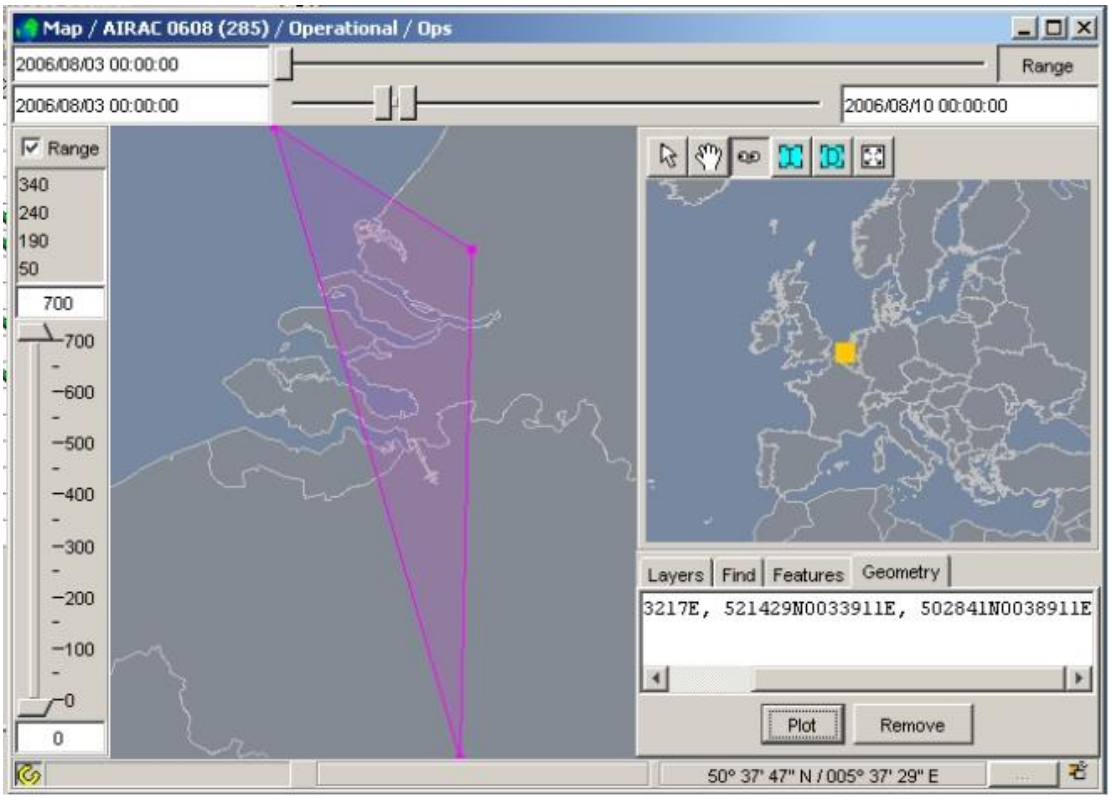


Figure 2-199: Geometry Drawing Example

(9) Click on the tab 'Layers'.

(10) A new layer is added to the map: Geometry.

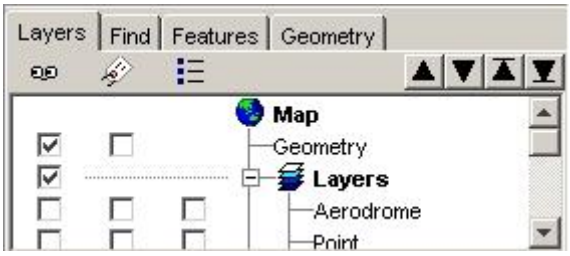


Figure 2-200: Geometry Layer

(11) Uncheck the layer.

(12) The triangle disappears.

(13) Check the layer again. Check also the labels property.

(14) The label of the triangle 'TRI1' is painted.

(15) The labels of the points are also painted.

(16) The point labels are the coordinates of the point.

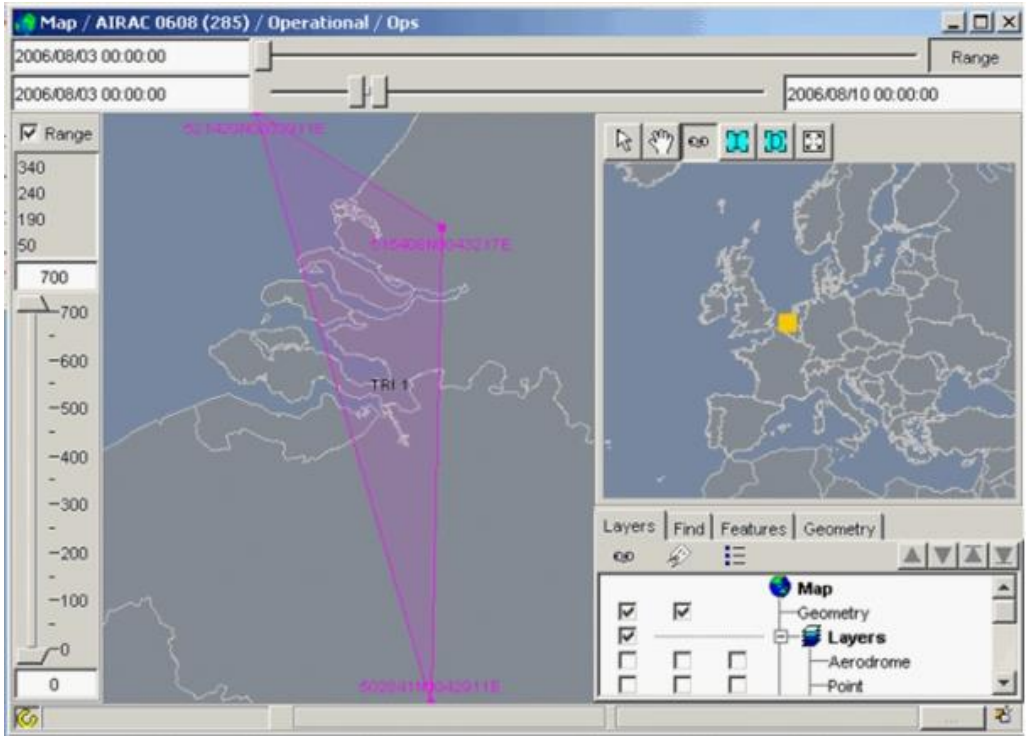


Figure 2-201: Geometry Drawing Example with Point Labels

(17) Select the 'Geometry' tab. Click on 'Remove'.

(18) The text area is cleared.

(19) The triangle is removed from the map.

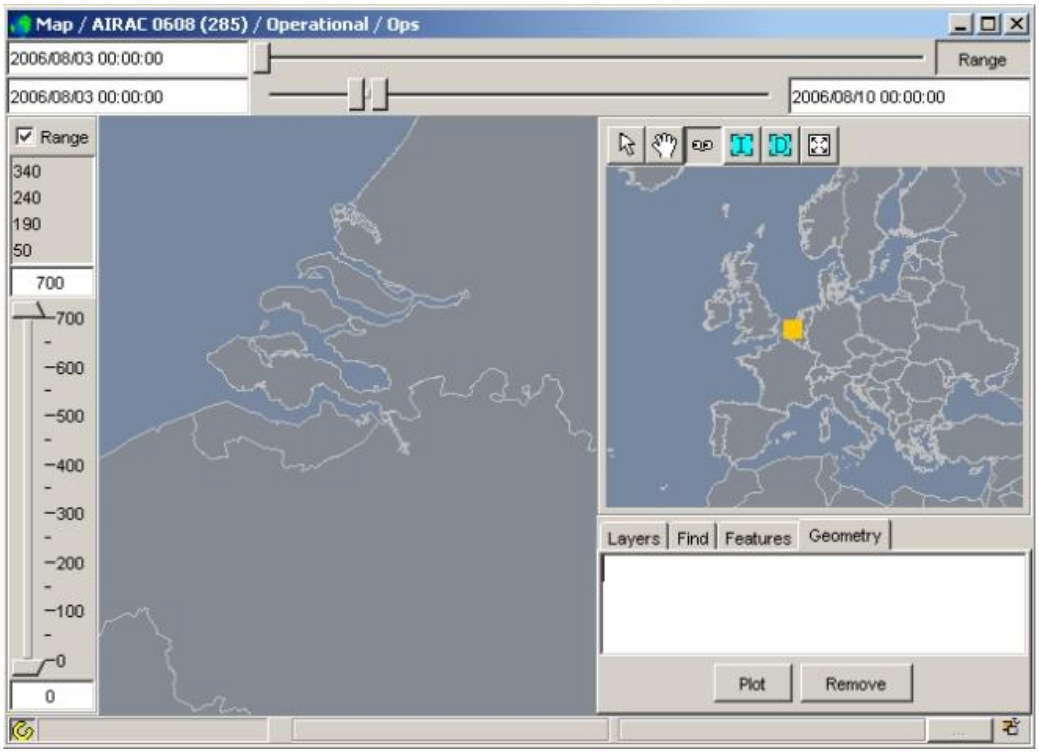


Figure 2-202: Geometry Remove Drawing Example

(20) **Coordinate Formats** shall support the following coordinates format:

a) L coordinates format:

i) The latitude direction is represented on the left of the latitude digits, and

ii) The longitude direction is represented on the left of the longitude digits
E.g. N063122 or W0093122

b) R coordinates format:

i) The latitude direction is represented on the right of the latitude digits, and

ii) The longitude direction is represented on the right of the longitude digits
E.g. 063122N or 0093122W

(21) In input:

a) The latitude digit representation is either:

i) ddmm: interpreted as ddmm00, or

ii) ddmmss: the latitude representation is then said to be complete

where:

iii) dd is in [00, 90]

iv) mm is in [00, 59]

v) ss is in [00, 59]

vi) If dd is 90, mm and ss are both 00

b) The longitude digit representation is either:

i) ddd: interpreted as ddd0000, or

ii) dddmm: interpreted as dddmm00, or

iii) dddmmss: the longitude representation is then said to be complete

where:

iv) ddd is in [000, 180]

v) mm is in [00, 59]

vi) ss is in [00, 59]

vii) If ddd is 180, mm and ss are both 00

(22) A coordinates characteristic defined as a character appearing within a latitude or a longitude representation, namely:

a) N or S, and

b) W or E, and

c) [0-9]

(23) **Coordinate Text.** As described below, a polygon can be input as a coordinates text:

A valid coordinates text in input is formatted as follows:

a) It is entirely consistent with either the L coordinates format or the R coordinates format, i.e. it does not contain some coordinates in the L format and others in the R format

b) It is *not* mandatory to have a separator between the latitude and the longitude of a point, or between the longitude of a point and the latitude of the next point (i.e. between points). If a separator is provided:

i) It may contain any number of characters, but

ii) These characters must be one of the defined delimiter characters and cannot be coordinates characters

(24) A few examples from NOM:

a) 5323N 00514W - 5318N 00446W - 5318N 00433W - 5243N 00116W - 5233N 00003W - 5157N 00208W - 5235N 00444W - 5305N 00511W - 5323N 00514W

[Note the loose separators]

b) 5522N 00841W - 5506N 00455W - 5445N 00544W - 5413N 00456W - 5357N 00305W - 5340N 00105W - 5135N 00045W - 5413N 00717W - 5522N00841W

[Note the loose separators]

c) 5243N 00853W - 5330N 00618W - 5150N 00829W - 5243N 00853W

d) 6416N01716W 6100N01930W 6100N02800W 7200N02800W 7800N01000W 7800N0L- 7230N0L 7300N00800W 6416N01716

e) N43 08 43 E020 40 05 N43 11 25 E020 40 04 N43 11 25 E020 44 17 N43 08 43 E020 44 18

f) 360000N 0060000W, 360000N 0054500W, 360126N 0054124W, 360322N 0053953W, 361220N 0053930W, 361430N 0053600W, 362200N 0054500W

g) 410000N0271914E 410033N0272038E 410420N0271832E 410315N0271455E

h) 473540N 0080000W - 465520N 0080000W - 464500N 0093000W

i) 4110N 03409W - 4200N 03137W - 4318N 03223W - 4225N 03456W - 4110N 03409W

[Note the presence of 7800N0L and 7230N0L: NOT supported]

(25) Multiple polygons can be plotted.

(26) The points of one polygon should be separated by one of the following characters or a space:"**,:-?!/()[];**"

(27) Polygons should be separated by at least one other characters.

(28) Enter in the text area:

(535408N0043217E)
(525408N0043217E) (531429N0033911E)
(515408N0043217E) (521429N0033911E) (502841N0038911E)
(495408N0043217E) (501429N0033911E) (482841N0038911E) (495408N0037217E)
(485408N0043217E)
(475408N0043217E) (481429N0033911E)
(465408N0043217E) (471429N0033911E) (452841N0038911E)
(445408N0043217E) (451429N0033911E) (432841N0038911E) (445408N0037217E)

(29) Click 'Plot'.

(30) The polygons are plotted.

(31) The names of the polygons are: POINT 1, LINE 1, TRI 1, POLY 1, POINT 2, LINE 2, TRI 2 and POLY 2.

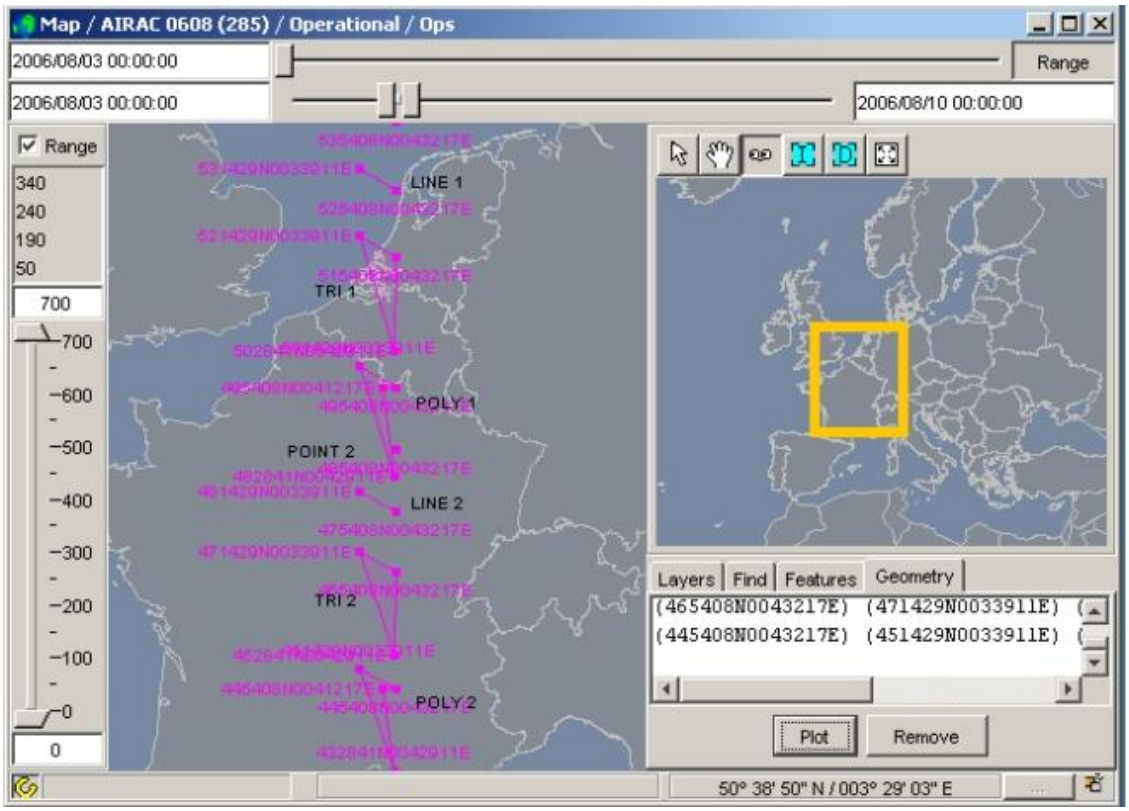


Figure 2-203: Geometry Polygons Drawing Example with Labels

(32) Open the 'Map Configuration'.

(33) Uncheck 'General/Paint/Geometry tab visible'.

(34) Click 'Apply'.

(35) The 'Geometry tab' is removed. The polygons are removed.

(36) **Feature to facilitate the input of a list of geo coordinates for airblock creation**

a) The intention is here to use the existing geometry tab which shall support the necessary lat/long formats.

b) Today it is already possible to cut and paste the text from the NOTAM into the geometry tab and plot it on the map.

c) It is further proposed to add a new button shown below that would take the polygon had been defined and populate the air block create window with that data as shown below.

d) Only one polygon at a time will be allowed by this mechanism.

e) One line in the geometry tab is treated as one polygon - as before.

f) The following formats shall be supported:

i) N43 08 43 E020 40 05 N43 11 25 E020 40 04 N43 11 25 E020 44 17 N43 08 43 E020 44 18

ii) 360000N 0060000W, 360000N 0054500W, 360126N 0054124W, 360322N 0053953W, 361220N 0053930W,361430N 0053600W, 362200N 0054500W

iii) 410000N0271914E 410033N0272038E 410420N0271832E 410315N0271455E

iv) 473540N 0080000W - 465520N 0080000W - 464500N 0093000W

v) - 4110N 03409W - 4200N 03137W - 4318N 03223W - 4225N 03456W - 4110N 03409W

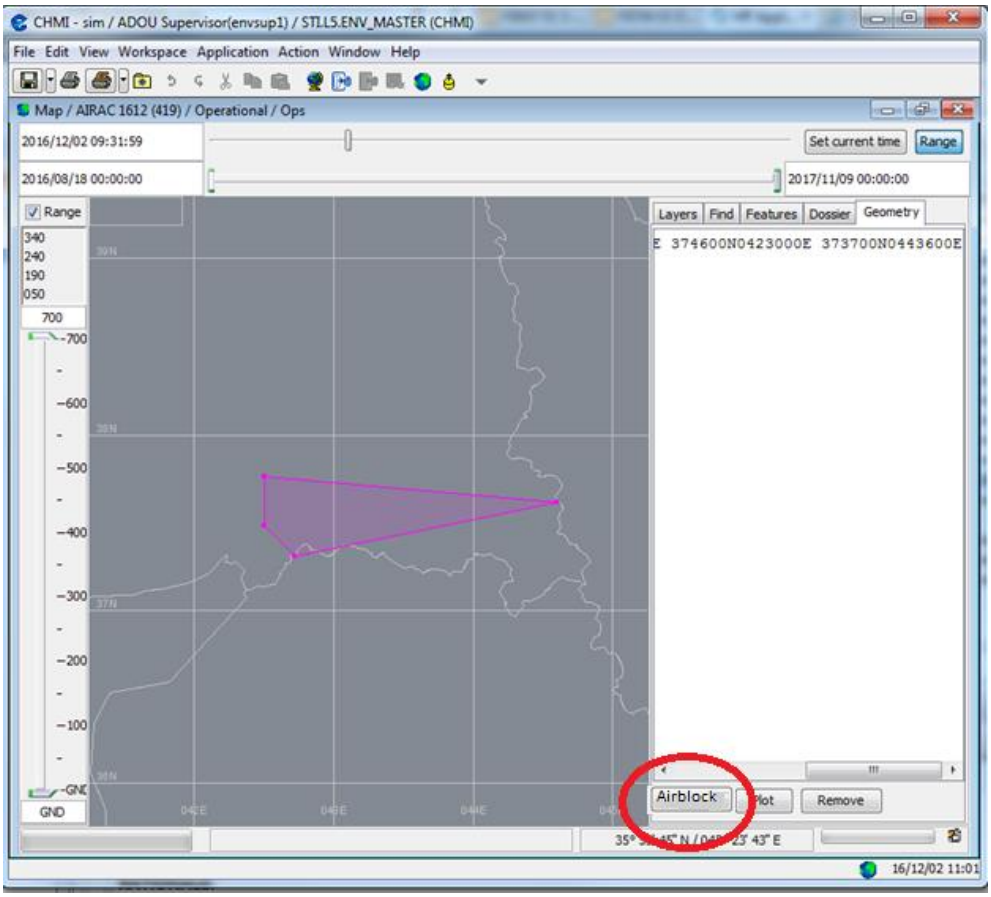


Figure 1: Airblock

(37) Selection of the air block button causes the polygon to be populated in the creator airblock window.

(38) The system will automatically reorder the vertices such that they are defined in a clockwise fashion such that they will pass the validation in the air block create.

(39) A new IDL method will be added to assist in this reordering. The intention is only to support ccw to cw re-ordering not general point re-ordering.

(40) The system shall default the Airblock Id field to the next available EU [???] id.

(41) All existing polygon validation will still apply.

(42) If the input text in geometry tab is not conformant to supported formats or contains unexpected characters, the system shall give the error to the user: "No polygons found in text, format error?"

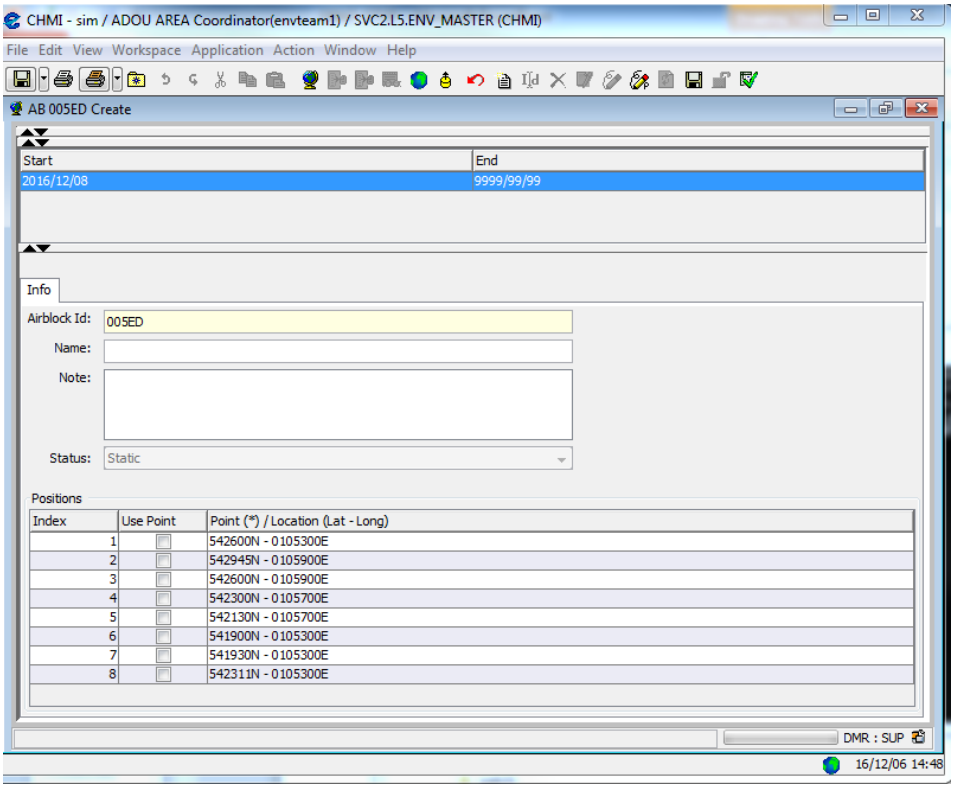


Figure 2

(43) Polygons that have edges that cross each other are not valid-see below.

(44) See AB creation rules in ENV section.

2.15.2 High zoom

(1) Sometimes there is a higher zoom needed to see the differences in geometries.

(2) Display the geometry tab again.

(3) Enter in the text area:
(515408N0043217E) (521429N0033911E) (502841N0038911E)
(515408.005N0043217E) (521429.005N0033911E) (502841.005N0038911E)

(4) Click plot.

(5) 2 Triangles are displayed.



Figure 2-204: Geometry High Zoom Example - Before

(6) Open the 'Map Configuration'.

(7) Make sure that 'General/Paint/Allow high zoom' is unchecked.

(8) Zoom in at the south point as deep as you can. At a certain moment, you can't zoom in deeper, but you almost can't see a difference between the 2 points. They differ 5 thousands of a second.



Figure 2-205: Geometry High Zoom Example - Without High Zoom

(9) Check in 'Map Configuration' the property 'General/Paint/Allow high zoom'.

(10) Click 'Apply'.

(11) Zoom now deeper into the map at the south points of the triangle. Now you can see the distance between the points.

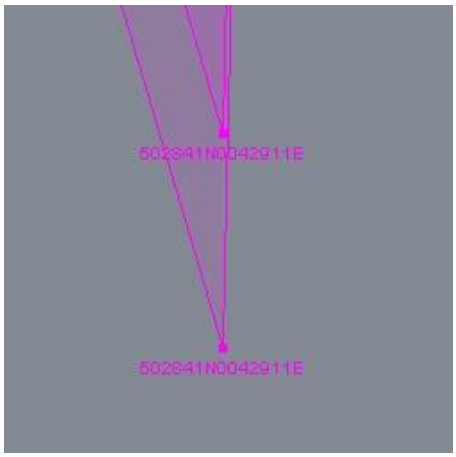


Figure 2-206: Geometry High Zoom Example - With High Zoom

2.16 Labelling

2.16.1 Introduction

(1) In this chapter we describe the labelling functionality.

2.16.2 Painting labels

(1) Labels of an object are painted when the layer is visible and the scale of the map is in the scale range for the layer and the scale of the map is in the scale range of the labels of the layer.

(2) Labels are always painted when the object is selected.

2.16.3 Label scale range

(1) Every layer has a scale range in which the objects are visible and a scale range in which the labels are visible.

(2) For most types those scale ranges are the same, but they can be different, for example for routes.

2.16.4 Always visible

(1) When a layer is visible and the map is in the scale range for painting the labels all the labels of every object of that layer should be visible.

(2) If a part of an object is outside the map, it is possible that the label is outside the bounds of the map. But for some types the label should be always visible if a part of the object is visible: route.

2.16.5 Label priority

(1) ATFCM objects have big labels. Some labels can be hidden under others.

(2) Some labels are indicating problems. They should be always visible. They should be painted above the less important labels.

(3) In this table you find the priorities. A label with a higher priority is always above a label with a lower priority.

Priority	Delay	Load
High	Above 45/red	Overload/red
Middle	Less than 45/yellow	High load/orange
Low	Less than 30/green	Low load/yellow
No	Less than 15/blue	normal load/green load undefined

Table 2-5: Label priority

(4) Labels of elements of the FMP monitor

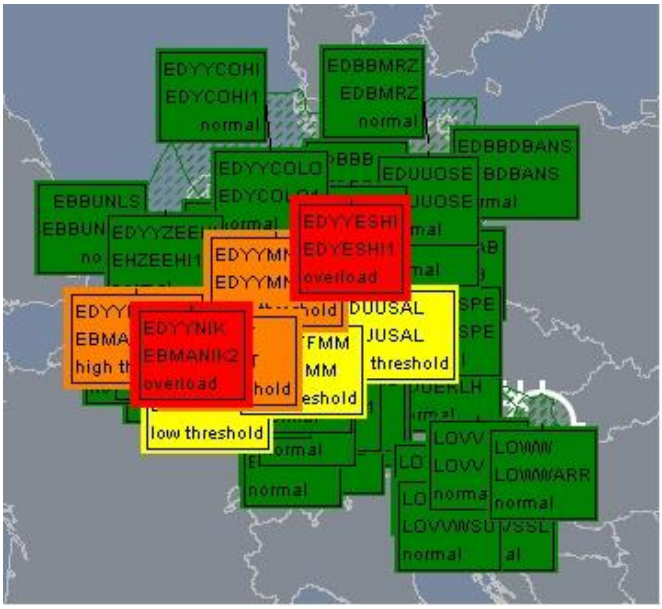


Figure 2-207: Labelling - Label priority of the FMP monitor

(5) Labels of elements of the Status monitor

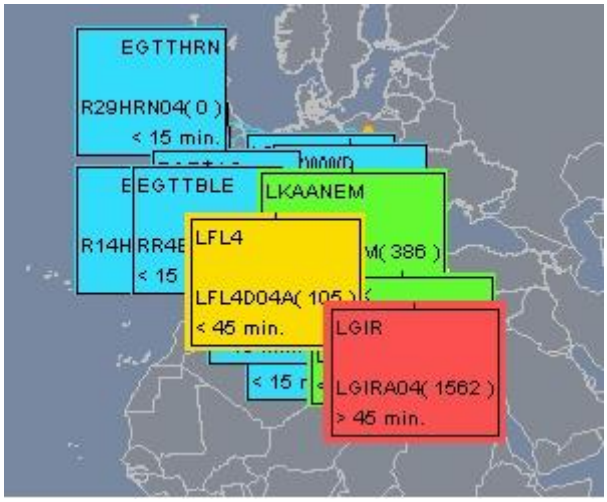


Figure 2-208: Labelling - Label priority of the Status monitor

2.16.6 Labelling segments

(1) Every segment of a route should have as label the name of the route.

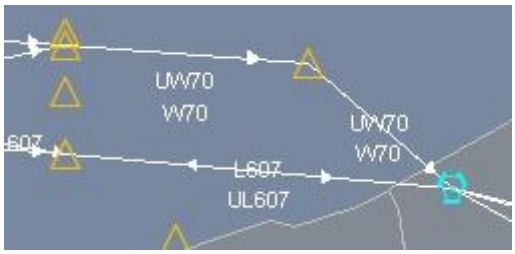


Figure 2-209: Labelling segments - Route Segments

(2) Every segment of every flow routing route of a restriction should have as label the name of the restriction.

(3) It is possible that there is multiple times the same label on a segment. This is because the same route segment is used multiple times as flow routing.



Figure 2-210: Labelling segments - Flow Routing Route of Restriction

(4) Not every segment of the flight should be labelled. There should be a label every 5 segments.



Figure 2-211: Labelling segments - Flight Route Segments

(5) If only a few segments of the flight are visible in the view, at least one segment should have a label.

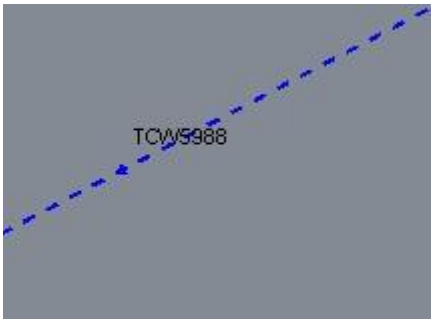


Figure 2-212: Labelling segments - Limited Flight Route Segments

2.16.7 Label displace

(1) Sometimes there are a lot of labels on the map. There are several mechanisms to make the overlap of labels minimal. We call this label displace.

2.16.8 Selecting

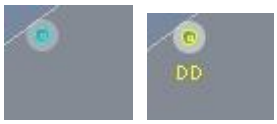
(1) When an object is selected, the object is painted in yellow. The labels of the object should be painted also in yellow.

(2) If the labels of the layer of the object are not visible, the labels will be painted if the object is selected.

(3) Aerodrome:



(4) Point:



(5) Route:

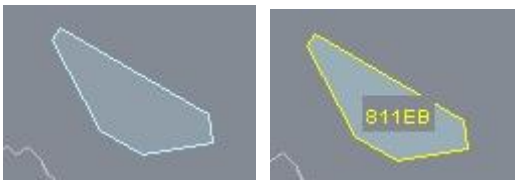
a) Points labelled:



b) Points not labelled:



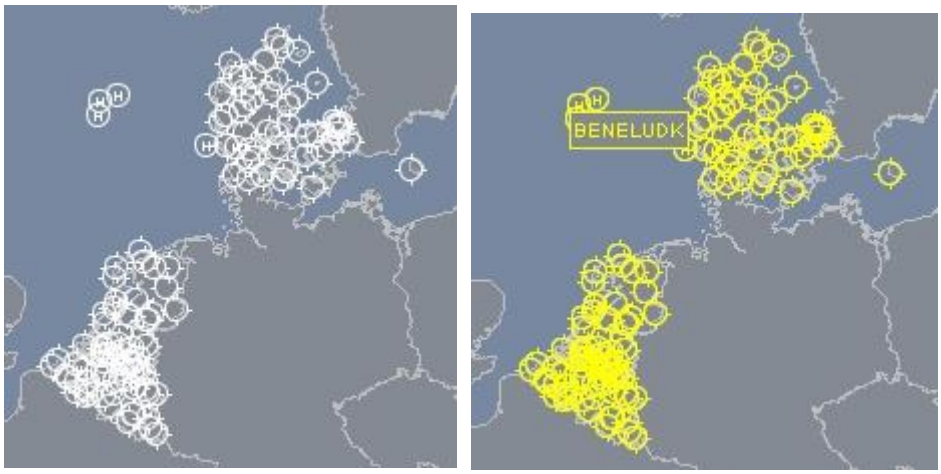
(6) Airblock:



(7) Airspace:



(8) Set of aerodrome:



(9) Restriction:



Figure 2-213: Labelling segments - Selected Object Labelling (Visible=OFF)

(10) The same should be valid for objects in the Layers layer group.

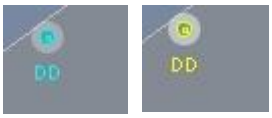
(11) If the labels of the layer of the object are visible, the label of the selected object should be painted on the same place as where they were painted unselected, only now they are painted in yellow.

(12) Objects in the Dossier layer group:

a) Aerodrome:



b) Point:



c) Route:

i) Points labelled:



ii) Points not labelled:



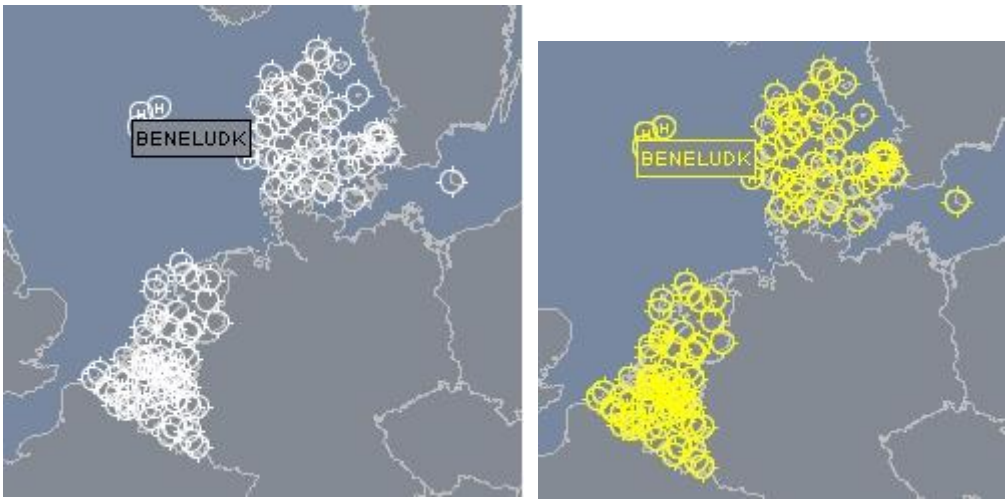
d) Airblock:



e) Airspace:



f) Set of aerodrome:



g) Restriction:



Figure 2-214: Labelling segments - Selected Object Labelling (Visible=ON)

(13) The same should be valid for objects in the Layers layer group.

2.16.9 Moving labels

(1) ATFCM objects have big labels. Some labels can be hided under others.

(2) It is possible to drag a label to another position.

(3) You can drag the label if you keep the control key pressed.

(4) If you pan the map afterwards, the labels stay in the same (panned) position.

(5) When you zoom, the labels return to their normal position.

(6) The labels before moving:

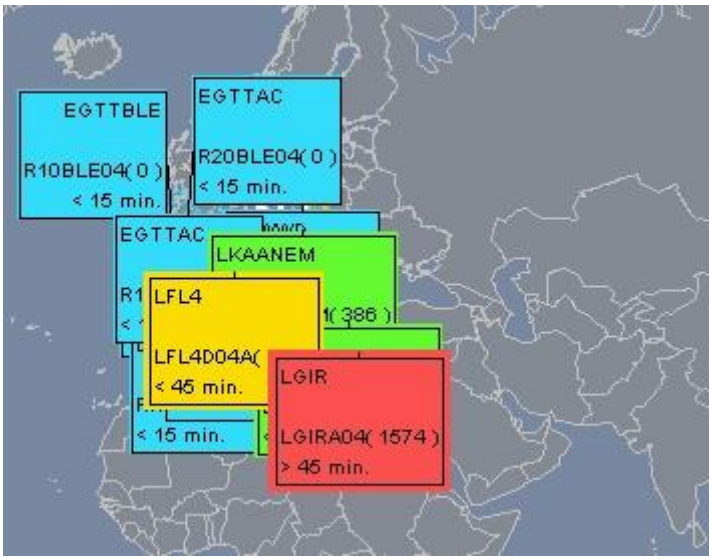


Figure 2-215: Labelling segments - Before Moving Labels

(7) Move the red label to the right by pressing the CTRL key and dragging the label:

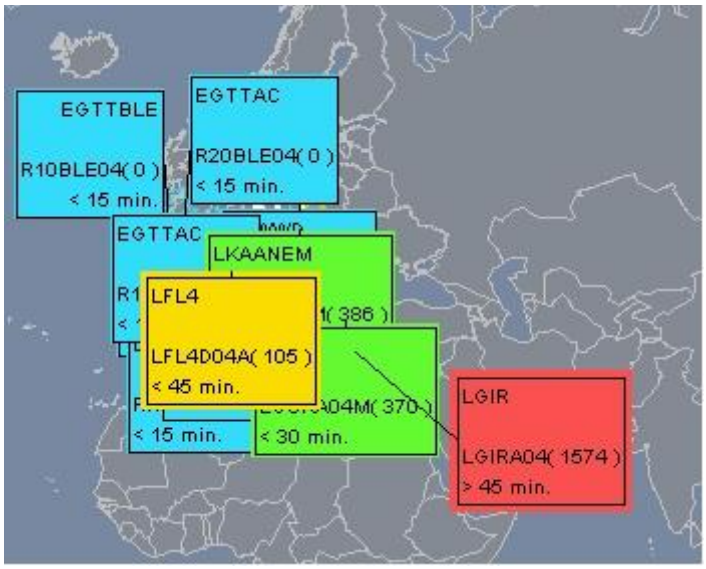


Figure 2-216: Labelling segments -Moving Labels

(8) Zoom out or in and the labels are painted normally:

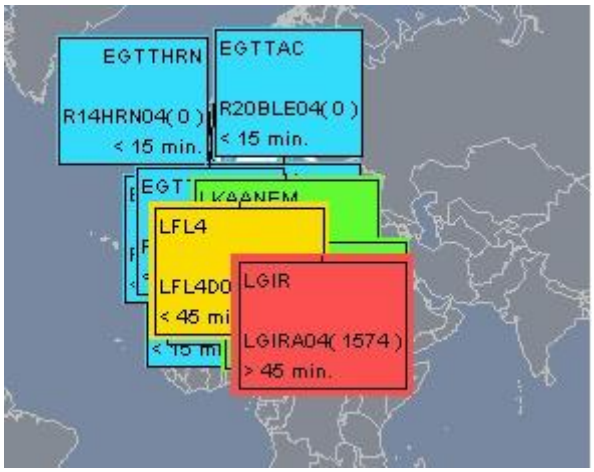


Figure 2-217: Labelling segments - Zoom After Moving Labels

2.16.10 One label for an object

(1) An object can be displayed multiple times in different layers. Every layer can display labels.

(2) For the same object the label should be displayed only one time.

(3) If the object is selected the label should be displayed also only one time.

(4) For routes, flights and routes in restriction should the label be displayed at least on one segment and not more than once on the same segment.

2.17 Painting one point and one label

(1) There are multiple layers that can paint the same point and the same point label. In this case only one layer will paint the point and only one layer will paint the label.

(2) There is an order of priority for paining the points and point labels. You can find the order in the next table.

(3) The layers above in the table have priority for painting a point or its label over layers under in the table.

Preview	Restriction
Preview	Set of aerodrome
Preview	Aerodrome
Preview	Point
Preview	Route
Dossier	Set of aerodrome
Dossier	Aerodrome
Dossier	Point
Dossier	Restriction
Dossier	Actual flight
Dossier	Calculated flight
Dossier	Estimated flight
Dossier	Route
Reference	Restriction
Reference	Route
Reference	Set of aerodrome
Reference	Aerodrome
Reference	Point
RVSM Entry Exit	Point

Table 2-6: Layer painting priority

3 Aerodrome Map Human Machine Interface

3.1 Aerodrome

3.1.1 Boundary class Aerodrome map

3.1.1.1 Geographical Display

3.1.1.1.1 Layout

a) International and Domestic aerodrome (INTL, DOM).



Figure 3-1: Aeronautical chart symbol for International and Domestic aerodrome (INTL, DOM)

b) Heliport (HELI)



Figure 3-2: Aeronautical chart symbol for heliport (HELI)

c) Non-ACTIVE aerodrome (NA)



Figure 3-3: Aeronautical chart symbol for non-ACTIVE aerodrome (NA)

3.1.1.1.2 Colours

(1) The colour of the aerodrome shall change when the aerodrome is selected.

(2) The default colour of the aerodrome does not change depending on criteria. It only changes when it is selected.

(3) If the aerodrome is part of another env object then it takes the colour specified in this env object.

		Black	Yellow
1	When Selected		**X**
2	Default	**X**	

Table 3-1: Aerodrome colours

(4) The first column on the table is the priority to determine the colour if different criteria are filled.

3.1.2 Labelling

(1) It shall be possible to show or hide the label.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the **corresponding** aerodromes.

3.1.3 Map behaviour with Aerodrome

(1) When selecting an aerodrome from the popup menu it must be possible to plot the restriction, the SID or/and the STAR of this aerodrome.

3.2 Set of Aerodromes

3.2.1 Boundary class Set of Aerodrome map

3.2.1.1 Geographical Display

3.2.1.1.1 Layout

(1) When a set of aerodrome is plotted on the map the system displays the list of aerodrome included in the set and the name of the set in the centre of gravity of a bounding box around the aerodromes.



Figure 3-4: Aeronautical chart symbol for Set of Aerodrome

3.2.1.1.2 Colours

(1) The colour of the aerodrome shall change when the aerodrome is selected.

(2) The default colour of the aerodrome does not change depending on criteria. It only changes when it is selected.

(3) If the aerodrome is part of another env object then it takes the colour specified in this env object.

		Black	**Yellow**
1	When Selected		**X**
2	Default	**X**	

Table 3-2: Aerodrome colours

(4) The first column on the table is the priority to determine the colour if different criteria are filled.

3.2.1.1.3 Labelling

(1) It shall be possible to show or hide the label.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the aerodromes.

4 Point Map Human Machine Interface

4.1 Point

4.1.1 Boundary class Point map

4.1.1.1 Geographical Display

4.1.1.1.1 Layout

(1) The default layout after a certain zoom factor is the following:

a) NVA: Navigation aid Point

i) VOR:



Figure 4-1: Aeronautical chart symbol for NVA: VOR

ii) DME:



Figure 4-2: Aeronautical chart symbol for NVA: DME

iii) VOR_DME:



Figure 4-3: Aeronautical chart symbol for NVA: VOR_DME

iv) TACAN:



Figure 4-4: Aeronautical chart symbol for NVA: TACAN

v) VORTAC:



Figure 4-5: Aeronautical chart symbol for NVA: VORTAC

vi) NDB, NDB_MKR, and MKR:



Figure 4-6: Aeronautical chart symbol for NDB, NDB_MKR and MKR

vii) ILS, MLS, TLS, LOC, DF, SDF, OTHER:



Figure 4-7: Aeronautical chart symbol for ILS, MLS and LOC

viii) LOC_DME, ILS_DME, MLS_DME:



Figure 4-8: Aeronautical chart symbol for LOC_DME and ILS_DME

ix) NDB_DME:



Figure 4-9: Aeronautical chart symbol for NDB_DME

b) PWP: Way Point (see Figure 4-10)



Figure 4-10: Aeronautical chart symbol for Way Point

c) ICP: Internal Point (see Figure 4-11)

d) GEO: Geo Point (see Figure 4-11)

e) RFP: Reference Point (see Figure 4-11)

f) RAD: Radar Point (see Figure 4-11)

g) TER: Terminal Point (see Figure 4-11)

h) BDY: Boundary Point (see Figure 4-11)



Figure 4-11: Aeronautical chart symbol for other points

(2) After a high level of zoom the points shall be represented as follow:

a) NVA: Navigation aid Point (see Figure 4-7)



Figure 4-12: Aeronautical chart symbol for ILS, LLZ and ILS/LLZ

b) PWP: Way Point (see Figure 4-10)



Figure 4-13: Aeronautical chart symbol for Way Point

c) ICP: Internal Point (see Figure 4-11)

d) GEO: Geo Point (see Figure 4-11)

e) RFP: Reference Point (see Figure 4-11)

f) RAD: Radar Point (see Figure 4-11)

g) TER: Terminal Point (see Figure 4-11)



Figure 4-14: Aeronautical chart symbol for other points

(3) It shall be possible to define in a parameter file the level of zoom where the point shall be displayed in a specific form.

(4) There is another special point type that does not exist in Environment. Those points are the transition points. They are given with the profile. Those points are located where the profile crosses an airspace. We can call those points TRN for transition. The layout of those points is the following:

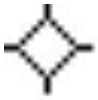


Figure 4-15: Aeronautical chart symbol for transition points

4.1.1.1.2 Colours

(1) The colour of the point shall change when the point is selected.

(2) The default colour of the point changes depending on different criteria describe in the following table:

		Blue	Yellow	Light blue	Dark blue	Orange	Pink	Green	Red
1	When Selected		**X**						
2	RS Rule followed							**X**	
3	RS Rule violated								**X**
4	rvsmEntryExit = true						**X**		
5	ifpsEntryExit = true		**X**						
6	OLD_ifpsInOut = out				**X**				
7	NVA			**X**					
8	PWP					**X**			
9	TRN		**X**						
10	All others	**X**							

Table 4-1: Point colours

(3) If the point is part of another env object then it takes the colour specified in this env object.

(4) The first column on the table is the priority to determine the colour if different criteria are filled.

4.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the **corresponding** points.

4.1.2 Map behaviour with Point

(1) When selecting a point in the popup menu it must be possible to plot the restriction, the route, the airspace, the airblock or the traffic volume that are using this point.

5 Airblock Map Human Machine Interface

5.1 Airblocks

5.1.1 Boundary class Airblock map

5.1.1.1 Geographical Display

5.1.1.1.1 Layout

a) Projected Airblock:



Figure 5-1: Aeronautical chart symbol for projected Airblock

b) Projected Airblock with coordinates:



Figure 5-2: Aeronautical chart symbol for projected Airblock with coordinates

c) Airblock layer and Airblock selected from layer:



Figure 5-3: Aeronautical chart symbol for Airblock layer and Airblock selected from layer

5.1.1.1.2 Colours

(1) The colour of the airblock shall change when the airblock is selected.

(2) The default colour of the airblock does not change depending on criteria. It only changes when it is selected.

(3) If the airblock is part of another env object then it takes the colour specified in this env object.

		Blue	**Yellow**
1	When Selected		**X**
2	Default	**X**	

Table 5-1: Airblock border

		Trans. Blue	**Trans. Yellow**
1	When Selected		**X**
2	Default	**X**	

Table 5-2: Airblock filling colours

(4) The first column on the table is the priority to determine the colour if different criteria are filled.

5.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding airblocks.

5.1.2 Map behaviour with Airblock

(1) When selecting an airblock in the popup menu there is no possible action.

6 Airspace Map Human Machine Interface

6.1 Airspace

6.1.1 Boundary class Airspace map

6.1.1.1 Geographical Display horizontal projection

6.1.1.1.1 Layout

(1) The horizontal projection of an airspace can be done with different level of definitions. The level 0 is the AS without any composing element. The level 1 is the AS with the first level of composing element. The composing element can be airspaces or airblocks. The last level will always be airblocks.

6.1.1.1.1.1 Level 0 show the Area AS

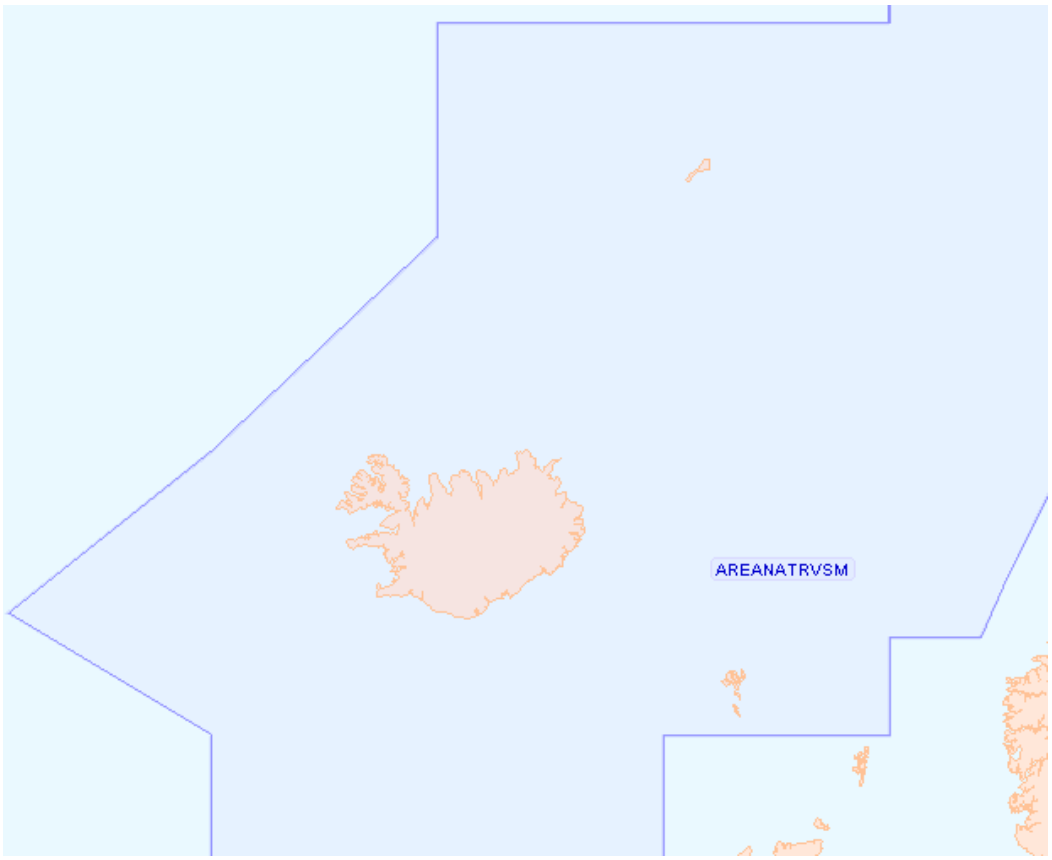


Figure 6-1: AS AREA Level 0

6.1.1.1.1.2 Level 1 show the Area AS and the NAS AS

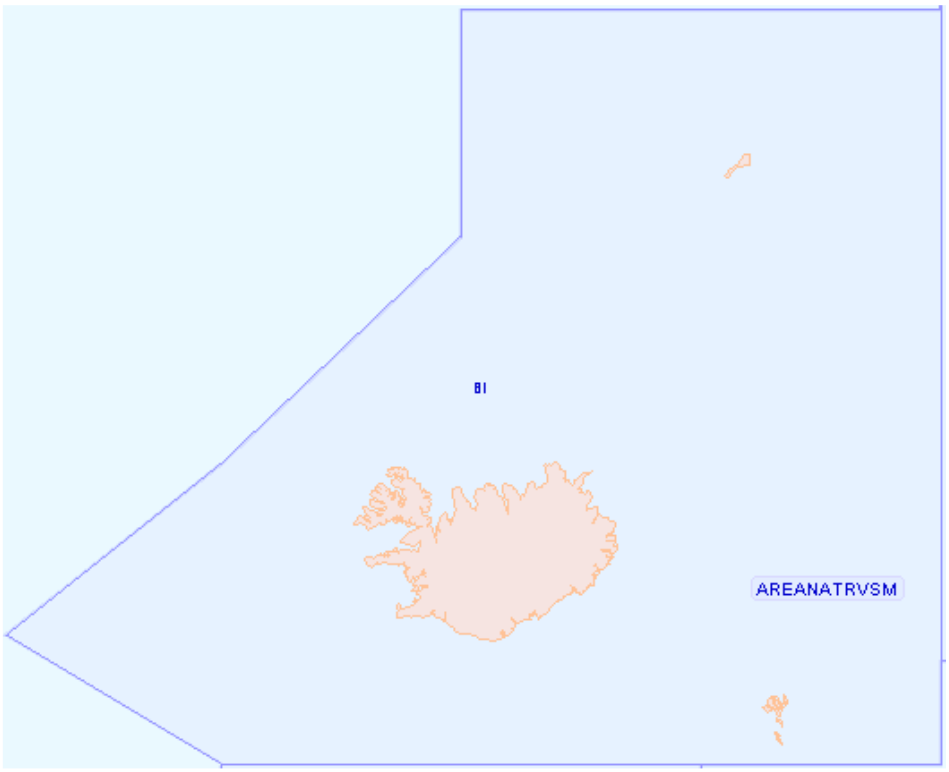


Figure 6-2: AS AREA Level 1

6.1.1.1.1.3 Level 2 show the Area AS and the FIR AS

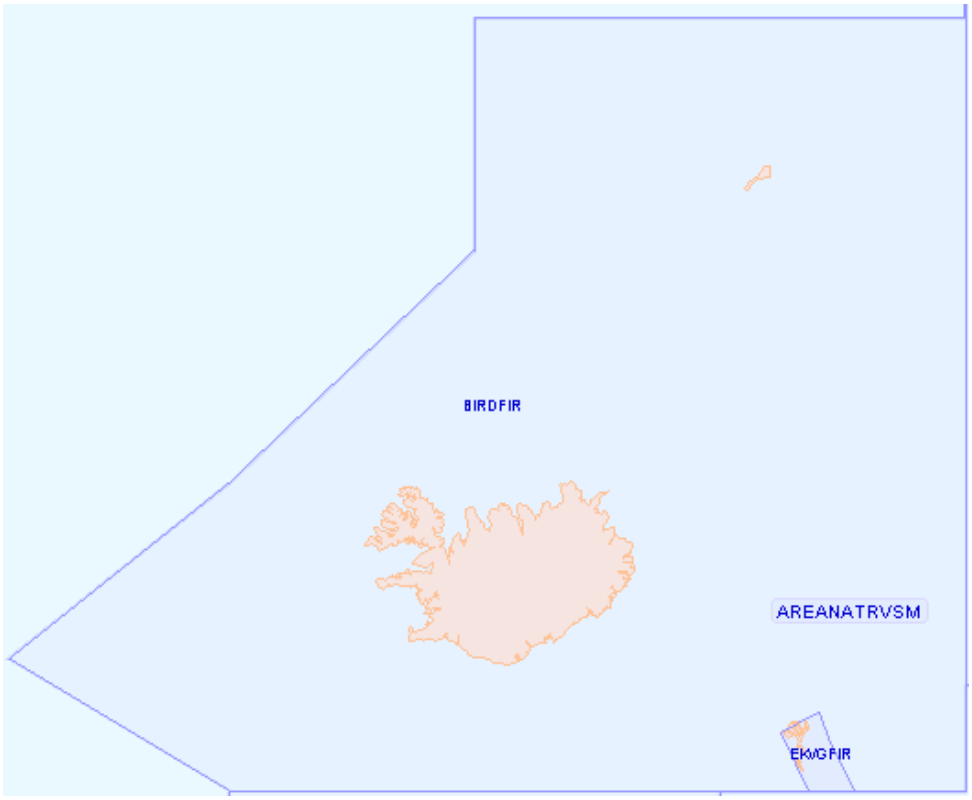


Figure 6-3: AS AREA Level 2

6.1.1.1.1.4 Level 3 show the Area AS and the AB

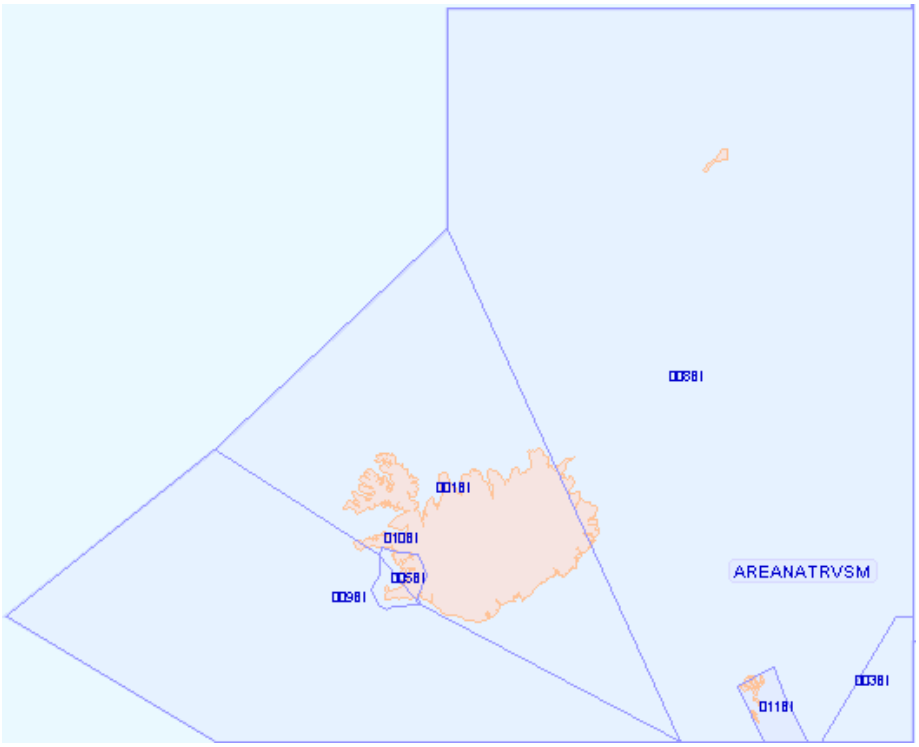


Figure 6-4: AS AREA Level 3

6.1.1.1.2 Colours

(1) The default colours are different for FMD operators, IFPS /RPL operators and ADS operators

(2) The default colour of the airspace border for the FMD Operators changes depending on different criteria describe in the following table:

		Blue	Green	Yellow	Orange	Red	Light grey
1	Airspace border only (layer)						**X**
2	Delays < 15 minutes	**X**					
3	Delays >= 15 < 30 minutes		**X**				
4	Delays >= 30 < 45 minutes			**X**			
5	Delays >= 45 minutes					**X**	
6	Capacity < 90%		**X**				
7	Capacity >= 90% < 100% minutes			**X**			
8	Capacity >= 100% < 110% minutes				**X**		
9	Capacity >= 110% minutes					**X**	

Table 6-1: Airspace border colours for FMD Operator

(3) The default colour of the airspace border for the ADS Operators changes depending on different criteria describe in the following table:

		Blue	Pink	Yellow	Grey	Light grey
1	Airspace border only (layer)					X
2	Administrative airspace				X	
3	Operational airspace	X				
4	Manageable airspace		X			
5	CDA			X		

Table 6-2: Airspace border colours for ADS Operator

(4) The first column on the table is the priority to determine the colour if different criteria are filled.

(5) The default colour of the airspace border for the IFPS /RPL Operators changes in sequence each time an airspace is displayed. The airspace border colour can be Grey, Blue, Orange or Green. When the layer is displayed the border colour is Light Grey.

(6) The default colour of the airspace filling for the FMD Operators changes depending on different criteria describe in the following table:

		Transp. Blue	Transp. Green	Transp. Yellow	Transp. Orange	Transp. Red	Transp.
1	Airspace border only (layer)						X
2	Delays < 15 minutes	X					
3	Delays >= 15 < 30 minutes		X				
4	Delays >= 30 < 45 minutes			X			
5	Delays >= 45 minutes					X	
6	Capacity < 90%		X				
7	Capacity >= 90% < 100% minutes			X			
8	Capacity >= 100% < 110% minutes				X		
9	Capacity >= 110% minutes					X	

Table 6-3: Airspace filling colours for FMD Operator

(7) The pattern is also different when the systems shows capacity or delays.

(8) The default colour of the airspace filling for the ADS Operators changes depending on different criteria describe in the following table:

		Transp. Blue	Transp. Pink	Transp. Yellow	Transp. Grey	Transp.
1	Airspace border only (layer)					X
2	Administrative airspace				X	
3	Operational airspace	X				

4	Manageable airspace		**X**			
5	CDA			**X**		

Table 6-4: Airspace filling colours for ADS Operator

(9) The first column on the table is the priority to determine the colour if different criteria are filled.

(10) The default colour of the airspace filling for the IFPS /RPL Operators changes in sequence each time an airspace is displayed. The airspace-filling colour can be Transp. Grey, Transp. Blue, Transp. Orange or Transp. Green. When the layer is displayed the colour is transparent.

(11) If the airspace is part of another env object then it takes the colour specified in this env object.

6.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the border of the corresponding airspace.

6.1.1.2 Geographical Display vertical projection

6.1.1.2.1 Layout

(1) The vertical projection of an airspace can only be done with a given route or flight profile.

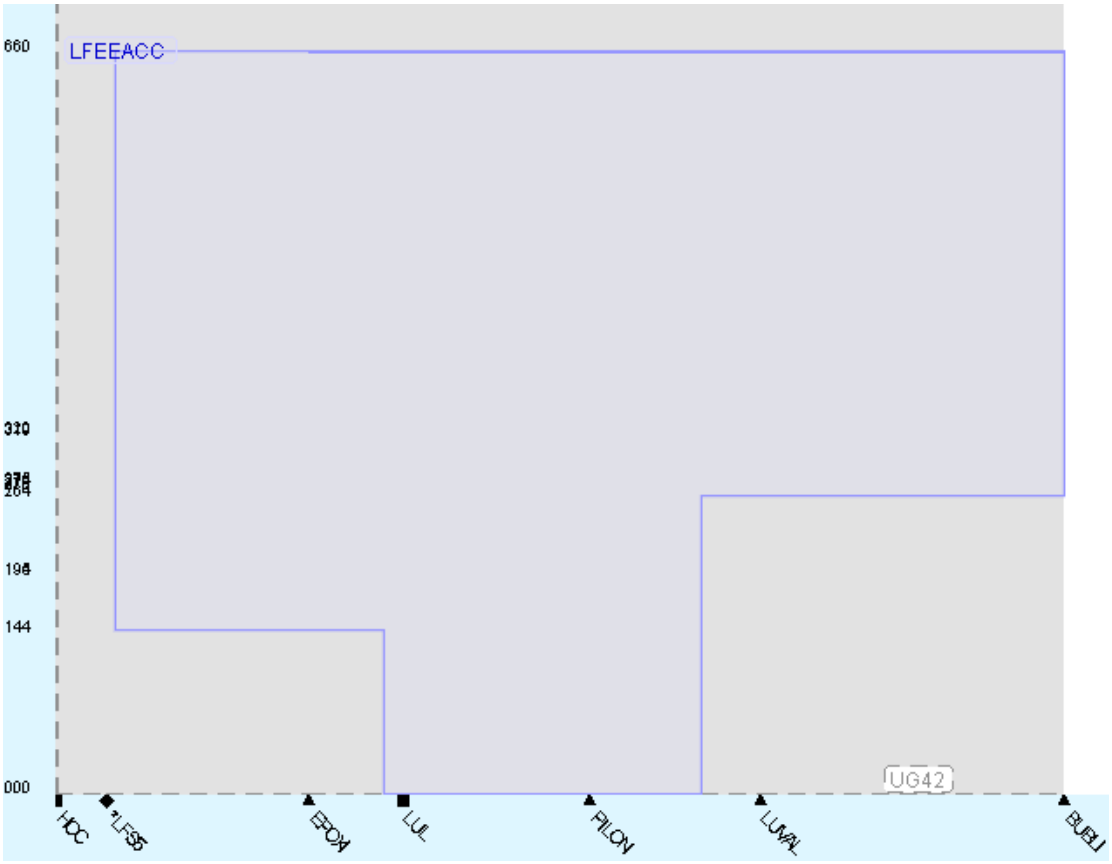


Figure 6-5: AS vertical projection

6.1.2 Map behaviour with Airspace

(1) When selecting an airspace in the popup menu there is no possible action.

7 Route Map Human Machine Interface

7.1 Route

7.1.1 Boundary class Route map

7.1.1.1 Geographical Display horizontal projection

(1) In all modes the routes will be displayed as followed:

a) A straight line colour, with a filled arrow symbol (▶) represents available routes.

b) A dotted line colour, with an empty arrow symbol (▷) represents unavailable routes.

c) Gaps are not represented.

7.1.1.1.1 General Mode

(1) This mode only shows the availability of routes, which is:

a) Available: full white, straight lines.

b) Not available: white, dashed line or optionally not visible.

c) Mixed CDR (one or more levels available AND one or more levels different (explicitly not available or undefined): white, dotted line.

(2) The General Mode is displayed after a certain level of zoom or on demand. This level of zoom must be defined in a parameter file. It must be possible to disable the display of the route network after a certain zoom level.

(3) The reference route layer can be displayed in General Mode. To change the mode of display, right click on the reference route layer in the layer control. Check the 'General Mode' property.

7.1.1.1.1.1 Layout

(1) Route shall be display with straight, dashed or dotted lines from the first point that composes it to the last point. There shall be an arrow between each segment depending on the possible direction of the route. There shall be a label between each segment if labelling is on.

a) Routes in General Mode picture:

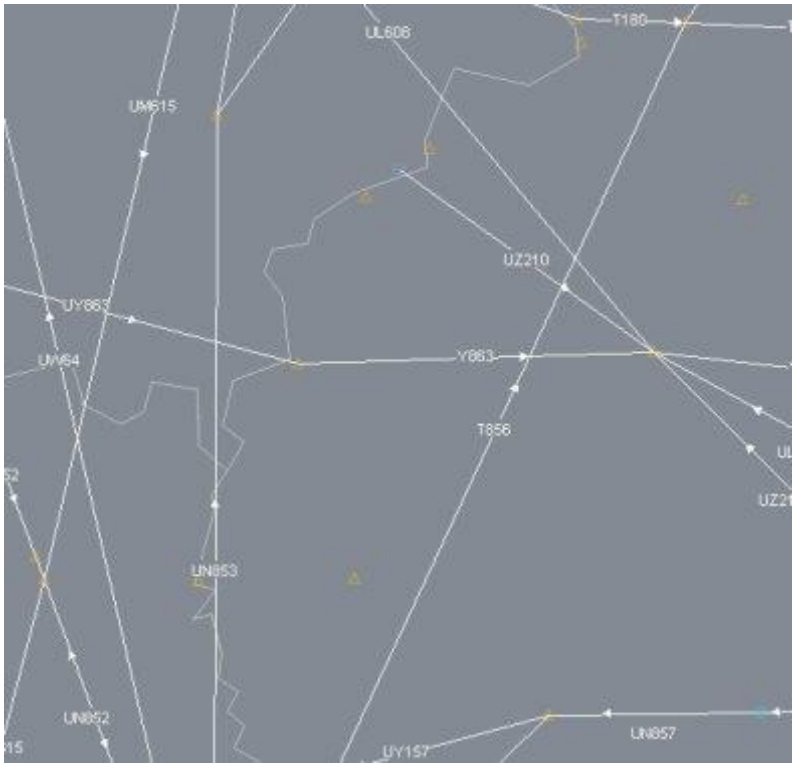


Figure 7-1: Aeronautical chart symbol for the routes in General Mode

b) STAR in General Mode picture:

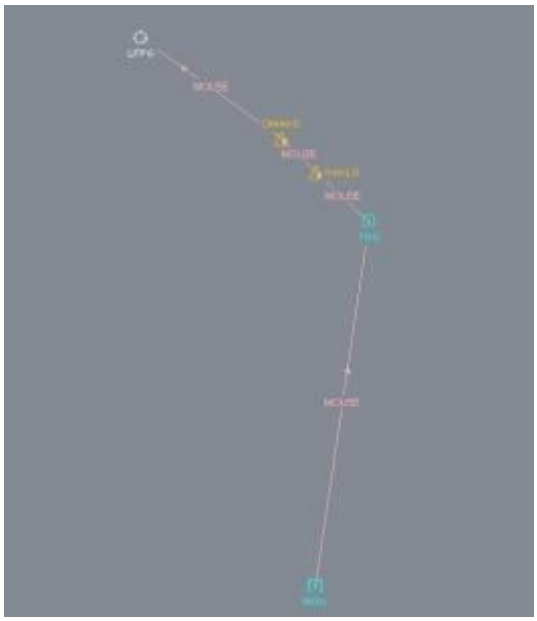


Figure 7-2: Aeronautical chart symbol for STAR in General Mode

i) If a point is the IAF of the STAR, it shall display a trigram above the point if defined in the preference.

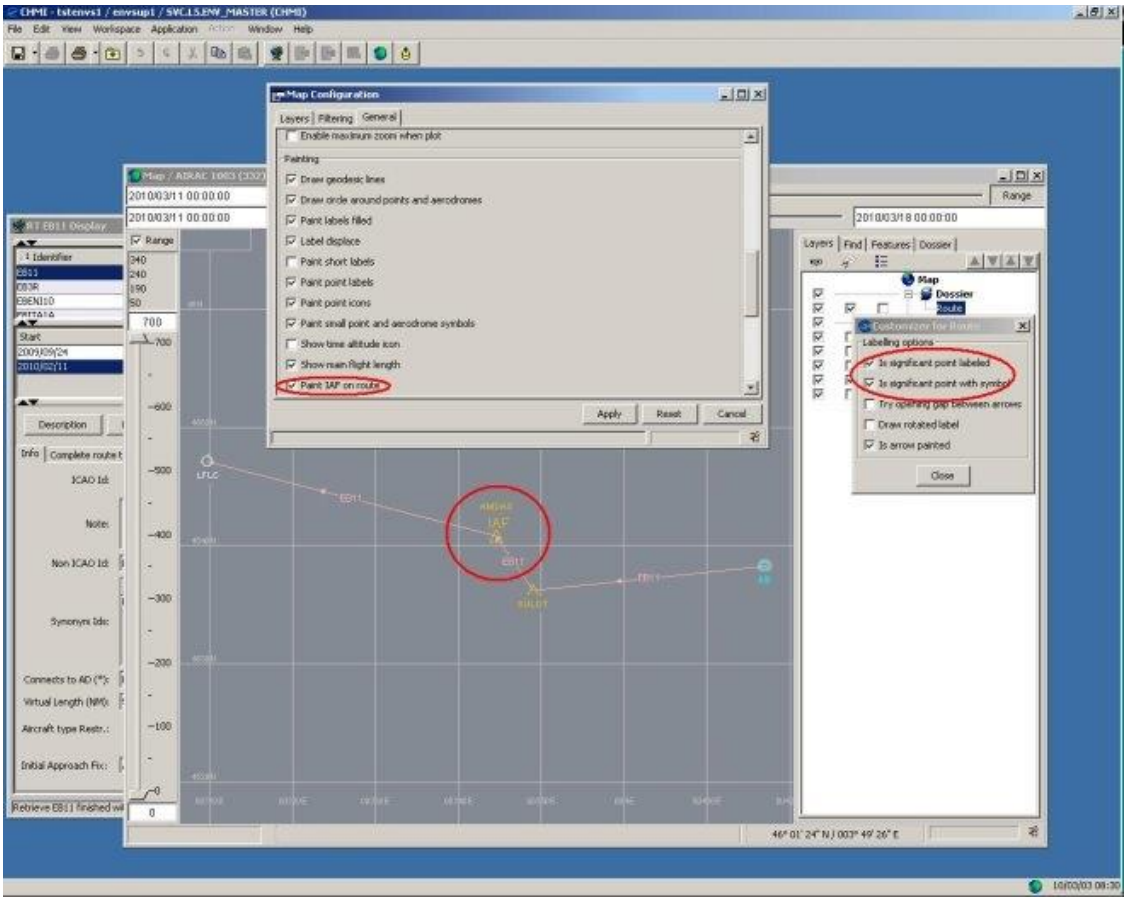


Figure 7-3: Aeronautical chart symbol for IAF of the STAR in General Mode

c) SID in General Mode picture:



Figure 7-4: Aeronautical chart symbol for SID in General Mode

7.1.1.1.1.2 Colours

(1) The colour of the route shall change when it is selected (Selected means selected in the popup menu).

(2) The default colour of the route change depending on criteria:

		Yellow	White	Pink	Blue
1	When Selected	**X**			
2	OtsRoute (otsDirection.West)			**X**	
3	OtsRoute (otsDirection.East)				**X**
4	SID				**X**
5	STAR			**X**	
6	Others		**X**		

Table 7-1: General Mode colours

(3) The first column on the table is the priority to determine the colour if different criteria are filled.

7.1.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label of the route and the label of the points that compose the route.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding route.

7.1.1.1.2 CAT Mode

(1) This is the mode where route categories are shown. Here they are:

a) CAT 1 is represented by a yellow colour.

b) CAT 2 is represented by a light blue colour.

c) CAT 3 is represented by a dark blue colour. Note that CAT 3 is never available.

d) CAT 0 (normal open ATS route) is represented by a green colour. This is always available.

e) CAT N (normal closed ATS route) is represented by a red colour. This is never available.

f) CAT U (route that is part of the time undefined) is represented by a purple colour or transparent (not displayed) according to a preference.

(2) Mixed route availability is represented by a dotted purple or magenta (@to be tested).

(3) This refers to portion where some ranges are available for flying while others are not (either explicitly unavailable for flying or undefined).

(4) « available » refers to:

a) Route portion category 0 without a NOTAM.

b) Route portion category 1 without a NOTAM.

c) Route portion category 2 with a CRAM.

(5) « Explicitly unavailable » refers to:

a) Route portion category N

b) Route portion category 0 with a NOTAM

c) Route portion category 1 with a NOTAM

d) Route portion category 2 without a CRAM

e) Route portion category 3

(6) « undefined » refers to:

a) Route portion not defined in the route definition. This corresponds to the levels always outside the vertical limits.

b) Route portion category U. This corresponds to the levels that are only part of the time undefined.

7.1.1.1.2.1 Layout

a) Routes in the CAT Mode Picture



Figure 7-5: Aeronautical chart symbol for Routes in the CAT Mode

7.1.1.1.2.2 Colours

(1) The colour of the route shall change when it is selected.

(2) The default colour of the route change depending on criteria and those criteria can change with the altitude and the time:

		Yellow	Light Yellow	Red	Dark Blue	Blue	Light Blue	Green	Light Green	*Purple*	*Magenta*
1	When Selected	X									
2	CDR0							Straight			
3	CDR0 NOTAM indicating it's closed								Dashed		
4	CDR1	Straight									
5	CDR1 NOTAM indicating it's closed		Dashed								
6	CDR2 CRAM indicating it's open.					Straight					
7	CDR2						Dashed				
8	CDR3				Dashed						
9	not Available			Dashed							
10	CDR U									Straight	
10	**Mixed** (2,4,6 and 3,5,7,8,9)										Dotted

Table 7-2: Route colours in the CAT Mode

(3) The first column on the table is the priority to determine the colour if different criteria are filled.

(4) If the route is part of another env object then it takes the colour specified in this env object.

7.1.1.1.2.3 Labelling

(1) It shall be possible to show or hide the label of the route and the label of the points that compose the route.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding route.

7.1.1.2 Geographical Display vertical projection

7.1.1.2.1 Layout

a) A route shall be displayed with a line for each flight level of the flight level series. The lines shall take a specific colour depending on the CDR category. If the route is open the lines shall be straight and the background shall be green. If the route is closed, the lines shall be dashed and the background shall be red. There shall be an arrow between each segment depending on the possible direction of the route.

b) The flight level series by id are described in BasicKinematics.doc.

c) The X axis represents the list of points, the Y axis the altitude. The space between each point shall be fix, dependent on the distance or dependent on the time when the route is displayed with a profile. It shall be possible to zoom in and out on the list of point and on the altitude by using the cursors.

d) It shall be possible to show or hide the flight level series and to show or hide the levels.

e) When the vertical view is open via the horizontal view, it must automatically zoom on the portion that was selected on the horizontal view. The portion before and after the selected portion must also be displayed.

f) On the top of the vertical view, the list of all points must always be visible (the controls to manipulate the direction and scale must be moved on the top/right corner). The visible part of the air route on the vertical view must be highlighted. It must be possible to change the visible part by selecting the points in the list.

g) The vertical size of the vertical view must be set to the range of flight levels defined.

h) The vertical view must give the possibility to request the window describing the route in details. The vertical view must also be accessible from the query panel i.e. it must be possible to directly plot a route on the vertical view without going through the horizontal view.

i) A vertical projection picture:

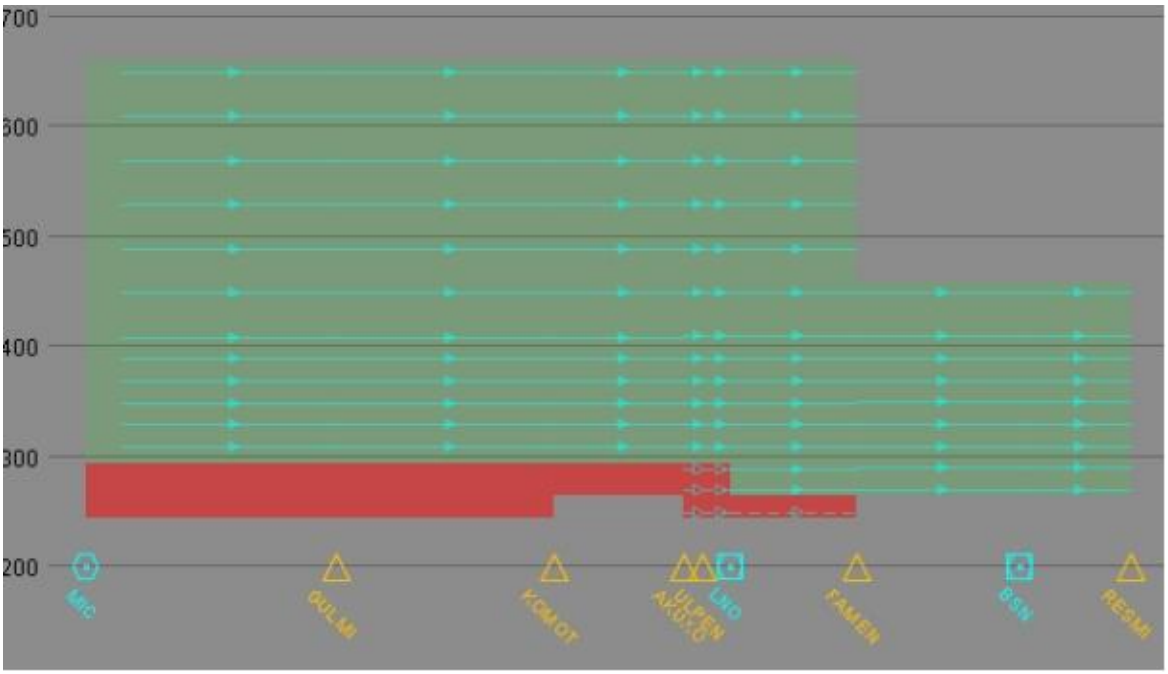


Figure 7-6: Aeronautical chart symbol for vertical projection of Route

7.1.1.2.2 Colours

1. The colour and style of the Route flight level series is defined by the following tables:

		Green	Yellow	Light Blue	Dark Blue	Red
1	CDR0	**X**				
2	CDR0 NOTAM indicates it's closed	**X**				
3	CDR1		**X**			
3	CDR1 NOTAM indicates it's closed		**X**			
4	CDR2			**X**		
4	CDR2 CRAM indicates it's open			**X**		
5	CDR3				**X**	
8	Not Available					**X**

Table 7-3: Route vertical line colours

		Straight	Dashed
1	CDR0	**X**	
2	CDR0 NOTAM indicates it's closed		**X**
3	CDR1	**X**	
4	CDR1 NOTAM indicates it's closed		**X**
5	CDR2		**X**
6	CDR2 CRAM indicating it's open	**X**	
7	CDR3		**X**
8	CDRU	**N/A**	**N/A**
9	Not Available	**N/A**	**N/A**
10	No definition	**N/A**	**N/A**

Table 7-4: Route vertical line styles

(1) The colour of the background depends on the state of the route. If the route is open, the background will be green, otherwise it will be red. If there is no route portion definition then there shall be no background. If the route is CDR U, the background will be purple or there shall be no background depending on the preference.

		Green	Red	Purple	No background
1	CDR0	**X**			
2	CDR0 NOTAM indicates it's closed		**X**		
3	CDR1	**X**			
4	CDR1 NOTAM indicates it's closed		**X**		
5	CDR2		**X**		
6	CDR2 CRAM indicating it's open	**X**			
7	CDR3		**X**		
8	Not Available		**X**		
9	CDRU			**X**	**X**
10	No definition				**X**

Table 7-5: Vertical view background colour

7.1.1.2.3 Labelling

(1) It shall be possible to show or hide the label of the route.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

7.1.2 Map behaviour with route

7.1.2.1 IFPS, RPL and TACT behaviour

7.1.2.1.1 Mode selection

(1) **IFPS, RPL & TACT** will be by default in the *General mode* display; and in this mode, non-available routes are NOT displayed, unless specified by user (map configuration menu).

(2) Then, if user selects (means released from the popup menu with shift key down) routes, this individual route will be automatically shown in the 'CAT' colour scheme, even though the general mode is still in force.

(3) Additionally, the user has the possibility to switch to *CAT mode* at any time. Then, all routes are displayed in colour.

7.1.2.1.2 Level band, time and direction selection

(1) **IFPS,RPL & TACT** maps will be by default set to the time, level(the highest level in the flight) & direction (the vector is from ADEP to ADES and the angle is 180) derived from the R/FPL.

(2) If the map does not derive from an R/FPL, the default level band is 000/999, time is actual time and direction is 360 degrees. The three parameters time, level, direction can be changed manually (e.g. via slide bars & vector & editable window).

7.1.2.2 ENV behaviour

7.1.2.2.1 Mode selection

(1) **ENV** will be by default in *CAT mode*. Though after a certain zoom out level (editable in user ID profile), the system will automatically turn into *General mode* display, for readability purposes. In this mode, labels disappear.

(2) Additionally, the user has the possibility to switch to general mode at any time.

7.1.2.2.2 Level band, time and direction selection

(1) **ENV** will not derive its time, level & direction from a FPL, but on the other hand, it needs access to different AIRAC's.

(2) Therefore, a map is always set to an AIRAC.

(3) Then, on the OPS release, default date & time is actual time. Direction is 360 degrees. Level band 000/999.

(4) On any other AIRAC, default time is the first day of the AIRAC at 00h01.

(5) Default level band is always 000/999. Direction is still 360 degrees.

8 Profile Map Human Machine Interface

8.1 Flight Profile

8.1.1 Introduction

(1) Flights from ATFCM, Predict, Archive and from IFPS can be plotted on the horizontal map and on the vertical map.

8.1.2 Plot a flight from ATFCM

8.1.2.1 Plot a flight on the horizontal map

(1) Select **Application/ ATFCM/ Flights/ Flight List** or **Application/ Predict/ Flights/ Flight List** or **Application/ Archive/ Flights/ Flight List**. The empty window for the flights is displayed.

(2) Enter i.e. in 'Where': Aerodrome and in 'Is': EBBR.

(3) Click on the 'Refresh' Button in the toolbar.

(4) The list with flights is displayed.

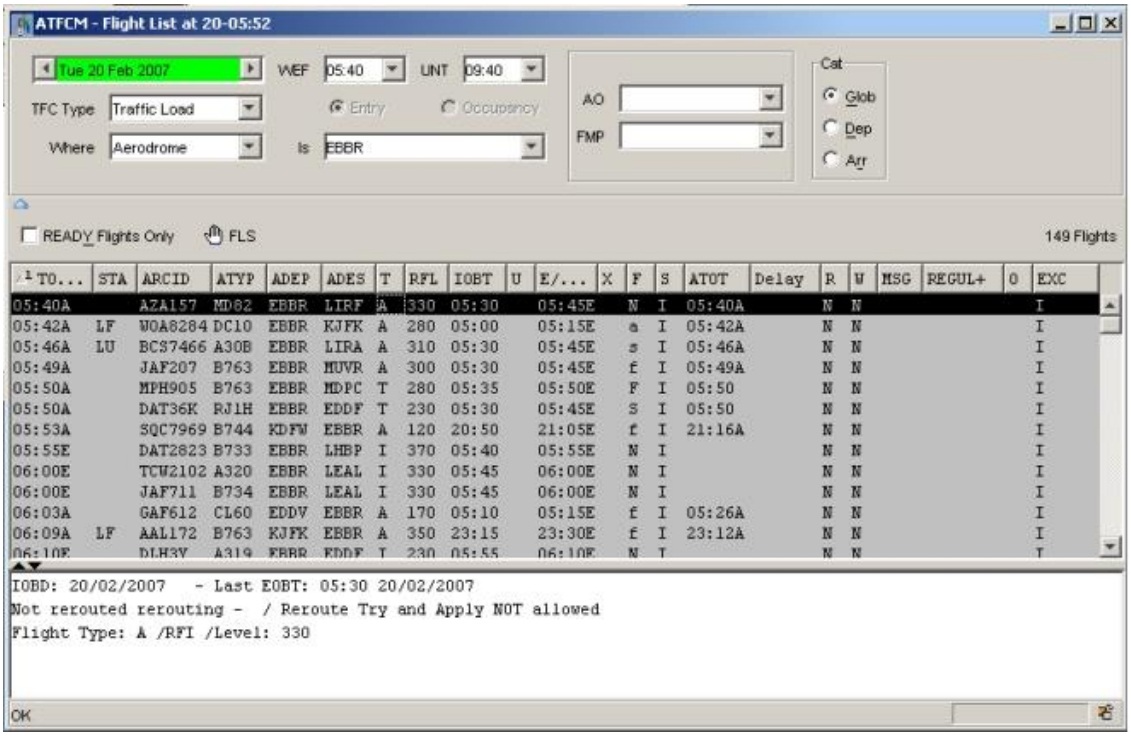


Figure 8-1: Plot flight on map - Flight List selection

(5) Right click on a flight. Select 'Flight Data'.

(6) The flight data of the selected flight is displayed.

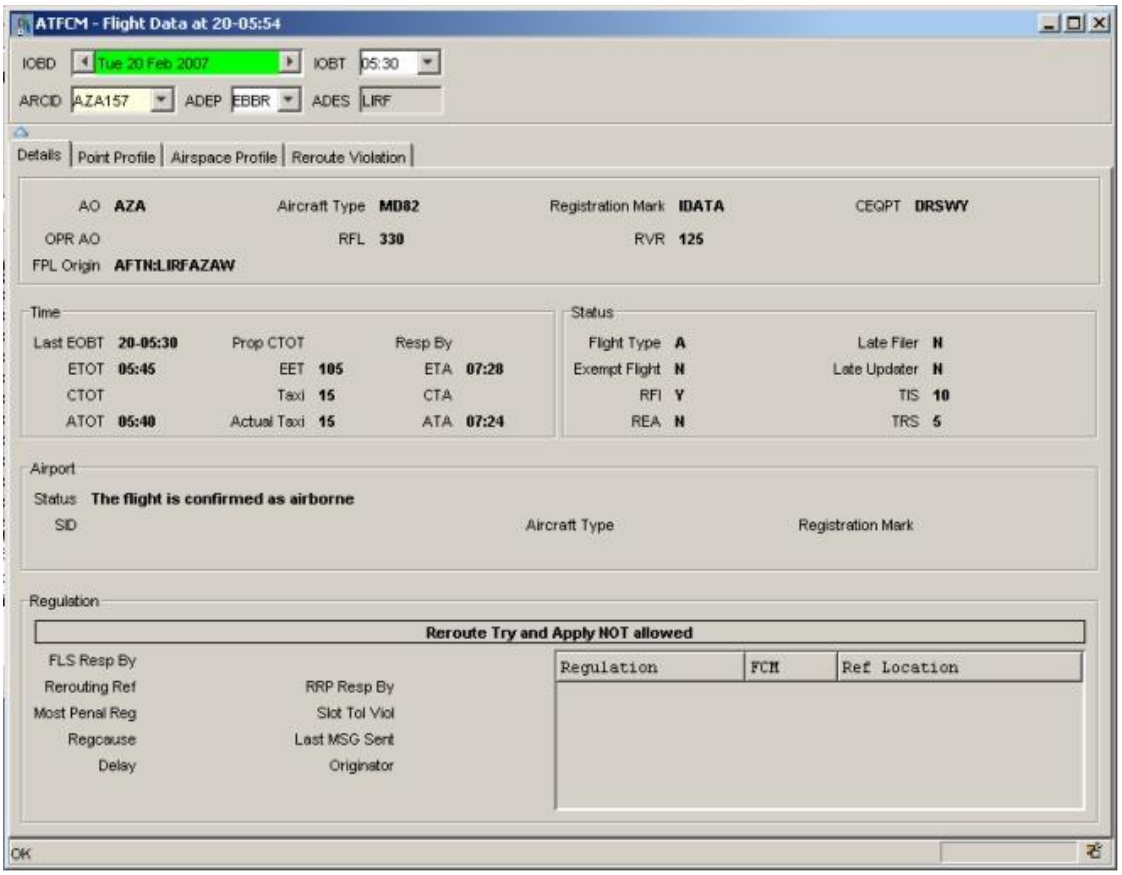


Figure 8-2: Plot flight on map from Flight Data

(7) There are 2 buttons in the toolbar.

a) The first to display the flight on the horizontal map.

b) The second to display the flight on the horizontal map and to open a vertical map for this flight.



Figure 8-3: Horizontal and vertical plot flight buttons

(8) Click on the Plot 'Horizontal Button' Button in the toolbar.

(9) The map is opened.

(10) The flight is displayed on the map.

(11) The lower map time boundary is set to the lowest value out of Initial or Latest OBD/T.

(12) The upper map time boundary is set to the highest value out of Initial or Latest OBD/T.



Figure 8-4: Plot flight on horizontal map

(13) In the 'Layers' tab, there are layers added to the 'Dossier'. 3 types of layers can be added in this order (if the flight has this kind of profile):

a) A layer 'Actual' that contains the actual flights. This flight is painted in white.

b) A layer 'Estimated' that contains the estimated flights. This flight is painted in blue.

c) A layer 'Calculated' that contains the calculated flights. This flight is painted in black.

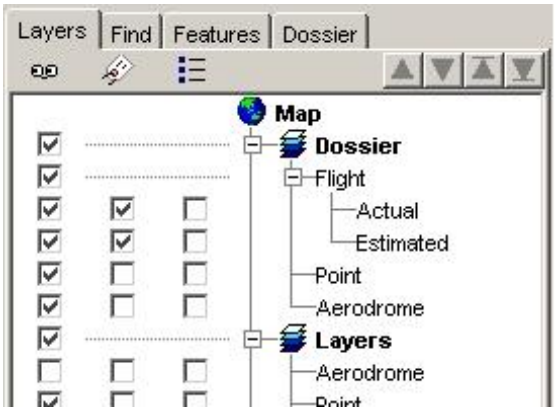


Figure 8-5: Layers added to map dossier

(14) The order of the layers can be changed, i.e. select the layer 'Estimated'. Click on the 'up' arrow. Now the blue estimated flight will be on top.

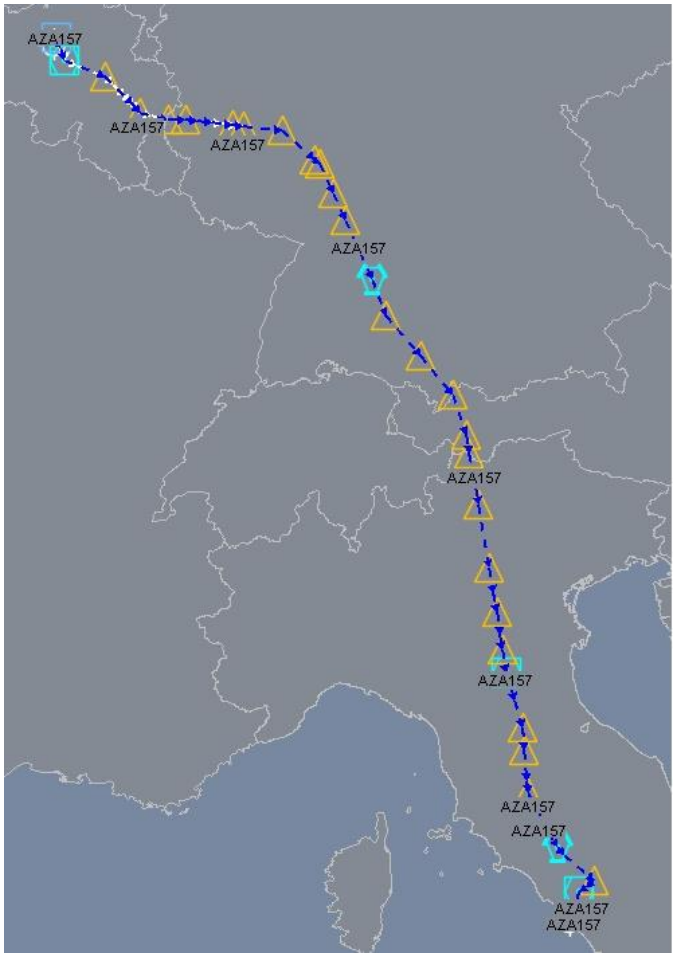


Figure 8-6: Layers added to map dossier - Changed order

(15) The flight appears also in the 'Dossier' tab.

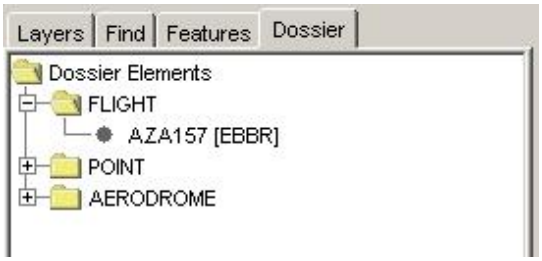


Figure 8-7: Flight added to Dossier tab

(16) Select again the 'Flight Data' window by clicking on it.

(17) The 'Plot Horizontal' button in the toolbar is pressed.



Figure 8-8: Horizontal and vertical plot flight buttons - Remove flight

(18) Click again on this button.

(19) The flight is removed from the map.

8.1.2.2 Plot a flight on the vertical map

(1) Show again the data of an ATFCM, Predict or Archive flight.

(2) Click on the 'Plot Horizontal and Vertical' Button in the toolbar.

(3) The map is opened.

(4) The flight is displayed on the horizontal map.

(5) A vertical view for the flight is also displayed.

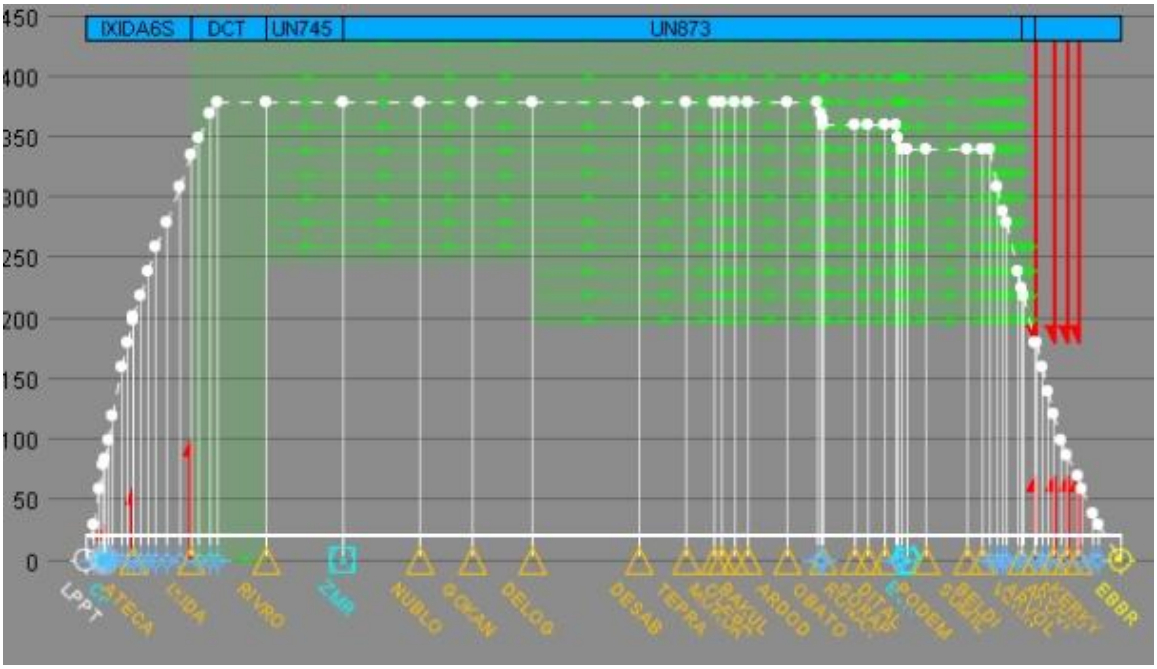


Figure 8-9: Plot flight on vertical map

(6) There are also different flight layers: 3 types of layers can be added in this order (if the flight has this kind of profile):

a) A layer 'Actual' that contains the actual flights. This flight is painted in white.

b) A layer 'Estimated' that contains the estimated flights. This flight is painted in blue.

c) A layer 'Calculated' that contains the calculated flights. This flight is painted in black.

(7) Select again the 'Flight Data' window by clicking on it.

(8) The 'Plot Horizontal' button and 'Plot Horizontal and Vertical' button in the toolbar are pressed.



Figure 8-10: Horizontal and vertical plot flight buttons - Plot both

(9) Click again on the 'Plot Horizontal and Vertical' button.

(10) The flight is removed from the map and the vertical map is closed.

8.1.3 Plot group of flight from ATFCM

(1) Select **Application/ ATFCM / Flights / Flight List** or **Application / Predict / Flight List** or **Application / Archive / Flight List**. The empty window for the flights is displayed.

(2) Enter i.e. in 'Where': Aerodrome and in 'Is': EBBR.

(3) Click on the 'Refresh' Button in the toolbar.

(4) The list with flights is displayed.

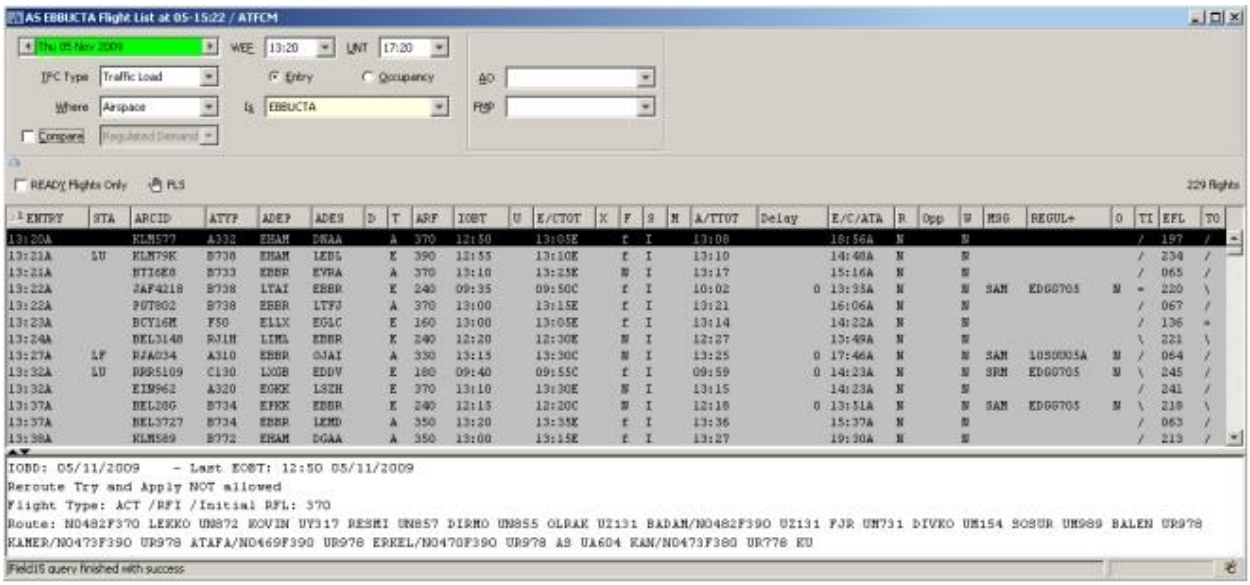


Figure 8-11: Plot group of flight on map - Flight List selection

(5) There are 2 buttons in the toolbar.

a) The first to plot all flights from the flight list on the horizontal map.

b) The second to plot the flights currently selected in the flight list on the horizontal map.



Figure 8-12: Plot All and Plot Selected flight buttons

(6) Click on the **Plot All** toggle Button **or** Click on the **Plot Selected** toggle Button in the toolbar.

(7) The map is opened (the **Plotted Flight List** is opened before if requested via **Plot Selected**)

(8) The group of flights are displayed on the map.

(9) The lower map time boundary is set to the lowest Initial or Latest OBD/T of the plotted flights.

(10) The upper map time boundary is set to the highest Initial or Latest OBD/T of the plotted flights.

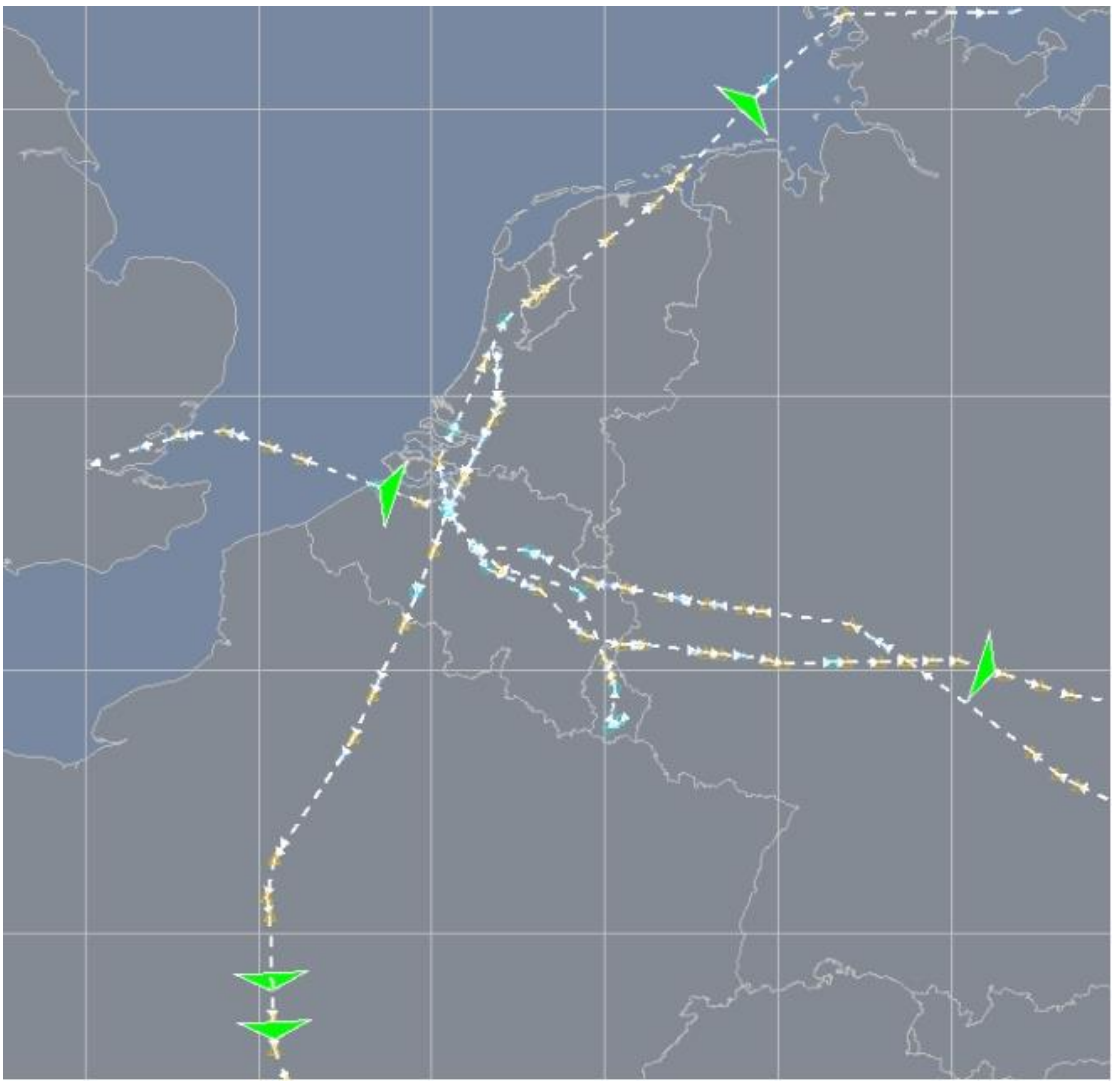


Figure 8-13: Plot group of flight on the map

(11) In the 'Layers' tab, there is a layers called *<title of the linked flight list>* added to the 'Dossier' in case of **Plot All** or in case of **Plot Selected**. In that layer, 4 types of layers can be added in this order:

a) A layer 'Latest' that contains the latest profile. In other words

i) Actual profile when it is available

ii) Calculated profile when Actual is not available

iii) Estimated profile when Actual and Calculated are not available

b) A layer 'Actual' that contains the actual flights. This layer can only be enabled when Latest is not selected.

c) A layer 'Calculated' that contains the calculated flights. This layer can only be enabled when Latest is not selected.

d) A layer 'Estimated' that contains the estimated flights. This layer can only be enabled when Latest is not selected.

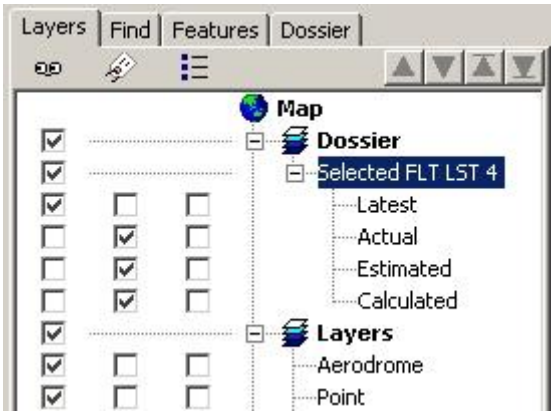


Figure 8-14: Layers added to map dossier

(12) The order of the layers can be changed, i.e. select the layer 'Estimated'. Click on the 'up' arrow. Now the blue estimated flight will be on top.

(13) The *<title of the linked flight list>* appears also in the 'Dossier' tab with the list of flights.

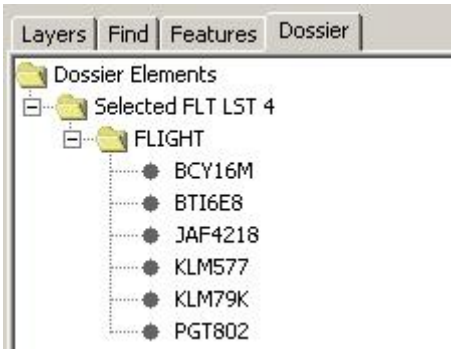


Figure 8-15: <title of the linked flight list> added to Dossier tab

(14) Select again the 'Flight List' window by clicking on it.

(15) The 'Plot All' or the 'Plot Selected' button in the toolbar is pressed.



Figure 8-16: Remove group of flights

(16) Click again on the pressed button.

(17) The group of flights is removed from the map.

(18) In the 'Layers' tab, the layers called *<title of the linked flight list>* is removed from the 'Dossier' layer.

8.1.4 Plot a route catalogue from ATFCM or Predict

8.1.4.1 Plot a route catalogue on the horizontal map

(1) Select Application/ ATFCM/ Environment/ Route catalogue or Application/ Predict/ Environment/ Route catalogue. The empty window for the route catalogue is displayed.

(2) Enter i.e. in 'Departure': EBBR and in 'Destination': LFPG.

(3) Click on the 'Send' Button in the toolbar.

(4) The list with route catalogue is displayed.

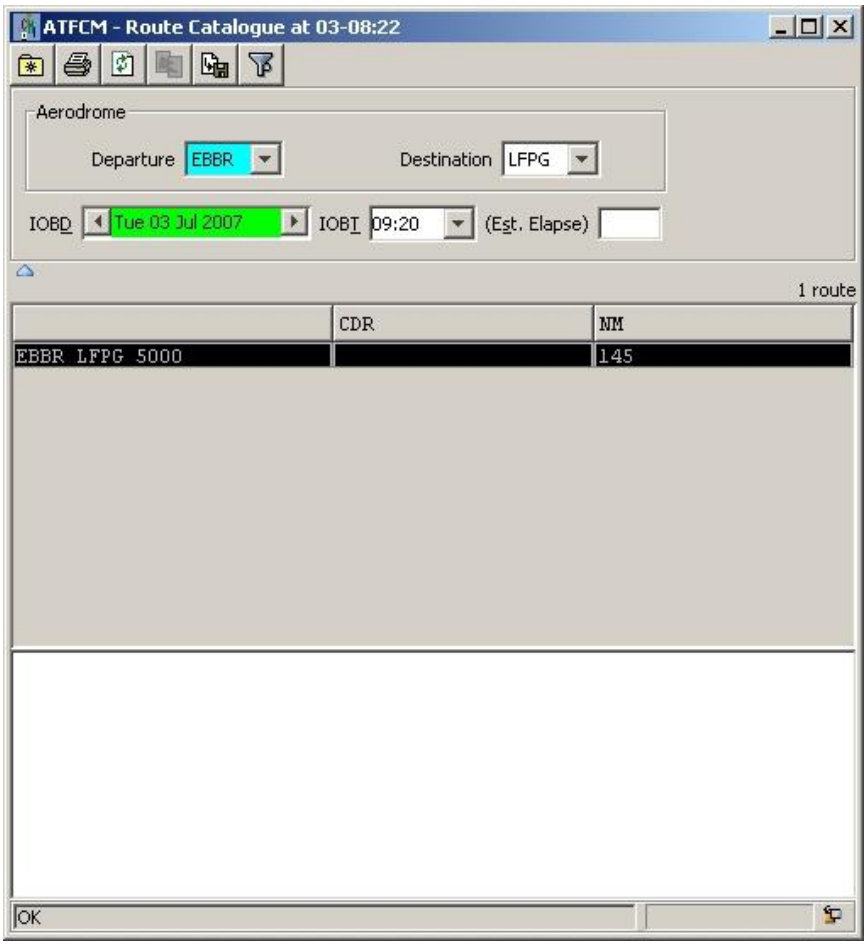


Figure 8-17: Plot route from Route Catalogue

(5) Right click on a route.

(6) 2 menu options appear in the popup menu.

(7) Click on the 'Plot Route' option.

(8) The map is opened.

(9) The route is displayed on the map.



Figure 8-18: Plot route on horizontal map

(10) In the 'Layers' tab, there is a layer added to the 'Dossier'. One type of layer is added:

a) A layer 'Route catalogue' that contains the actual route.

(11) The route from the catalogue is painted in green.

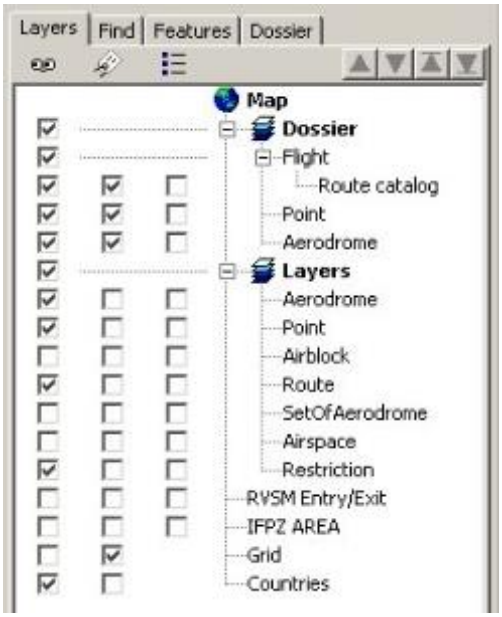


Figure 8-19: Layer flight / Route catalogue added to map dossier

(12) The route appears also in the 'Dossier' tab.

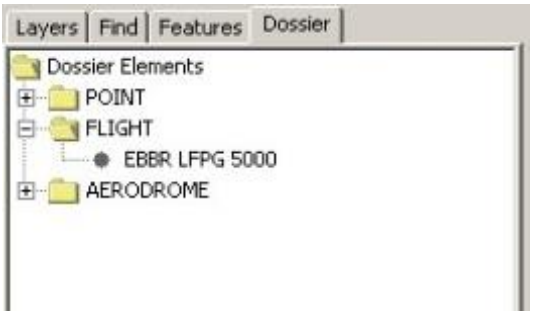


Figure 8-20: Flight route added to Dossier tab

8.1.4.2 Plot a route catalogue on the vertical map

(1) Right click on the plotted route select 'Dossier/Flight/<name of route catalogue>/vertical view'.

(2) The vertical view is opened.

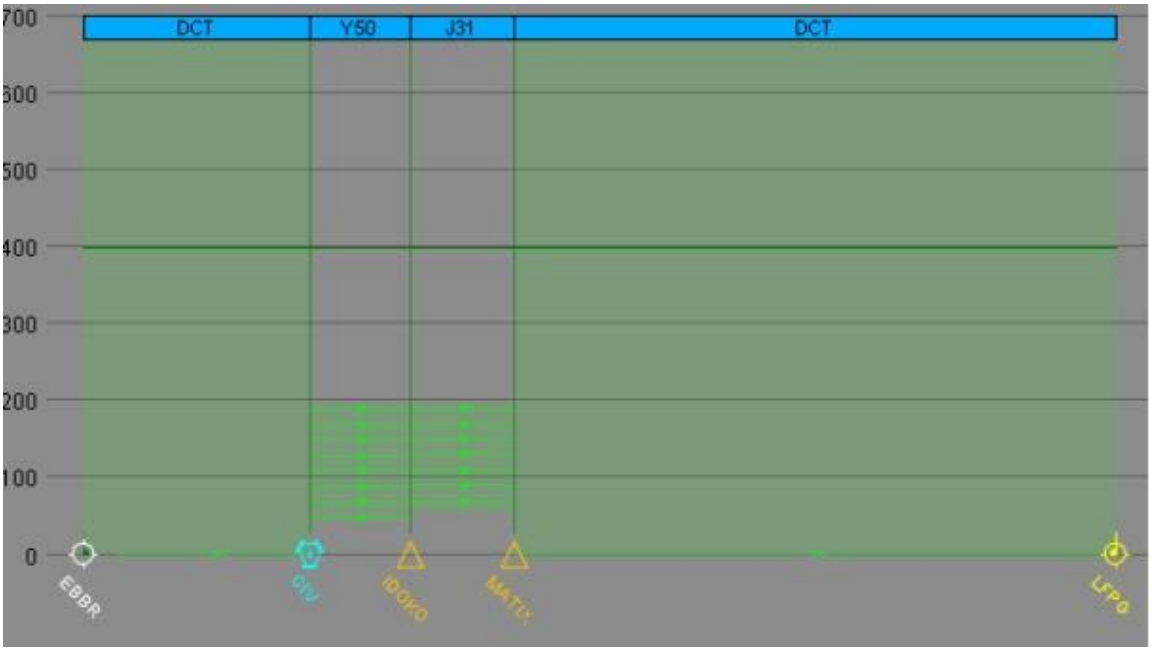


Figure 8-21: Plot route on vertical map

(3) There is one flight layer:

a) A layer 'Route Catalogue' that contains the actual route.

(4) This route is painted in green.

8.1.5 Plot an alternate route from ATFCM

8.1.5.1 Plot an alternate route on the horizontal map

(1) Select Application/ ATFCM/ Flights/ Flight List. The empty window for the flights is displayed.

(2) Enter i.e. in 'Where': Aerodrome and in 'Is': EBBR.

(3) Click on the 'Refresh' Button in the toolbar.

(4) The list with flights is displayed.

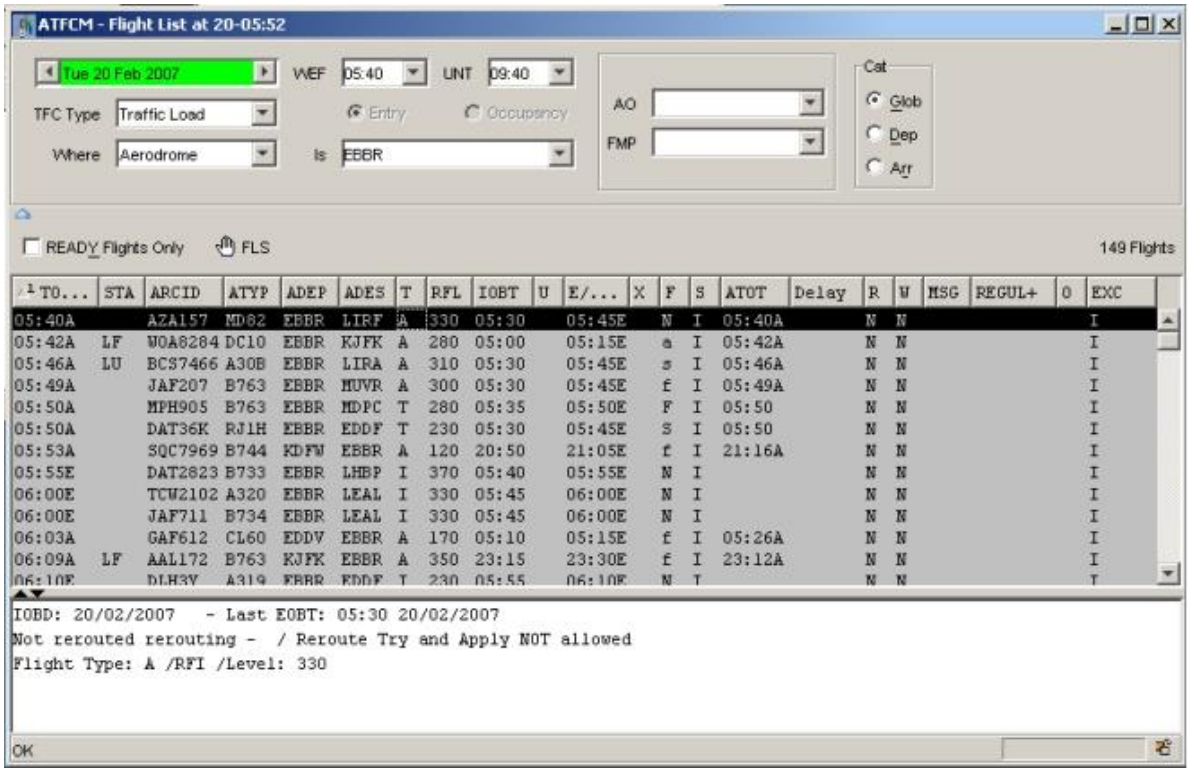


Figure 8-22: Plot alternate route on map - Flight List selection

(5) Right click on a flight. Select 'Alternate Routes'.

(6) The list with alternate routes is displayed.

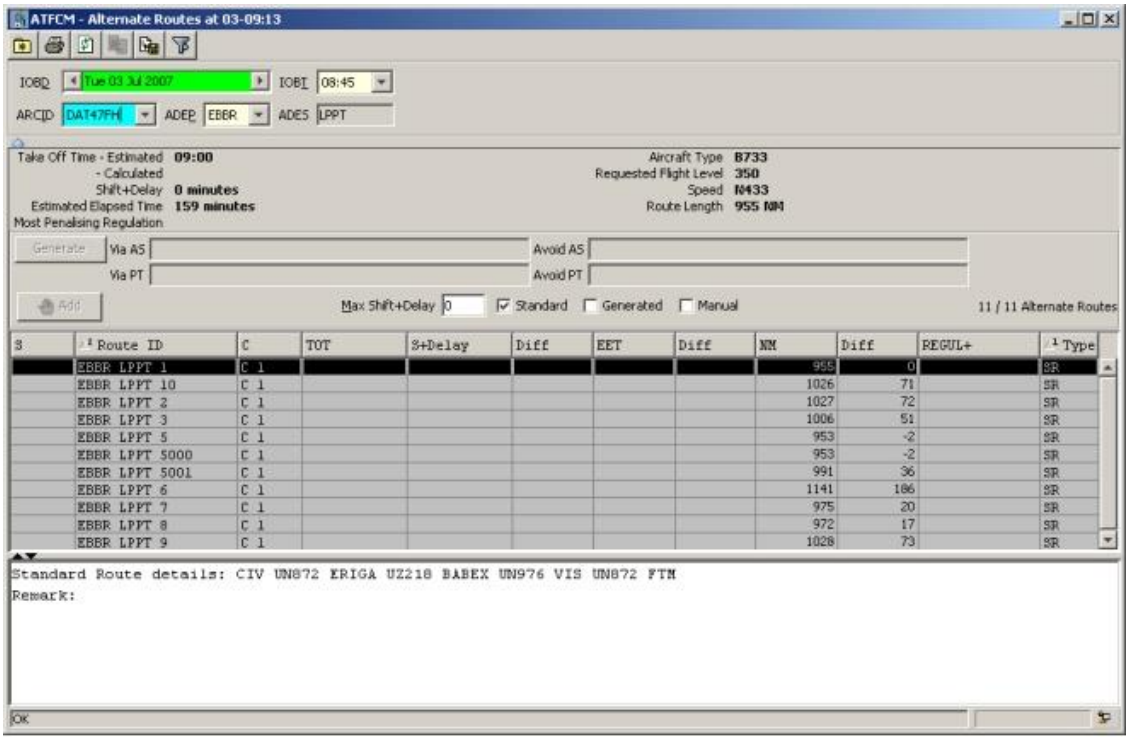


Figure 8-23: Plot alternate route on map from alternate route list

(7) Right click on a route.

(8) 2 menu options appear in the popup menu.

(9) Click on the 'Plot Route' option.

(10) The map is opened.

(11) The route is displayed on the map.

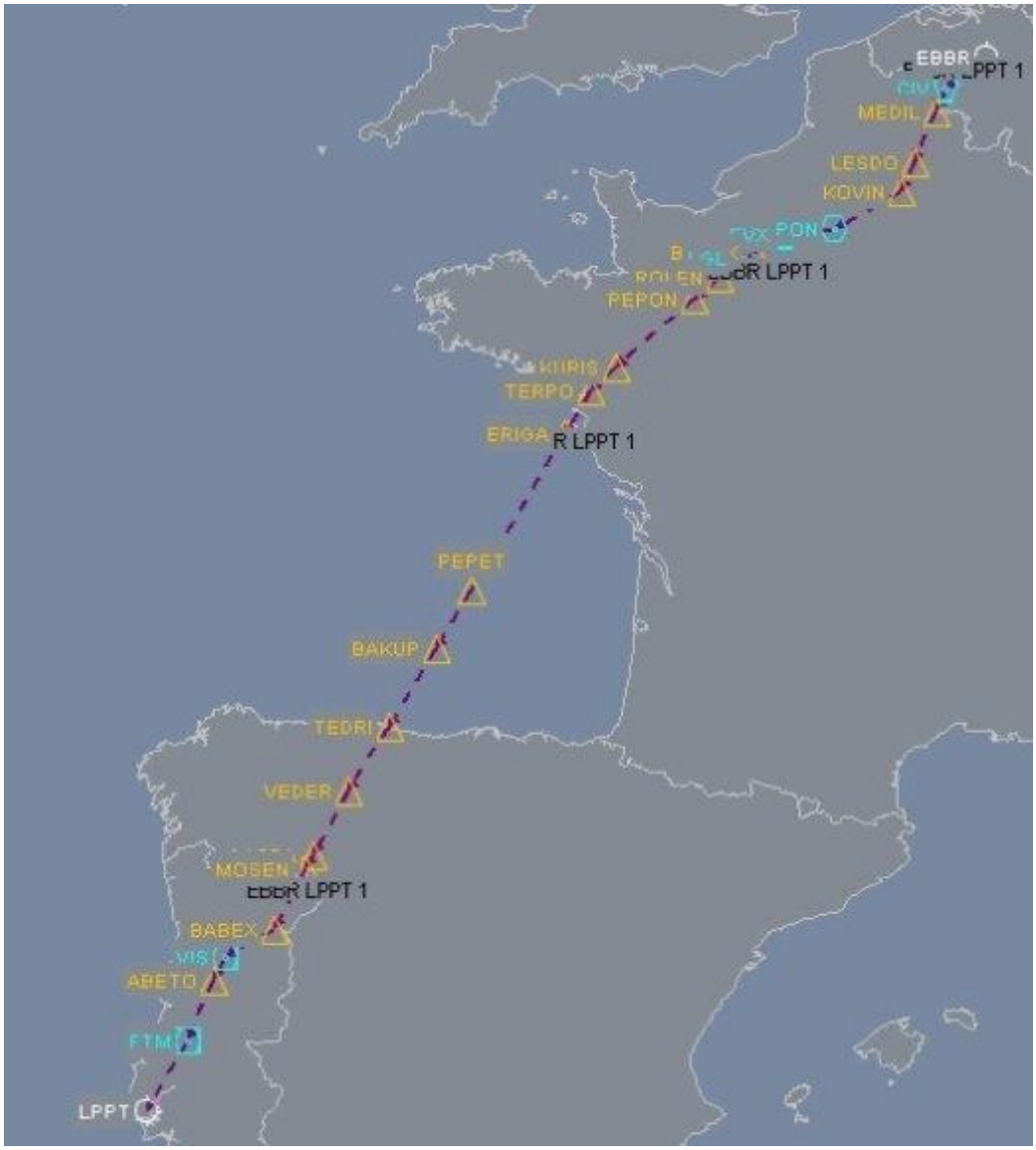


Figure 8-24: Plot alternate route on horizontal map

(12) In the 'Layers' tab, there is a layer added to the 'Dossier'. One type of layer is added:

a) A layer 'Route catalogue' that contains the actual route.

(13) The route from the catalogue is painted in green.

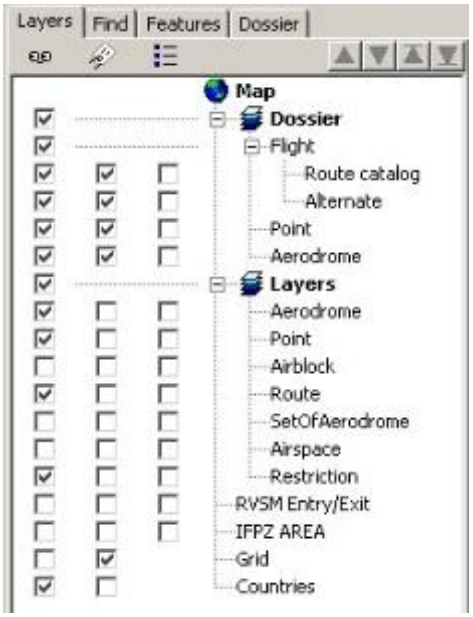


Figure 8-25: Layer flight / Alternate added to map dossier

(14) The route appears also in the 'Dossier' tab.

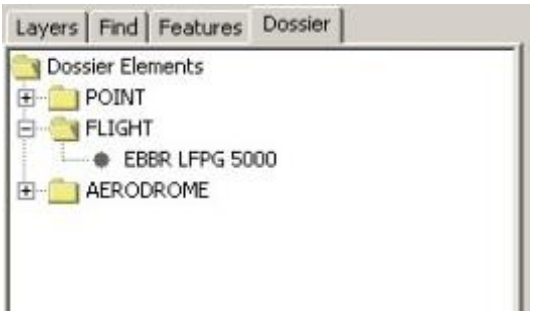


Figure 8-26: Flight alternate route added to Dossier tab

8.1.5.2 Plot a route catalogue on the vertical map

(1) Right click on the plotted route select 'Dossier/Flight/<name of alternate route>/vertical view'.

(2) The vertical view is opened.

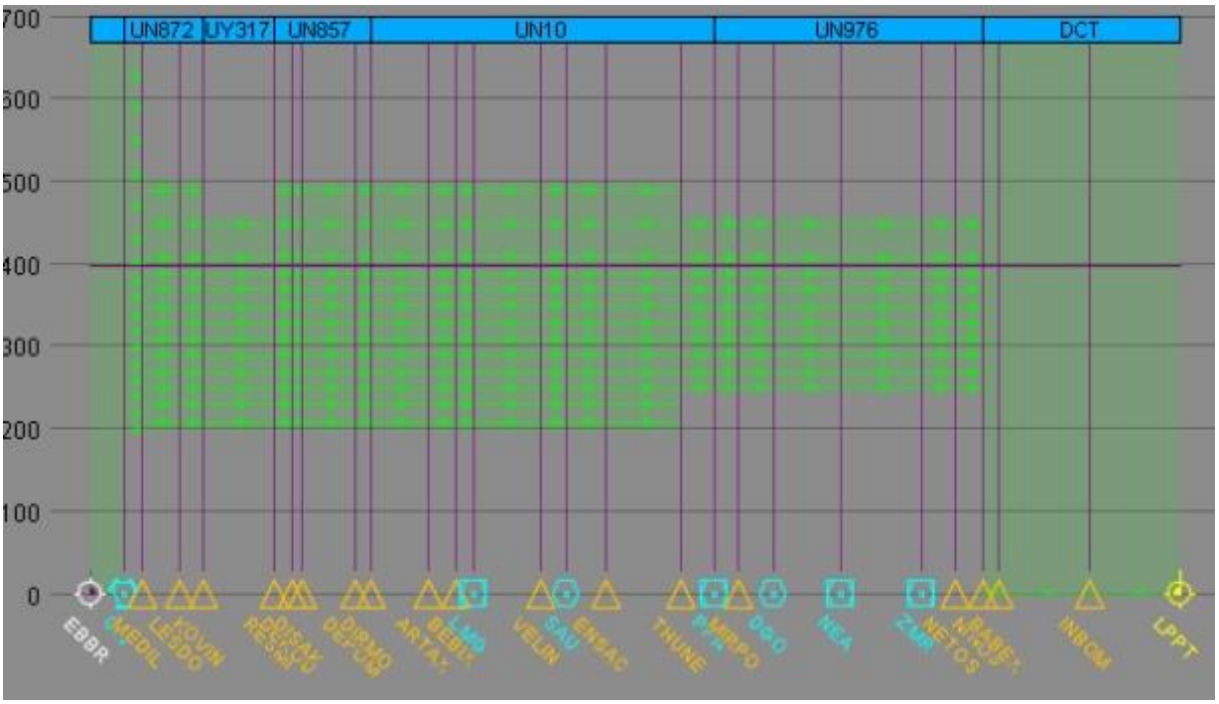


Figure 8-27: Plot alternate route on vertical map

(3) There is one flight layer:

a) A layer 'Alternate Route' that contains the actual route.

(4) This route is painted in purple.

8.1.6 Plot a flight from IFPS

8.1.6.1 Plot a flight on the horizontal map

(1) There are different possibilities to plot a flight from IFPS. We show here one way.

(2) Select 'Application/ Ifps/ Create New FPL'.

(3) Enter i.e. in the text area:
(FPL-AIN788-IN
-DC86/H-SDGHWY/C
-EGVN1500
-N0468F230 DCT MALBY L9 CPT DCT BPK/N0470F350 UM185 CLN UL620 BASNO
UL603 TEGBA UL605 BULEN/N0470F350 UL605 BAG UL614
BULUT/N0470F370 UL614 SRT UR21 KABAN/N0472F370 R784 NOLDO UP975
ILMAP DCT
-ORMM0545 OKBK
-EET/EHAA0033 EDVV0054 EDUU0103 LOVV0136 LHCC0155 LRBB0220
LBWR0252 LTBB0314 LTAA0328 ORBB0439
REG/ZSOSI SEL/AQCJ)

(4) Click on 'test'.

(5) If the flight plan can be plotted, the 'Plot' button is enabled.

(6) Click on 'Plot'.

(7) The map is opened or brought to the front, if it was already open.

(8) The flight is displayed on the horizontal map.

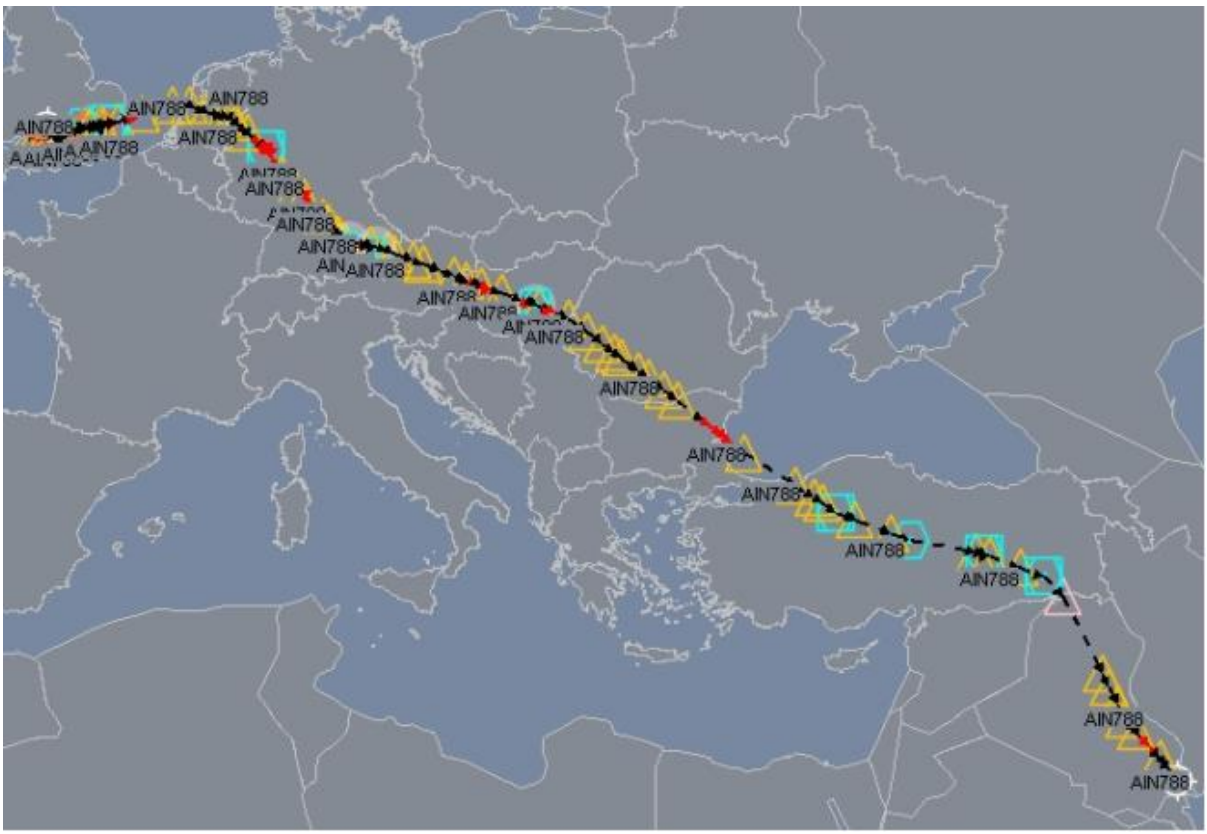


Figure 8-28: Plot IFPS flight on horizontal map

(9) The flight is added to the layer with the estimated flights.

(10) The flight is painted in black.

(11) The flight can be removed by removing all the elements from the dossier. Click 'Clear All Dossier Elements.'



Figure 8-29: Clear All Dossier Elements button

8.1.6.2 Plot a flight on the vertical map

(1) Do the same as in section 8.1.6.1 until point 11.

(2) Right click on the flight. Select 'Dossier/ Flight/ <name of flight>Estimated/ Vertical View.

(3) The vertical view for the flight is opened.

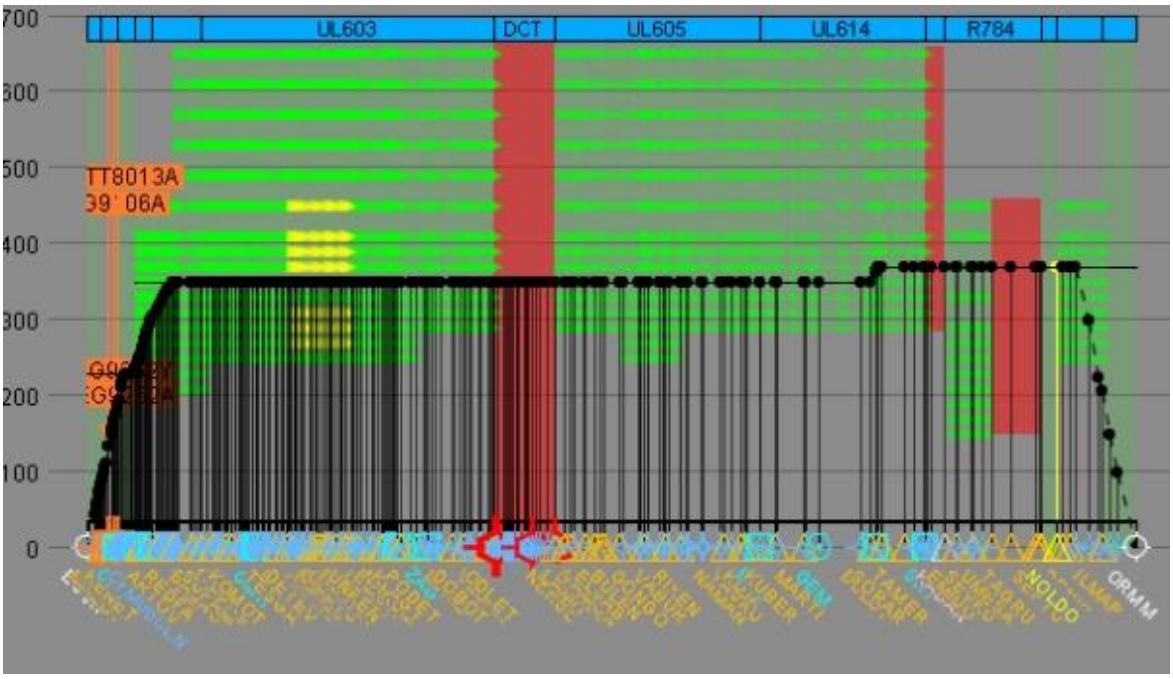


Figure 8-30: Plot IFPS flight on vertical map

8.1.7 Plot a proposed route from IFPS

8.1.7.1 Plot a proposed route on the horizontal map

(1) Do the same as in section 8.1.6.1 until point 4.

(2) Click on 'Propose Routes'.

(3) The list with proposed routes is displayed.

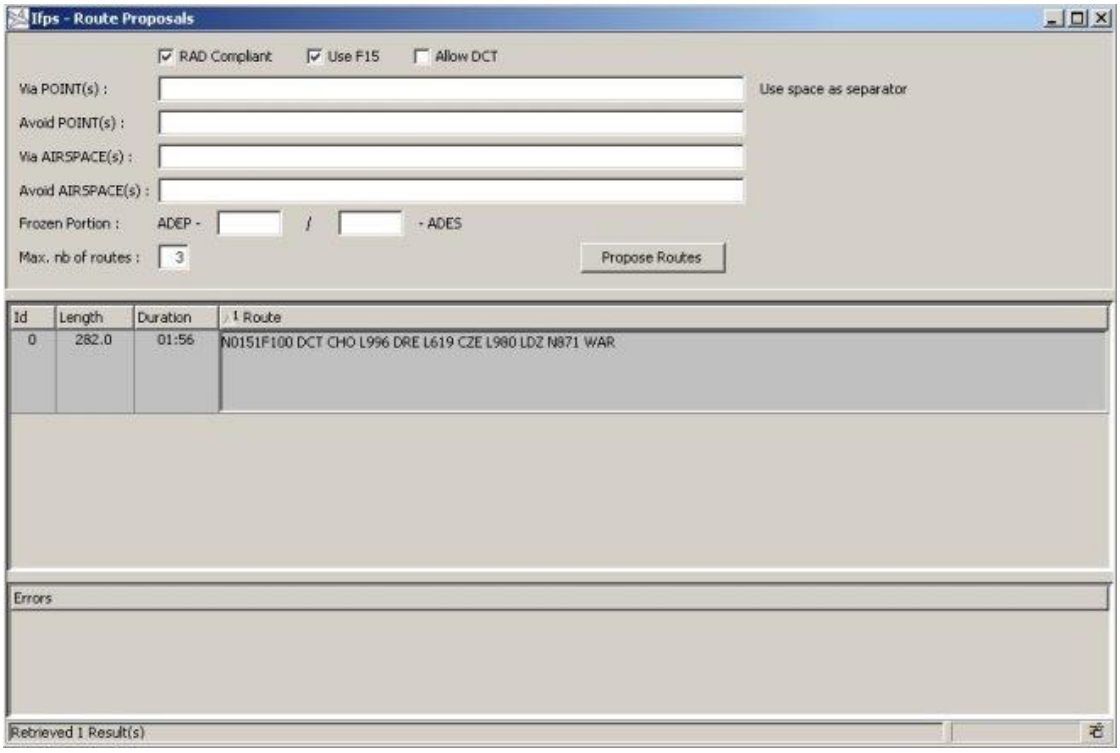


Figure 8-31: Plot route on map from IFPS route proposal

(4) Right click on a route.

(5) 2 menu options appear in the popup menu.

(6) Click on the 'Plot selected Route' option.

(7) The map is opened.

(8) The selected route and the flight are displayed on the map.

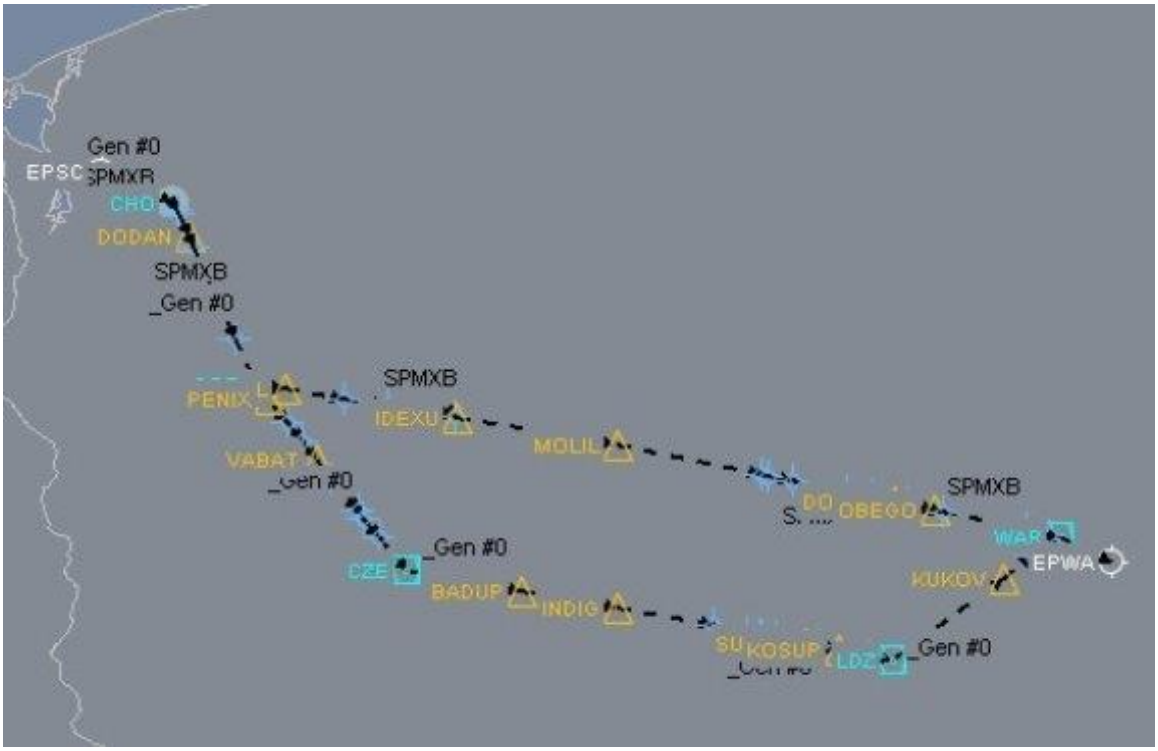


Figure 8-32: Plot IFPS route proposal on horizontal map

(9) The proposed route is added to the layer with the proposed.

(10) The proposed route is painted in black.

(11) The proposed route can be removed by removing all the elements from the dossier. Click 'Clear All Dossier Elements.'



Figure 8-33: Clear All Dossier Elements button

8.1.7.2 Plot a proposed route on the vertical map

(1) Do the same as in section 8.1.6.1 until point 10.

(2) Right click on the flight. Select 'Dossier/ Flight/ <name of proposed route>/ Vertical View.

(3) The vertical view for the proposed route is opened.

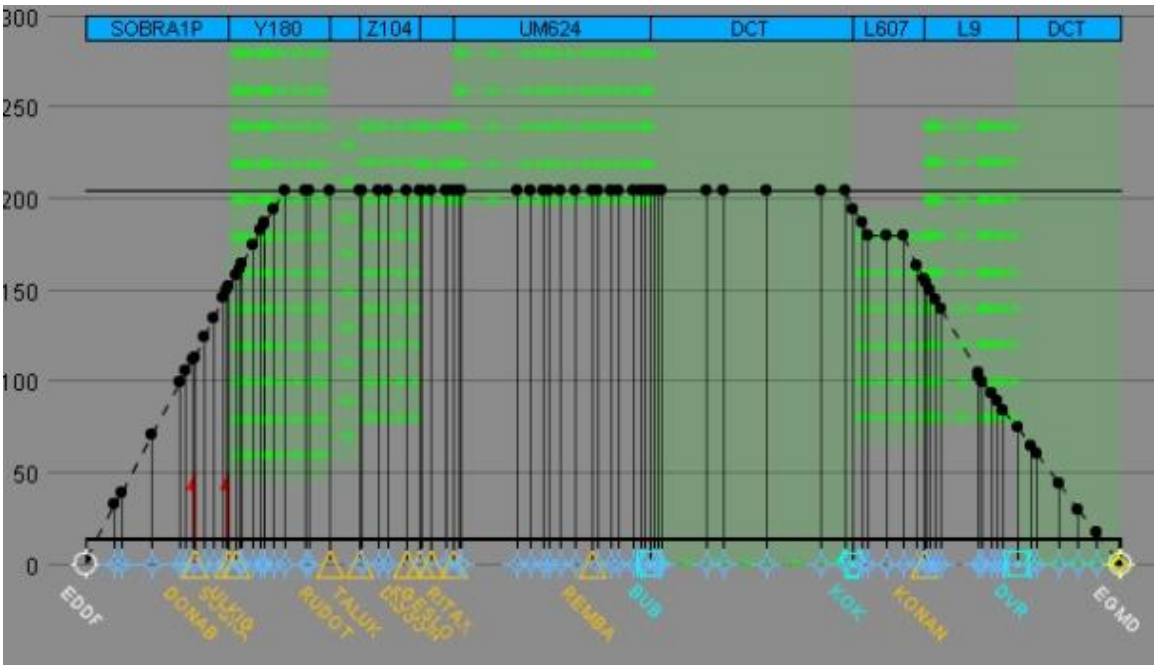


Figure 8-34: Plot IFPS route proposal on vertical map

8.1.8 Types of a flight

8.1.8.1 Introduction

(1) There are 6 types of flight data layers (The drawing sequence may be adjusted by the user):

8.1.8.2 Actual flight.

(1) By default, the actual flight data layer is drawn on top.

(2) The colour is white.

8.1.8.3 Estimated flight.

(1) The estimated flight data layer

(2) The colour is

a) blue if the flight is retrieved from ATFCM, Predict or Archive and

b) black if the flight is retrieved from IFPS.

8.1.8.4 Calculated flight.

(1) The calculated flight data layer

(2) The colour is black.

8.1.8.5 Route catalogue.

(1) The Route catalogue data layer

(2) The colour is green.

8.1.8.6 Alternate route.

(1) The alternate route data layer

(2) The colour is purple.

8.1.8.7 Proposed route.

(1) By default, the proposed route data layer is inserted above the estimated layer.

(2) The colour is:

a) Black when the proposed route has the same length as the current one.

b) Green when the proposed route is shorter than the current one.

c) Red when the proposed route is longer than the current one.

8.1.9 Settings for flights on the map

(1) Select the map by clicking on it.

(2) Click on the 'Map Configuration…' Button in the toolbar.

(3) The 'Map Configuration' dialog is displayed.

(4) Select the tab 'General'.

(5) The second group of settings are for flights.

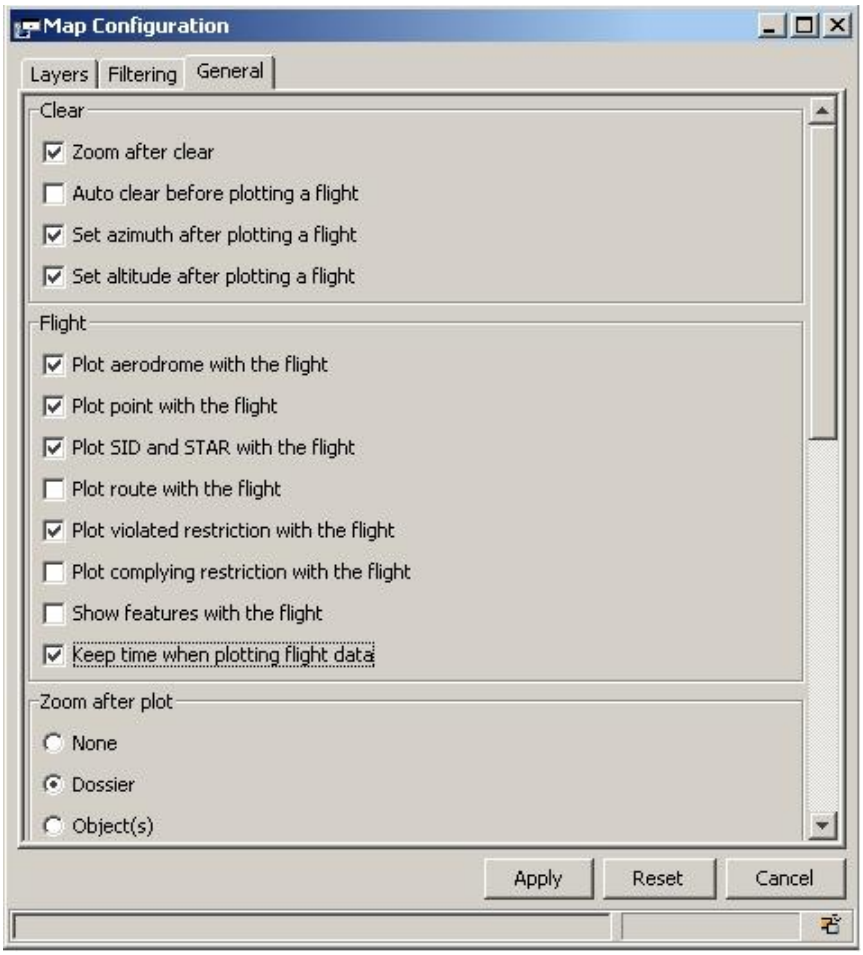


Figure 8-35: Map configuration - Flight settings

(6) The 'Plot aerodrome with the flight' setting: when true, the departing and arriving aerodromes of the flight are also added to the dossier of the map.

(7) The 'Plot point with the flight' setting: when true, the points of the flight are also added to the dossier of the map.

(8) The 'Plot SID and STAR with the flight' setting: when true, the SID and STAR of the flight are also added to the dossier of the map.

(9) The 'Plot route with the flight' setting: when true, the routes of the flight are also added to the dossier of the map.

(10) The 'Plot violated restriction with the flight' setting: when true, the violated restrictions of the flight are also added to the dossier of the map.

(11) The 'Plot complying restriction with the flight' setting: when true, the complying restrictions of the flight are also added to the dossier of the map.

(12) The 'Show features with the flight' setting: when true, the 'Features' tab is opened with the features of the flight, when the flight is plotted on the map.

8.1.10 Settings for group of flights on the map

(1) Right click on the layer called *<title of the linked flight list>* in the 'Dossier'.

(2) The flight group customizer is opened. This window contains the settings how the flights are painted and labelled.



Figure 8-36: Flight group customizer

(3) See point 8.1.8 for flight and group of flights common customizing.

(4) The '*Paint Flight Route*' setting: when true, the route is displayed. If false only the position of the aircraft is displayed. Default is false.

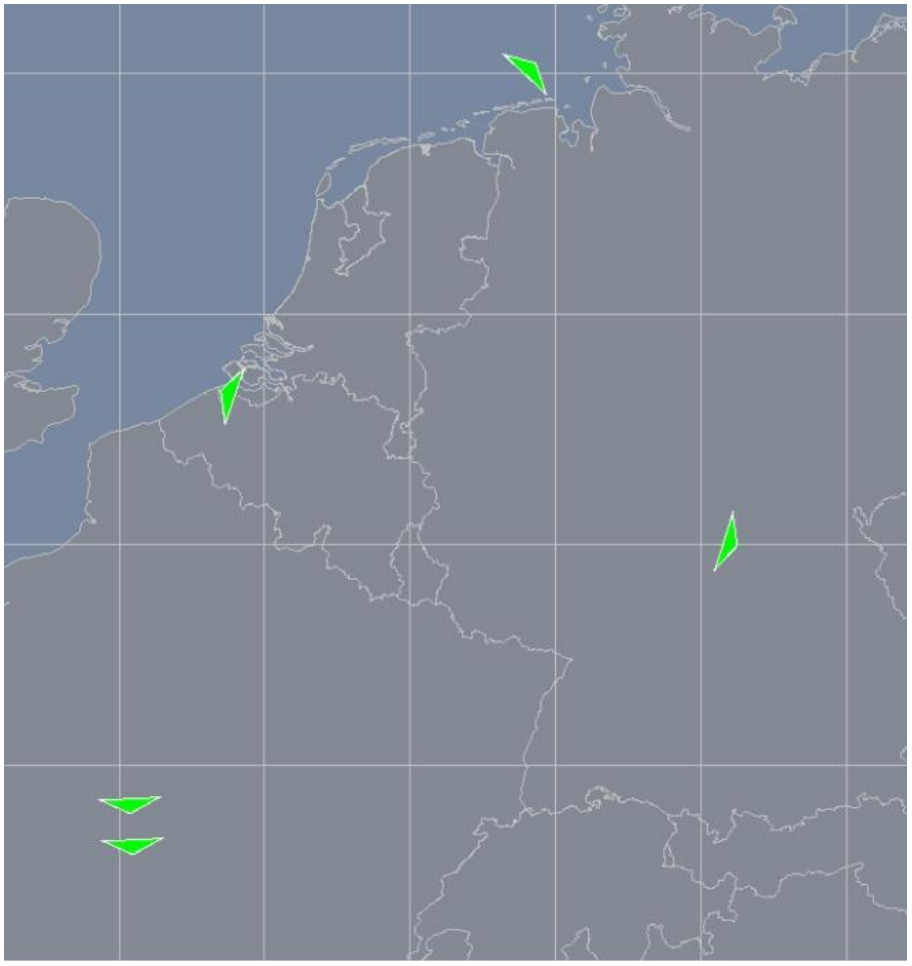


Figure 8-37: Flight customizer - Paint flight route

(5) The 'Aircraft Symbol Inner Colour' can be selected out of a list of 8 predefined colours. The default colour is the first free colour available from that list.

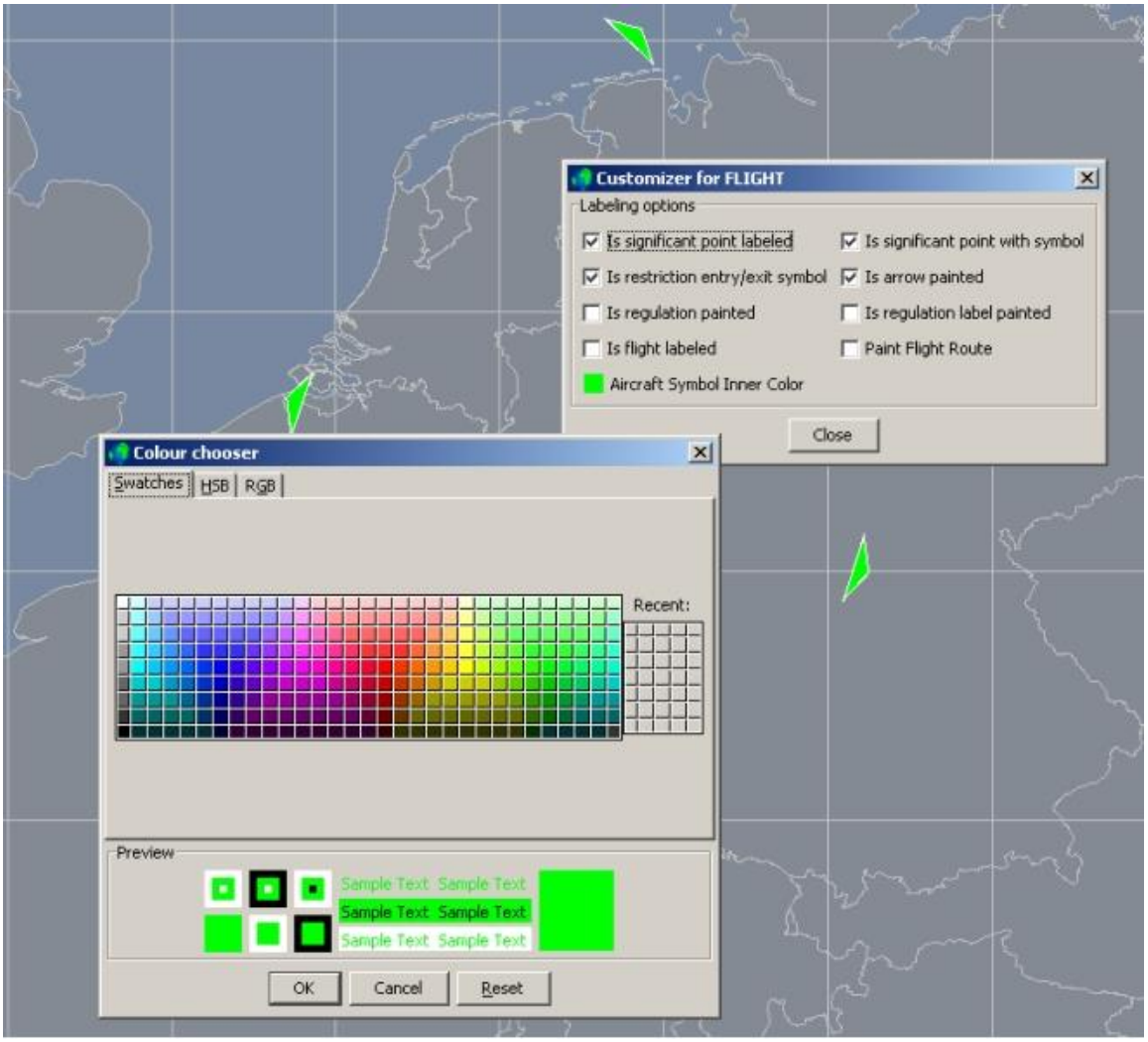


Figure 8-38: Flight customizer - Aircraft symbol inner colour

8.1.11 Other settings for flights on the map

(1) Right click on the Flight group layer in the 'Dossier' tab.

(2) The flight customizer is opened. This window contains the settings how a flight is painted and labelled.



Figure 8-39: Flight customizer

(3) The 'Is significant point labelled' setting: when true, the significant points of the flight have a label, if the flight layer is labelled.



Figure 8-40: Flight customizer - Is significant point labelled

(4) The 'Is significant point with symbol' setting: when true, the significant points of the flight are painted on the flight.



Figure 8-41: Flight customizer - Is significant point with symbol

(5) The 'Is restriction entry/exit symbol' setting: when true, the restriction entry or exit symbols are painted on the flight.

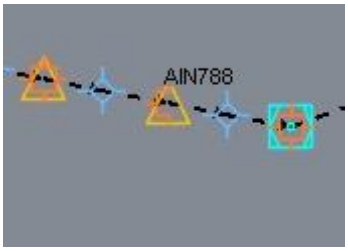


Figure 8-42: Flight customizer - Is restriction entry/exit symbol

(6) The 'Is arrow painted' setting: when true, the arrow showing the direction of the flight is painted in every segment, when the segment is not too small.

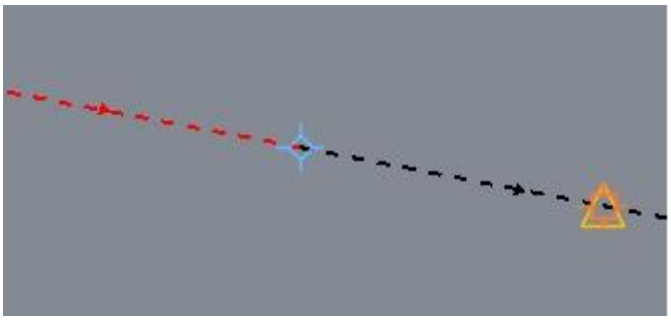


Figure 8-43: Flight customizer - Is arrow painted

(7) The 'Is regulation painted' setting: when true, the regulations of the flight are painted in the flight.



Figure 8-44: Flight customizer - Is regulation painted

(8) The 'Is regulation label painted' setting: when true, the labels of the regulations of flights are painted, if the flight is labelled.

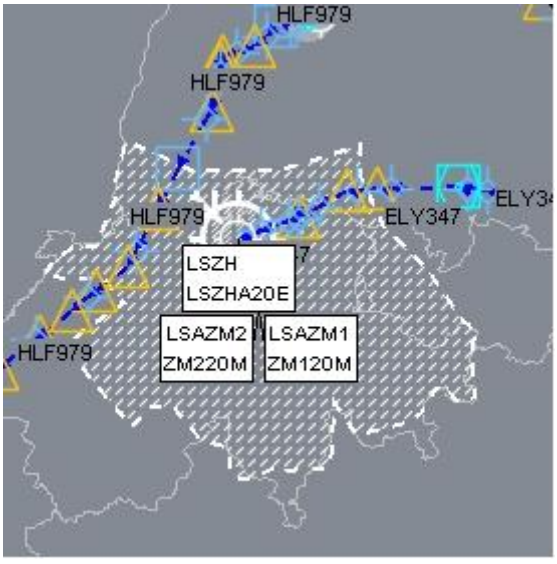


Figure 8-45: Flight customizer - Is regulation label painted

8.1.12 Elements of a flight

8.1.12.1 Points and aerodromes

(1) Flights are a list of points and aerodromes.

(2) The points are connected in the horizontal view by a dashed line.

(3) The point or aerodrome icons can be painted.

(4) The labels of the point and aerodromes can also be painted.

(5) The colour is black.

8.1.12.2 Route portions and DCT segments

(1) Flights are a list of route portions and DCT segments.

(2) On the vertical projection, route portions are displayed as described in Part 7: Route Vertical Projection.

(3) On the vertical projection, DCT segment are displayed in green or depending on the DCT restriction as described in Part 9.2: DCT restriction linked with profile.

8.1.13 Labels of a flight

(1) Flights are labelled in black.

(2) The label is the ARCID of the flight.

(3) Every 5 segments a label is painted on the segment.

(4) When you zoom in on a flight, the flight label should always be visible.

(5) The different elements of a flight (points, regulations) can also be labelled.

(6) The labels should be displaced, when the label displace setting is enabled.

(7) Points of a flight that are also displayed in other layers, should only be labelled once.

8.1.14 Aircraft position indicator

(1) On every flight profile the position of the aircraft at the time of the map is indicated with a triangle.

(2) The border colour of the triangle has the same colour than the route. The Inner colour of the triangle is the one defined in 'Aircraft Symbol Inner Colour' option of the flight customizer.

9 Restriction Map Human Machine Interface

9.1 Traffic Flow and Profile Tuning Restrictions

9.1.1 Boundary class Traffic Flow and Profile Tuning Restriction map

9.1.1.1 Geographical Display horizontal projection

9.1.1.1.1 Basic representation

9.1.1.1.1.1 Layout

(1) The basic representation of the restriction will be the reference(s) location(s) of the restriction with the RS Business Id and the textualDescription. The RS Business Id and the textualDescription will be located at the centre of gravity of the reference location. If there is more than one reference location, the position of the RS Business Id and the textualDescription will be at the centre of gravity of a bounding box around all the reference location. It will be possible with a right button mouse click to open the detail window of the restriction. From this detail window it will be possible to display all or the selected element of the restriction for the detailed representation (cf. 9.1.1.1.2 Detail representation). It will also be possible with a right button mouse click to display directly the FlowRoutingsElements and the FlowConditionsElements.

a) Basic representation picture when the reference location is a point:



Figure 9-1: Aeronautical chart symbol for the restriction Basic representation

b) List of options in the popup menu:

i) Restrictions-> <Name of the restriction> ->Show detail: will open the detail data window of the restriction.

ii) Restrictions-> *<Name of the restriction>* ->Plot FlowroutingsElements: will display all FlowroutingsElements on the map.

iii) Restrictions-> *<Name of the restriction>* ->Plot FlowConditionsElements: will display all FlowConditionsElements on the map.

iv) Restrictions-> *<Name of the restriction>* ->FlowroutingsElements-> *<Name of the element>*: will display the elements on the map.

v) Restrictions-> *<Name of the restriction>* -> FlowConditionsElements: *<Name of the element>*: will display the elements on the map.

9.1.1.1.1.2 Colours

(1) The colour of the basic representation of the restriction shall change to Yellow when it is selected.

(2) The default colour of the basic representation of the restriction is grey.

		Yellow	Orange
1	When Selected	**X**	
2	Default		**X**

Table 9-1: Restriction basic representation colours

9.1.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label of the restriction and the text associated with the restriction.

(2) The labels and text shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction.

9.1.1.1.2 Detail representation

9.1.1.1.2.1 Layout

(1) When an element or a group of element of the restriction are selected to be displayed, they shall be represented as follow:

a) FlowRoutingsElements:

i) If the FlowRoutingsElement is not an airspace, it shall be displayed with dotted lines between points, between points and aerodromes, between points and route portion. There shall be an arrow between each segment depending on the possible direction of the flow. If the flow is bi-directional, no arrow shall be displayed. The route portion shall be displayed with straight lines. If the route portion is routeIdRelevent then the Id of the route will be displayed on each segment otherwise the Id of the route shall not be displayed. It shall be possible to show or hide those elements.



Figure 9-2: Aeronautical chart symbol for the restriction detail representation of FlowRoutings

ii) If the FlowRoutingsElement is an airspace the airspace shall be displayed as followed:

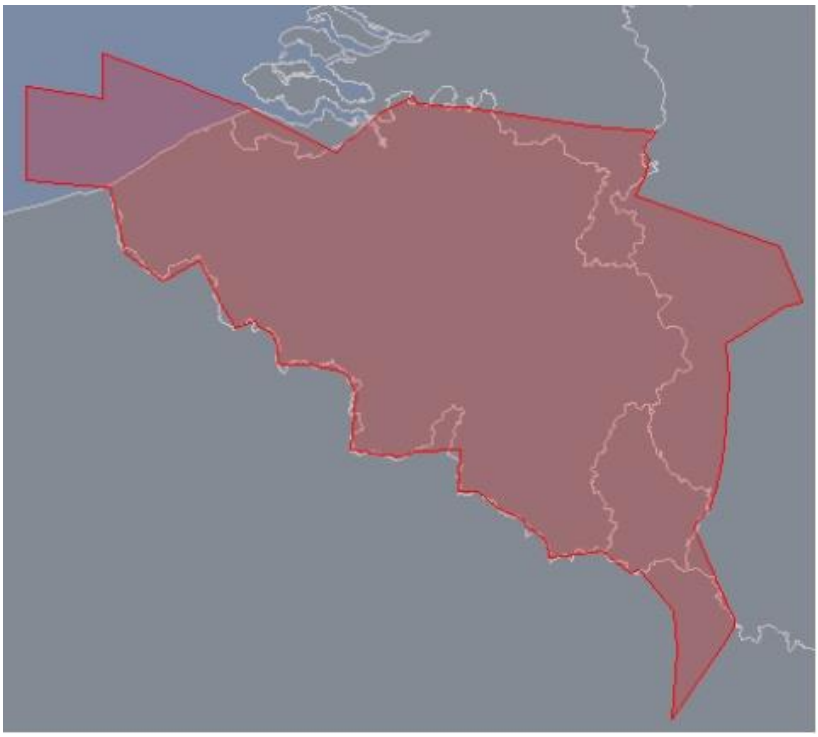


Figure 9-3: Aeronautical chart symbol for the restriction detail representation of FlowRoutings airspace

b) FlowConditionsElements:

i) On the map only the following elements are displayed: aerodromeTL, significantPointTL, setOfAerodromeTL, AirspaceTL, AirspaceCL, Airspaceborder, routePortionTL, and terminalProcedureTL. The representation of those elements shall be based on how those environment objects are displayed in standalone. The differences are the colours and the patterns filling them depending on attributes (cf. 9.1.1.1.2.2 Colours and pattern). It shall be possible to show or hide those elements.

ii) Arrival Aerodrome include representation:



Figure 9-4: Aeronautical chart symbol for the restriction detail representation of FlowConditions included arrival aerodrome

iii) Departing Aerodrome not include representation:

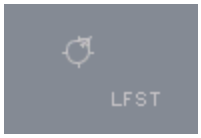


Figure 9-5: Aeronautical chart symbol for the restriction detail representation of FlowConditions not included departing aerodrome



Figure 9-6: Aeronautical chart symbol for the restriction detail representation of FlowConditions included crossing airspace

iv) An airspace that is the reference location:

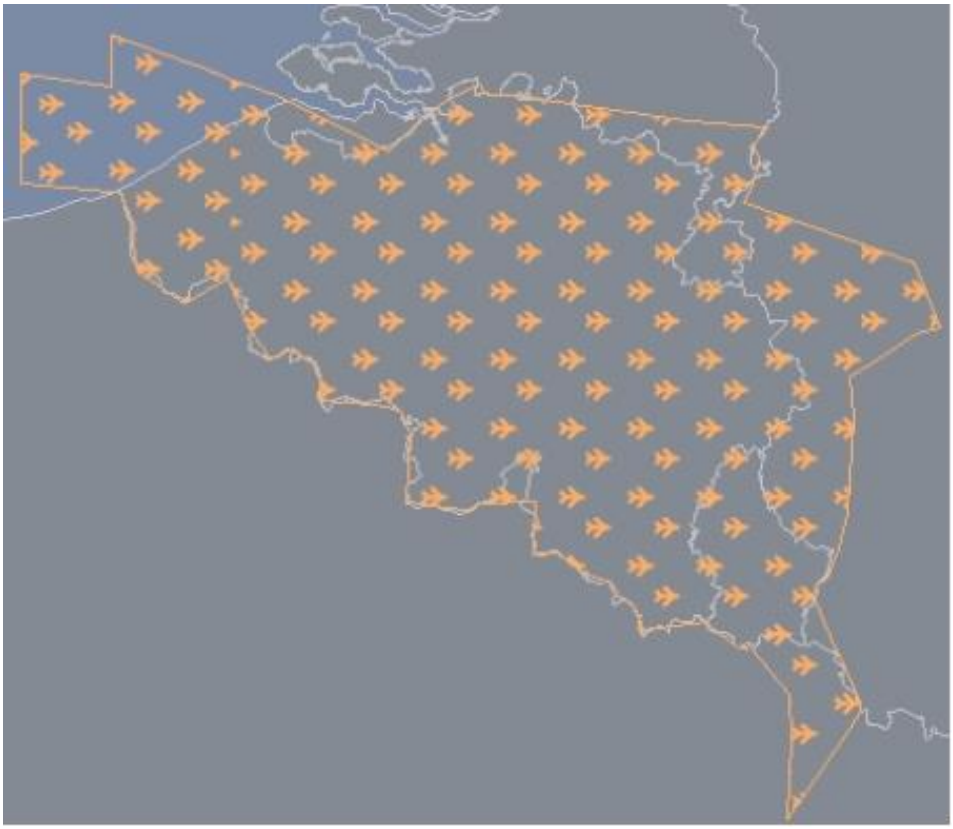


Figure 9-7: Aeronautical chart symbol for the restriction reference location representation of an airspace

v) A point that is the reference location:



Figure 9-8: Aeronautical chart symbol for the restriction reference location representation of a point

(2) FUA Restrictions:

a) FUA Restrictions are HS TrafficFlow Restrictions with an RSA Airspace as reference location. Those Restrictions can have an OverallApplicability including vertical limits.

b) If the Restriction OverallApplicability contains vertical limits, then the Restriction shall be only displayed if the altitude slider is positioned within the applicability vertical limits.

9.1.1.1.2.2 Colours and pattern

(1) The colour of the restriction and all elements that composes it shall change when it is selected. All element of the restriction shall change when selected.

a) FlowRoutingsElements:

		Yellow	Green	Red
1	When Selected	**X**		
2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-2: Restriction detail representation colours of FlowRoutings

b) FlowConditionsElements:

i) aerodromeTL, significantPointTL, setOfAerodromeTL, AirspaceTL, AirspaceCL, Airspaceborder, routePortionTL, and terminalProcedureTL colours are depending on the fact that they included or not and if they are the reference location.

		Yellow	Orange	White	Light Grey
1	When Selected	**X**			
2	Is reference location		**X**		
3	Included			**X**	
4	not Included				**X**

Table 9-3: Restriction detail representation colours of FlowConditions

ii) aerodromeTL, significantPointTL, setOfAerodromeTL, AirspaceTL, AirspaceCL, AirspaceED and Airspaceborder filled in pattern are depending on the fact that they are departing, arrival, departing and arrival or crossing and for airspace if they are exiting. For airspace border crossing, the common border is drawn in thicker width whose line style is the crossing border pattern defined in Table 9-4. If the

crossing border overlap in the selected flight level range the overlapping portion is filled-in by the defined pattern. The colour of the pattern changes following the rules defined in Table 9-3. The routePortionTL, and terminalProcedureTL don't have any filled in pattern.

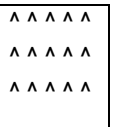
							
1	Departing and Arrival	**X**					
2	Departing		**X**				
3	Arrival			**X**			
4	Crossing Border				**X**		
5	Crossing					**X**	
6	Exiting						**X**

Table 9-4: Restriction detail representation pattern of FlowConditions

9.1.1.1.2.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element.

9.1.1.1.3 Traffic Flow and Profile Tuning Restriction linked with flight profile

9.1.1.1.3.1 Layout

(1) Only the FlowRoutings in the context of the flight profile shall be displayed.

a) FlowRoutingsElements:

i) If the FlowRoutingsElement is not an airspace, it shall be displayed with dotted lines between points, between points and aerodromes, between points and route portion. There shall be an arrow between each segment depending on the possible direction of the flow. If the flow is bi-directional, no arrow shall be displayed. The route portion shall be displayed with straight lines. If the route portion is routeIdRelevent then the Id of the route will be displayed on each segment otherwise the Id of the route shall not be displayed. It shall be possible to show or hide those elements.

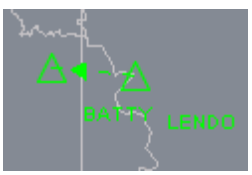


Figure 9-9: Aeronautical chart symbol for the restriction detail representation of FlowRoutings

ii) If the FlowRoutings is an airspace the airspace shall be displayed as followed:

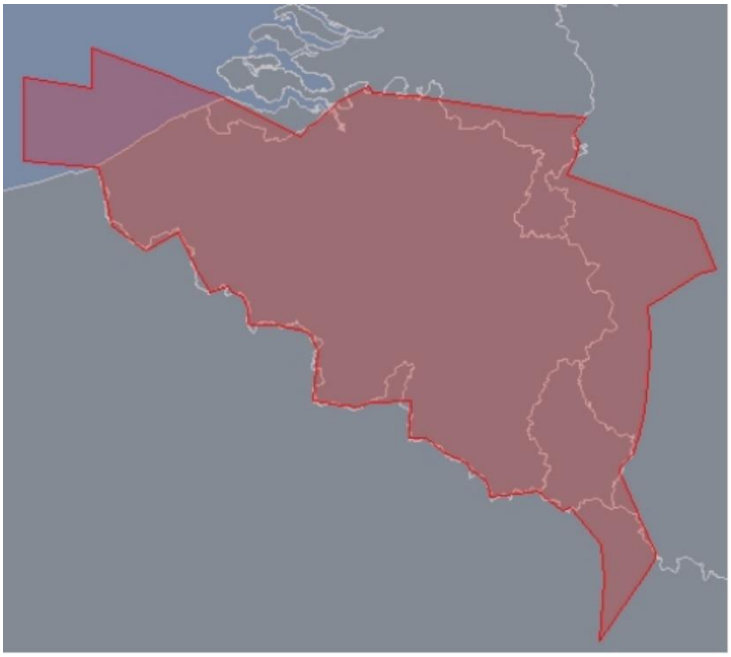


Figure 9-10: Aeronautical chart symbol for the restriction detail representation of FlowRoutings airspace

9.1.1.1.3.2 Colours

(1) The colour of the FlowRoutingElements shall change when it is selected.

		Yellow	Green	Red
1	When Selected	**X**		
2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-5: Restriction representation colours of FlowRoutings with flight profile

9.1.1.1.3.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element

9.1.1.2 Geographical Display vertical projection

9.1.1.2.1 Restriction linked with flight profile

9.1.1.2.1.1 Layout

(1) It shall be possible to display or hide restrictions violated by the flight profile.

(2) It shall be possible to display or hide restrictions compliant with the flight profile.



Figure 9-11: Violated/compliant Restrictions

Note: only restrictions with an airspace as reference location will be shown on the vertical view, except for the PTRs (see types below).

Note: Profile Tuning Restrictions will always be shown, independent of whether they are violated or compliant.

(3) It shall be possible to select the type of restrictions in the Layers.

Types:

- HS : Hard/Soft TrafficFlow Restrictions
- PT : Profile Tuning Restrictions

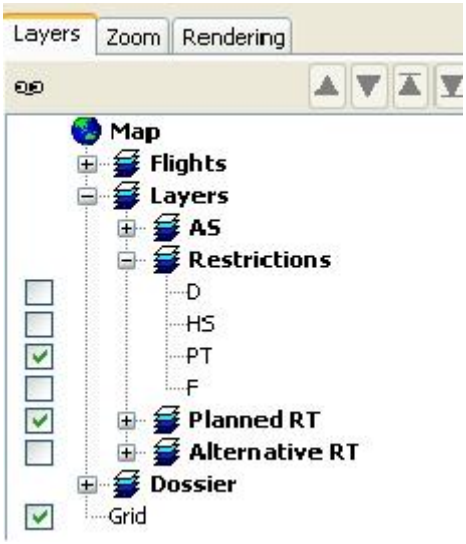


Figure 9-12: Restriction Layers

(4) It shall be possible to display or hide restrictions from the dossier.

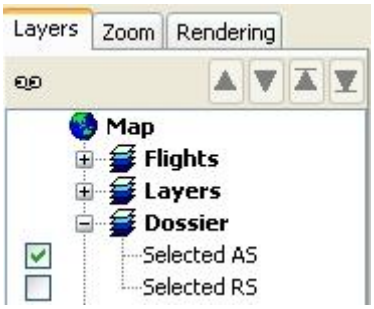


Figure 9-13: Restriction Dossier

(5) Airspace elements for Profile Tuning Restrictions

a) They shall be displayed on the vertical projection as shown in the figure below. The restricted area shall be indicated as a rectangle filled with vertical orange lines. The start and end point of the restriction shall be defined by an orange icon.

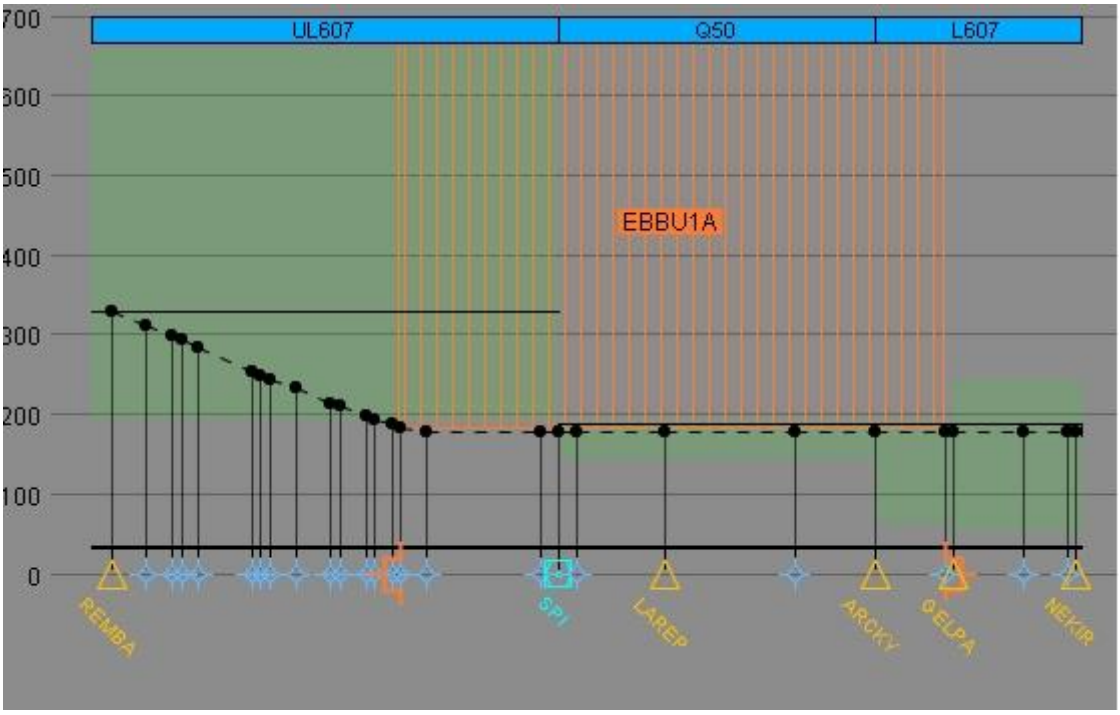


Figure 9-14: Restriction linked with a flight profile on an airspace (forbidden FLR-UNL)

b) The special altitude values GND (ground) and UNL (unlimited) shall be handled as respectively the lower and upper bounds of the vertical projection.

c) The special altitude values FLR (floor) and CEL (ceiling) shall be handled as respectively the lower and upper bounds of the airspace.

d) Both mandatory and forbidden restrictions are allowed. Note that, in both cases, the restricted area shall be displayed on the vertical projection.

(6) Airspace elements for FUA Restrictions

a) These are HS TrafficFlow Restrictions with an RSA Airspace as reference location.

b) They shall be displayed on the vertical projection as shown in the figure below. The reference location shall be shown as an orange rectangle filled with airplanes. The restricted area shall be indicated as a filled red rectangle. The start and end point of the restriction shall be defined by red icon.

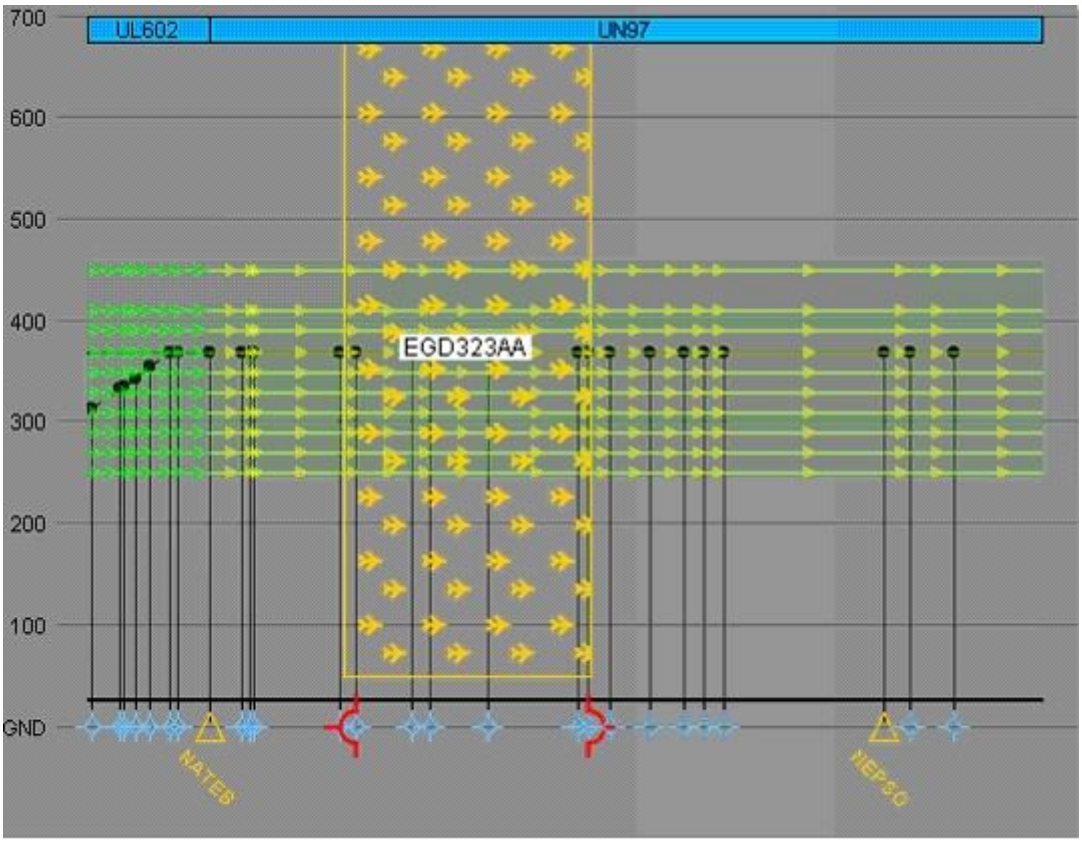


Figure 9-15: Restriction linked with a flight profile on an airspace (forbidden FLR-UNL)

c) If the Restriction OverallApplicability contains vertical limits, then the reference location airspace shall be displayed only for the applicability vertical limits and not for the complete airspace vertical limits.

(7) Route elements

a) They shall be displayed on the vertical projection as shown in the figure below. The restricted area shall be indicated as a rectangle filled with vertical orange lines. The start and end point of the restriction shall be defined by an orange icon.

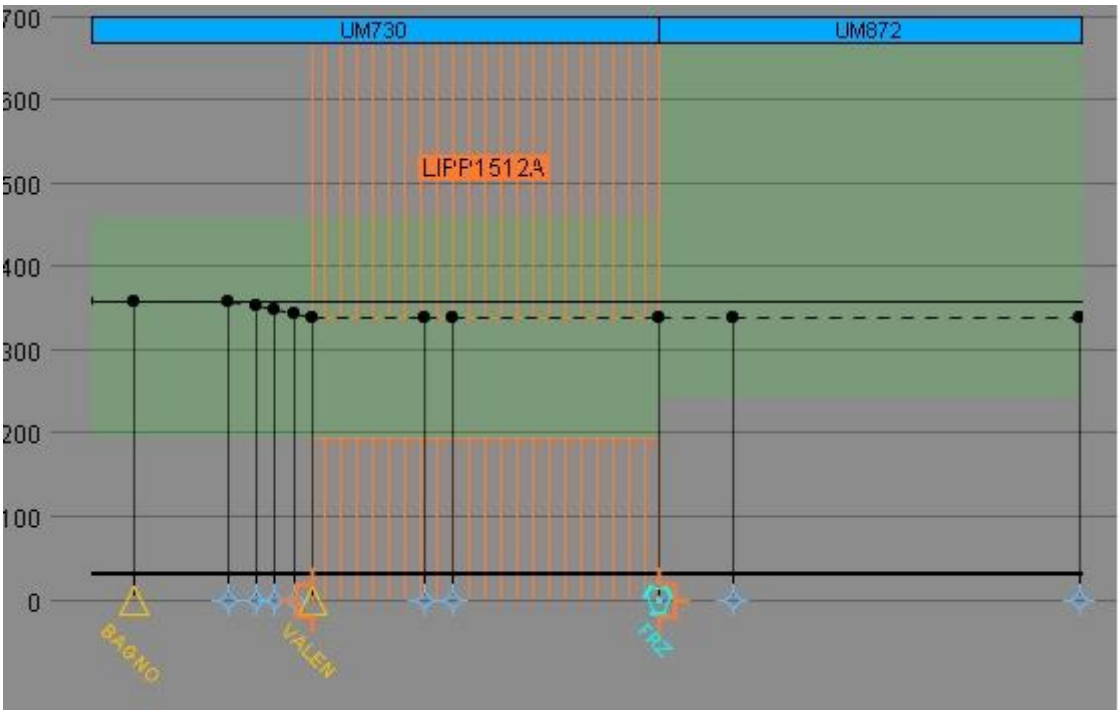


Figure 9-16: Restriction linked with a flight profile on a route (mandatory 195-340)

b) The special altitude values GND (ground) and UNL (unlimited) shall be handled as respectively the lower and upper bounds of the vertical projection.

c) The special altitude values FLR (floor) and CEL (ceiling) are forbidden.

d) Both mandatory and forbidden restrictions are allowed. Note that, in both cases, the restricted area shall be displayed on the vertical projection.

(8) Point elements

a) They shall be displayed on the vertical projection as shown in the figure below. The restricted area shall be indicated as an orange line. The start and end point of the restriction shall be indicated with an orange icon. Note that, in this case, the start and end point shall be the same point.

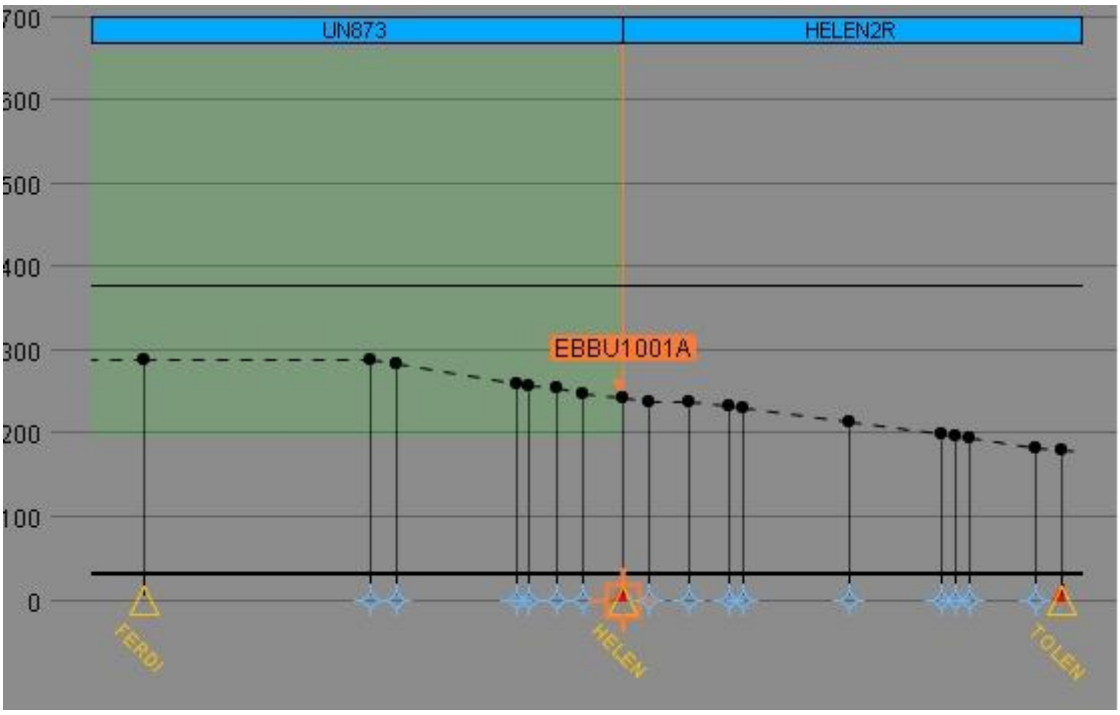


Figure 9-17: Restriction linked with a flight profile on a point (forbidden 245-UNL)

b) The special altitude values GND (ground) and UNL (unlimited) shall be handled as respectively the lower and upper bounds of the vertical projection.

c) The special altitude values FLR (floor) and CEL (ceiling) are forbidden.

d) Both mandatory and forbidden restrictions are allowed. Note that, in both cases, the restricted area shall be displayed on the vertical projection.

(9) List of options in the popup menu:

a) Restrictions-> <Name of the restriction> ->Show detail: will open the detail data window of the restriction.

9.1.1.2.1.2 Colours

(1) The colour of the restriction element shall be orange.

9.1.1.2.1.3 Labelling

(1) There shall be no label collision.

(2) The label background shall have the same colour as the corresponding restriction element.

9.2 DCT Limit Restrictions

9.2.1 Boundary class DCT Limit Restriction map

9.2.1.1 Geographical Display horizontal projection

9.2.1.1.1 Basic representation

9.2.1.1.1.1 Layout

(1) The basic representation of the restriction will be the reference location of the restrictions GroupId (Aerodrome or airspace) with the RS Business Id, the textualDescription and with the VerticalLimit when the reference location is an airspace. The RS Business Id, the textualDescription and the VerticalLimit will be located at the centre of gravity of the reference location. If the reference location is an aerodrome the ExceedingLimit is displayed as a red circle around the Aerodrome with a range equal to the value of the ExceedingLimit. It will be possible with a right button mouse click to open the detail window of the restriction. From this detail window it will be possible to display all or the selected element of the restriction for the detailed representation (cf. 9.1.1.1.2 Detail representation). It will also be possible with a right button mouse click to display directly the FlowRoutingsElements and the FlowConditionsElements.

a) Basic representation picture for DctLimit restriction with an airspace as reference location:

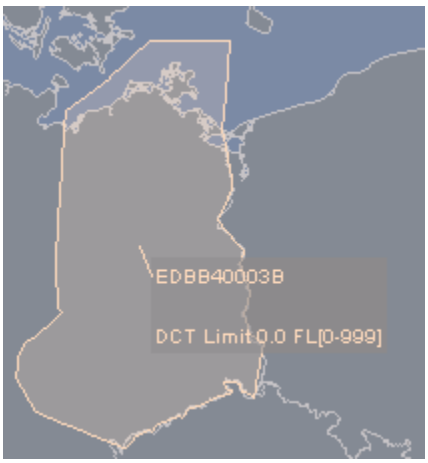


Figure 9-18: Aeronautical chart symbol for Airspace DCTLimit restriction Basic representation

b) Basic representation picture for DctLimit restriction with an aerodrome as reference location:

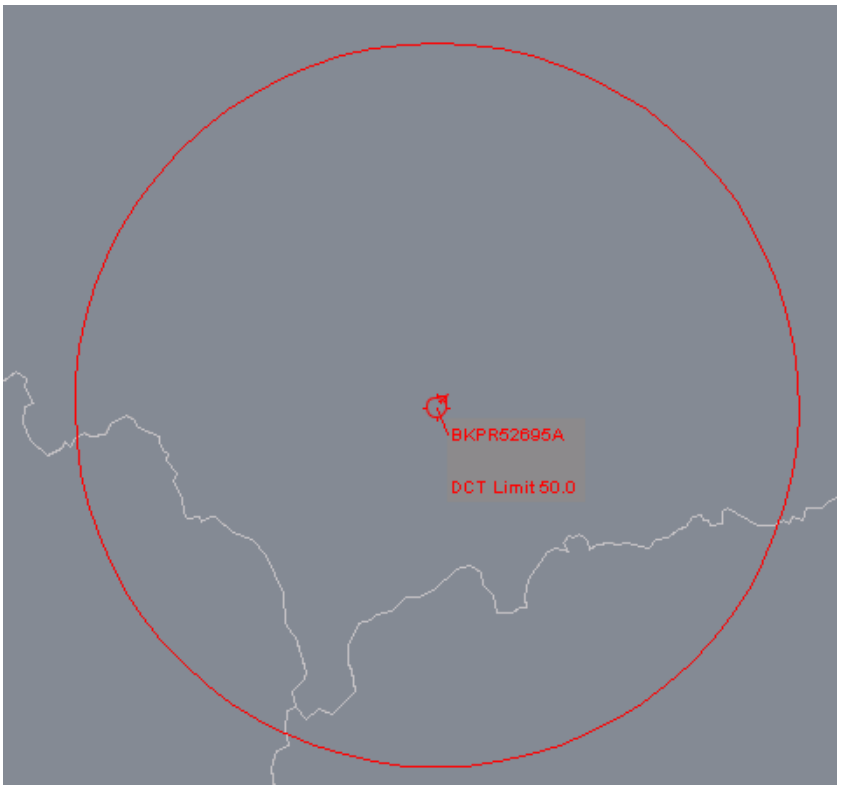


Figure 9-19: Aeronautical chart symbol for Aerodrome DCTLimit restriction Basic representation

c) List of options in the popup menu:

i) Restrictions-> <Name of the restriction> ->Show detail: will open the detail data window of the restriction.

ii) Restrictions-> *<Name of the restriction>* ->Plot FlowroutingsElements: will display all FlowroutingsElements on the map.

iii) Restrictions-> *<Name of the restriction>* ->Plot FlowConditionsElements: will display all FlowConditionsElements on the map.

iv) Restrictions-> *<Name of the restriction>* ->FlowroutingsElements-> *<Name of the element>*: will display the elements on the map.

v) Restrictions-> *<Name of the restriction>* -> FlowConditionsElements: *<Name of the element>*: will display the elements on the map.

9.2.1.1.1.2 Colours

(1) The colour of the restriction shall change when it is selected.

(2) For DctRestrictions on an Airspace: The default colour of the ReferenceLocation changes depending on the Cross border DCT allowed flag of the airspace which serves as ReferenceLocation:

		Yellow	Green	Red
1	When Selected	**X**		
2	Cross Border DCT allowed		**X**	
3	Cross Border DCT not Allowed			**X**

Table 9-6: DCT Restriction on Airspace basic representation colours

(3) For DctRestrictions on an Aerodrome: The default colour of the referenceLocation will be Red.

9.2.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label of the restriction and the text associated with the restriction.

(2) The labels and text shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction.

9.2.1.1.2 Detail representation

9.2.1.1.2.1 Layout

(1) When an element or a group of element of the restriction are selected to be displayed, they shall be represented as follow:

a) FlowRoutingsElements:

i) The FlowRoutingsElement are always DctSegments. It shall be displayed with dotted lines between points and between points and aerodromes. There shall be an arrow between each segment depending on the possible direction of the flow. It shall be possible to show or hide those elements.

ii) The FlowRoutings from the MandatoryRestrictions of the RestrictionGroupId shall be displayed in Green. The FlowRoutings from the ForbiddenRestrictions of the RestrictionGroupId will be displayed in Red.

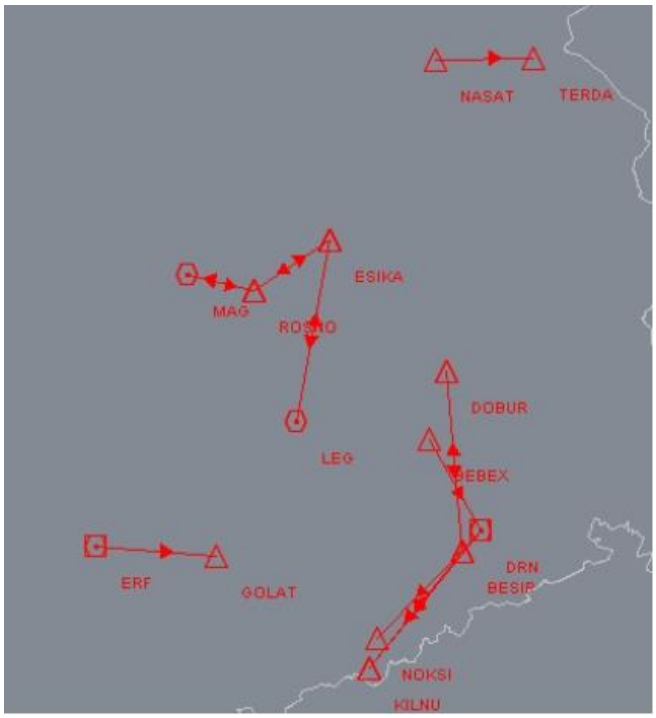


Figure 9-20: Aeronautical chart symbol for the restriction detail representation of FlowRoutings

b) FlowConditionsElements (ReferenceLocation):

i) The FlowConditionsElements of the restriction will be like the reference location of the restrictions GroupId (Aerodrome or airspace) with the RS Business Id, the textualDescription and with the VerticalLimit when the reference location is an airspace. The RS Business Id, the textualDescription and the VerticalLimit will be located at the centre of gravity of the reference location. If the reference location is an aerodrome the ExceedingLimit is displayed as a red circle around the Aerodrome with a range equal to the value of the ExceedingLimit. It will be possible with a right button mouse click to open the detail window of the restriction. From this detail window it will be possible to display all or the selected element of the restriction for the detailed representation (cf. 9.1.1.1.2 Detail representation). It will also be possible with a right button mouse click to display directly the FlowRoutingsElements and the FlowConditionsElements.

ii) Aerodrome representation:

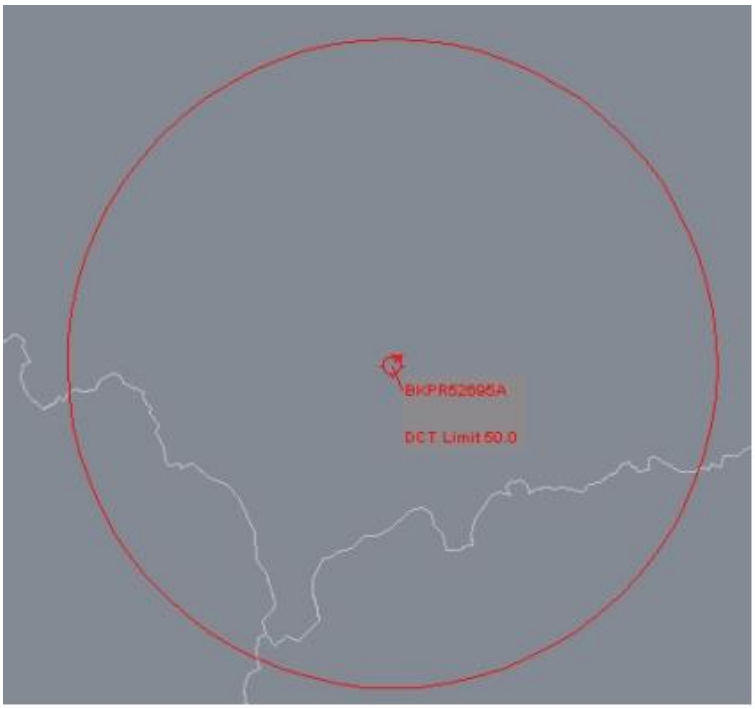


Figure 9-21: Aeronautical chart symbol for the restriction detail representation of FlowConditions included arrival

iii) Airspace representation:



Figure 9-22: Aeronautical chart symbol for the restriction reference location representation of an airspace.

9.2.1.1.2.2 Colours and pattern

(1) The colour of the restriction and all elements that composes it shall change when it is selected. All element of the restriction shall change when selected.

a) FlowRoutingElements

i) The FlowRouting is always a DctSegment it changes depending on a flag Mandatory/Forbidden:

		Yellow	Green	Red
1	When Selected	**X**		

2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-7: Restriction detail representation colours of DctLimitations FlowRoutinElements

b) FlowConditionsElements (ReferenceLocation):

i) If the DctLimit Restriction ReferenceLocation is an airspace, the default colour of the airspace depend on the flag CrossBorderDctAllowed.

The actual ExceedingLimit value is a red Label:

		Yellow	**Green**	**Red**
1	When Selected	**X**		
2	ExceedingLimit ValueLabel			**X**
3	CrossBorderDctAllowed		**X**	
4	CrossborderDctNotAllowed			**X**

Table 9-8: Restriction detail representation colours of DctLimitations

ii) If the DctLimit Restriction ReferenceLocation is an Aerodrome then the colour will always be red.

9.2.1.1.2.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element.

9.2.1.1.3 DCTLimit Restriction linked with flight profile

9.2.1.1.3.1 Layout

(1) Only the FlowRoutings in the context of the profile shall be displayed.

a) FlowRoutingsElements:

i) The FlowRoutingsElement are always DctSegments. It shall be displayed with dotted lines between points and between points and aerodromes. There shall be an arrow between each segment depending on the possible direction of the flow. It shall be possible to show or hide those elements.

ii) The FlowRoutings from the MandatoryRestrictions of the RestrictionGroupId shall be displayed in Green. The FlowRoutings from the ForbiddenRestrictions of the RestrictionGroupId will be displayed in Red.

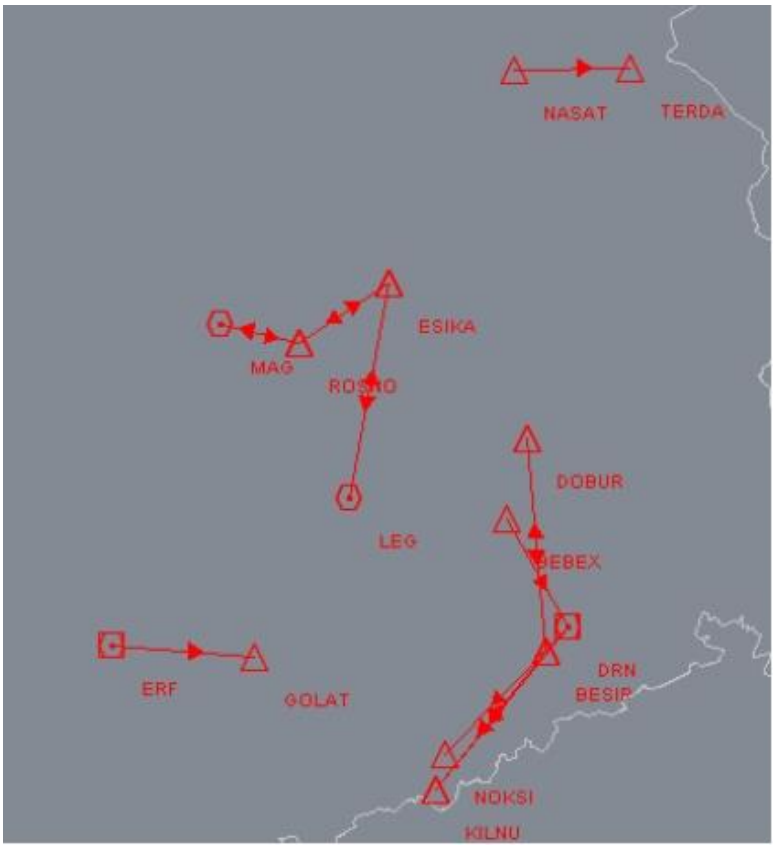


Figure 9-23: Aeronautical chart symbol for the restriction detail representation of FlowRoutings

9.2.1.1.3.2 Colours

(1) The colour of the restriction and all elements that composes it shall change when it is selected. All element of the restriction shall change when selected.

a) FlowRoutingElements

i) The FlowRouting is always a DctSegment it changes depending on a flag Mandatory/Forbidden:

		Yellow	Green	Red
1	When Selected	**X**		
2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-9: Restriction detail representation colours of DctLimitations FlowRoutinElements

9.2.1.1.3.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element.

9.2.1.2 Geographical Display vertical projection

9.2.1.2.1 Restriction linked with flight profile

9.2.1.2.1.1 Layout

(1) It shall be possible to display or hide restrictions violated by the flight profile.

(2) It shall be possible to display or hide restrictions compliant with the flight profile.



Figure 9-24: Violated/compliant Restrictions

Note: only restrictions with an airspace as reference location will be shown on the vertical view, except for the PTRs (see types below).

(3) It shall be possible to select the type of restrictions in the Layers.

Types:

- D: DCT Restrictions

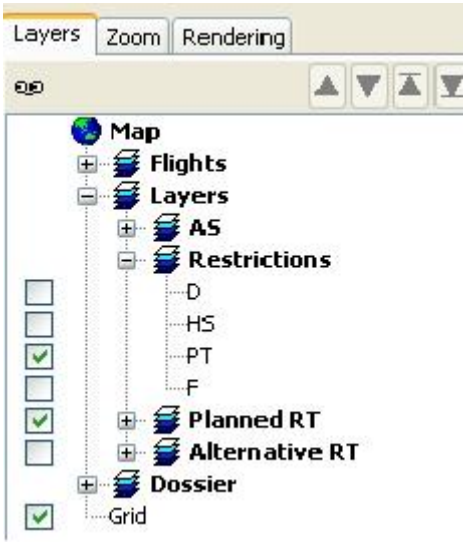


Figure 9-25: Restriction Layers

(4) It shall be possible to display or hide restrictions from the dossier.

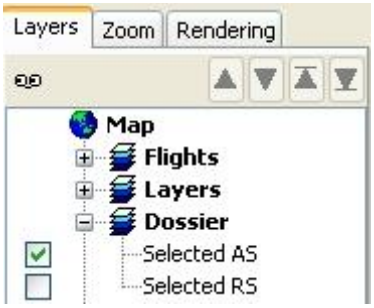


Figure 9-26: Restriction Dossier

(5) Complying restrictions shall be displayed on the vertical projection as shown in the figure below.

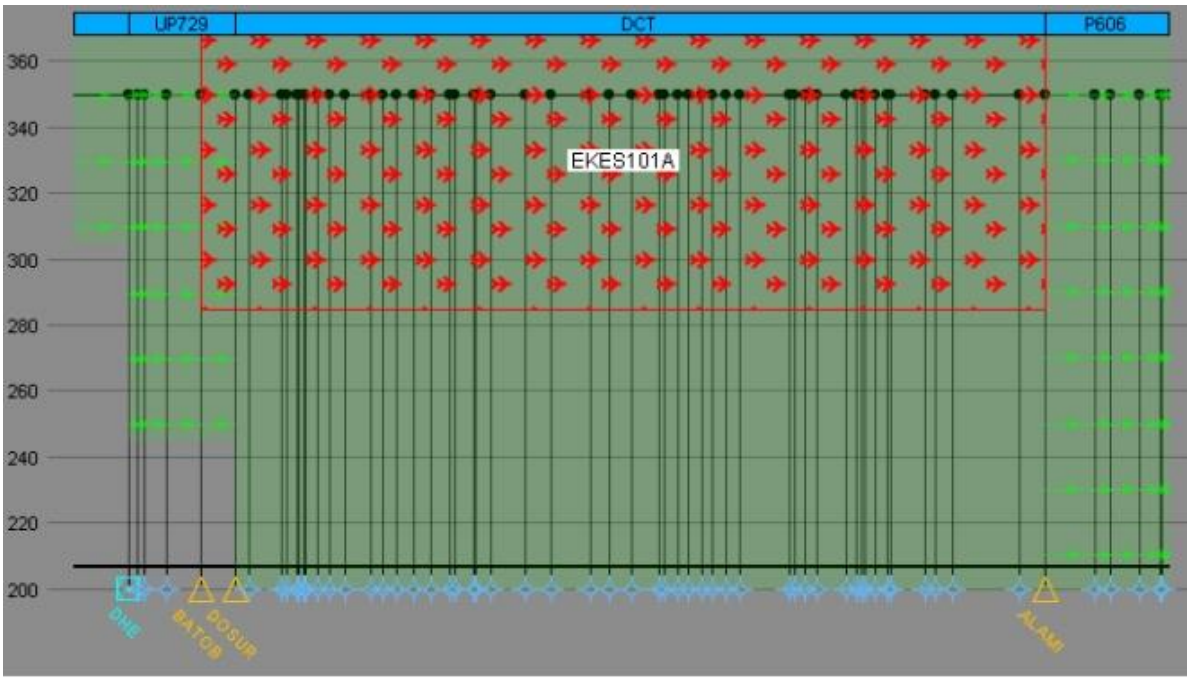


Figure 9-27: Restriction linked with a flight profile (complying DCT)

(6) Violating DCT restrictions shall be displayed on the vertical projection as shown in the figure below. The start and end point of the restriction shall be defined by a red icon.

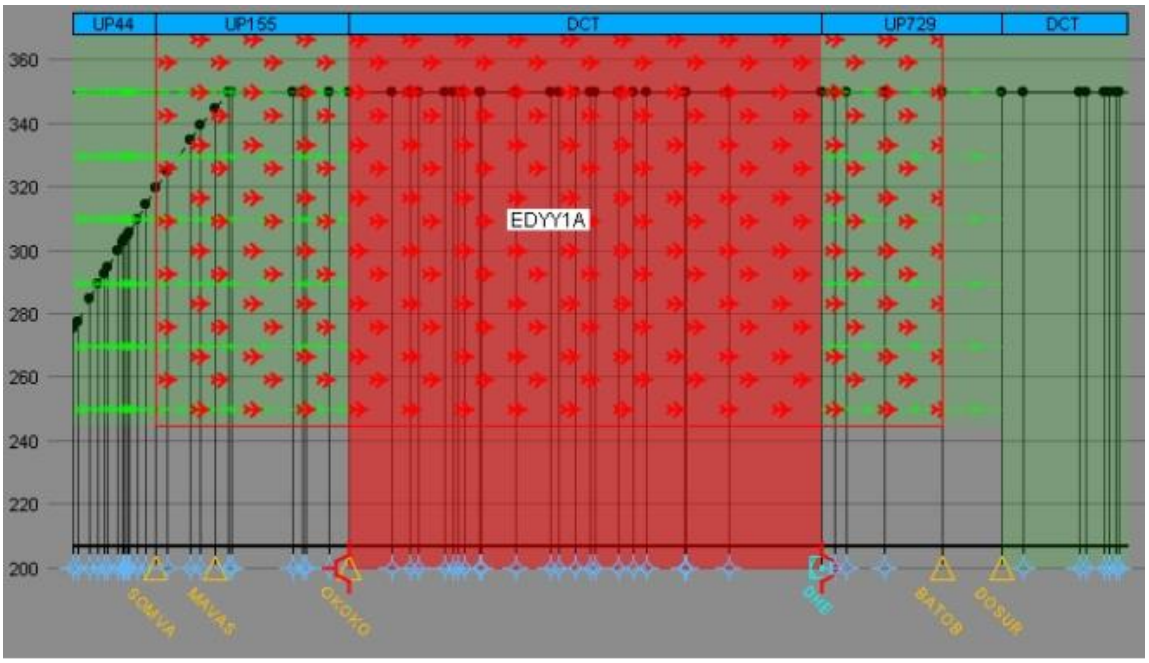


Figure 9-28: Restriction linked with a flight profile (violating DCT)

9.2.1.2.1.2 Colours

(1) The background of the DCT segment shall be coloured green.

(2) The background of the DCT segment shall be coloured red.

9.2.1.2.1.3 Labelling

(1) There shall be no label for DCT restrictions.

9.3 Fra Dct Restrictions

9.3.1 Boundary class Fra Dct Restriction map

9.3.1.1 Geographical Display horizontal projection

9.3.1.1.1 Basic representation

9.3.1.1.1.1 Layout

(1) The basic representation of the restriction will be the reference location of the restrictions (airspace or airspaceBorder) with the RS Business Id. The RS Business Id will be located at the centre of gravity of the airspace, when the reference location is an airspace, or at the centre of gravity of one of the two involved airspaces when the reference location is an airspaceBorder. It will be possible with a right button mouse click to open the detail window of the restriction. From this detail window it will be possible to display all or the selected element of the restriction for the detailed representation (cf. **9.1.1.1.2 Detail representation**). It will also be possible with a right button mouse click to display directly the FlowRoutingsElements and the FlowConditionsElements.

a) Basic representation picture for Fra Dct restriction with an airspace as reference location:

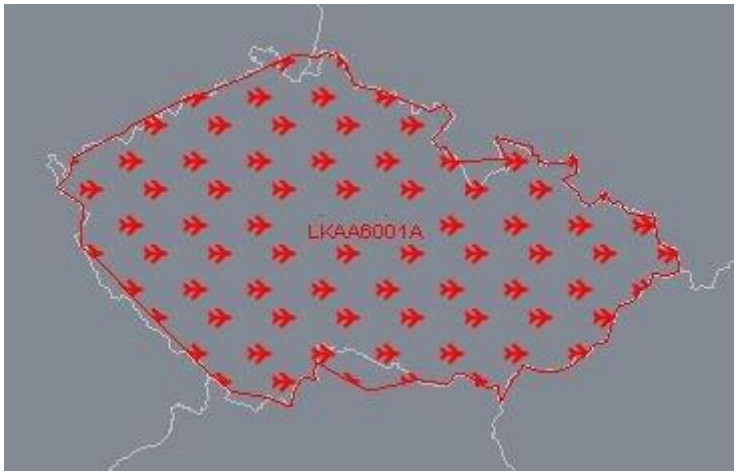


Figure 9-29: Aeronautical chart symbol for the Fra Dct Restriction Basic representation

i) airspaceBorder Representation: Both Airspaces involved must be displayed. Overlap of the 2D projections is possible as there will be horizontal borders. The layout shall allow the user to see the underlying airspace.

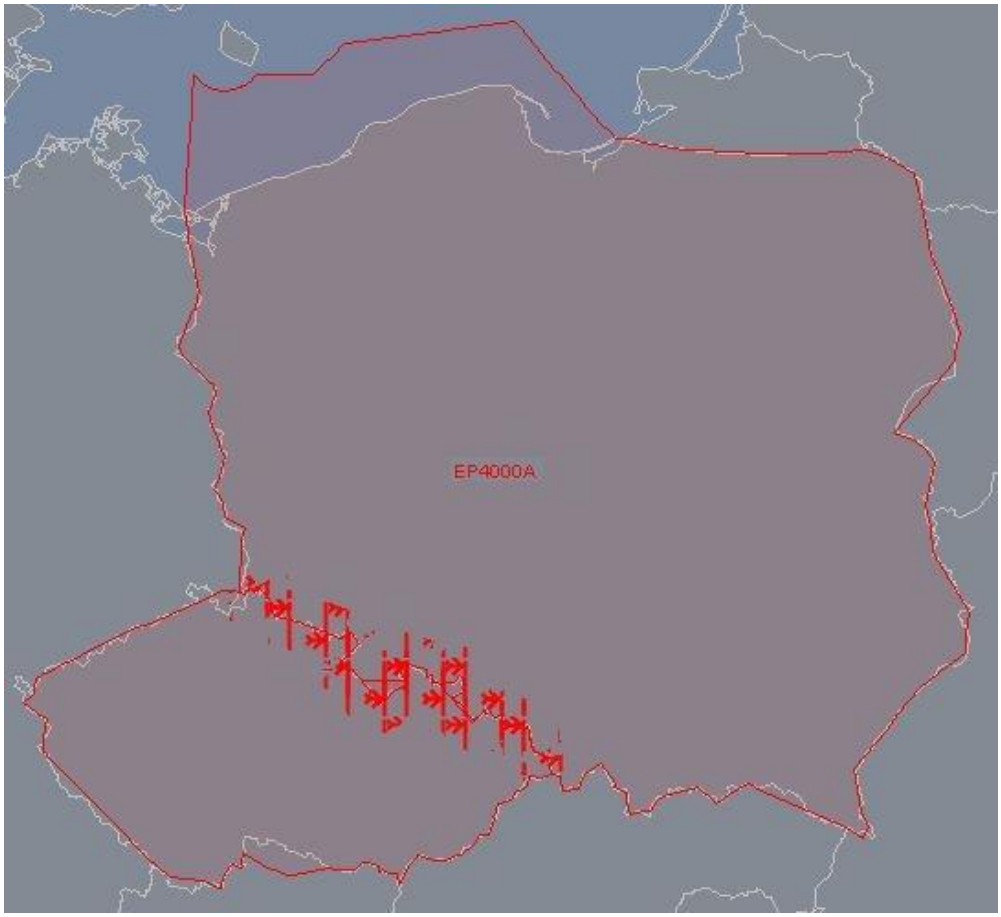


Figure 9-30: Aeronautical chart symbol for the restriction reference location representation of an FRA AirspaceBorder

b) List of options in the popup menu:

i) Restrictions-> <Name of the restriction> ->Show detail: will open the detail data window of the restriction.

ii) Restrictions-> *<Name of the restriction>* ->Plot FlowroutingsElements: will display all FlowroutingsElements on the map.

iii) Restrictions-> *<Name of the restriction>* ->Plot FlowConditionsElements: will display all FlowConditionsElements on the map.

iv) Restrictions-> *<Name of the restriction>* ->FlowroutingsElements-> *<Name of the element>*: will display the elements on the map.

v) Restrictions-> *<Name of the restriction>* -> FlowConditionsElements: *<Name of the element>*: will display the elements on the map.

vi) Restriction [F] -> *<Name of the restriction>* -> Associated Restrictions: *<Name of the associated FRA Restrictions>*: will display the restrictions on the map.

– For a FRA Border Restriction, this are the associated FRA Airspace Restriction(s),

– For a FRA Airspace Restriction, this are the associated FRA Border Restriction(s).

9.3.1.1.1.2 Colours

(1) The colour of the restriction shall change when it is selected.

(2) For DctRestrictions on an Airspace: The default colour of the ReferenceLocation changes depending on the Cross border DCT allowed flag of the airspace which serves as ReferenceLocation:

		Yellow		Red
1	When Selected	**X**		
2	ReferenceLocation			**X**

Table 9-10: Restriction basic representation colours

9.3.1.1.1.3 Labelling

(1) It shall be possible to show or hide the label of the restriction and the text associated with the restriction.

(2) The labels and text shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction.

9.3.1.1.2 Detail representation

9.3.1.1.2.1 Layout

(1) When an element or a group of element of the restriction are selected to be displayed, they shall be represented as follow:

a) FlowRoutingsElements:

i) A SignificanPointFRE shall be represented by the point symbol and its business id.



Figure 9-31: Aeronautical chart symbol for the restriction FlowRoutingElement representation of a point

ii) If the FlowRoutingsElement is an FRA (Free Route Airspace) entry-exit point, the point shall be displayed as follows:



Entry Exit Entry-Exit

Figure 9-32: Aeronautical chart symbol for the restriction detail representation of FRA entry-exit point FlowRoutings

iii) If the FlowRoutingsElement is an FRA intermediate point, the point shall be displayed as follows:



Figure 9-33: Aeronautical chart symbol for the restriction detail representation of FRA intermediate point FlowRoutings

iv) DctSegments shall be displayed with dotted lines between points and between points and aerodromes. There shall be an arrow between each segment depending on the possible direction of the flow. It shall be possible to show or hide those elements.

v) FlowRoutings from the MandatoryRestrictions of the RestrictionGroupId shall be displayed in Green. The FlowRoutings from the ForbiddenRestrictions of the RestrictionGroupId will be displayed in Red.

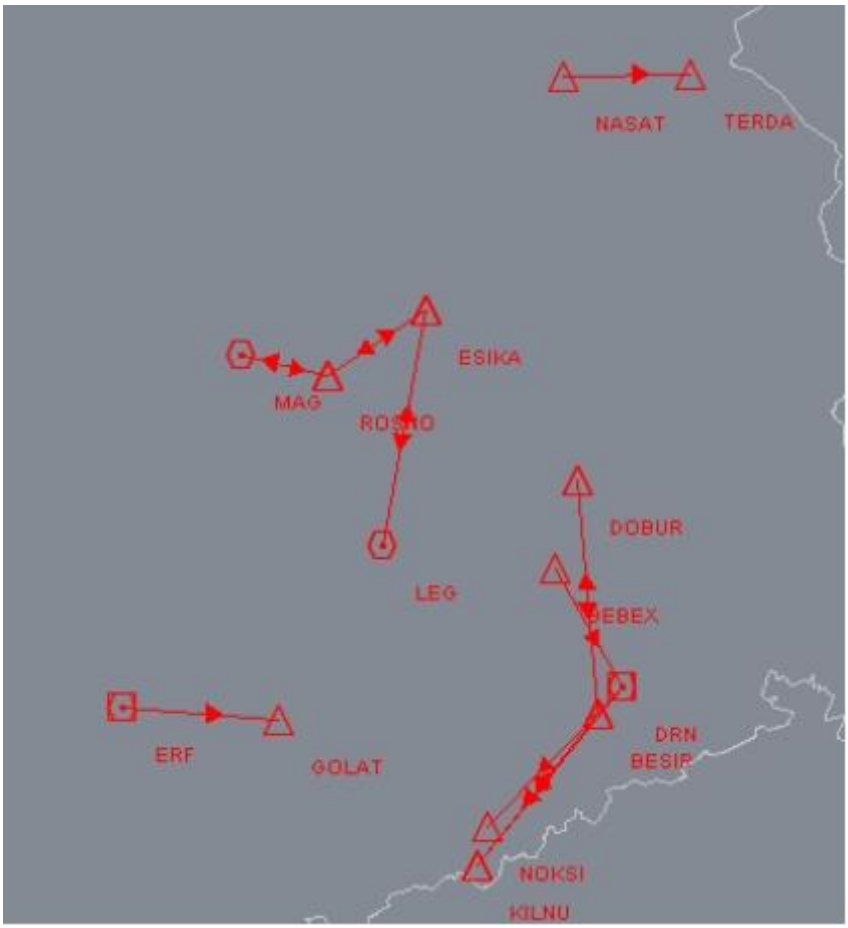


Figure 9-34: Aeronautical chart symbol for the restriction detail representation of forbidden FlowRoutings

vi) FlowRoutings of a FRA border restriction show all the allowed cross border segments and the allowed FRA entry-exit border points in the direction of the border (in the example from EDUUUTA to DKSEFAB).

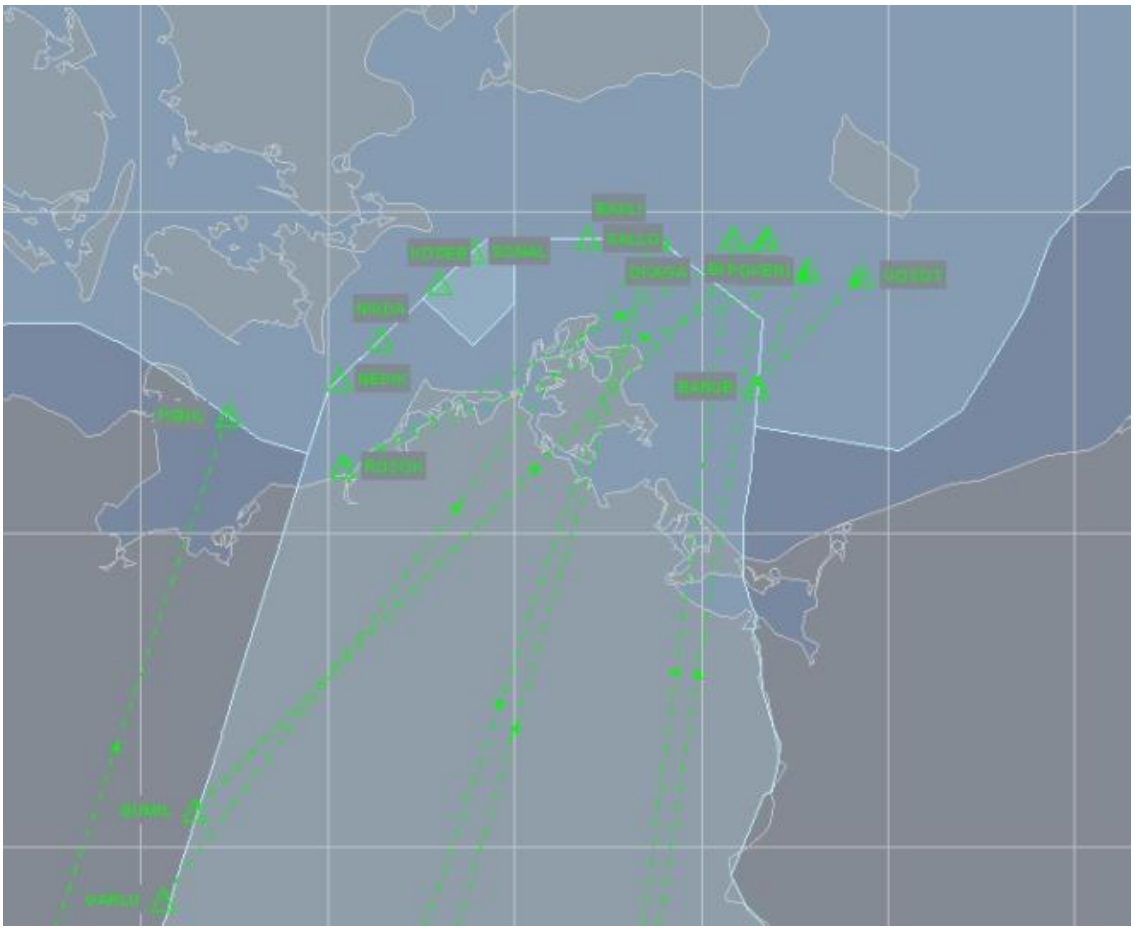


Figure 9-35: Aeronautical chart symbol for the restriction detail representation of allowed FlowRoutings of a FRA border Restriction

vii) FlowRoutings of a FRA airspace restriction include all the allowed cross border segments and crossing segments and all the allowed FRA entry-exit and intermediate points. The direction of the segments and the entry-exit point is given relative to the FRA airspace (DKSEFAB in the example).

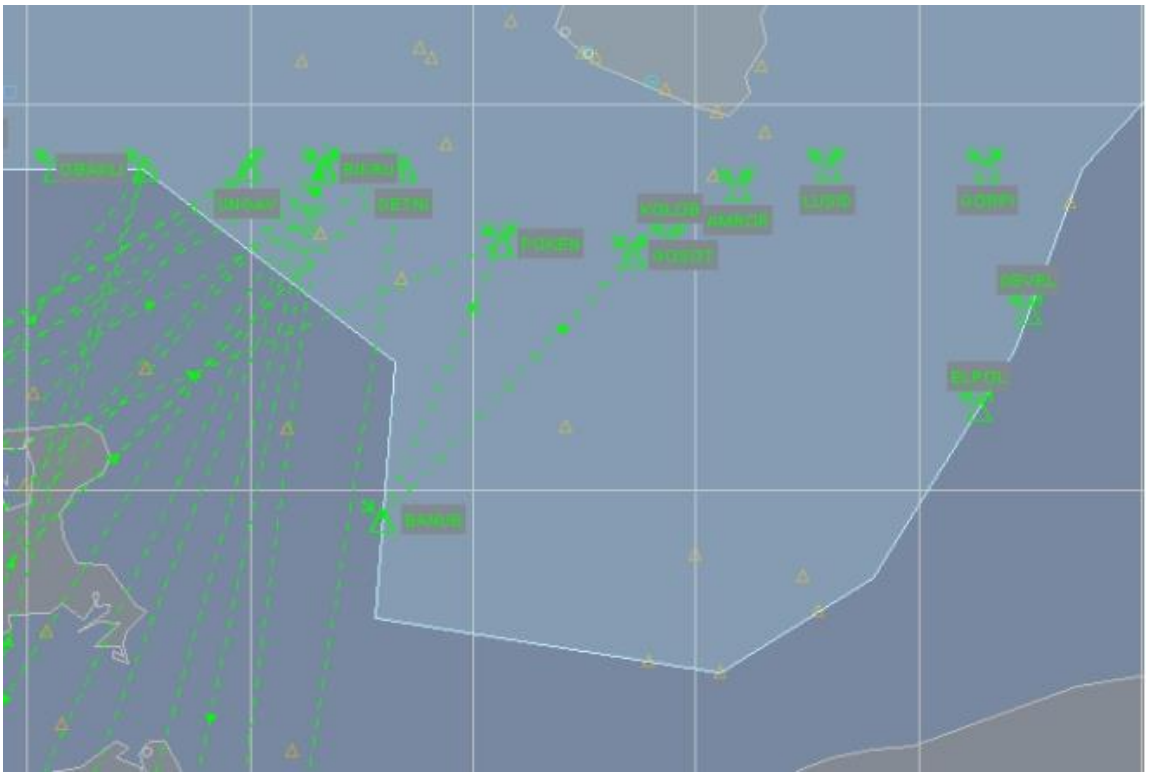


Figure 9-36: Aeronautical chart symbol for the restriction detail representation of allowed FlowRoutings of a FRA Airspace Restriction

b) FlowConditionsElements (ReferenceLocation):

- The FlowConditionsElements of the restriction will be displayed like the reference location of the restrictions (airspace or airspaceBorder): with the RS Business Id. The RS Business Id will be located at the centre of gravity of the airspace, when the FlowConditionElement is an airspace, or at the centre of gravity of one of the two involved airspaces when the FlowConditionElement is an airspaceBorder. It will be possible with a right button mouse click to open the detail window of the restriction. From this detail window it will be possible to display all or the selected element of the restriction for the detailed representation (cf. **9.1.1.1.2 Detail representation**). It will also be possible with a right button mouse click to display directly the FlowRoutingsElements and the FlowConditionsElements.

i) Airspace representation:

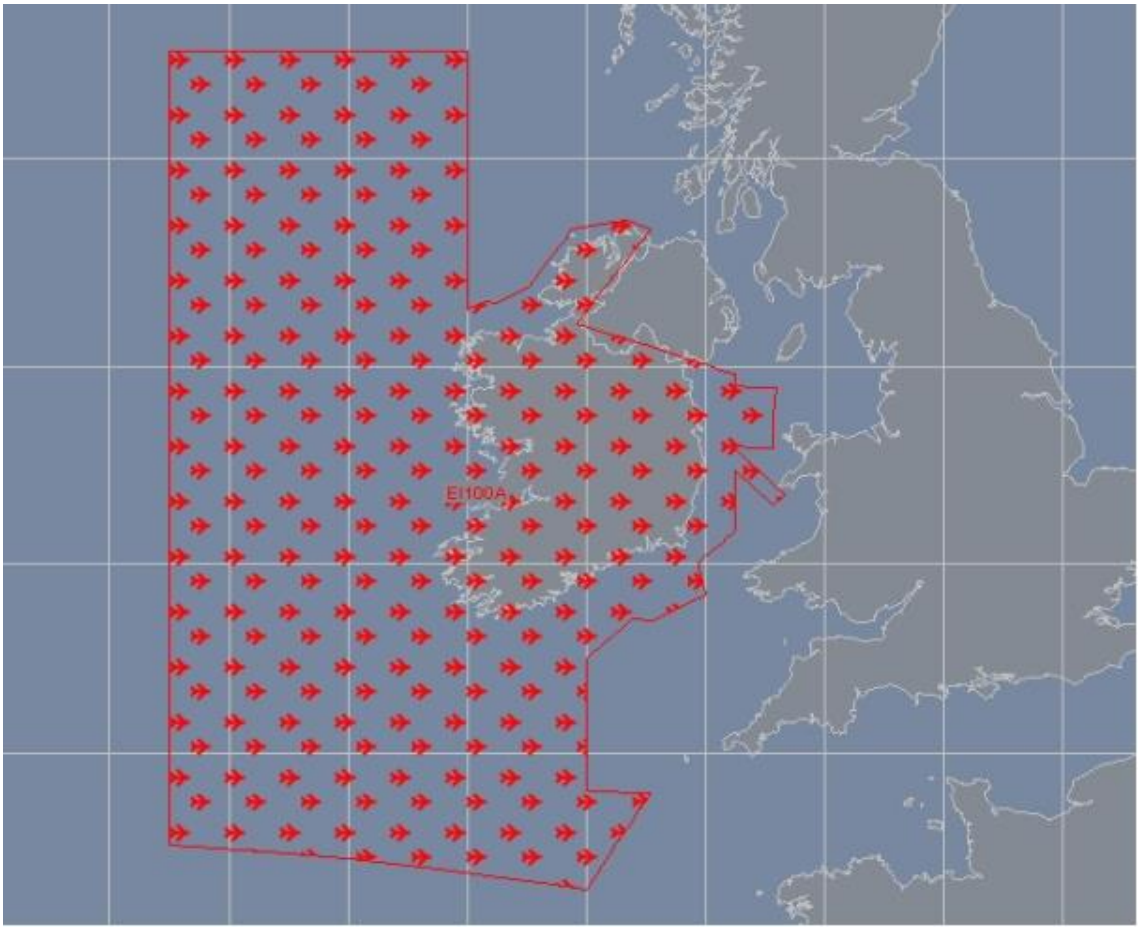


Figure 9-37: Aeronautical chart symbol for the restriction reference location representation of an airspace

ii) AirspaceBorder Representation:

iii) Both Airspaces involved must be displayed.

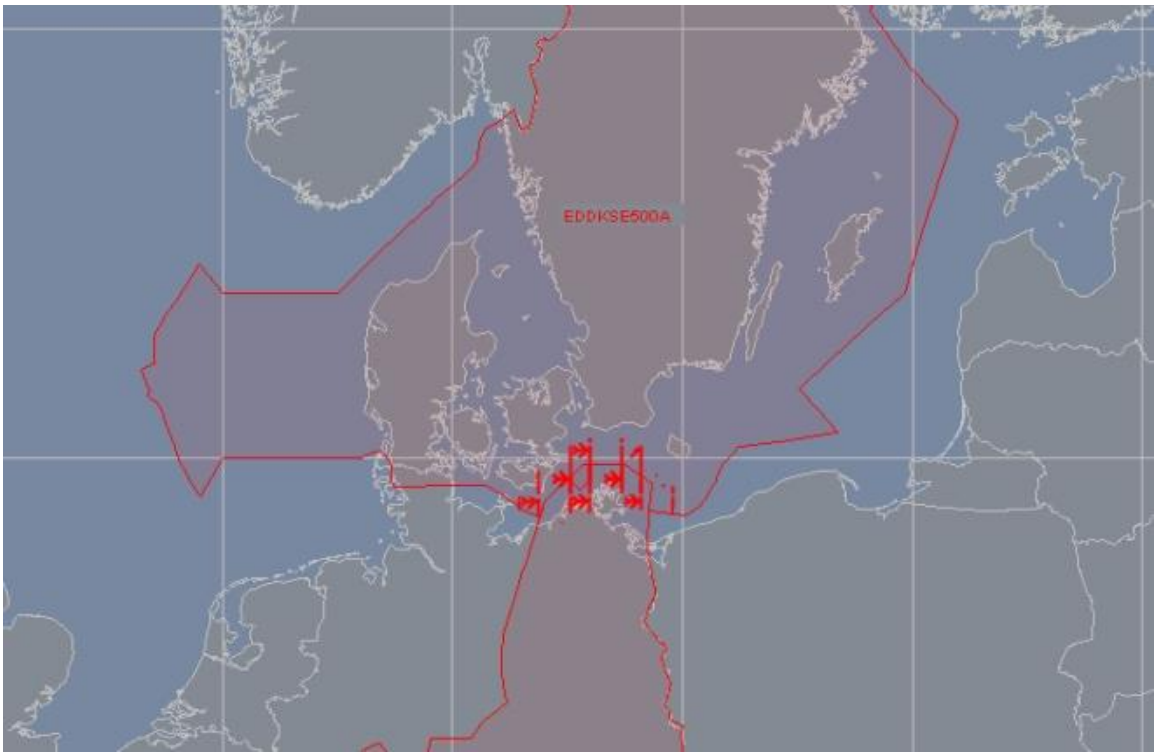


Figure 9-38: Aeronautical chart symbol for the restriction reference location representation of an AirspaceBorder

iv) Please note that a border can be lateral or horizontal. The following example is a mixed border, containing a lateral and a horizontal part. In case of a horizontal border, the 2D projections of the airspaces will overlap. The layout shall allow the user to see the underlying airspace.

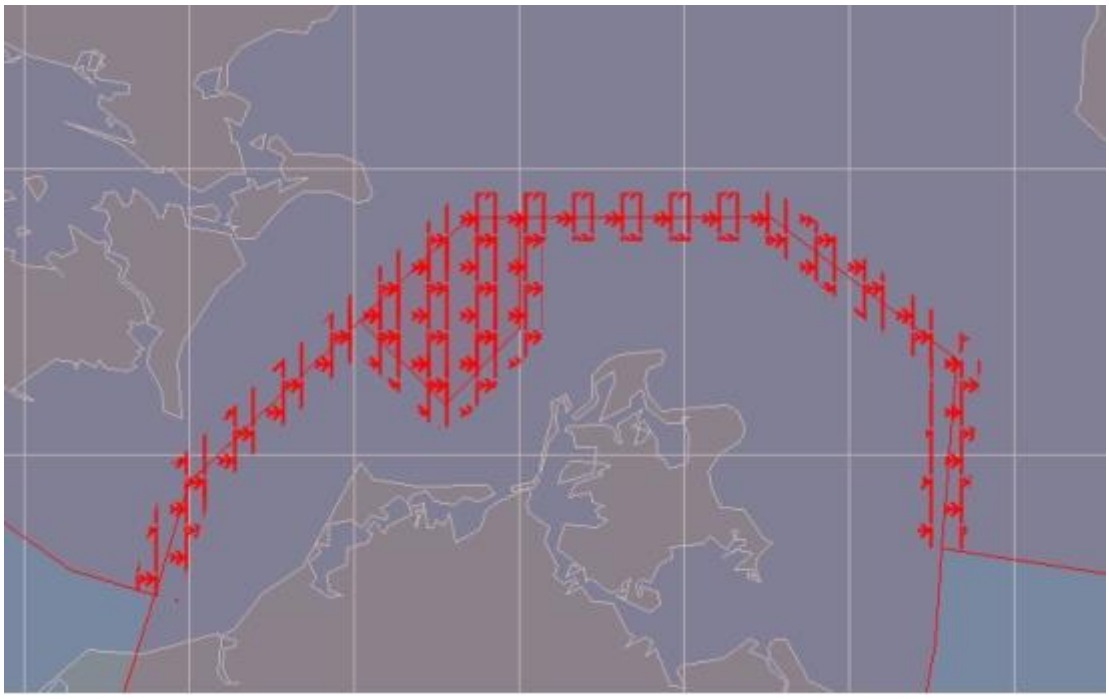


Figure 9-39: Aeronautical chart symbol for the restriction reference location representation of an airspace border (detail)

9.3.1.1.2.2 Colours and pattern

(1) The colour of the restriction and all elements that composes it shall change when it is selected. All element of the restriction shall change when selected.

a) FlowRoutingElements

i) The FlowRouting changes depending on a flag Mandatory/Forbidden:

		Yellow	Green	Red
1	When Selected	**X**		
2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-11: Restriction detail representation colours of DctLimitations FlowRoutinElements

b) FlowConditionsElements (ReferenceLocation):

i) The default colour of the airspace or airspace border is red. The actual ExceedingLimit value is a red Label:

		Yellow		Red
1	When Selected	**X**		
2	ExceedingLimit ValueLabel			**X**
3	Reference Location			**X**

Table 9-12: Restriction detail representation colours of DctLimitations

9.3.1.1.2.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element.

9.3.1.1.3 Fra Dct Restriction linked with flight profile

9.3.1.1.3.1 Layout

(1) Only the FlowRoutings in the context of the profile shall be displayed.

a) FlowRoutingsElements shall be displayed as described in section 9.3.1.1.2.1 (1) a).

9.3.1.1.3.2 Colours

(1) The colour of the restriction and all elements that composes it shall change when it is selected. All element of the restriction shall change when selected.

a) FlowRoutingElements

i) The FlowRouting changes depending on a flag Mandatory/Forbidden:

		Yellow	Green	Red
1	When Selected	**X**		
2	Mandatory		**X**	
3	Forbidden			**X**

Table 9-13: Restriction detail representation colours of DctLimitations FlowRoutinElements

9.3.1.1.3.3 Labelling

(1) It shall be possible to show or hide the label of the restriction element.

(2) The labels shall change depending on the zoom level. This shall be defined in a parameter file.

(3) There shall be no label collision.

(4) The label shall have the same colour as the corresponding restriction element

9.3.1.2 Geographical Display vertical projection

9.3.1.2.1 Restriction linked with flight profile

9.3.1.2.1.1 Layout

(1) It shall be possible to display or hide restrictions violated by the flight profile.

(2) It shall be possible to display or hide restrictions compliant with the flight profile.



Figure 9-40: Violated/compliant Restrictions

Note: only restrictions with an airspace as reference location will be shown on the vertical view, except for the PTRs (see types below).

(3) It shall be possible to select the type of restrictions in the Layers.

Types:

- F: FRA Restrictions

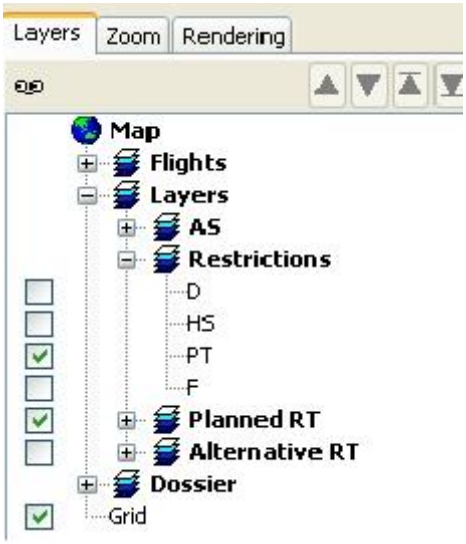


Figure 9-41: Restriction Layers

(4) It shall be possible to display or hide restrictions from the dossier.

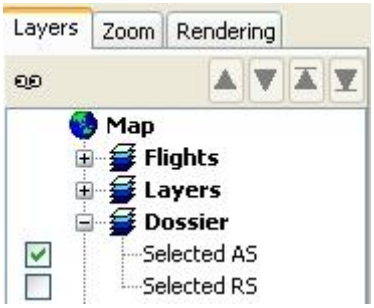


Figure 9-42: Restriction Dossier

(5) Complying restrictions shall be displayed on the vertical projection as shown in the next figure.

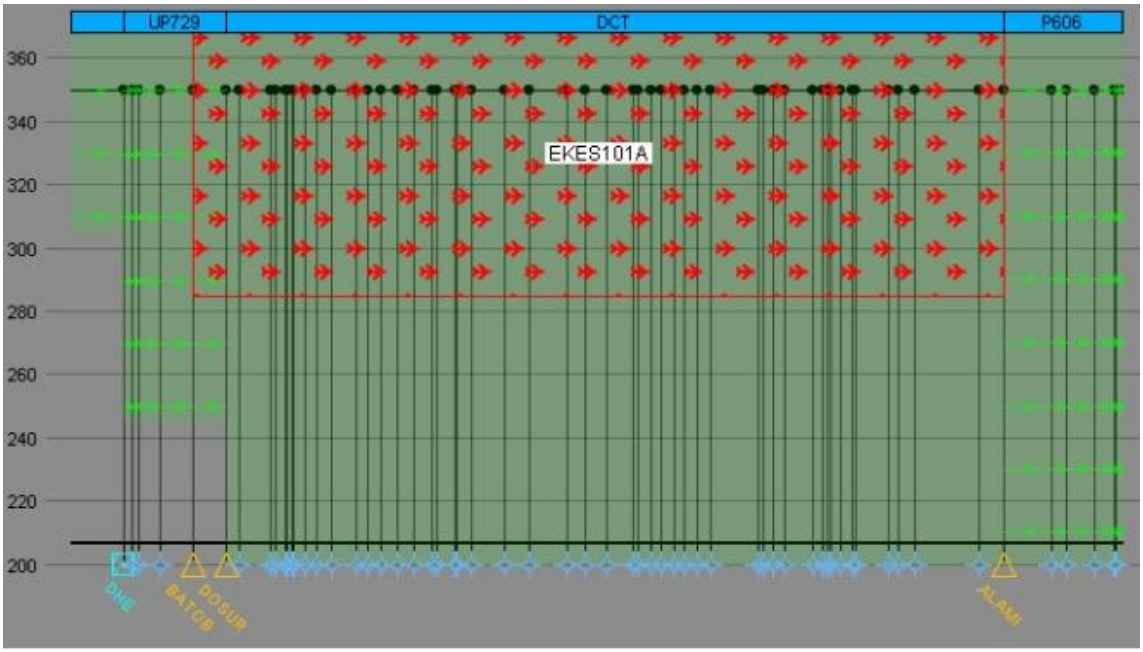


Figure 9-43: Restriction linked with a flight profile (complying DCT)

(6) Violating FRA DCT Airspace restrictions shall be displayed on the vertical projection as shown in the next figure. The start and end point of the restriction shall be defined by a red icon.

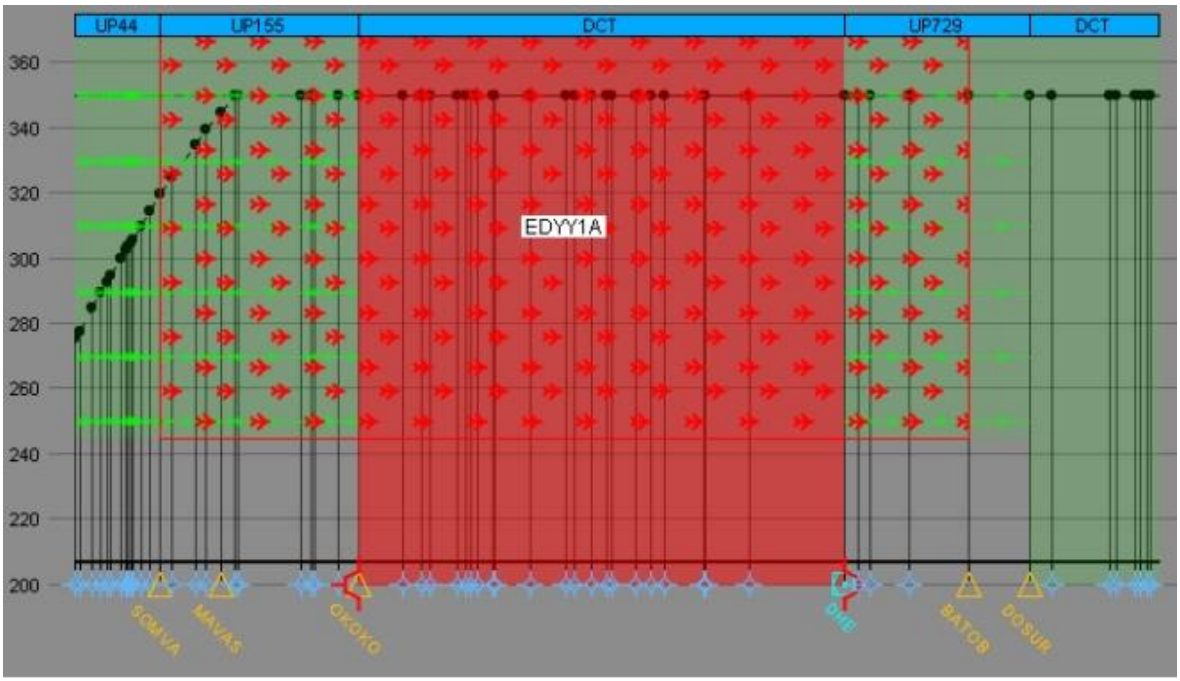


Figure 9-44: Restriction linked with a flight profile (violating DCT)

(7) Violating FRA DCT border restrictions shall be displayed on the vertical projection as shown in the next figure. The point where the flight profile crosses the FRA border shall be defined by a red icon.

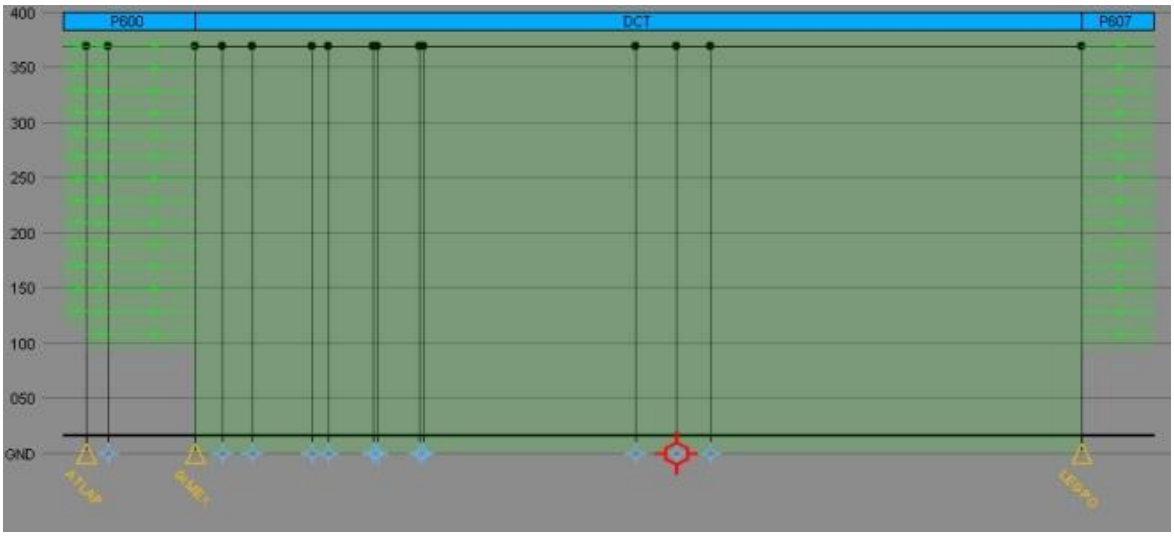


Figure 9-45: Restriction linked with a flight profile (violating FRA border)

9.3.1.2.1.2 Colours

(1) The background of the allowed FRA DCT segment shall be coloured green.

(2) The background of the forbidden FRA DCT segment shall be coloured red.

9.3.1.2.1.3 Labelling

(1) There shall be no label for FRA DCT restrictions.

(2) The user must have access to a preference which will determine whether Restrictions are displayed on the Vertical View for the ATFCM Flight list / ARCD list. This preference by default will be set to now display them.

10 Traffic Volume Map Human Machine Interface

10.1 Traffic Volume

10.1.1 Boundary class Traffic Volume map

10.1.1.1 Geographical Display

10.1.1.1.1 Basic representation

10.1.1.1.1.1 Layout

(1) Display of the REF LOC. Display of arrows representing the FWs. Display of the name of the traffic volume top left (when requested). Display of the names of the flows. Flows are represented by arrows in the TV. Within the RL area, all arrows should intersect at the same point; this being the gravity centre of the RL The arrow crosses the Ref Loc. The flows are oriented from the gravity centre of the upstream element to the gravity centre of the downstream element.

Seven cases:

a) No flow, means all flow. Display label "ALL FWs" at the centre of gravity of the reference location if there are no flows.

b) Either the flows cross the RL. Meaning that the upstream element is outside the RL AND the downstream element is at the other side of it. The arrow crosses the Ref Loc. The flows are oriented from the gravity centre of the upstream element to the gravity centre of the downstream element.

c) Either the flow 'lands' IN the RL. The arrow ends in the RL, just across the boundary. For readability, it does not go to the centre. It is oriented from the gravity centre of the upstream element to the gravity centre of the RL.

d) Either the flow 'takes off' FROM the RL. Opposite of 'the flow 'lands' IN the RL'.

e) Both the upstream and the downstream element are outside the RL but on 'the same side' of it. To determine what 'on the same side' is, the rule is: If the gravity centres of upstream and downstream elements stand within an angle off 90 degrees from the centre of the Reference Location, then the upstream and downstream elements are considered 'neighbours'. The way to represent that would be an arrow in a U shape, slightly penetrating the Reference Location, the tail coming from the upstream element and the head going to the downstream element.

f) Either both conditions are inside the Reference Location (or underneath). A curved arrow is shown inside the RL.

g) Either the flow has only an upstream or a downstream definition only. An arrow is represented that points from / to the (gravity centre of the) flow

element towards/outwards the (gravity centre of the) RL, but does not penetrate it.

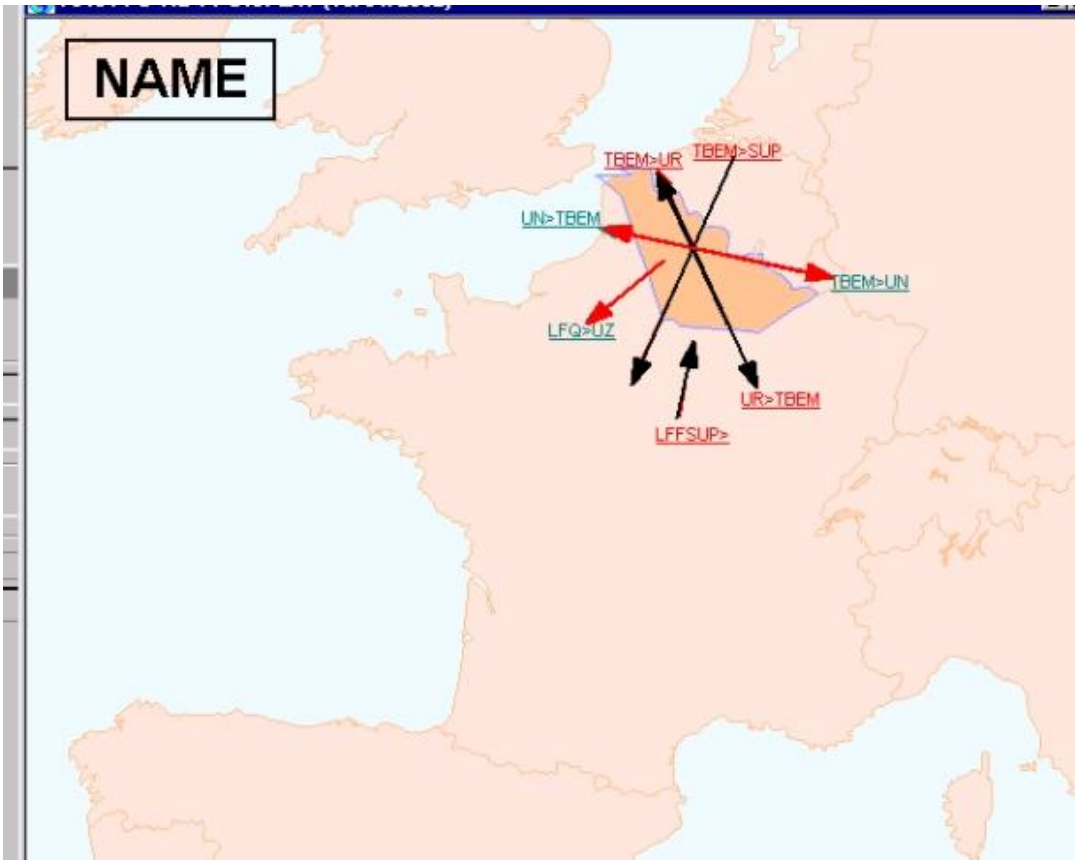


Figure 10-1: Aeronautical chart symbol for Traffic Volume Basic representation

h) List of options in the popup menu:

i) Traffic Volume-> <*Name of the TV*> ->Show detail: will open the detail data window of the Traffic Volume.

ii) Traffic Volume -> <*Name of the TV*> ->Plot ReferenceLocation: will display the ReferenceLocation on the map.

iii) Traffic Volume -> <*Name of the TV*> ->Plot Flows: will display all Flows on the map.

iv) Traffic Volume -> <*Name of the TV* > ->Flows-> <*Name of the Flow*>*:* will display the flow on the map.

v) Traffic Volume -> <*Name of the TV* > ->Flows-> <*Name of the Flow*> ->*Upstream*: will display the upstream elements on the map.

vi) Traffic Volume -> <*Name of the TV* > ->Flows-> <*Name of the Flow*> ->*Downstream*: will display the downstream elements on the map.

10.1.1.1.1.2 Colours

(1) The colour of the Traffic Volume Basic representation should change when it is selected.

		Yellow	Orange	Black	Red	Green	Blue
1	When Selected	**X**					
2	Reference Location		**X**				
3	Flow Included			**X**			
4	Flow Excluded				**X**		
5	Flow Included/Exempted					**X**	
6	Flow Exempted						**X**

Table 10-1: Traffic Volume basic representation colours

(2) The first column on the table is the priority to determine the colour if different criteria are filled.

10.1.1.1.1.3 Labelling

(1) It should be possible to show or hide the label.

(2) The labels should change depending on the zoom level. This should be defined in a parameter file.

(3) There should be no label collision.

(4) The label should have the same colour as the Traffic volume or flow.

10.1.1.1.2 Detail representation

10.1.1.1.2.1 Layout

(1) The same representation then the basic one but with the representation of the flow

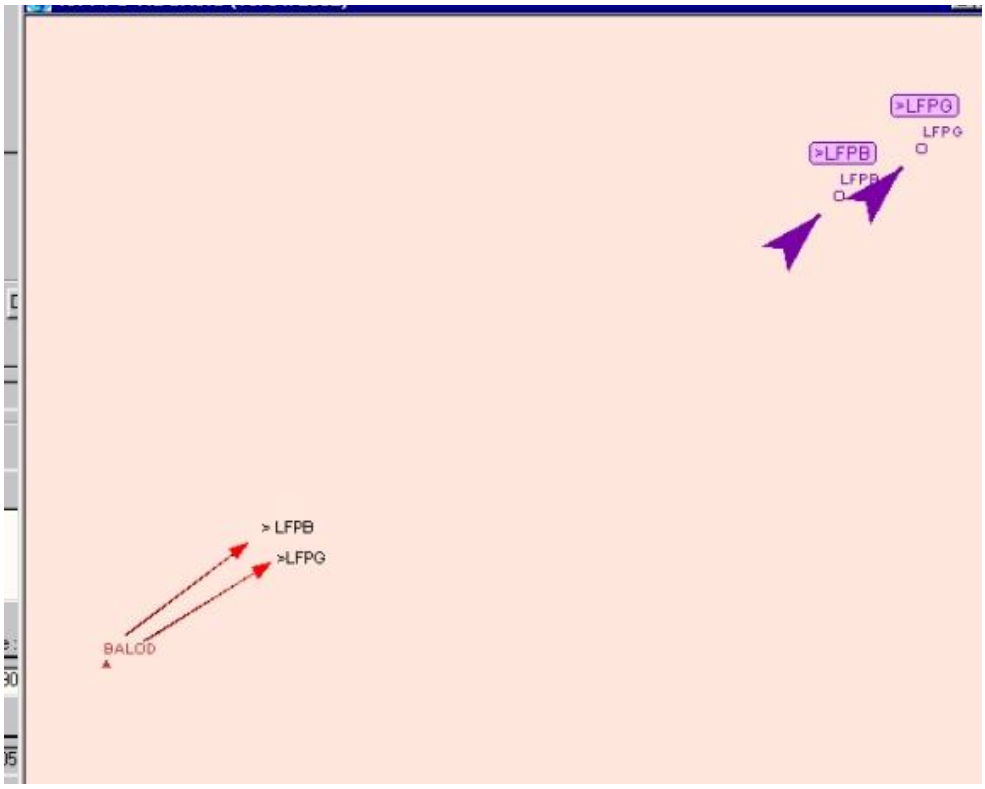


Figure 10-2: Aeronautical chart symbol for Traffic Volume Detail representation

a) List of options in the popup menu:

i) Flow-> <*Name of the Flow*> ->Show detail: will open the detail data window of the Traffic Volume.

ii) Flow -> <*Name of the FW*> -> *Upstream*: will display the upstream elements on the map.

iii) Flow -> <*Name of the FW*> -> *Downstream*: will display the downStream elements on the map.

10.1.1.1.2.2 Colours

(1) The colour of the Traffic Volume detail representation should change when it is selected.

		Yellow	Orange	Black	Red	Green	Blue	Purple
1	When Selected	**X**						
2	Reference Location		**X**					
3	Flow Included			**X**				
4	Flow Excluded				**X**			
5	Flow Included/ Exempted					**X**		
6	Flow Exempted						**X**	
6	Flow							**X**

Table 10-2: Traffic Volume detail representation colours

(2) The first column on the table is the priority to determine the colour if different criteria are filled.

10.1.1.1.2.3 Labelling

(1) It should be possible to show or hide the label.

(2) The labels should change depending on the zoom level. This should be defined in a parameter file.

(3) There should be no label collision.

(4) The label should have the same colour as the traffic volume or flow.

10.1.1.1.3 Flow representation

10.1.1.1.3.1 Layout

(1) The Flow is displayed by an arrow going from the upstream elements centre of gravity to down stream elements centre of gravity. The name of the flow and the name of the elements are displayed.



Figure 10-3: Aeronautical chart symbol for Flow

10.1.1.1.3.2 Colours

(1) The colour of the Flow representation should change when it is selected.

		Yellow	Purple
1	When Selected	**X**	
2	Default		**X**

Table 10-3: Flow representation colours

(2) The first column on the table is the priority to determine the colour if different criteria are filled.

10.1.1.1.3.3 Labelling

(1) It should be possible to show or hide the label.

(2) The labels should change depending on the zoom level. This should be defined in a parameter file.

(3) There should be no label collision.

(4) The label should have the same colour as the flow.

10.1.1.1.4 Reference Location representation

10.1.1.1.4.1 Layout

(1) The Reference location is displayed in orange. The name of the reference location and the name of the element are displayed.

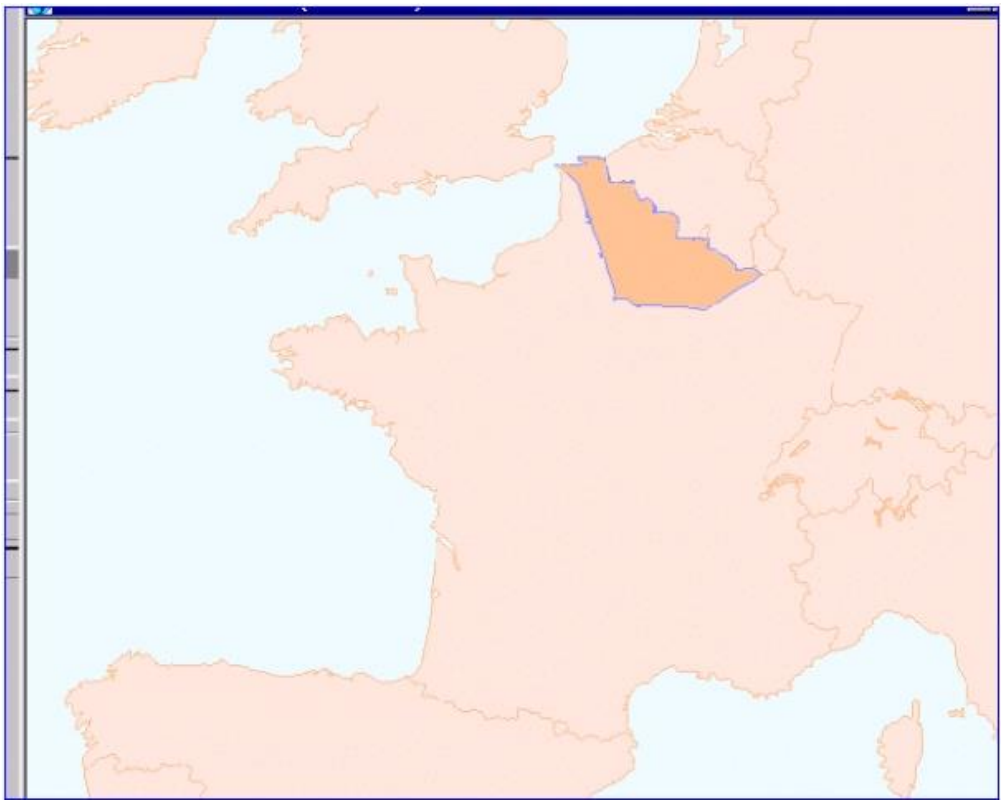


Figure 10-4: Aeronautical chart symbol for Traffic Location

10.1.1.1.4.2 Colours

(1) The colour of the Flow representation should change when it is selected.

		Yellow	Orange
1	When Selected	**X**	
2	Default		**X**

Table 10-4: Flow representation colours

(2) The first column on the table is the priority to determine the colour if different criteria are filled.

10.1.1.1.4.3 Labelling

(1) It should be possible to show or hide the label.

(2) The labels should change depending on the zoom level. This should be defined in a parameter file.

(3) There should be no label collision.

(4) The label should have the same colour as the flow.

10.1.2 Map behaviour with traffic volume

(1) When selecting a traffic volume in the popup menu it must be possible to select the flows and the reference location. When selecting the flows it must be possible to select the upstream and down stream elements.

11 Monitor

11.1 Common Requirements

(1) Current map time:

a) When a Monitor is plotted on the map, a white line with a globe at the top is drawn on the monitor horizontal bar chart that represents the current time of the map.

b) You can change the current map time by dragging the globe at the top of a Monitor.

c) When the time of the monitor is changed using the globe, the current time of the map should align to the same time.

d) Depending on the selected time, different Monitor details can be displayed and the colour of these details can change.

(2) When Monitor details have no colour displayed for the selected time (i.e. no load state defined), then these Monitor details are not visible on the map.

(3) The colour representation per load state of the Monitor details is defined by user adjustable ATFCM preferences. Following default values are used:

a) Red Overload

b) Orange High threshold

c) Yellow Low threshold

d) Green Normal Load

e) Magenta Undefined load because of unknown capacity

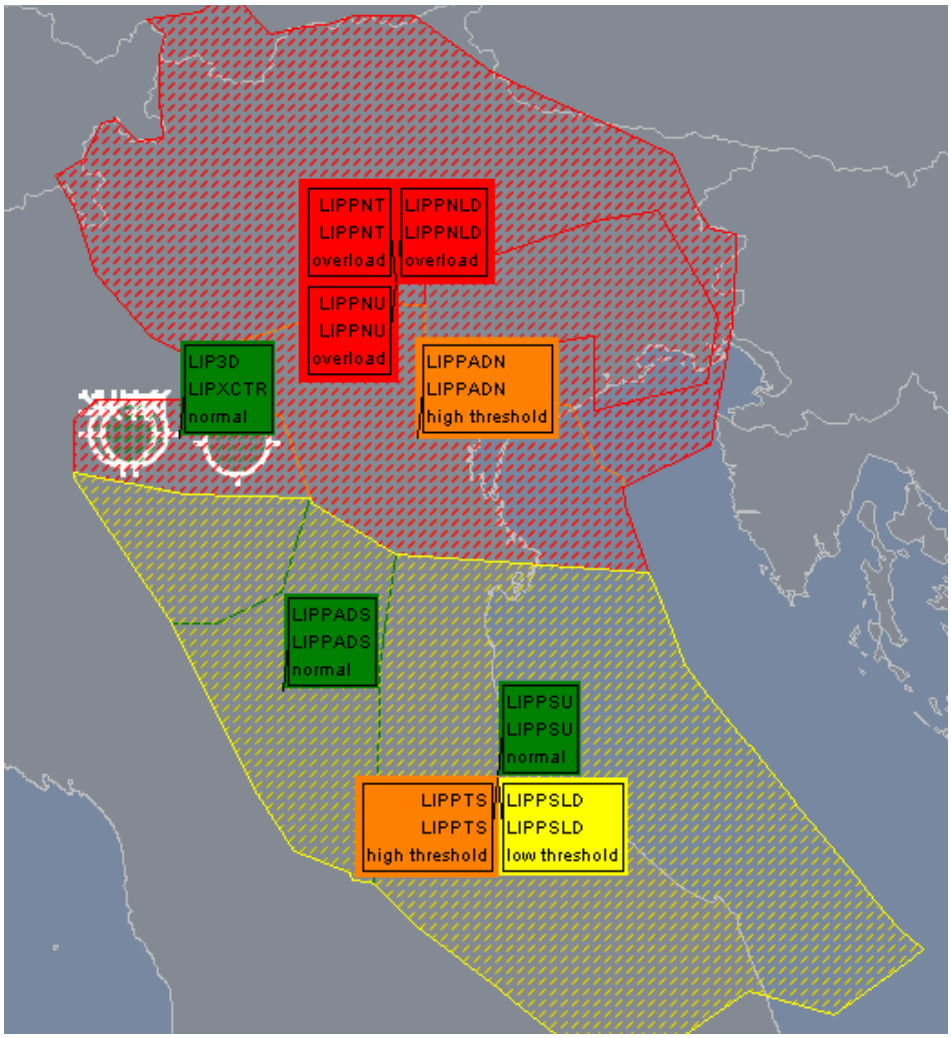


Figure 11-1: Map load state colours

(4) The labels of load state details with a higher load are painted above the other labels (i.e. the red overload labels are always on top).

(5) When the CTRL key is pressed, the labels can be moved to different positions.

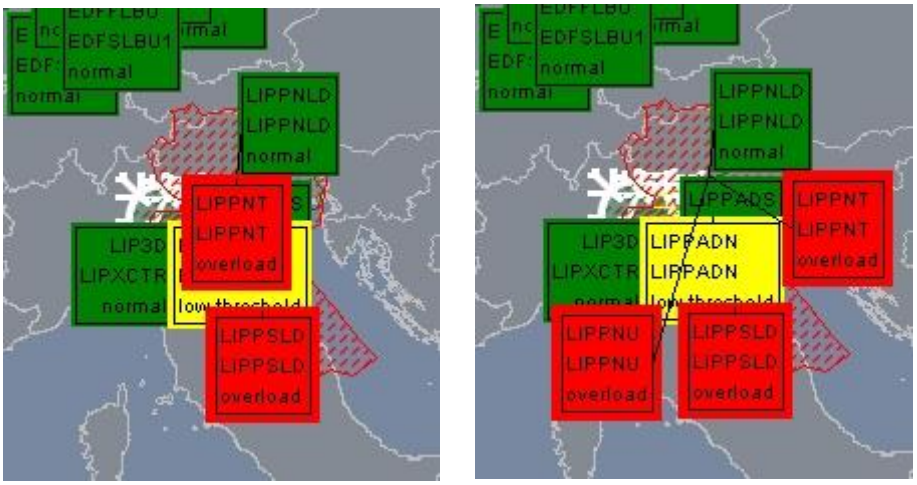


Figure 11-2: Load labels hierarchy and moved

(6) Delay states associated to a regulation are geographically represented on the map, by the protected location (arrival airport) of the regulated traffic volume, when applicable, otherwise by the regulated reference location.

(7) The regulated location and the associated label shall NOT be drawn after regulation cancel time, if any.

(8) The colour representation per delay state of the Monitor details is defined by user adjustable ATFCM preferences. Following default values are used:

a) Red >45

b) Yellow <45

c) Green <30

d) Blue <15

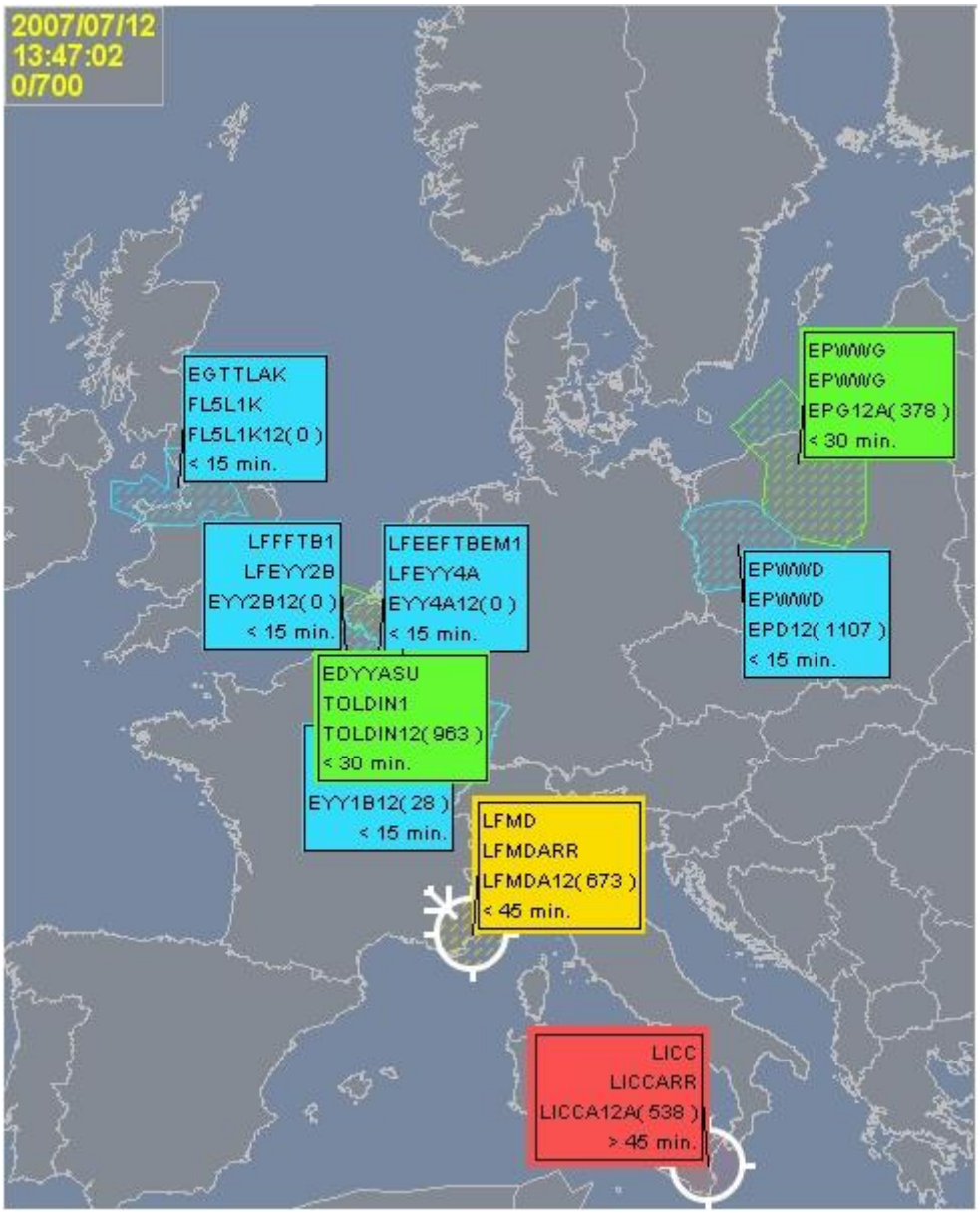


Figure 11-3: Map delay state colour

(9) The labels of delay state details with a higher delay are painted above the other labels (i.e. the red overload labels are always on top).



Figure 11-4: Delay labels hierarchy

(10) When the CTRL key is pressed, the labels can be moved to different positions.

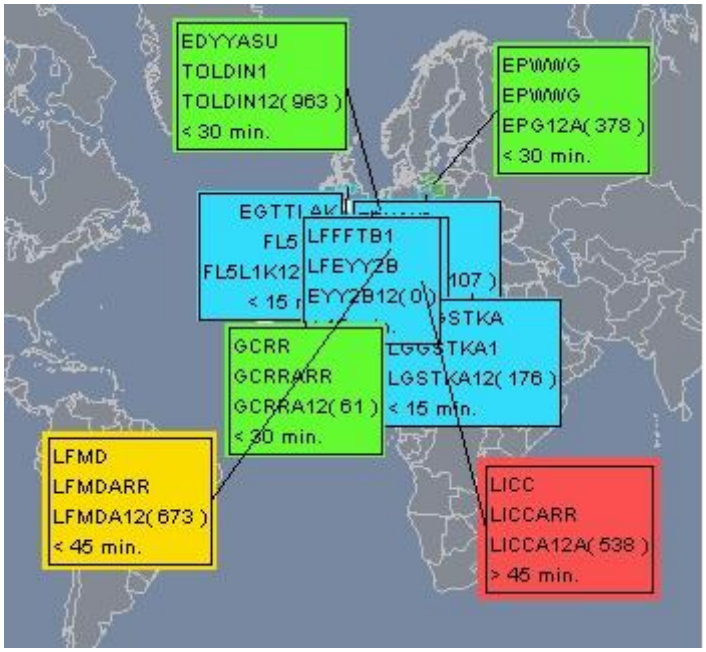


Figure 11-5: Moved delay labels

(11) In the overview and properties pane that may be displayed on the right side of the map, there is an **ATFCM** tab.

(12) It contains:

a) A **Load** tree to list Monitor items for which Load States are plotted

b) A **Regulation** tree to list Monitor items for which Delay States are plotted

(13) The level of the tree below **Load** and **Regulation** identifies the source Monitor window.

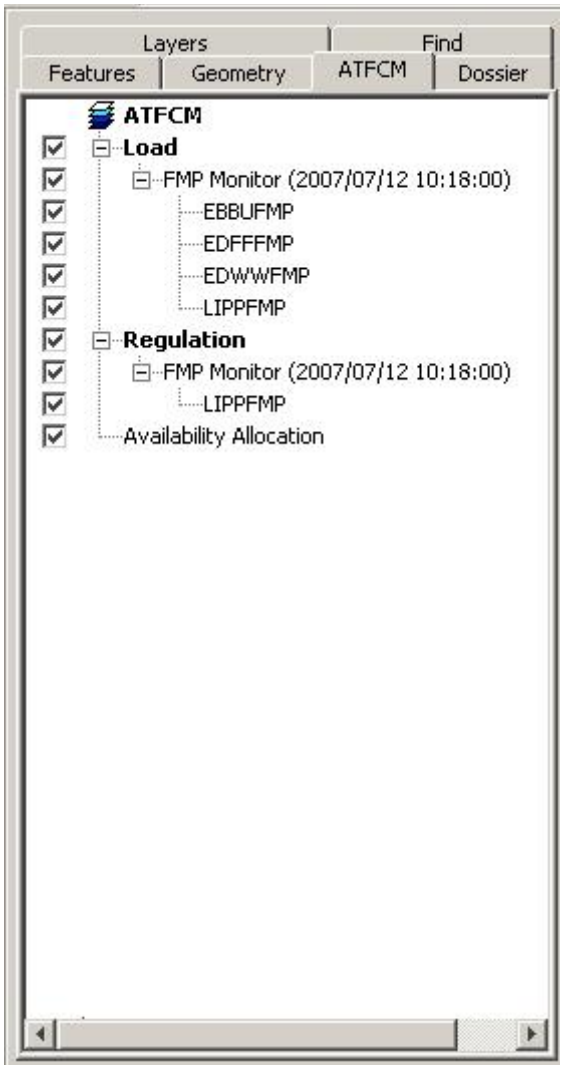


Figure 11-6: Monitor in ATFCM tab

(14) The layer control contains also a new layer group: **ATFCM**.

(15) This group may contain 3 subgroups:

a) **Load**

b) **Delay**

c) **Availability Allocation**

(16) Each layer group can be made visible or invisible.

(17) The corresponding labels can be displayed or not.

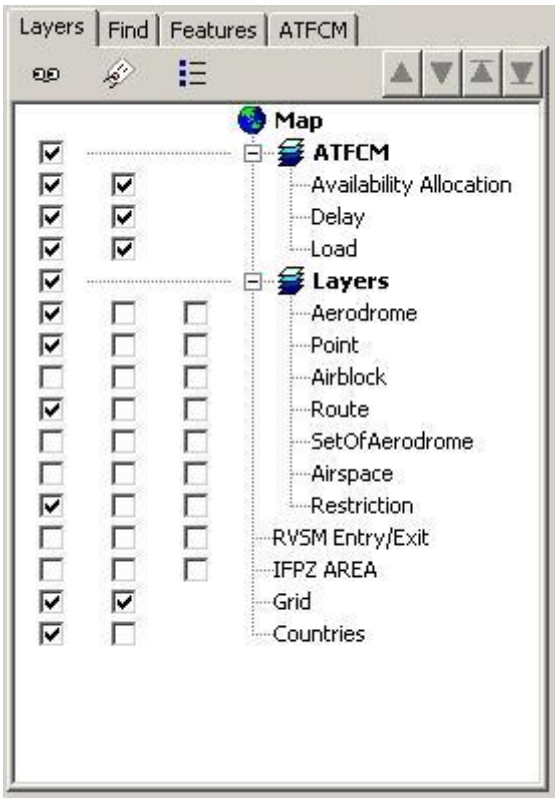


Figure 11-7: ATFCM map layer pane

(18) When the currently active window is a Monitor which is plotted on the map, then the Plot button is displayed as pressed.

(19) When the Plot button is un-pressed, the Monitor objects are removed from the map.

11.2 FMP Monitor

(1) Open the FMP monitor:

a) Start the CHMI ATFCM.

b) Select one of the following menu items:

i) Application / ATFCM / Traffic/Monitor / FMP Monitor

ii) Application / Predict / Traffic / Monitor / FMP Monitor

iii) Application / Archive / Traffic / Monitor / FMP Monitor

c) The window for the FMP monitor is shown.

d) Select some FMPs.

e) Refresh the monitor.

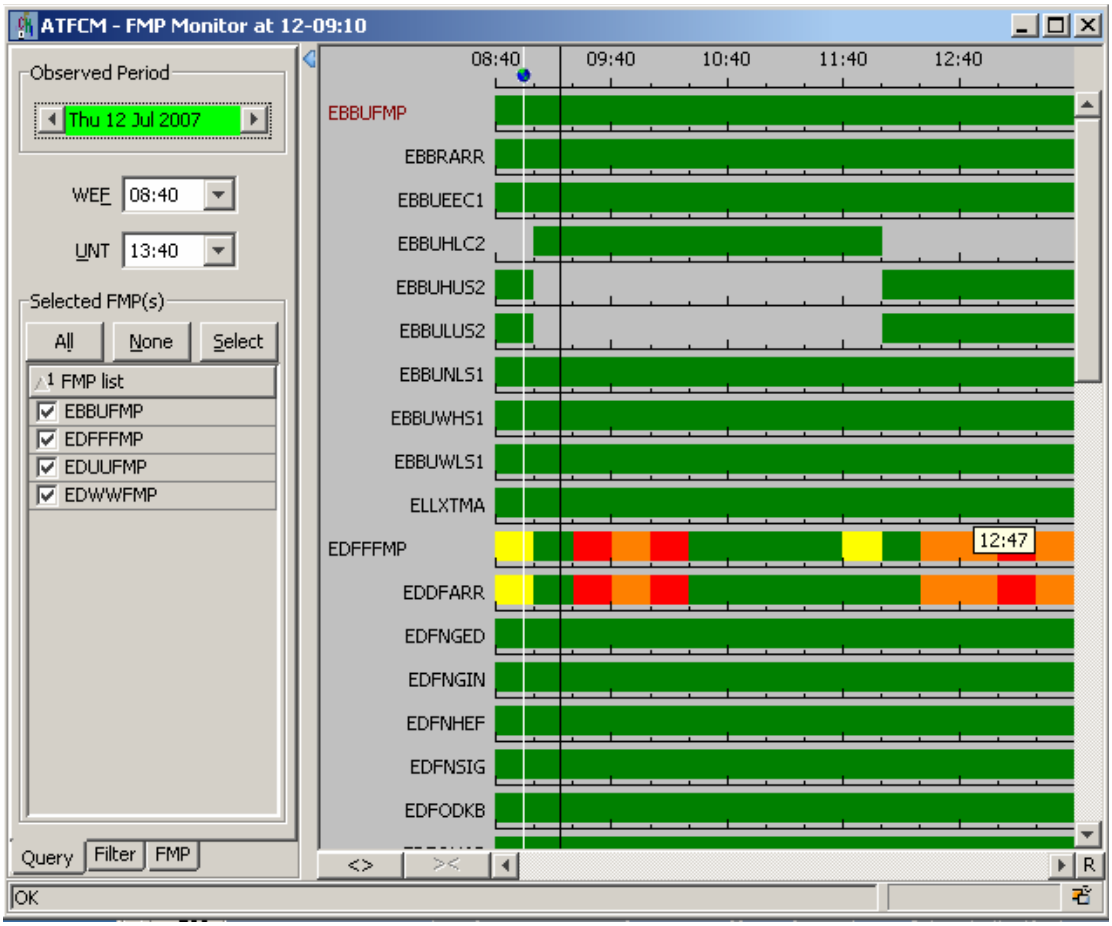


Figure 11-8: FMP monitor

(2) Click on the Plot button in the tool bar.

a) The map is opened, if it was not already opened.

b) It is brought to front, if it was already open.

c) The FMP Monitor details are displayed on the map.

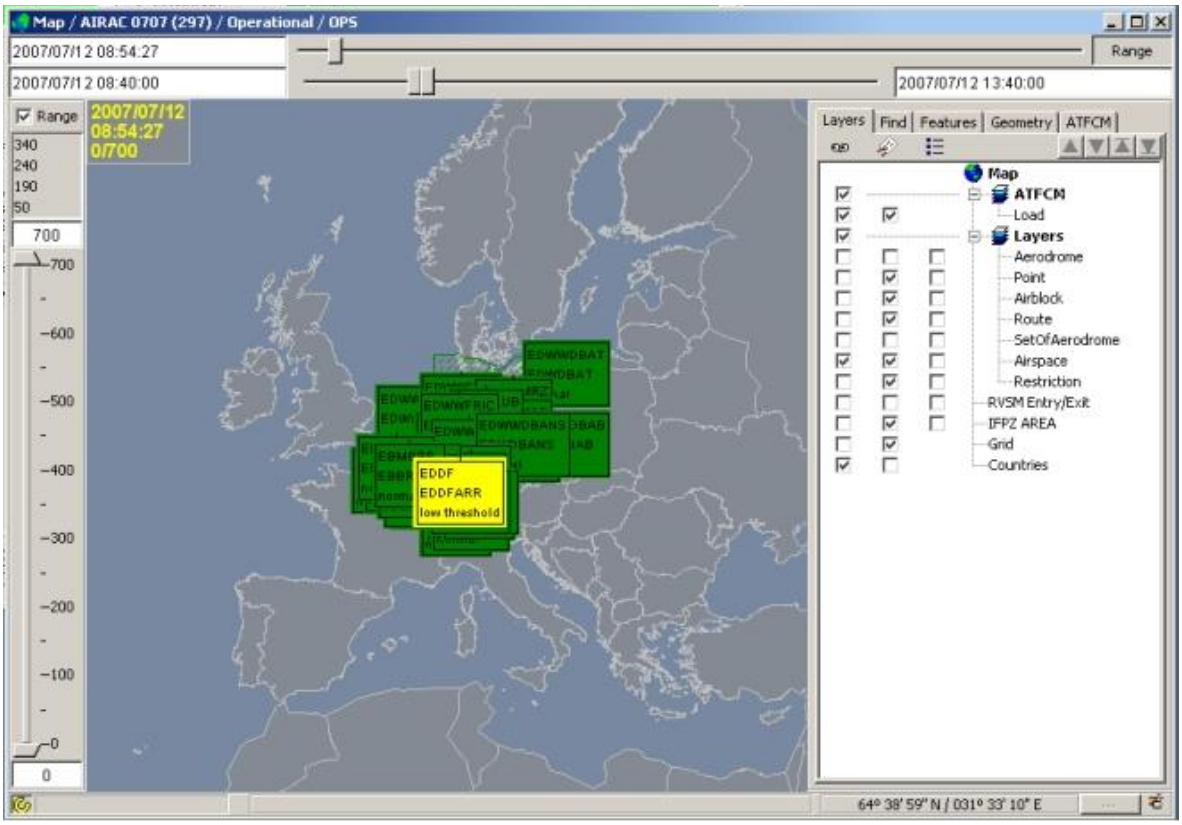


Figure 11-9: FMP monitor on the map

(3) FMP Monitor details can be:

a) Airspaces

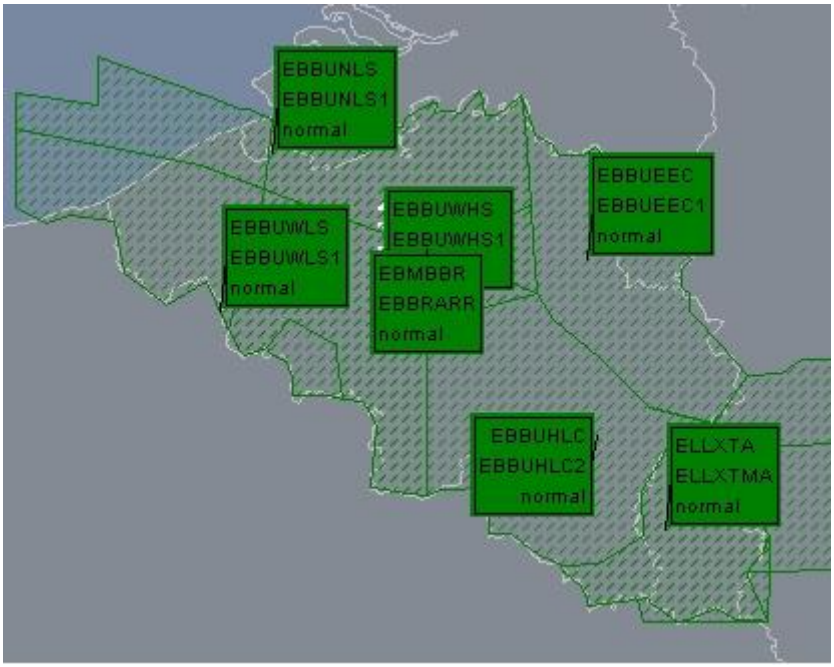


Figure 11-10: FMP airspaces reference location

b) Aerodromes



Figure 11-11: FMP aerodrome reference location

c) Set Of Aerodromes

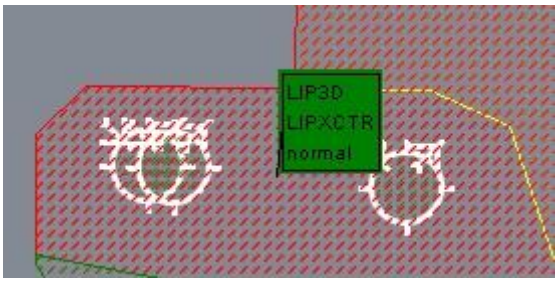


Figure 11-12: FMP set of aerodrome reference location

d) Significant Points

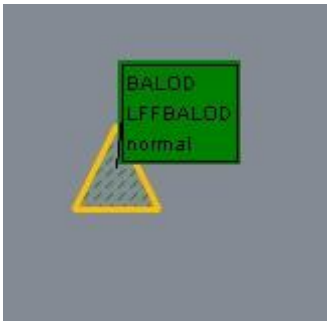


Figure 11-13: FMP point reference location

(4) On the **ATFCM** tab of the overview and properties pane, below **Load** and **Regulation:**

a) An FMP Monitor source window is identified by the string **FMP Monitor (***yyyy/mm/dd hh:mm-hh:mm***)** where *yyyy/mm/dd hh:mm-hh:mm* is the query period

b) The nested tree level identifies the FMPs for which details have been plotted

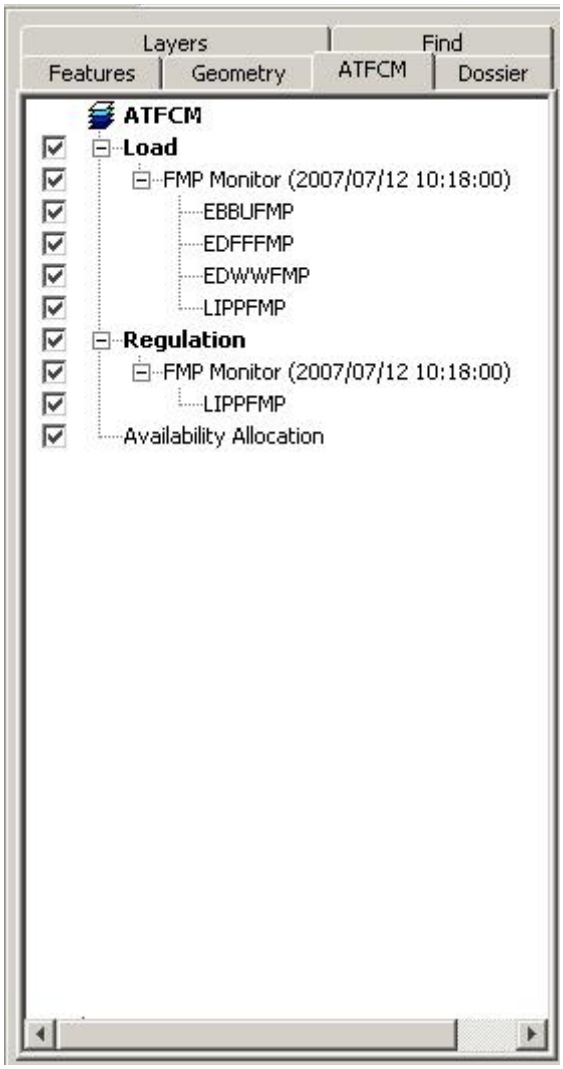


Figure 11-14: FMP monitor in ATFCM tab

11.3 ATC Airspace Monitor

(1) Open the ATC Airspace Monitor:

a) Start the CHMI ATFCM.

b) Select one of the following menu items:

i) Application / ATFCM / Traffic / Monitor / ATC Airspace Monitor

ii) Application / Predict / Traffic / Monitor / ATC Airspace Monitor

c) The window for the ATC Airspace Monitor is shown.

d) Enter an ATC Unit Airspace: EBBUCTA.

e) Chose a configuration.

f) Refresh the monitor.

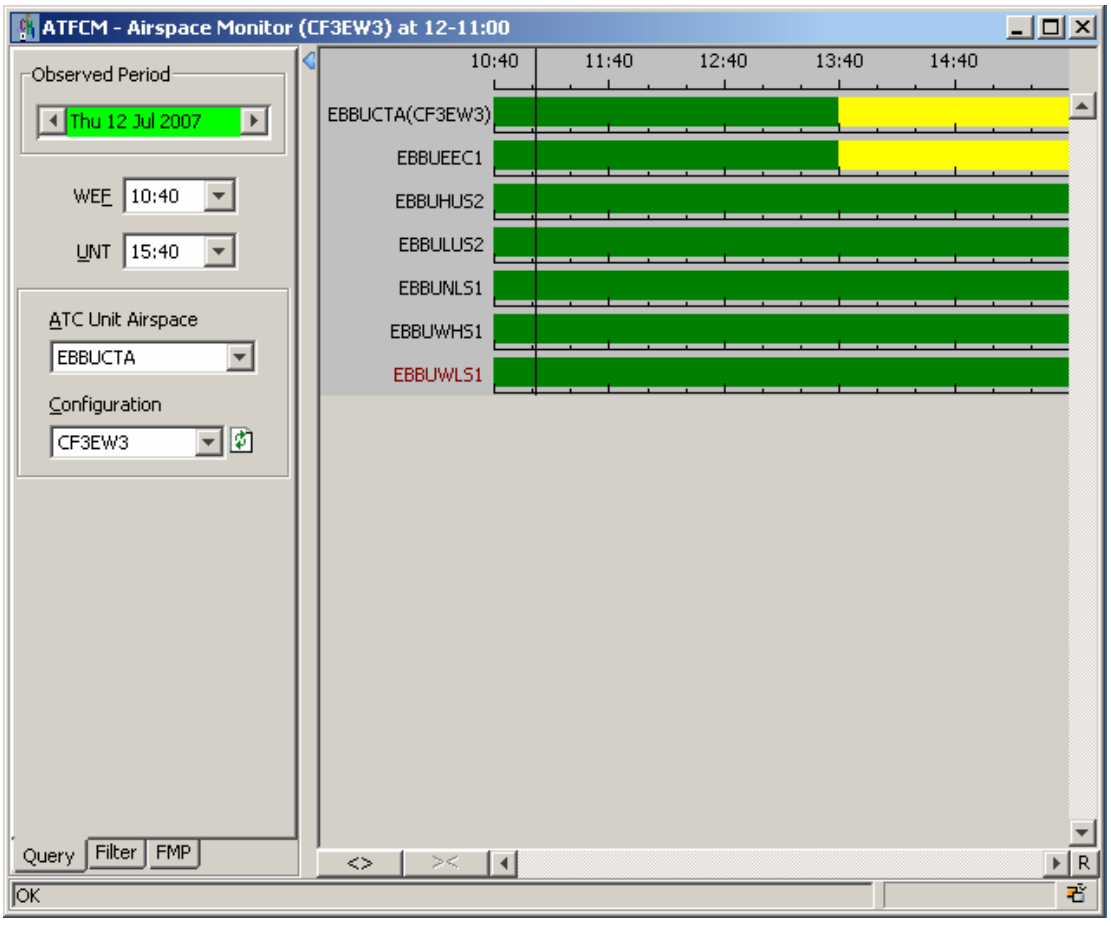


Figure 11-15: Airspace monitor

(2) Click on the Plot button in the tool bar.

a) The map is opened, if it was not already opened.

b) It is brought to front, if it was already open.

c) The ATC Airspace Monitor details are displayed on the map.

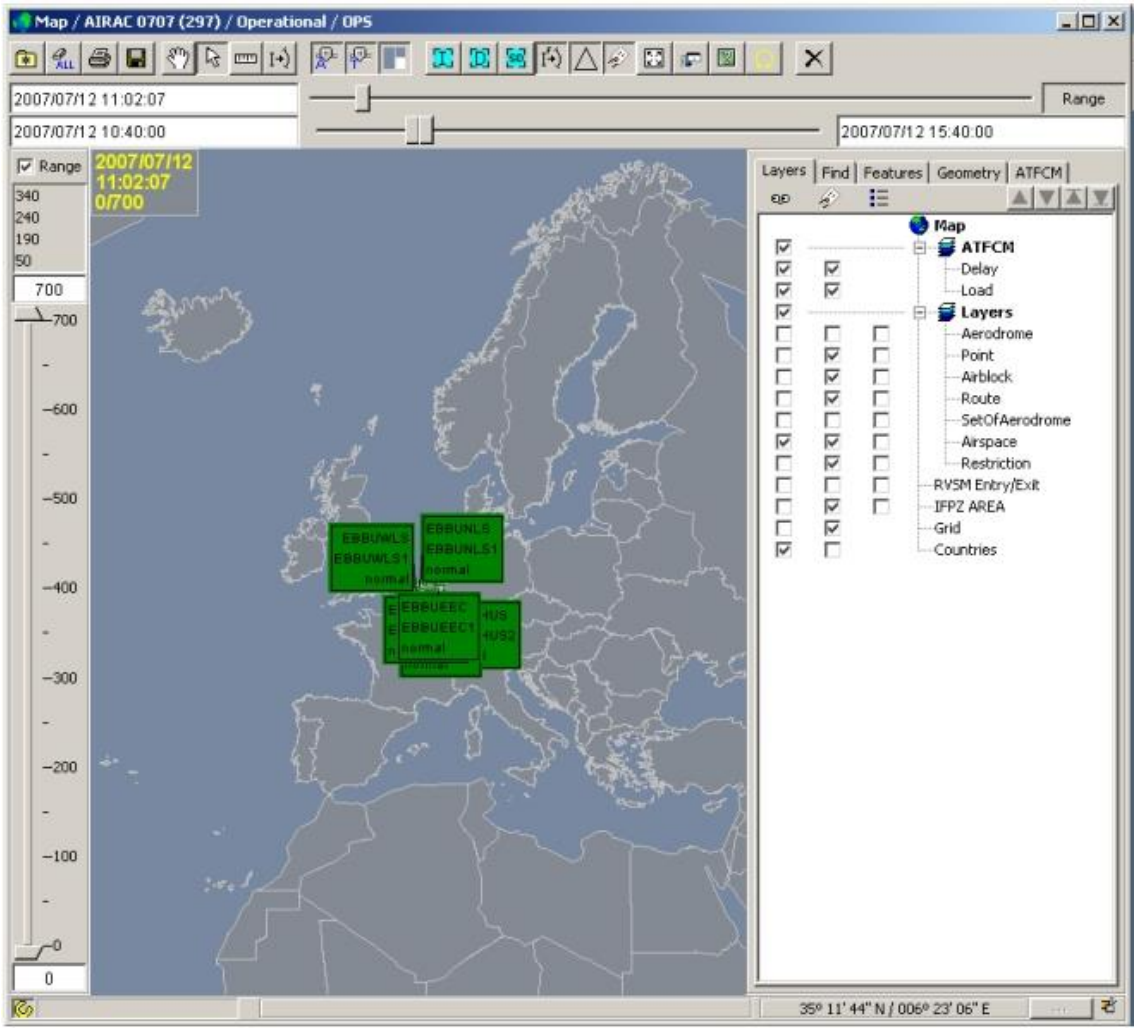


Figure 11-16: Airspace monitor on the map

(3) The ATC Airspace Monitor only contains airspaces:

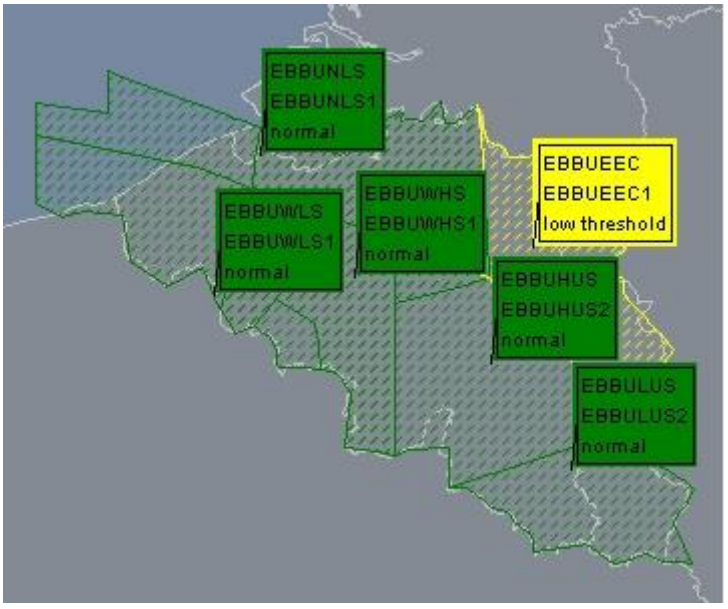


Figure 11-17: Airspace monitor map colour

(4) On the **ATFCM** tab of the overview and properties pane, below **Load** and **Regulation:**

a) An ATC Airspace Monitor source window is identified by the string **ATC Airspace Monitor (***yyyy/mm/dd hh:mm-hh:mm***)** where *yyyy/mm/dd hh:mm-hh:mm* is the query period

b) The nested tree level identifies the ATC Unit Airspace and the associated Configuration for which details have been plotted

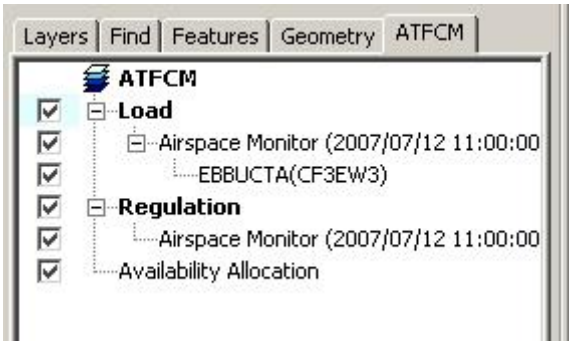


Figure 11-18: ATC Airspace Monitor in ATFCM tab

11.4 ATFCM Monitor

(1) Open the ATFCM Monitor:

a) Start the CHMI ATFCM.

b) Select one of the following menu items:

i) Application / ATFCM / Measures / ATFCM Monitor

ii) Application / Predict / Measures / ATFCM Monitor

c) The window for the ATFCM Monitor is shown.

d) Refresh the monitor.

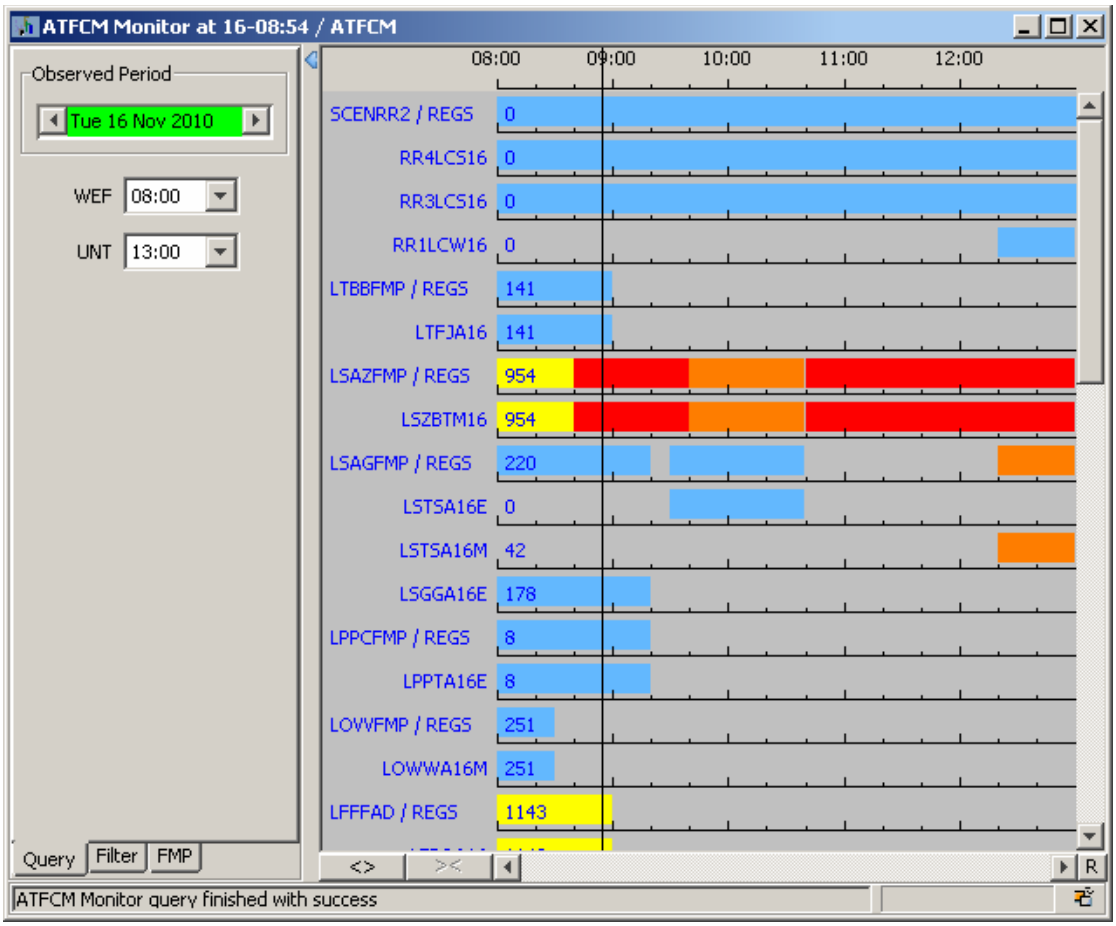


Figure 11-19: ATFCM Monitor

(2) Click on the Plot button in the tool bar.

a) The map is opened, if it was not already opened.

b) It is brought to front, if it was already open.

c) The ATFCM Monitor details are displayed on the map.

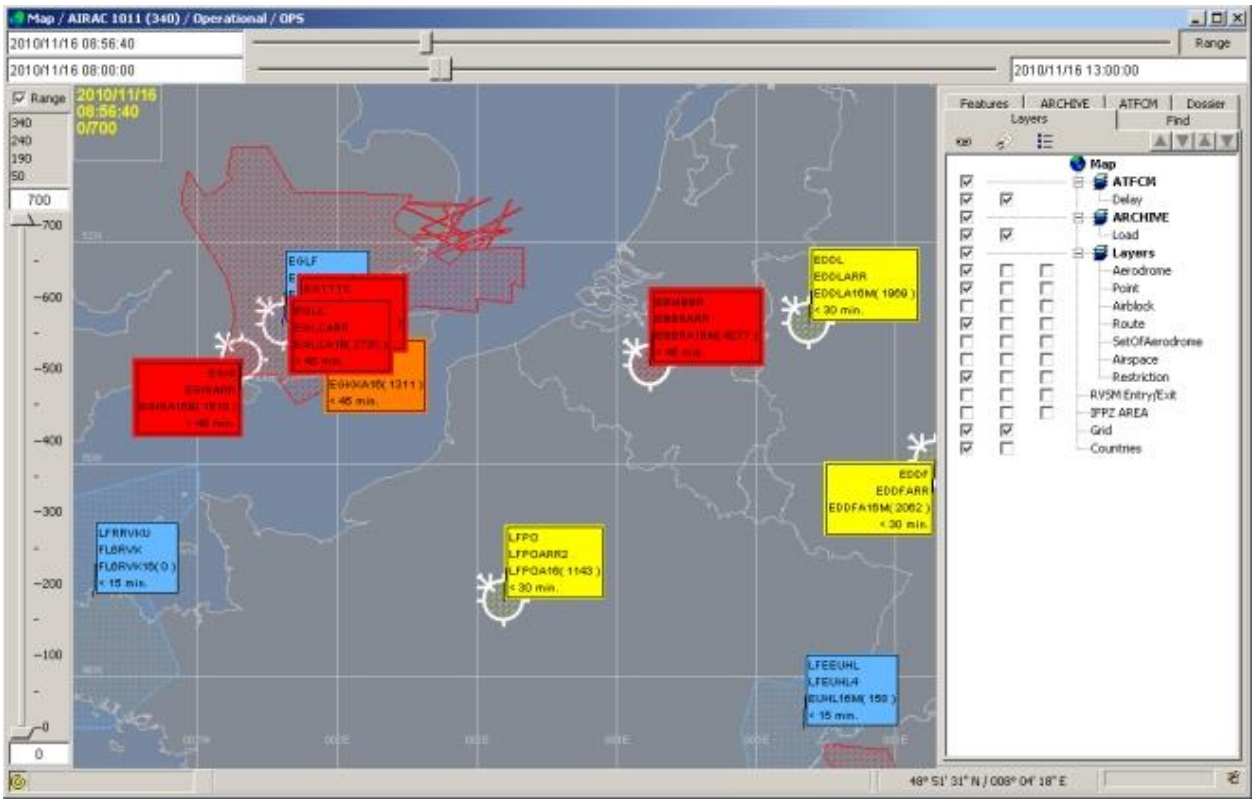


Figure 11-20: ATFCM Monitor on the map

(3) ATFCM Monitor details can be:

a) Airspaces



Figure 11-21: ATFCM Monitor airspaces reference location

b) Aerodromes



Figure 11-22: ATFCM Monitor aerodrome reference location

c) Set Of Aerodromes

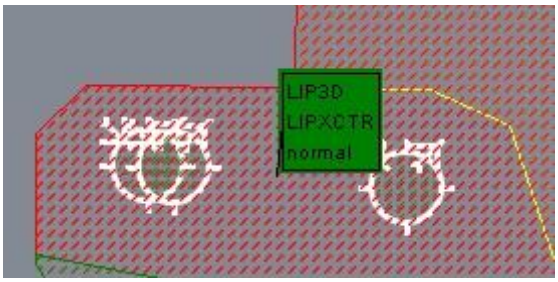


Figure 11-23: ATFCM Monitor set of aerodromes reference location

d) Significant Points

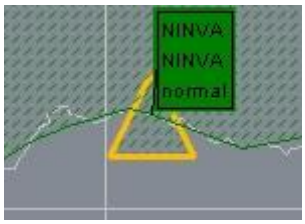


Figure 11-24: ATFCM Monitor point reference location

(4) On the **ATFCM** tab of the overview and properties pane, below **Regulation:**

a) An ATFCM Monitor source window is identified by the string **ATFCM Monitor (***yyyy/mm/dd hh:mm-hh:mm***)** where *yyyy/mm/dd hh:mm-hh:mm* is the query period

b) The nested tree level identifies the FMPs for which details have been plotted

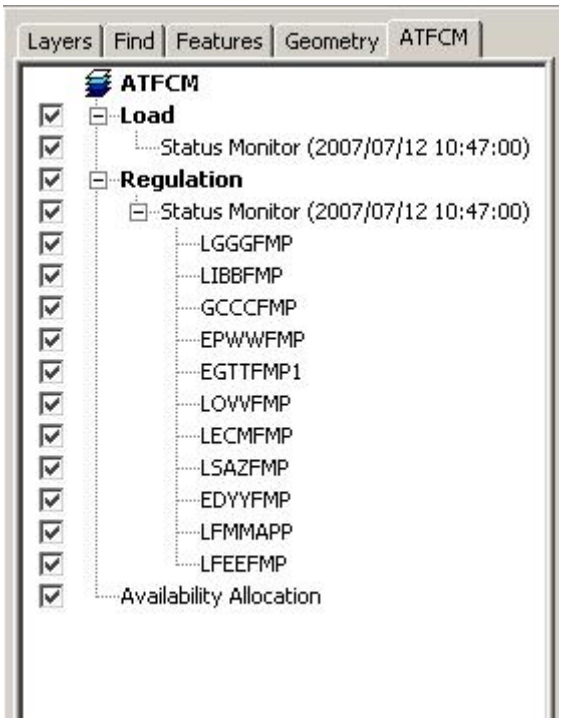


Figure 11-25: ATFCM Monitor in ATFCM tab

(5) There is no **Load** tree node inserted because the ATFCM Monitor does not display any load states but only regulation delay states.

11.5 CIAM Airspace Status Monitor

(1) Open the CIAM Airspace Status Monitor:

a) Start the CHMI FUA.

b) Select the menu item Application / FUA / ASM Monitor.

c) The window for the CIAM Airspace Status Monitor is shown.

d) Select the desired AMCs or FMPs.

e) Refresh the monitor.

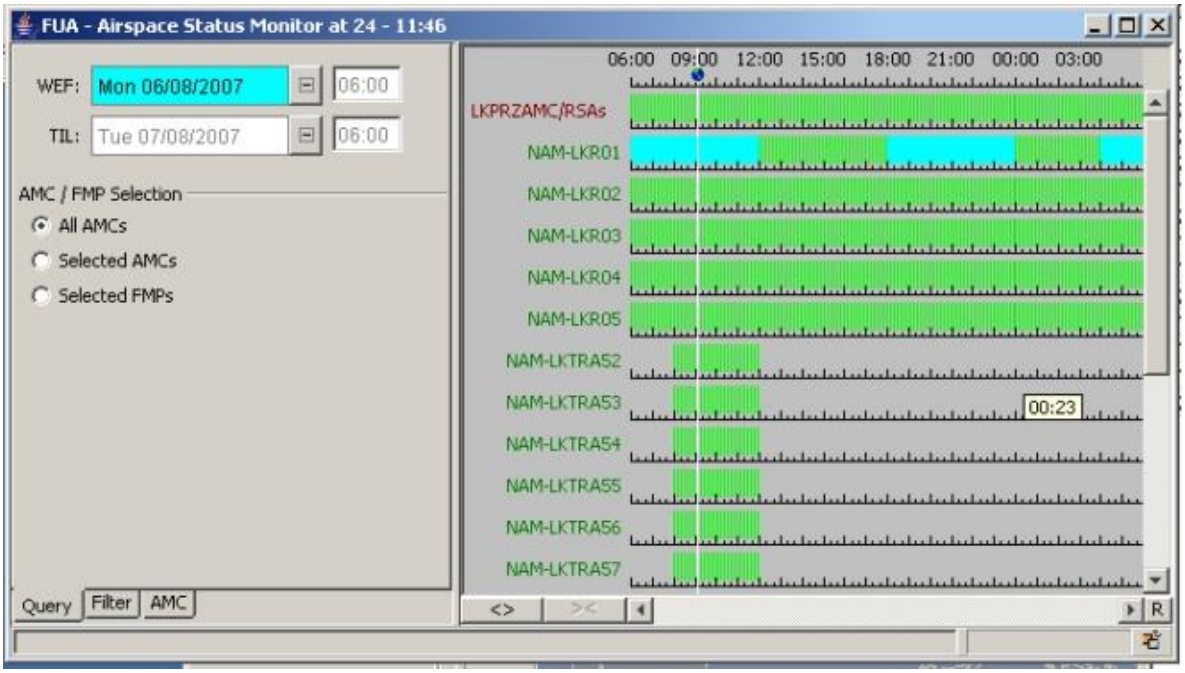


Figure 11-26: CIAM Airspace Status Monitor

(2) Click on the Plot button in the tool bar.

a) The map is opened, if it was not already opened.

b) It is brought to front, if it was already open.

c) The CIAM Airspace Monitor Status details are displayed on the map.

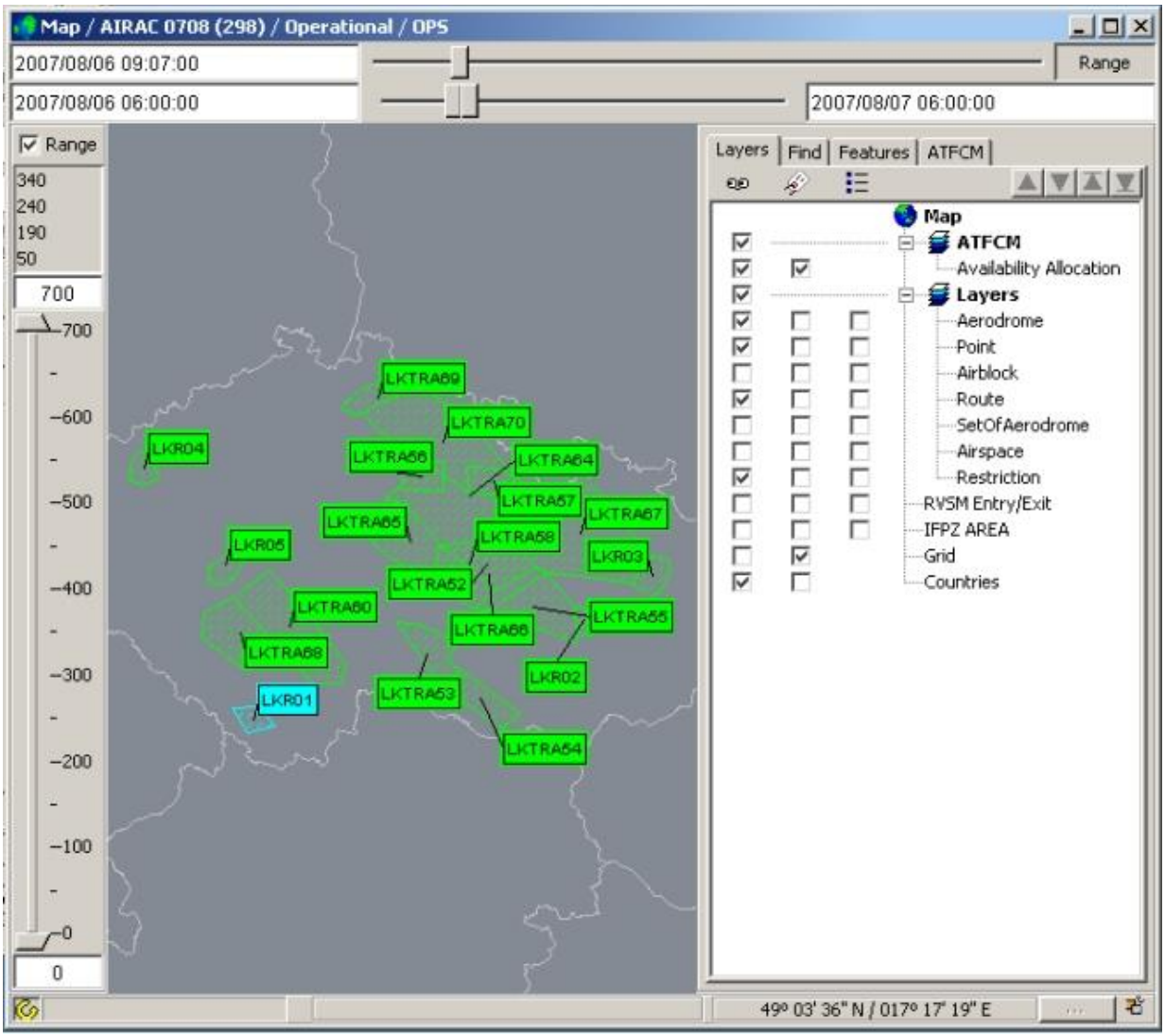


Figure 11-27: CIAM Airspace Status Monitor on the map

(3) CIAM Airspace Status Monitor details can be only RSA airspaces.

(4) The colour representation per RSA availability and allocation state of the CIAM Airspace Status Monitor details is defined by user adjustable FUA preferences.

(5) Following default colour values are used:

a) RGB= 0, 51, 255 AMA-NAM Availability

b) RGB= 0, 153, 0 AMA-NAM Allocation

c) RGB= 0, 153, 0 RCA Allocation

d) The **AMA-NAM allocation colour** takes precedence over **AMA-NAM availability colour** when they overlap.

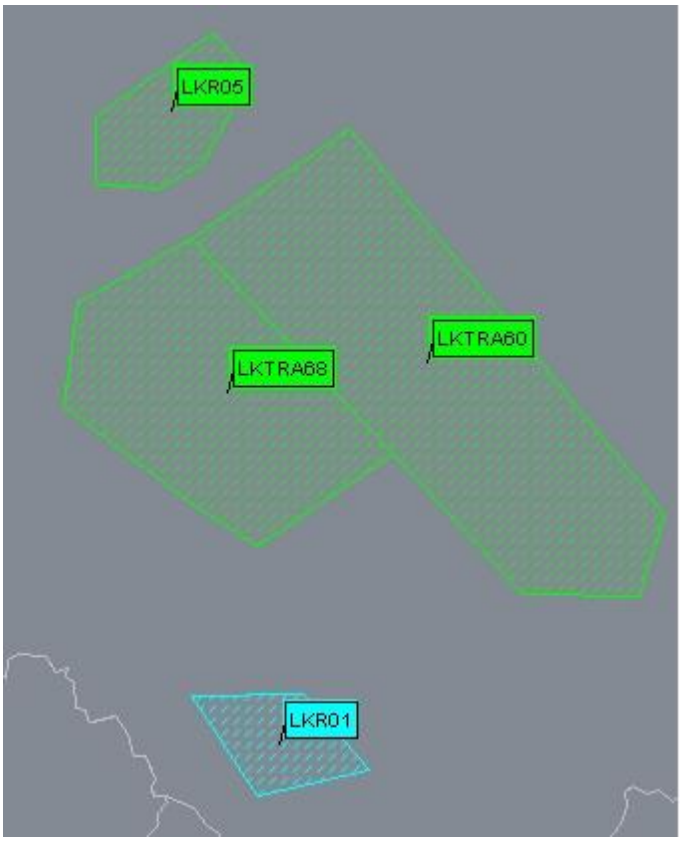


Figure 11-28: CIAM Airspace Status Monitor label

(6) When the CTRL key is pressed, the labels can be moved to different positions.

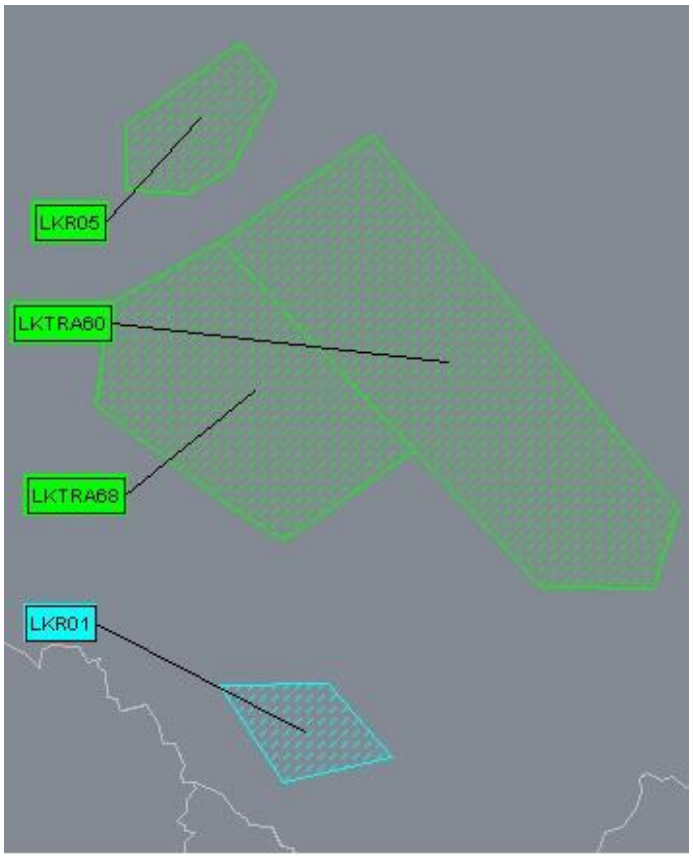


Figure 11-29: CIAM Airspace Status Monitor label hierarchy

(7) On the **ATFCM** tab of the overview and properties pane, following details are inserted below the **Availability Allocation** tree node**:**

a) A CIAM Airspace Status Monitor source window is identified by the string **Airspace Status Monitor (***yyyy/mm/dd hh:mm-hh:mm***)** where *yyyy/mm/dd hh:mm-hh:mm* is the query period

b) The nested tree level identifies the AMCs or the FMPs for which details have been plotted

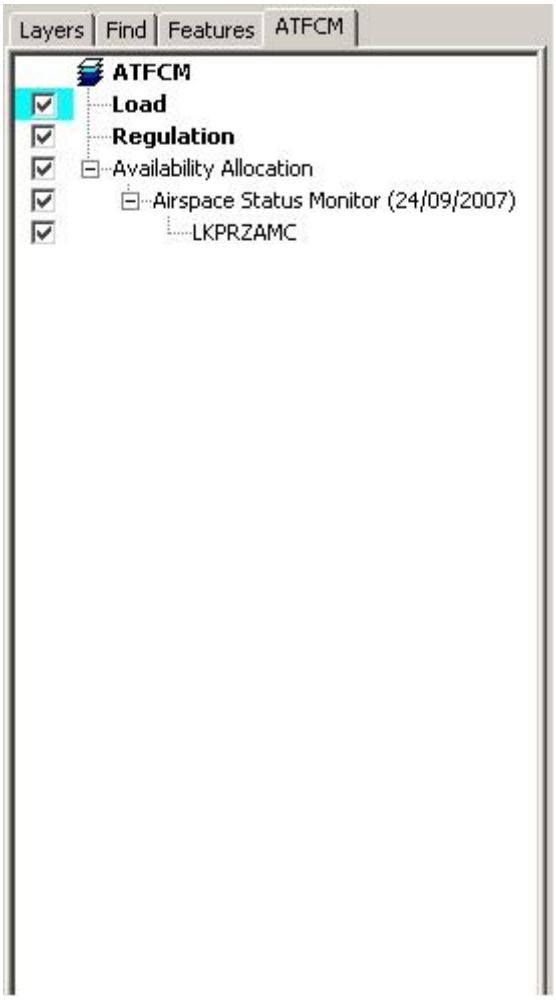


Figure 11-30: CIAM Airspace Status Monitor in ATFCM tab

(8) There is no **Load** nor **Delay** tree node inserted because the ATFCM Monitor does not display any load states or regulation delay states.

Abbreviations

Abbreviations and acronyms used in this document are available in the EUROCONTROL Air Navigation Inter-site Acronym List (AIRIAL) which may be found online:

http://www.eurocontrol.int/airial/definitionListInit.do?skipLogon=true&glossaryUid=AIRIAL.

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